

The Swords of Armageddon

U.S. Nuclear Weapons Development since 1945

VOLUME V

Chuck Hansen, ed.

Release 01

October 1995

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VOLUME V CONTENTS

PART III: THERMONUCLEAR WEAPONS DEVELOPMENT, 1942 - Present

3. BUILDING A BETTER BOMB, 1955 - 1959

- Weapons for the Stockpile, V-3
- Clean and Dirty Bombs, V-10
- The Shepley-Blair Book, V-28
- Improving Yield-to-Weight Ratios, V-53
- Planning for Operation TEAPOT, V-55
- Concerns About the UCRL Program, V-60
- TEAPOT Plans, August 1954, V-63
- The Killian Committee Report, February 1955, V-80
- Operation REDWING, V-85
 - REDWING and "Clean" Weapons, V-87
 - U.S. -- U.K. Cooperation: To Be or Not to Be?, V-90
 - Objectives of REDWING, V-93
 - REDWING Planning and Organization, V-99
 - Spring 1955: A Possible Test Moratorium Looms, V-105
 - Operation TEAPOT: Prelude to REDWING, V-110
 - Planning for REDWING Continues, V-115
 - REDWING and "Clean" Weapon Tests, V-120
 - Revising the Test Schedule, V-135
 - REDWING Test Operations, V-180
 - Preparations for REDWING, V-185
 - Results of REDWING, V-219
- Operation HARDTACK (Phase I), V-238
 - Objectives and Planning, V-238
 - Plans for "Clean" Weapon and High-Altitude Tests, V-245
 - Planning for Operation PILGRIM, V-254
 - Plans for HARDTACK Underwater Tests, V-266
 - Early 1957: Pressure Builds for a Test Moratorium, V-272
 - The HARDTACK Shot Schedule: LASL vs UCRL, V-287
 - Consideration of an Alternate Test Site, V-292
 - More Controversy over "Clean" Weapons, V-305
 - UCRL Plans for "Clean" Tactical Weapons, V-333
 - Plans for the "Open" Shot, V-356

INDEX, V-359

3. BUILDING A BETTER BOMB, 1955 - 1959

Weapons for the Stockpile

Even before CASTLE was conducted, the Department of Defense was looking ahead to more-deliverable and more-versatile thermonuclear warheads beyond those that might become available after CASTLE as a result of the short-term "emergency capability" program.

In mid-November 1953, DOD had provided to the AEC informal guidance as to three general types of thermonuclear weapons it desired:

1. A weapon weighing not more than 8,500 lbs. and having a yield of one to three megatons is needed for delivery by a variety of aircraft and missiles.
2. A weapon weighing not more than 20,000 lbs. and having a yield of around five megatons is needed for delivery by the B-47.
3. A weapon weighing not more than 50,000 lbs. and having a yield of the order of 10 megatons is needed for delivery from the B-36 and B-52.

In February 1954, the MLC added to this list a request for a high-yield thermonuclear weapon weighing not more than 3,000 lbs.¹

At that time, the JCS "shopping list" was as follows:

The Joint Chiefs of Staff consider that thermonuclear (TN) weapons, which will be available for use in the event of an emergency in 1954, will substantially enhance the military posture of this nation.

In order to further our TN capability, it is appropriate to state some of the operational needs visualized for these

¹"Comments Concerning the H-Bomb Development," Department of Defense, Military Liaison Committee to the Atomic Energy Commission, 21 April 1954; Memorandum for the Chairman dated November 17, 1953 from Col. Vincent G. Huston, USAF, Acting Director of Military Application, USAEC, Subject: Development of Thermonuclear Weapons.

weapons which will permit development effort to be applied immediately. This action will provide operational weapons through 1957 in addition to or as replacements for those now providing an emergency capability.

In the implementation of war plans, very high-yield weapons will reduce delivery costs appreciably under many circumstances. The delivery systems available through 1957 have been studied, and general characteristics of a family of very high-yield weapons in consonance therewith are stated below:

a. A thermonuclear weapon with the maximum practical yield but not less than 20 megatons. This weapon may weigh as much as 50,000 lbs. but a lighter weight is desirable.

Such weapon should be compatible with the B-36 and B-52 aircraft. Exact bomb characteristics will be determined when weight, yield, and cost relationship become available.

b. A TN weapon with a yield of at least five megatons and a maximum weight of 23,000 lbs. If these design factors can be improved, priority should be given first to reducing the weight to 15,000 lbs. while holding the yield at five megatons, and then to holding the weight at 15,000 lbs. and increasing the yield to 10 megatons or more.

This weapon must be compatible with the B-47 aircraft.

c. A thermonuclear weapon with a yield of two to 10 megaton weighing not more than 8,500 lbs. This weapon should be compatible with the B-47, B-66, A3D, and AJ aircraft and applicable missiles.

d. A weapon with a yield from one-quarter to three megatons weighing not more than 3,000 lbs. This weapon must be adaptable for internal or external (carriage) and for release or

separation at subsonic and supersonic speeds.

Such weapon will be compatible with bomber and fighter-bomber type aircraft and applicable missiles.

With respect to items (c) and (d) above, the Joint Chiefs of Staff believe that [deleted] Further the weights listed for the weapons in paragraphs (c) and (d) above are not intended to limit the efforts of the Atomic Energy Commission (AEC) to reduce such weight when practicable.

It is not believed that the results of the tests of CASTLE will invalidate any of these requirements. Instead, CASTLE will provide data refining the characteristics of these weapons and will provide guidance for judicious selection of weapons compatible with future weapon systems developments.

The Joint Chiefs of Staff consider that the desired distribution of weapons by 1 July 1957 should be in consonance with the tentative guidance recently furnished (to) the AEC. In this regard a 3-to-1 ratio was established for large weapons to small weapons.

Within the large weapon category, a 5-to-1 ratio should prevail for weapons of the 23,000 lb. class to the 50,000 lb. class, respectively.

Within the small weapon category, it is desired that the ratio of 3,000 to 8,500 lb. weapons as of 1 July 1957 be approximately 3-to-1, but not to the exclusion of achieving the total number of small thermonuclear weapons stated in our memorandum for the Secretary of Defense, dated 15 December 1953, subject: "Desired Composition of the Atomic Stockpile," as approved by the Secretary of Defense on 19 December 1953. This ratio of 3-to-1 for weapons within the small category is expected to increase significantly in subsequent years.

In reply to a request from the AEC concerning the weaponization of the CASTLE

test version of the TN weapon [deleted], the Joint Chiefs of Staff believe that in light of their stated requirements for TN weapons, this course of action should be followed. However, it is pointed out that it is not the desire of the Joint Chiefs of Staff to exclude the development of other weapons in the same general weight and yield category and such development should be continued by the AEC.

It is requested that you forward these requirements to the AEC, informing them that the Joint Chiefs of Staff desire:

- a. The development of weapons in subparagraphs 3a, 3b, 3c, and 3d.
- b. That they be kept informed of the estimated complete design release date and stockpile entry data.²

To satisfy the 8,500 lb. weapon requirement, plans were underway to weaponize the ZOMBIE device and have the first weapon available early in 1955. The weaponized ZOMBIE would weigh 7,000 to 8,000 lbs. and give a yield in excess of a megaton.

Another possibility related the requirement for an 8,500 lb. warhead was to design a system employing the principles of the ALARM CLOCK; one variation being investigated would weigh around 4,000 lbs. A weapon of this type might be available by the end of 1955, depending on how quickly the necessary engineering and testing could be done. The yield of the device could be increased by raising its weight.

With regard to the 20,000 lb. weapon requirement, the SHRIMP would weigh around 25,000 lbs. as presently configured and would therefore not meet the stated specification. The extent to which the weight could be reduced depended upon results of the SHRIMP test in CASTLE, and detailed design could not be started until CASTLE was over.

A reduced-scale SHRIMP having a weight between 15,000 and 20,000 lbs. could probably be available by the end of 1955. The SHRIMP had demonstrated that thermonuclear fuels only partially enriched in lithium-6 were useful, and that light-

²Memorandum for the Chairman, Military Liaison Committee to the Atomic Energy Commission dated 5 February 1954 from Brig. Gen. Edwin H. J. Carns, USA, Secretary, The Joint Chiefs of Staff, subject: Development and Production Guidance Concerning Thermonuclear Weapons.

weight aluminum cases were practicable, leading to weapons in a 20,000 lb. class.³

If the SHRIMP were to be weaponized, that work could start immediately, proceeding concurrently with the ZOMBIE, and both weapons could probably be made available by the middle of 1955. If DOD desired that this be done, then the AEC wanted to be notified as soon as possible.

Yet another possibility that might meet this requirement would be to scale-down the JUGHEAD; still another was a further scaling-down of the ALARM CLOCK to a design that would weigh around 13,000 lbs. The latter possibility rested largely upon the desirability of bomb weights considerably less than 20,000 lbs.

As far as the 50,000 lb. weapon requirement was concerned, an improved version of one of the first "emergency capability" designs would probably suffice. The best line of approach could not be selected until CASTLE was concluded. The major question here was whether to reduce the amount of critical materials while maintaining the yield, or to increase yield at the expense of critical materials.

The AEC realized that the entire subject of thermonuclear weapons, including the "emergency capability" program, would have to be re-examined after CASTLE. However, some weapon design would have to be accomplished before CASTLE to guarantee rapid progress.⁴

In February 1954, the JCS had stated that thermonuclear weapons were mandatory if the U.S. were to maintain its military superiority over the USSR, and that the production of these weapons was the cheapest method to obtain high yield weapons and improved destructive capability. The JCS also believed that production of large numbers of thermonuclear weapons would be cheaper than increasing fission weapon and delivery vehicle manufacture.⁵

To meet this requirement, the AEC would have had to triple

³Atomic Energy Commission, Status Report for June 30, 1954, Part III - Weapons, July 22, 1954, pp. 1, 3.

⁴Memorandum for the Chairman dated November 17, 1953 from Col. Vincent G. Huston, USAF, Acting Director of Military Application, USAEC, Subject: Development of Thermonuclear Weapons.

⁵Letter dated 5 February 1954 to The President, The White House, from Lewis L. Strauss, Chairman, USAEC, and Charles E. Wilson, Secretary of Defense.

its lithium-6 production quota and devote at least two Hanford and four Savannah River reactors to tritium production.⁶

Before CASTLE ended, and after the first two shots of the SHRIMP and the RUNT gave yields far above prediction, LASL believed that an ALARM CLOCK-type device using lithium enriched to 95% Li-6 could yield 30 MT, and that a smaller SHRIMP could be built also using 95% enriched lithium.⁷

By mid-May 1954, Dr. Edward Teller of UCRL stated his hopes for both lightweight and very low-yield thermonuclear weapons:

So many new facts have become known in connection with the Eniwetok test that we feel that we need some time to think things over and proceed in the best of new directions. As I mentioned to you in Washington, we are looking with particular interest at small, easily-deliverable megaton (yield) weapons which we hope we can make in a weight class of 3,000 pounds and less.

We are looking with equal interest at the small-yield bombs which are to produce explosions of one or two kilotons or even less. You know that I have felt for a long time that such weapons can be produced in great numbers and might be decisive in peripheral wars.⁸

Late in May, Gen. K. E. Fields described at an AEC meeting "a family of thermonuclear weapons being produced or developed

⁶Letter dated February 5, 1954 to Robert LeBaron, Chairman, MLC, from Lewis L. Strauss, Chairman, USAEC.

⁷Minutes of the 39th Meeting of the General Advisory Committee to the USAEC, March 31, April 1 and 2, 1954, p. 23.

⁸Letter dated May 13, 1954 to Lewis Strauss, Chairman, USAEC, from Edward Teller. In this letter, which was written after Teller testified against Oppenheimer during the latter's security clearance/loyalty hearing, and before the AEC handed down its final decision to deny clearance to Oppenheimer, Teller states: "I feel very certain that Oppenheimer's continued interference was exceedingly harmful. I hope I can speak about this interference in the past tense." Teller may or may not have known at this time (from Strauss) what the predetermined outcome of the hearing was to be.

ranging in yield from 250 KT to more than 20 MT." Fields also commented upon the implications of thermonuclear weapons for air defense and intercontinental missiles.⁹

After CASTLE concluded in May, four classes of thermonuclear weapons, weighing 43,000, 17,500, 8,000, and 3,000 lbs. were under consideration by the AEC.

The 3,000 lb. weapon would have to be tested; UCRL director Dr. Herbert York believed that the smaller the weapon, the greater the number of tests required. York said that while the 3,000 lb. weapon was a goal, a 1,500 lb. thermonuclear warhead was also a possibility.

At a GAC meeting late in May 1954, in response to a question from Dr. John von Neumann about the likelihood of a small 10 KT weapon, York said that Livermore believed that such a weapon could be made using uranium hydride, i.e., uranium deuteride. Some hydride chemical studies were in progress at UCRL. York discussed the characteristics of three small-diameter, low-yield warheads; UCRL proposed to test one or two next spring in Nevada during Operation TEAPOT, the next scheduled continental U.S. nuclear test series.¹⁰

LASL was also looking at small weapons with yields in excess of one kiloton, mainly in spherical implosion geometries for air defense warheads. The lab had no recent experience with linear implosion; however, Livermore was conducting non-nuclear linear implosion experiments at Inyokern, California, and at "S" site at LASL.¹¹

Also at this time, even in spite of the planned construction of a third ADP, the AEC would not be able to fulfill military requirements for Li-6 production. The AEC did not expect to be able to use Li-7 in its weapons in large quantities until 1957; even then, there would still probably be a requirement for lithium enriched to 40% in the Li-6 isotope in addition to normal Li-7. Use of any amount of 40% enriched lithium would result in a substantial increase in yield.¹²

⁹Minutes of 990th AEC Meeting, May 24, 1954.

¹⁰Minutes of the Fortieth Meeting of the General Advisory Committee to the USAEC, May 27-29, 1954, pp. 19, 20.

¹¹Minutes of the 40th Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, May 27-29, 1954, pp. 23, 24.

¹²Minutes of the 40th Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, May 27-29, 1954, p. 6.

Clean and Dirty Bombs

Early in July 1954, the AEC was still considering "doping" thermonuclear weapons with materials that would generate highly-radioactive contaminating fallout after an explosion.

In September 1947, Dr. Edward Teller had noted that "by absorbing the neutrons (produced by fusion) in appropriate materials and generating (radio)activities of the right (half life), one might obtain from the Super many times the radioactive effect produced by an Alarm Clock."¹³

This particular virtue was also raised in December 1949 during the deliberations of the National Security Council that led to President Truman's decision to proceed with the "superbomb" program.

One contaminating material considered, at least by the popular press, was cobalt. As early as the fall of 1953, beginning in the London Express newspaper, a number of articles had examined the possible consequences of spreading radioactive cobalt isotopes via fallout from a hydrogen bomb. In April 1954, the AEC concluded that if a one-megaton bomb produced 2.5×10^{27} neutrons, and if every neutron were to be captured by a cobalt-59 nucleus, then about 550 lbs., or 250 kilograms, of cobalt-59 (with a volume of about a cubic foot) would be required.

Much more likely, due to the low cross-section of cobalt for neutron capture, and the likely dispersal of the cobalt by the explosion, then less than 0.1% of the neutrons released would be absorbed by the cobalt.

At best, a one-megaton bomb might produce about 270,000 curies of cobalt-60. Assuming that 25% of this material fell to the ground over a 5,000 square mile area, then the initial dose rate from the cobalt-60 would be about 0.15 milliroentgens per hour. This low dosage would have insignificant health effects compared to the radiation dose rate from the other fission products of a nuclear weapon for many months after its detonation.¹⁴

¹³"On the Development of Thermonuclear Bombs," Edward Teller, LA-643, Los Alamos Scientific Laboratory, February 16, 1950, p. 25.

¹⁴Memorandum dated April 12, 1954 to John C. Bugher, Director, Division of Biology and Medicine, USAEC, from Gordon W. Dunning, Health Physicist, Biophysics Branch, Division of Biology and Medicine, USAEC, subject: Summary of Unclassified Calculations on Production of Cobalt-60 in Fission Devices;

Another possible contaminant was tantalum. A February 1954 report concluded that a "rigged" fission weapon would contaminate a very small area to a dose-rate level of military significance. A cobalt- or tantalum-rigged thermonuclear weapon detonated at the surface would contaminate to a significant level a large area that would not otherwise be affected by blast, prompt nuclear, or thermal radiation.

Such a widespread high level of contamination would deny re-entry, reconstruction, and re-occupation or inhabitation of this area for an "exceedingly" long period. Appropriate targets for this kind of land denial included vital communication centers, railroad marshaling yards, harbors, etc. which had to be re-occupied and reconstructed even after destruction by atomic or thermonuclear weapons.

The report used all up-to-date information available on the characteristics of the SAUSAGE device, fission and thermonuclear weapons neutron flux and energy spectrum, capture cross-sections of tantalum and cobalt, and fallout patterns of surface-detonated weapons.¹⁵ The results of the disastrous CASTLE Bravo shot served to emphasize the potential military utility of high levels of extremely radioactive fallout generated by "dirty" or "rigged" bombs.

In a letter to the AEC in the spring of 1954, Dr. J. Carson Mark, Theoretical Division leader at LASL, stated that there were no particular advantages to be gained from using either cobalt or tantalum, versus the contamination provided by normal fission byproducts, and suggested instead the use of isotopes with moderately-long half lives:

This is a reply to inquiry as to the amount of additional radioactivity which might be obtained by placing cobalt or other materials in a thermonuclear weapon. ...

It would seem useful first to indicate the nature and magnitude of the effects which could be obtained under favorable conditions.

In the devices we have tested, we assume that

memorandum for Dr. H. S. Craig dated October 21, 1953 from N. C. Debevoise, subject: Cobalt Bomb.

¹⁵Memorandum dated 10 February 1954 to Commanding General, Chemical Corps Research and Engineering Command, Army Chemical Center, Maryland, from Col. Fred J. Delmore, subject: Transmittal of Report on "Potentialities of Rigged Atomic Weapons."

the interaction between the fuel and a considerable mass of tuballoy [deleted]

It is assumed that the neutron absorber is in some fashion incorporated into a bomb of present design without having an important effect on overall weight, size, or performance. [deleted]

[deleted] The effects of such additional radioactivity can be considered from a couple of different points of view. On the one hand, it increases the total dose across time at any point. Assuming one activated atom per fission, the total doses after five hours would be approximately doubled (provided the isotope added had a half-life on the order of 12 hours or longer), and the doses after five or six days would be approximately quadrupled (provided the half-life were about two weeks or longer).

One can also inquire whether, by selecting an isotope of a particular lifetime, it would be possible to maintain some desired level of contamination at some selected spot over usefully longer times than the fission products themselves would provide. The time at which a particular activity is most prominent in comparison with fission product decay is about a mean life (of the radioisotope) after (it is created).

Both much before and much after that time, the contamination level is very little affected by the presence of the isotope. A set of curves is attached showing the ratio of activity between an isotope of indicated mean life and the fission products....

A curve is also attached showing the enhancement of radioactivity produced by one activated atom of tantalum per fission. At a time around 150 days (after activation), the (radio)activity is at a level about eight times what it would otherwise be, and the period through which this activity holds above the level otherwise provided by fission products is extended by a factor of about five for all times between 20 and 100 days (after its creation). Before 10 days and after 1,000, the tantalum makes no difference.

The above considerations raise the question of whether cobalt offers anything of interest. The effects from one activated cobalt atom per fission can be pictured from the attached curves. (These effects) would not be felt until after 150 days and would continue till about 20,000 days (Author's note: approximately 54.75 years). Even three activated cobalt atoms, which could only be obtained by firing the bomb in a very large cobalt or cobalt-bearing vessel, would not be felt until after about 50 days.

Both for the purpose of maintaining some high level of contamination over a stated period and also for the purpose of increasing the total radiation dose, the cobalt lifetime would seem to be much too long. For this latter use, incidentally, it may be noted that only half of the specific contribution of the cobalt will have been delivered at the end of five years.

Lifetimes of the order of one day or shorter, on the other hand, would appear to be too short to give any useful effect. This leaves the region of lifetimes from a few days to a few hundred days as the only ones of possible interest. On a casual inspection, tantalum was the only element that suggested itself, so that it has been specifically considered here.

It might further be noted that the fact that only a mild enhancement of radiation levels appear possible with present thermonuclear bombs [deleted] The situation would be very different in the case of an explosion along the lines of the "classical super" where induced radioactivity might be pushed to somewhat higher levels, but more particularly would heavily dominate the radioactivity otherwise provided.

Mark also offered a few comments about how much material might be required to cause a specific level of enduring post-blast radiation:

We come now to the amount of material which might be required to achieve a stated effect. The first point is that the absorption cross-section at thermal (neutron) energies are irrelevant for this consideration in conjunction with weapons designed along

present lines [deleted]

There seems no obvious way of degrading these (fusion-generated 14 MeV) neutrons to thermal (neutron) energy in the close neighborhood of the explosion unless the explosion should be in the sea, for example. For the neutrons available, then, the absorption cross-sections are not likely to be very large and are certainly very poorly known.

The masses (of special contaminating materials) which could conceivably be accommodated without leading to essential perturbation in present (thermonuclear bomb) designs would be: [deleted]

[deleted]

The fraction of neutrons which would be absorbed in a layer of such materials is, of course, proportional to the product of the absorption cross-section and the moles of atoms of the material per square centimeter through which the neutrons must pass.

In order to absorb a third of the neutrons, for example, it would be necessary to have about a sixth of a mole of atoms per square centimeter of a material with an absorption cross-section of three barns. Though not really known, it seems possible that tantalum has a cross-section of the order of a few barns in the relevant energy region. The amount of tantalum required would consequently be of the order of 30 grams per square centimeter.

Mark closed his paper by noting that there were still many unknowns in this area, even for a material such as tantalum:

Before being able to make much more specific statements than above, it will be necessary to obtain more precise absorption cross-section data for various materials in the kilovolt (neutron energy) region. For example, the estimate given of the sort of effect available by the use of tantalum is based on the assumption that the normal tantalum isotope Ta-181 is converted to Ta-182.

The efficiency of conversion which might be obtained [deleted] is estimated to be in the

general order [deleted] on the basis of an absorption cross-section of a few barns. However, it is known that Ta-182 has an extremely large absorption cross-section at thermal (neutron) energies.

If it should also happen to have a relatively large absorption (cross-section) at a few kilovolts, a noticeable part of the Ta-182 would be converted to Ta-183, which has something like a six day half-life. It is thus possible that the tantalum contamination could differ from that indicated here, by being somewhat lower at later times but becoming appreciable somewhat sooner, while the total (radioactivity) of the tantalum effect would perhaps be a little smaller than shown.

Even granting uncertainties of the sort just mentioned, it would appear to us that the present picture (regarding contamination) is good enough to serve as a basis for decisions as to whether or not further detailed consideration is necessary. It also appears that the (contaminating) effects available are of modest, if not dubious, value; so that we propose not to pursue the subject unless there appears to be some need to strive for a particular effect within the range pictured above.¹⁶

In a letter dated July 1, 1954 to a member of the JCAE, AEC Chairman Lewis Strauss commented on the military utility of "dirty" bombs, including those that might incorporate cobalt:

Additional information comes from Operation CASTLE this Spring (sic), where high-yield shots were fired on a coral reef, on a coral island, and on barges in shallow water. The radioactivity produced (by the shots) came predominantly from the reactions of neutrons with uranium in the devices. The only extraneous materials added were tracers for diagnostic purposes, and these did not contribute significantly to the radioactivity produced.

....

¹⁶Letter dated April 20, 1954 to Brig. Gen. K. E. Fields, Director of Military Application, USAEC, from Carson Mark, T Division, LASL.

... An analysis of fallout measurements (from the first CASTLE shot) ... indicates that, if the radioactive debris had fallen on land instead of mostly into the open sea, an area on the order of 5,000 square miles would probably have been rendered hazardous to human life for a time. During the first few days, there would have been a high percentage of fatalities for personnel in the open, unshielded by buildings or even rough terrain.

....

The primary purposes of employing low-altitude or surface bursts of weapons in military operations would be to create a crater and to increase the area of blast damage. The effects of radiological contamination might be to hinder repair of the facilities or to increase the number of casualties. Where radiological contamination is not desired, for example, because of hazards to friendly personnel, the height of burst must be chosen in such a way that the ball of fire does not touch the ground.

The possible military usefulness of radiological contamination raises the question as to whether the effects can be intensified or prolonged by adding special materials to weapons. This has been suggested especially in connection with thermonuclear weapons because of the great production of neutrons, some of which could be absorbed in added materials to make them radioactive.

The results to be obtained in this way have to be compared with the already very high initial intensity of the radioactive products formed from uranium, which is presently a normal constituent of these weapons (Author's note: in the fusion tamper around the secondary).

Adding a material which would be converted to a radioactive product having a half life of several days might increase the area of personnel hazard or force personnel to stay in shelters for a longer time. This possibility is under consideration but has not yet been fully evaluated.

The much mentioned possibility of adding cobalt to weapons is of doubtful military importance. The radioactive cobalt-60 that would be formed has a half life of five years. Its intensity would be very low compared with the initial intensity of the radioactive products of uranium. It would be a matter of months before the intensity of the (radiation of the) uranium products dropped to the level of cobalt-60 and, except possibly near the detonation point, this level would not be of military significance.

If radiological contamination by high-yield weapons were used for military purposes, its distribution would be difficult to control for individual detonations. The mechanism of distribution is sensitive to wind directions and velocities up to the top of the mushroom-shaped cloud.

Accurate prediction of the location of the hazardous area with respect to the detonation point would require extensive wind data at altitudes up to about 100,000 feet, and such data are difficult to obtain, especially for points in enemy territory.

However, if a number of high-yield weapons were detonated within a region, the region could be generally contaminated without accurate knowledge of wind conditions for each detonation. The situation in regard to such indiscriminate dissemination of radioactive materials would be different for targets deep in enemy territory than for targets in enemy-occupied territory or near the front lines where there would be hazards to friendly personnel.

In summary, the detonation of high-yield weapons of present design at or near the surface will, by radiological contamination, extend the anti-personnel effects of such weapons over considerable areas that would otherwise not be affected.¹⁷

¹⁷Letter dated July 1, 1954 to Senator John W. Bricker, Chairman, Military Applications Subcommittee, JCAE, from Lewis L. Strauss, Chairman, USAEC.

Even without any special "doping," fallout from high-yield weapons could, depending on wind and weather, be far-reaching and deadly. Had Washington D.C. been ground zero for a bomb comparable to the CASTLE Bravo (SHRIMP) device, and had fallout been distributed to the northeast in a pattern identical to its distribution to the east of Bikini, then lethal radiation would have extended into New York city and high dosages well into New England almost to the Canadian border.¹⁸

A later post-CASTLE Air Force study stated that if a 15 MT bomb were detonated either at ground level or was buried 500 feet underground, an area ranging between 5,000 and 10,000 square miles would be dusted with fallout of such intensity that human habitation or traversal would be impossible from 12 to 48 hours. By such a means, atomic weapons could be used to deny a relatively large area to an enemy.¹⁹

By July, the two-stage TX-14, -16 and -17 bombs constituted the U.S. "emergency thermonuclear capability;" the TX-14 and -16 were to be retired (the TX-14s were out of stockpile by October; the TX-16s were removed even sooner). A program was underway to develop a retarding parachute for the TX-17 to shorten its time-of-fall from its present 120 seconds, and an automatic nuclear insertion system was also being designed. Contact fuzing, for surface bursts, was also being studied; however, it would not be available for at least two years (the MK 17 was originally limited to airburst fuzing only).²⁰

These weapons all used kilogram quantities of lithium deuteride, with varying degrees of enrichment in the lithium-6 isotope, and uranium enriched to both 37.5% and 93.5% in U-235.

Also following CASTLE, the DOD created specific thermonuclear weapon classes. Class "A" weapons, including the MKs 17 and 24, Mods 0 and 1, weighed 50,000 lbs. or less, with a minimum yield of 20 MT; Class "B" weapons, currently only TX-21 SHRIMP, weighed 23,000 lbs. (to be reduced to 15,000 lbs.) and yielded a minimum of five MT.

¹⁸Hewlett and Holl, pp. 181, 182.

¹⁹RADIATION HAZARDS DURING ATOMIC WARFARE, Lt. Col. N. M. Lulejian, Headquarters, Air Research & Development Command, Baltimore, Maryland, November 1954, p. 35.

²⁰Minutes of the Forty-First Meeting of the General Advisory Commission to the U.S. Atomic Energy Commission, July 12-15, 1954, p. 5.

Class "C" weapons, now limited to the TX-15, weighed 8,500 lbs. or less, with a minimum yield of two to 10 MT; and Class "D" weapons, which were currently only on the drawing board, weighed 3,000 to 4,000 lbs., with a minimum yield of one-third to three megatons.

The MK 17 Mod 0 and MK 24 Mod 0 were already in stockpile, with the Mod 1 versions of these bombs expected in December 1954. The TX-15 might be ready in the spring of 1955, with the MK 21 entering stockpile in August 1955. A Class "D" weapon was not expected before 1956.²¹

The TX-15 was being considered to fulfill the Class "C" thermonuclear requirement. Sandia had assumed responsibility for the detailed internal engineering of this weapon, thus becoming involved in nuclear design for the first time. The Sandia program was supervised by Los Alamos.

The first delivery of a MK 15 to stockpile was scheduled for April 30, 1955. The bomb was engineered for storage as a completely assembled unit, except for its tail fins. The MK 15 was equipped with barometric and proximity fuses; there was a strong desire on the part of the Air Force for additional contact fuzing. However, contact fuzing for two-stage weapons was difficult; how to do it was currently unknown.

There was not enough room in the bomb bays of available delivery aircraft to place a fuze probe on the front of the bomb, to allow a detonation signal to reach the high explosives before the rapidly-falling bomb was crushed by impact with the ground.

The 17,400 lb. TX-21, a scaled-down SHRIMP, was in its infancy and a lighter-weight version might eventually replace the TX-15 as a Class "C" bomb. The TX-21 appeared to be compatible with the externally-carried B-58 weapons pod.²²

By this time, the higher-than-predicted yields of most of the CASTLE shots were known to be attributable to nuclear reactions of lithium-7, which had earlier been assumed to be

²¹Minutes of the Forty-First Meeting of the General Advisory Committee to the USAEC, July 12-15, 1954, pp. 13, 14; Atomic Energy Commission, Status Report for June 30, 1954, Part III - Weapons, July 22, 1954, p. 5.

²²Minutes of the Forty-First Meeting of the General Advisory Committee to the USAEC, July 12-15, 1954, pp. 5, 6, 10. At this meeting, the GAC saw a special classified exhibit at Sandia in Albuquerque of mock-ups of various warheads and missile mountings, including a full TX-15 assembly.

a much less useful fuel than either lithium-6 or deuterium. This meant that unenriched, naturally-occurring lithium would be an important component of future thermonuclear weapons.²³

At the other end of the U.S. nuclear weapons spectrum, small tactical weapons were also under development, in 30, 22, and 15 inch diameters. Exact specifications of military requirements for these devices were still somewhat uncertain. There seemed to be little military interest in the 30-inch warhead, which could be made with existing implosion techniques. If DOD were to become interested in this warhead, a considerable savings of fissionable materials for the desired yield could be achieved.

The greatest DOD interest in the 15" device was for use as an air-to-air rocket warhead, and in the 22" device for use in the TALOS-W surface-to-air missile and as a potential tactical bomb.²⁴

Near the beginning of September 1954, Operations TEAPOT and DIXIE were planned for Nevada in the spring and fall of 1955, respectively, and Operation REDWING was to be the next thermonuclear test series in the Pacific in the spring of 1956. On September 2, the MLC "urgently request(ed)" that the AEC consider advancing the one-point detonation test from DIXIE to TEAPOT in order to determine the inherent nuclear safety of the new generation of "sealed pit" primaries and fission weapons, some of which were slated for test during TEAPOT.²⁵

Operation JULEP, the next Nevada series, was still slated for the fall of 1956. TEAPOT was to be conducted between February 1 and April 30, 1955, and would include two air drops, a subsurface shot, five 300-foot tower shots, and four 500-foot tower shots.²⁶

²³Minutes of the Forty-First Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, July 12-15, 1954, p. 15.

²⁴Minutes of the Forty-First Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, July 12-15, 1954, pp. 18, 19, 20, 21.

²⁵Memorandum for the Chairman, USAEC, dated September 2, 1954 from Herbert B. Loper, Chairman, MLC, subject: Department of Defense Support for Operation TEAPOT.

²⁶Unaddressed note dated September 8, 1954 by Lewis L. Strauss, Chairman, USAEC; Operation TEAPOT, Report of the Test Manager, Joint Test Organization, Nevada Test Site, Spring 1955, pp. 7, 8.

By this time, LASL's GMX Division was still supporting high explosive implosion system tests for UCRL. The latter lab lacked both facilities and broad competence in this field, and LASL was manufacturing, developing, and test firing HE systems for the new Livermore facility. UCRL was building a test firing site at the Nevada Test Site.

Nonetheless, UCRL expected LASL to continue supplying all HE components for these experiments indefinitely.²⁷ Although the second weapons lab had been in operation for almost two years by now, it still depended heavily on Los Alamos for many things. This dependency, unknown to the public and to most of the government outside the AEC, would soon be ironically overshadowed by a slanderous book that almost destroyed LASL's reputation and which deliberately overstated UCRL's thermonuclear accomplishments and contributions to the nation's weapons stockpile.

In mid-September, DOD was evaluating its possible requirements for both high-yield "salted" and "clean" thermonuclear weapons:

Consideration of the effects of nuclear weapons, particularly those with high yields, indicates that radioactive fall-out and contamination are among their most important characteristics, and indeed may be one of the factors governing their use in some situations.

It also appears that it may be possible to control to a considerable extent, by different weapon designs, the relative amounts of long-lived radioactive products per megaton of total energy yield. It is understood that the AEC is gathering information as to special materials which could be added or substituted to enhance or diminish the radioactive contamination.

For your information, the Department of Defense is considering the broad aspects of the contamination problem as well as the more detailed operational factors which would influence the degree of contamination.

²⁷Letter dated September 22, 1954 to Brig. Gen. K. E. Fields, Division of Military Application, USAEC, from Dr. Herbert F. York, Director, UCRL-Livermore.

These studies, when combined with results of your investigation, should enable the Department of Defense to decide whether or not there is need for high yield weapons with different contaminating characteristics than those for which military characteristics are now specified.

Cooperation of the AEC is requested in the investigation of this problem. The immediate need is for a better understanding of all the pertinent facts.²⁸

LASL was still having problems working with UCRL. The two nuclear weapons labs now competed for Sandia's resources, and for both nuclear and non-nuclear testing time and facilities at Nevada and in the Pacific. The forthcoming Operation TEAPOT in Nevada, an objective of which was the development of primaries and other components for the next generation of thermonuclear weapons, brought matters to a head when it appeared that UCRL might require HE fabrication and testing support from the U.S. Navy's Naval Ordnance Laboratory (NOL) in Corona, California.

On September 22, LASL director Norris Bradbury, in a letter to Brig. Gen. Fields, seized upon this possibility as an opportunity to suggest that UCRL be relegated to a wholly subordinate role to LASL, and that the California lab not produce any complete nuclear weapons designs. This was not an idea that would make either Herbert York or Edward Teller happy, and this subordination did not occur; UCRL eventually established its own remote explosives test facility, Site 300, near Livermore.

Bradbury noted that under appropriate circumstances, LASL could take over and finish the development of a small and relatively-inefficient linearly-imploded tactical nuclear weapon system for artillery shells and small missile warheads now under study at UCRL. The older New Mexico lab was distinctly uncomfortable with the fact that it now had to fight for future contracts:

A number of recent queries and discussions indicate the resurgence of an old administrative problem which may arise to confront the AEC. This problem has its origin in the fact that two weapon research

²⁸Letter dated September 17, 1954 to Lewis L. Strauss, Chairman, USAEC, from Donald A. Quarles, Assistant Secretary of Defense for Research and Development, DOD.

and development laboratories exist; that these laboratories work in the same fields and towards the same objectives; and that only one ordnance engineering laboratory, Sandia, exists.

The first area in which this problem may appear is typified by the UCRL linear implosion system. This system is not new -- it certainly was discussed at LASL at least as early as 1944 -- and represents an obvious interpolation of [deleted] It is presumably of interest in tactical atomic weapon applications, but its high cost in plutonium makes it of very dubious early applicability in any area where, if there is to be any real military significance, large numbers are required.

UCRL has raised the question as to how their development of this device can be expedited. They currently have a contract with the Stanford Research Institute for certain HE firings; the LASL is firing a certain amount of experimental HE shots for them to assist in their preparations for TEAPOT; they are planning one or two shots for TEAPOT; and they have asked to introduce portions of this program into the NOL. In considering the latter point, the Division of Military Application indicated some reluctance "if the task could be conducted more quickly and efficiently at the LASL."

Undoubtedly, if this device had an appropriate priority, and if there were any reasonable prospect of its entering the atomic stockpile in large numbers in the near future, the relevant development work could be done at the LASL. The task involved is straightforward, although not particularly exciting.

Devices of this nature have seemed to us to be more appropriate to the 1960-65 epoch when, it might be guessed, the stockpile philosophy and related active material availability would suggest a number of systems which were tactically attractive even if high in unit cost.

We would see (at this time) no problem in introducing such a system into normal LASL

development in time for substantial numbers to appear by 1960 and thereafter. However, it is our belief that there are many more important devices to occupy our attention for the next few years.

Specifically, in this connection, we see no harm in the NOL doing whatever UCRL wishes within the present facilities of NOL and the finances available to UCRL or the Division of Military Application. Although the HE fabrication and firing facilities at NOL are rather limited, we suppose that the (AEC) would be and should be reluctant to use this project as a basis for giving support, financial and otherwise, to the duplication at NOL of the kind of HE firing and fabrication capabilities which exist at LASL.

Should specific weapon designs be required for a system of this nature in the next three to four years, we would recommend that the LASL (and probably Sandia) have nothing to do with it and that other arrangements be made.

On the other hand, if the present activities at Livermore are directed only towards (determining) the fundamental capabilities of systems of this nature, and actual weapon development of the most appropriate tactical systems is not required until substantial active material can be allocated, then it is probable that the LASL can undertake the nuclear and HE development for whatever system or systems are of most interest, and, working with Sandia, produce an actual weapon of families of weapons.

Such a development probably should not be suggested for the LASL until 1957 or 1958, and the UCRL system is only one of a number of types which will have to be considered when active material for such military purposes becomes relatively plentiful.

Bradbury also claimed that competing new Class "D" thermonuclear weapon projects at UCRL and LASL were redundant, costly in terms of testing time, and not likely to produce design breakthroughs in the near future. Bradbury suggested that even if a Livermore design were successful, it

should be "weaponized" under LASL auspices:

The second area in which difficulties can possibly arise in the field of small, two-stage weapons. Since, in the larger systems at least, the application of [deleted] has been proven to be unnecessary, UCRL has naturally turned to the Class D type. As was inevitable, no responsible group of people is going to continue attempting to exploit systems which do not offer any obvious advantage.

Consequently, as could have been (and was) predicted, the Livermore designs for small two-stage systems are similar in size, weight, cost, and yield to Los Alamos systems. They differ in detail in internal geometry, but it is possible that Nature is reasonably tolerant of various geometries in these systems, although in the end, one or another may be slightly superior with respect to a given size, weight, or cost.

Again, there is certainly no objection (other than the length of the test series and the fallout complications) of the introduction of the UCRL (bomb) case test into TEAPOT. All too little is yet known about the two-stage assembly process and it would be difficult to predict the behavior of the UCRL case device from observation of the LASL test and vice versa, so both tests are of technical interest.

However, it should be clearly recognized that what the AEC now has in its LASL-Livermore weapons research complex is almost precisely identical with the "competition" set up for aircraft manufacturers in the design of a new plane. All the designs submitted are basically the same because aerodynamics is a science and not an art, and no manufacturer will propose to produce an airplane which will fly twice as far, twice as fast, for half the weight his competitor will propose. (Author's note: in the example of the W-47 between 1956 and 1958, the latter is exactly what Livermore tried to do: make a "quantum leap" in thermonuclear weapons design and get out from behind LASL's shadow once and for all.)

Out of the TEAPOT series will have to come a specific design for one, and perhaps two, Class D weapons. If there are two, it will be because the practical limitations imposed for Class D weapons for missile warheads are different from those imposed on Class D weapons (when used as) free-fall bombs.

It may be best to optimize in each area (and at the price of two systems) rather than impose the necessary limitations of one delivery system on the other. (Author's note: this is not what finally happened: both LASL's W/MK 28 Class D system and UCRL's W/MK 27 Class D weapon were eventually used as both bombs and missile warheads.)

However, whether a Los Alamos system or a Livermore system is used will possibly turn out to result in a trivial nuclear difference. The Livermore case test device [deleted] differs from LASL systems in the radiation engineering. At best, it will be little better than LASL systems; if it is worse, Livermore will certainly come even closer to Los Alamos designs in future tests.

The specific point that (needs to be made) here is that the two laboratories are not working towards widely-different practical objectives; they are working towards the same overall objective and although the approaches differ in a number of significant details, it would seem unlikely that the resultant weapons will be widely different from a practical viewpoint.

The Class D bomb may be slightly better in consequence; it will not come any sooner, and it will not open up any new vistas of warfare not originally foreseen in the two-stage weapon area.

After TEAPOT, however, actual ordnance engineering of Class D systems will have to occur. At that time, the LASL will probably recommend that the best features of the LASL and Livermore systems be combined, or that the best system be selected for Class D applications, and that the ordnance development proceed between Los Alamos and

Sandia in the normal fashion.

We will probably not recommend that two slightly different nuclear systems be engineered at Sandia to satisfy both the LASL and UCRL, and we presume that the distance between UCRL and Sandia will make detailed collaboration there difficult. (Author's note: in 1956, Sandia opened a new office at Livermore to support UCRL systems weaponization in the same fashion that Sandia in Albuquerque supported LASL's weapons programs.)

Bradbury now further suggested that UCRL's entire weapons development role be examined, and that perhaps the new California lab's work might be ancillary to the role of LASL. The LASL director also indirectly rebuked some of his critics who had claimed that the older New Mexico research lab was too conservative in its approaches to weapons design:

We would like to take this occasion to suggest that the objectives of the whole Livermore weapon program be carefully explored after TEAPOT. The original concept for the Livermore Laboratory was that it should explore ideas or systems not receiving attention at Los Alamos. It was also believed in some quarters that brilliant new ideas would flow from the establishment of competition.

Both the [deleted] and [deleted] are old ideas which had not yet seemed sufficiently urgent or promising for the LASL to exploit in (the face of more urgent) weapon demands. The two-stage area is now one in which Livermore and Los Alamos differ primarily in details of radiation engineering. The brilliant new ideas have not appeared. (Author's note: this latter statement turned out to be rather premature.)

It appears that a perfectly possible and appropriate, permanent role for the Livermore Laboratory is to supply Los Alamos and the AEC data and information on the capabilities of various nuclear systems in which they are interested. Such data and information may certainly involve such things as full-scale nuclear tests.

However, having obtained such information, its comparison and integration, as appropriate, into other systems, and the final design of any resultant weapon, should be the responsibility of Los Alamos, working ultimately with the Sandia Corporation to obtain an actual device.

Just as (nuclear) cross-section information is obtained in many laboratories and supplied to Los Alamos, and just as the (Project) Matterhorn group supplied calculational data and ideas to us, so could Livermore supply even more extensive informational material relevant to nuclear explosions without themselves becoming involved in the problems of actual ordnance engineering of specific weapons. Unless and until some really new ideas appear, present AEC facilities utilized in this manner appear to us to be adequate.²⁹

Bradbury noted in closing that his letter had "not been coordinated with the Livermore Laboratory" and that "copies of this letter have not been sent to Livermore." This was understandable: if Teller or York had gotten wind of Bradbury's suggestions, they would have made vehement reply.

The Shepley-Blair Book

In the fall of 1954 came yet another blow to LASL. A new book, The Hydrogen Bomb: The Men, The Menace, The Mechanism, written by Clay Blair, Jr. and James Shepley, two Washington, D.C. area staff writers for Henry Luce's TIME-LIFE-FORTUNE organization, charged that LASL and the GAC, specifically, its chairman, J. Robert Oppenheimer, had deliberately delayed work on the hydrogen bomb between 1946 and 1950. Lewis Strauss and Edward Teller were painted as the heroes of this struggle, and the claim was made again that the Soviets, by their explosion of August 1953, had "beaten" the U.S. in the race for a deliverable hydrogen bomb.

The book, like the May 1953 FORTUNE and the April 1954 TIME articles, was rife with errors, speculation, innuendoes, and carefully-marshaled evidence to support preconceived ideas. The book also served to drive the final nails into Oppenheimer's public reputation, going far beyond the findings of the security review that had revoked Oppenheimer's "Q"

²⁹Letter dated September 22, 1954 to Brig. General K. E. Fields, Director of Military Application, USAEC, from N. E. Bradbury, Director, LASL. Emphasis in original.

security clearance at the end of June 1954.

The book also charged that "essentially nothing had, in fact, been done about thermonuclear weapons development between June 1946 and January 31, 1951 (sic);" that Norris Bradbury and LASL were Oppenheimer's puppets and that both were intractably opposed to hydrogen bomb development; that Los Alamos was "loaded with Communists and former Communists;" that the GAC had, in 1948, recommended against development of the so-called "fractional-crit" bombs; that LASL was more interested in developing fission weapons than fusion bombs; and that Teller had been "forced out" of LASL because of his advocacy of the Super.³⁰

A lengthy excerpt from the book appeared in the September 24, 1954 issue of U.S. NEWS AND WORLD REPORT magazine; the magazine was released to the press on September 21, about a week and a half before scheduled publication of the book.

Teeming with errors and calumny, the article was filled with deliberate or unintentional misstatements of fact which were at wide variance with reality. A few of them are listed below; "SB" refers to the article:

³⁰THE HYDROGEN BOMB: THE MEN, THE MENACE, THE MECHANISM, James Shepley and Clay Blair, Jr., David McKay Company, New York, 1954, pp. 94, 95, 96, 118-120, 140. All of these claims are either distortions or outright falsehoods. For example, in February 1948, the GAC explicitly recommended the development of smaller atomic bombs (see MK 5 history in Volume VI). Edward Teller left LASL when the laboratory chose not to appoint him director of the Family Committee and after he made an arbitrary demand that the IVY Mike shot schedule be advanced by three months. A great deal of theoretical and experimental work was done at LASL on thermonuclear weapons between June 1946 and January 31, 1950, the date of Truman's famous directive. In addition, Norris Bradbury and his LASL colleagues, although perhaps more plodding and methodical at times than Teller might have wished, were no opponents of thermonuclear weapons development. Despite Teller's impatience with the pace of hydrogen bomb development at Los Alamos, the establishment of the second weapons lab at Livermore in the fall of 1952 did not measurably accelerate deployment of U.S. thermonuclear weaponry: by the time UCRL had its first successful large-scale fusion test in the summer of 1956, LASL had already deployed five thermonuclear weapon designs (the MKs 14, 15, 17, 21, and 24); UCRL's first design, the W-27, did not go into production until September 1958, two years after the summer 1956 test and six years after the establishment of the second lab. By then, LASL had also fielded the MK 28.

SB

U.S. thermonuclear weapons development was stalled for four years "while German scientists captured by the Communists worked feverishly in Soviet laboratories"

J. Robert Oppenheimer "almost single-handedly stopped the (U.S.) H-bomb program in its tracks"

The Soviets had a deliverable hydrogen bomb six months before the U.S.

The Soviet atomic device detonated in September 1949 "yielded six times" the force of the U.S. bomb dropped on Hiroshima

During WW II, Edward Teller was "denied a specific job in connection with the atomic bomb"

By 1946, "Teller was at least within sighting distance of a successful H-bomb" and that the 1946 Super conference at LASL "described in detail how a hydrogen bomb could be assembled within two years time"

Reality

The Russian thermonuclear bomb was an all-Soviet effort and did not use the services of any captured Germans (although Germans had assisted the Soviet atomic bomb project)

Oppenheimer was not the only or strongest opponent of U.S. H-bomb development

The August 1953 Soviet test was of a submegaton, single-stage advanced high-yield fission bomb similar to a September 1947 idea by Teller; the Soviet device was not even as powerful as the stockpiled U.S. MK 18 "Super Oralloy Bomb," a 500-kiloton weapon

The first Soviet nuclear detonation yielded at most about 20 kilotons, the yield of the first U.S. test in New Mexico

Teller worked on implosion calculations for the second wartime bomb design (the FAT MAN) and his contributions to the Manhattan project were limited only by his preoccupation with the Super

The 1946 Super design was unworkable and while the purpose of the 1946 Super conference had not been to specify actual test device designs, two concepts were proposed: one was simple and relatively amenable to

analysis but did not immediately appear to be feasible; a second concept, of unknown feasibility, was considered too complex for reliable quantitative analysis

At the time of the first Soviet atomic test in September 1949, LASL "was then engaged exclusively in fission bomb research"

Paper studies and theoretical research on the Super had been underway for more than four years by September 1949

That LASL was "hostile" to the H-bomb

LASL's thermonuclear study programs followed the policy directives of the U.S. Atomic Energy Commission

The number of theoretical physicists at LASL declined during 1950

The number of theoretical physicists working on the Super grew from 20 in the summer of 1949 to 38 by January 1951.

Stan Ulam had mentioned only "in passing" something that "turned on a small light somewhere in Teller's storehouse of ideas" that led to a successful U.S. thermonuclear weapon design

The application of the "three concepts" was far more complex than portrayed by Shepley and Blair

By June 1952, after work had started at Eniwetok to prepare for Operation IVY, "progress on the thermonuclear bomb still lagged"

Work on the H-bomb was going ahead at LASL and UCRL about as fast as time and money allowed

Los Alamos was "energized" by the creation of the Livermore laboratory and only then "threw its full energies and very great talents into the thermonuclear program"

Work on the H-bomb proceeded as planned at LASL after UCRL was established in September 1952 and LASL-designed bombs became the first stockpiled U.S. thermonuclear weapons

The November 1, 1952 Mike test yielded only three megatons

IVY Mike shot yield was 10.4 megatons

The August 1953 Soviet thermonuclear test yielded a megaton

The Soviet test yielded around 470 kilotons, less than the 540 kiloton yield of the U.S. MK 18 "Super Oralloy Bomb"

By 1949, "while Washington debated whether or not to launch a thermonuclear program as a result of (the first Soviet atomic test), the Soviets were plugging away on their own thermonuclear program and had amassed a dangerously long lead (over the U.S.)"

The Soviet H-bomb program had begun in earnest in 1948, only a year earlier, and the U.S. quest for a true multistage H-bomb based on the "three concepts" was always ahead of the Soviet program: the Soviets did not test a device embodying the "three concepts" until November 1955, three years after IVY and a year and a half after Operation CASTLE, when the U.S. successfully tested deliverable thermonuclear weapons

A test of a 45 megaton hydrogen bomb had been scheduled as the second CASTLE test and had later been removed from the CASTLE schedule and "postponed for a later test in a greater expanse of the Pacific"

No such test was ever contemplated during or after CASTLE

General Curtis LeMay assumed command of the U.S. Air Force's Strategic Air Command after the first Soviet atomic test in September 1949

LeMay took over SAC on October 19, 1948, nearly a year before the Soviet test

Due to the delaying efforts of unnamed "nuclear physicists", no new USAF fighter prototypes were flown between June 1947 and June 1950

F-86 first flight: October 1, 1947; F-89 first flight: August 16, 1948; F-94 first flight: July 1, 1949; first flights of XP-85, XF-87, XF-88, XF-90, XF-91, XF-92, XF-93, XF-96A, and XF-97A all between February 1948 and June

Like the book from which it was extracted, the article was probably the most deliberately dishonest and misguided history of the U.S. hydrogen bomb ever written, and was quite obviously a vicious attack on both Oppenheimer and LASL. It is possible that had the AEC not lifted Oppenheimer's security clearance earlier in 1954, in effect excommunicating him, the book was intended to destroy his reputation.

The AEC had reviewed galleys of the book for security breaches and had exchanged correspondence with its authors, citing "changes requested by the AEC in the interest of security." At least one AEC letter warned of the commission's misgivings "regarding the general detrimental effect to the (U.S.) atomic energy project it is felt will be caused by the book's discussion of some of the controversial items." The same letter also called the authors' attention to information which the AEC "believes may be held classified by the Department of Defense and/or the Central Intelligence Agency."³²

³¹"Inside Story of the Hydrogen Bomb: How the U.S. Almost Lost It," James R. Shepley and Clay Blair, Jr., U.S. NEWS AND WORLD REPORT, September 25, 1954, pp. 58, 60, 68, 72, 94, 101, 102, 103, 104. LeMay SAC command date is from THE DEVELOPMENT OF STRATEGIC AIR COMMAND, 1946-1973, John T. Bohn, Command Historian, Office of the Historian, Headquarters, Strategic Air Command, Offutt AFB, Nebraska, 19 September 1974, p. 9; fighter prototype flight dates are from ENCYCLOPEDIA OF U.S. AIR FORCE AIRCRAFT AND MISSILE SYSTEMS, VOLUME 1: POST-WORLD WAR II FIGHTERS, 1945-1973, Marcelle Size Knaack, Office of Air Force History, Washington, D. C., 1978, pp. 53, 84, 101, 313, 315, 317, 319, 321, 323, 325, 327. Clay Blair, Jr. was described on the book's endpaper as a Pentagon correspondent for TIME magazine; as such, he seemed to be remarkably uninformed about contemporary aerospace developments. Most of the aircraft prototype flights cited were noted in various issues of AVIATION WEEK magazine in the late 1940s. For other discrepancies between Shepley-Blair and the truth, see Bethe, "Comments on the History of the H-Bomb." In their defense regarding the interaction between LASL and UCRL, Shepley and Blair did concede that "in the end, the more mature staff and the more experienced atomic scientists at Los Alamos significantly outpaced the new laboratory at Livermore in providing the hardware that became the hydrogen bomb."

³²Letter dated August 13, 1954 to James R. Shepley, TIME, Inc., from Shelby Thompson, Acting Director, Division of Information Services, USAEC. Even as recently as the spring of 1979, an AEC successor organization, the Department of Energy, was

Reaction to the book was swift and damning. Former AEC Chairman Gordon Dean, under whose stewardship many of the alleged misdeeds occurred, stated in a review published in the November 1954 issue of the BULLETIN OF THE ATOMIC SCIENTISTS that the Shepley-Blair book was "vicious," "untrue," and a "horrifying combination of little knowledge, outright untruths, and questionable motives." He cited as an example of the author's misunderstanding the original Super design (largely the handiwork of Teller) which Dean characterized as "a bomb, which as then conceived, should never have been built -- and never was built."³³

By coincidence, LASL had been rewarded in July 1954 with a Presidential Citation for its work on fission and fusion weapons. This citation was issued in part to allay unrest among the staff of the laboratory following the Oppenheimer hearings; the citation was cynically referred to as "Operation Butterup" by the Santa Fe New Mexican, a newspaper serving the lab community.³⁴

urging one writer to send to them for review his thoughts on hydrogen bomb designs.

³³Review by Gordon Dean of Shepley-Blair book in BULLETIN OF THE ATOMIC SCIENTISTS, November 1954, pp. 357, 362. On September 10, 1954, Dean wrote to Lewis Strauss and urged that "in view of your key position from which to refute these (false references to Dean's participation in the U.S. thermonuclear weapons program), I feel very strongly that you have a duty either to see that the book is not published or that the false statements concerning me are corrected." Dean warned of a "virtual revolution" at LASL if the book were published, and that the only possible result of its publication would be a further serious split between the scientific and non-scientific people in the atomic energy program. In a letter dated September 23, 1954 from Dean to former Secretary of State Dean Acheson, Dean characterized the book as "a vicious book; a great libel on the people associated with the Atomic Energy (sic) program in those days; and it does a cruel injustice to Norris Bradbury and his Los Alamos staff." Lewis Strauss wrote to the editors of LIFE magazine on September 24, 1954 and stated that Gordon Dean, while an AEC commissioner in 1949 when the Super was debated, had not opposed the H-bomb but was instead a "vigorous proponent" of it. Shepley and Blair had stated in their book that Dean was an opponent of the H-bomb.

³⁴Hewlett and Holl, p. 111; citation from the White House dated 8 July 1954. This citation had been recommended by the GAC in mid-June 1954. (Letter dated June 3, 1954 to Lewis L. Strauss, Chairman, USAEC, from I. I. Rabi, Chairman, GAC.)

In response to the book, Bradbury, Teller, and other weapons scientists, issued several statements. Teller noted that the Shepley-Blair book was an "irrelevant discussion." Enrico Fermi noted that not only Teller, but the entire staff of LASL had been responsible for bringing the hydrogen bomb to fruition.

New Mexico Senator Clinton P. Anderson, who was also a member of the JCAE, quickly came to the defense of LASL. In a stinging attack, Anderson decried current AEC secrecy, questioned the scientific savvy of Dr. Edward Teller, noted Ulam's contribution to the "three concepts," and highlighted UCRL's poor performance of nuclear tests to date:

Regardless of how much value there may be in the book, THE HYDROGEN BOMB, by James R. Shepley and Clay Blair, Jr., which is condensed in the current issue of U.S. News and World Report, the fact remains that it libels Los Alamos Scientific Laboratory and misleads the American people.

Los Alamos is in the State (sic) I represent but it is the strong defensive arm of the whole country and this nation cannot afford to see it slandered. I do not intend to remain silent when confronted with an attack upon the American scientific community in general and upon the Los Alamos Laboratory in particular. This book can only serve to divide us when the urgent need of the hour is for unity.

As a member of the Joint Committee on Atomic Energy, it has been my privilege to be fully informed on the highly classified facts of our weapons development program. Possessed of this knowledge, I can state that the Shepley-Blair book does violence to the truth. It misstates facts; it distorts facts; and it omits facts.

While the book may say that but for the activities of certain individuals there would be no hydrogen bomb, I will say that but for the (Los Alamos) scientific laboratory and its staff, the United States would not have today this weapon.

Like the story of every other great scientific development, the history of the hydrogen bomb program is a complex story. For comparison, one need only look at the difficulties and delays encountered in

bringing new type aircraft from the germ of an idea into production. Some day I hope that the requirements of security will permit the full hydrogen bomb program to be told, with the benefit of official information.

When this happens, I fervently hope it will be told with dignity, accuracy, and balance. I do not regard this as a proper subject for the kind of gossip treatment it has received in this book.

The Los Alamos Scientific Laboratory needs no defenders. Its magnificent accomplishments speak for themselves -- in the incredible development of atomic and hydrogen weapons, and in the ever-increasing effectiveness and versatility of our nuclear stockpile. No writer can obscure the successes achieved by the dedicated men and women who work at Los Alamos under their superbly qualified director, Dr. Norris Bradbury.

In this connection, I want to associate myself with the sentiments President Eisenhower and Admiral Strauss expressed in the Presidential citation of August 2, 1954: "The (Los Alamos Scientific) Laboratory's momentous success in the field of fission weapons has been followed by equal accomplishments in the fusion field. These achievements are the result of a remarkable group endeavor and the devout and skillful individuals of the staff of the laboratory." The (Shepley-Blair) book tried to make a mockery of that well-deserved citation.

Despite allegations in the book to the contrary, I may say this most unequivocally: except for the Los Alamos Scientific Laboratory, our nation would not possess hydrogen weapons. Every single nuclear bomb that has gone off has been designed at Los Alamos. I wish that the Atomic Energy Commission would remove from its classified character the report of what has happened to the bombs designed by Dr. Teller and the Livermore Laboratory.

The authors of this book on the hydrogen bomb leave the impression that Teller is

the great genius and that everything he does is magic. They might have asked the Atomic Energy Commission how his thermonuclear bombs have thus far exploded.

It is not my purpose to take part in a squabble between (Dr. J. Robert) Oppenheimer and Teller but there is no reason for putting into a book the suggestion that many "United States physicists were paralyzed by the guilt of Los Alamos." The glory of Los Alamos is that it has produced a deliverable hydrogen bomb and has done it by a remarkable display of genius.

I resent particularly the suggestion that after orders had been given to develop the hydrogen bomb, "the Los Alamos scientists seemed willing to work on any other project except the hydrogen bomb." There follows in the book a reference to "fractional-crit" bombs.

Dr. Bradbury and his great team of scientists should get credit for realizing that they had to develop a triggering device to such a high degree of potency that it could make the hydrogen bomb workable. The effort they put into "fractional-crit" bombs not only increased the value of our atomic bomb program but was a significant step toward the success of the H-bomb.

I regret that the book carries this statement, "The AEC scientists at Los Alamos continued to show little interest in (Teller's) new concept and saw no need for taking any special action or reorganizing the thermonuclear program." Actually, I think the record will show that two men came up with a good idea and Teller was one of them. I wish the press could examine the memos at Los Alamos and see if another scientist was not working in the same path that Teller finally followed.

Significantly, given a good idea, the Livermore Laboratory under Dr. Teller and the Los Alamos Laboratory under Dr. Bradbury tried to develop workable deliverable hydrogen bombs. After this slander on Los Alamos it would seem only fair that the

Atomic Energy Commission release the batting averages. Let's see who makes the hits and runs and let's see who produced the duds.

The book, as condensed on page 104 of (U.S. News & World Report), tells of the March 1 shot and claims that it exploded with the incredible force of 15 megatons. Whose design was it? A few paragraphs ahead, the article said that Los Alamos made the greater contribution. Why could it not have added when it mentioned the explosion with incredible force that the bomb came from Los Alamos and was quite different from the designs of Dr. Teller?

There is a heading reading, "Less Than Candid." That characterizes the book. There is reference to new calculations which "pushed the giant bomb near the practicable limit of thermonuclear weapons." I believe those calculations referred to the weapons developed by Los Alamos. New Mexico would have appreciated enough candor on the parts of the authors to have had them assert that, if it's true.

The book is designed to build up Teller, but in doing so, Teller tries to move up on the skulls of Dr. Bradbury and his able crew at Los Alamos. It seems to me that General (Leslie R.) Groves, as he retired, said something about letting the second team take over. Well, the second team did take over at Los Alamos and the United States has achieved its greatest measure of defense because of their remarkable accomplishments. We should be singing their praises and let Edward Teller wait until he can demonstrate his ability to devise and construct a hydrogen bomb.³⁵

³⁵Statement of Senator Clinton P. Anderson, Member of the Joint Committee on Atomic Energy with reference to THE HYDROGEN BOMB by James R. Shepley and Clay Blair, Jr. The first six and a half paragraphs of Anderson's statement was drafted by Francis Cotter, a JCAE staff aide; the original draft was much stronger in its condemnation of the book and called it "a discredit to the honorable profession of journalism;" an "irresponsible volume." (Teletype dated September 23, 1954 to Honorable Clinton P. Anderson from Francis P. Cotter.) The Shepley-Blair book was one of the first titles in a long

On September 21, the same day that excerpts from the book appeared in U.S. NEWS & WORLD REPORT, Brigadier General Kenneth E. Fields, AEC's Director of Military Application, prepared a draft of a proposed open letter to Norris Bradbury which Fields suggested that AEC Chairman Strauss send to LASL and release to the press:

Dear Dr. Bradbury:

This is in reference to the book THE HYDROGEN BOMB by James R. Shepley and Clay Blair, Jr., to be published September 30, 1954.

As you know so well, the public reporting of classified work is a difficult and well-nigh impossible undertaking, particularly for a subject so complex as that of the development of the hydrogen bomb. This fact is unfortunately apparent in the above book in that we see implications drawn with respect to you personally and to the Los Alamos Scientific Laboratory with which we can not agree.

Lest there be any doubt in this respect, the Commission wants to assure you and the Laboratory that this view is properly reflected in the recent Presidential citation to the Laboratory, which the Commission recommended to the President and which he saw fit to award.³⁶

The following day, Strauss wrote to Bradbury and claimed that he (Strauss) "had no influence to prevent (the) appearance (of the Shepley-Blair book)." Strauss also stated that the Presidential Citation to LASL reflected the AEC's opinion of the weapons lab. There is evidence that Strauss had seen and reviewed at least one advance draft of the book several months before it was finally published, and that he may have been a strong force behind its publication and the attack on Oppenheimer in the book, in effect acting as an uncredited

series of panegyric hagiographies about Dr. Teller, including later books by Messrs. Blumberg and Owens, Blumberg and Panos, and Wood and Mark.

³⁶Attachment to Memorandum for the General Manager, USAEC, dated September 21, 1954 from K. E. Fields, Brigadier General, USA, Director of Military Application, USAEC. Some parts of this statement were eventually released by Strauss.

third author.³⁷

On September 24, Bradbury, with AEC approval, convened a remarkable press conference in his office on "The Hill." LASL was still a "closed" city at this time, and this was only the second time that Bradbury had met with the press since early 1946. The lab director defended at length, and in as much detail as he could without violating security, LASL's performance of the thermonuclear weapons program since the end of World War II.

That this press conference was even held at all was quite remarkable. The AEC was loath to wash its dirty laundry in public, especially when classified information might accidentally be released. However, in the later words of Dr. York, "Bradbury just exploded; (the press conference) was abnormal. That was the first time anybody ever really said anything about what the laboratories were doing, and that's cause Norris just wouldn't tolerate it anymore."³⁸

Bradbury was still handicapped by security restraints, a problem not shared by Shepley and Blair. To fully refute all the falsehoods and errors in the book would have required massive disclosure of highly-classified information.

Nonetheless, Bradbury made an impassioned speech on behalf of LASL and his colleagues, taking pains to point out that it was LASL, and not the new Livermore lab, that "developed every successful thermonuclear weapon that exists today" and that work on thermonuclear weapons had never ceased between 1946 and the present, citing as examples several portions of LASL's technical programs for 1950 and 1951.

Bradbury also noted that had fission weapon research stopped in 1946 in favor of only fusion research, "the essential fission techniques required for practical thermonuclear weapons would not have been developed."³⁹

³⁷Letter dated July 10, 1954 to Lewis Strauss from Pearl Carroll, TIME Incorporated; this letter includes an attachment in Strauss' handwriting that lists suggested changes to and questions about the manuscript.

³⁸Interview of Dr. Herbert York by Chuck Hansen in La Jolla, California, September 29, 1993.

³⁹Statement made by Norris E. Bradbury at LASL on September 24, 1954. This statement was circulated widely by the AEC and was reprinted in the BULLETIN OF THE ATOMIC SCIENTISTS, November 1954, pp. 358 and 359, and in part in LOS ALAMOS SCIENCE, Vol. 4 No. 7, Winter/Spring 1983, pp. 27, 28.

Perhaps most importantly, Bradbury refuted at length the Shepley-Blair claim that thermonuclear research had lagged at LASL between 1945 and 1952, and that Edward Teller and the new Livermore lab deserved all the credit. Bradbury emphasized that "thermonuclear (research) never stopped." The Shepley-Blair book implied that no work had been done in the U.S. on the hydrogen bomb between the end of World War II and President Truman's order at the end of January 1950.

The Los Alamos lab and the AEC felt that while the usual reaction to uncleared comments on the U.S. nuclear weapons was "no comment," the book required an exceptional and lengthy response:

The reason for the conference this morning is, as I think all of you are aware, quite specific. Ordinarily the Laboratory does not talk about its program, does not talk about its accomplishments, does not try to take credit for these accomplishments. This is in accordance with the requests of the Atomic Energy Commission, and with the policy and with the philosophy of the Laboratory.

However, there has recently appeared a rather extraordinary document by a pair of relatively unknown individuals, in which such extraordinary and fantastic comments, imputations, speculations about Los Alamos and its activities, its accomplishments, and its people are made, that I felt that we could not rely upon our traditional "no comment" and that it was important for you, as representatives of the press, and for the Laboratory that the facts in this particular matter be set forth and set forth correctly insofar as they can be within the requirements of the Atomic Energy Commission personnel and news personnel.⁴⁰

The LASL director also noted that security requirements placed stringent limitations on what he and other lab employees could say publicly, and that these restrictions were not necessarily shared by other persons with political axes to grind:

⁴⁰Transcript of Press Conference by Norris Bradbury, et. al. at Los Alamos, New Mexico, September 24, 1954. This transcript includes a long prepared statement by Bradbury followed by several pages of questions and answers.

... On the one hand, the loyal citizen with access to classified information is in an impossible situation. He cannot talk about those things which might be exciting but which are classified. The only people therefore who can talk, basically, are people who do not know the facts, who imagine the facts, who speculate and surmise, who depend upon leaks, rumors and information that may or may not be malevolently inspired.

One of the topics that arose during questions at the press conference was the current state of relations between two U.S. nuclear weapons labs. The Shepley-Blair book implied that the creation of UCRL goaded LASL into harder work on the hydrogen bomb. Like many other Shepley-Blair statements, this was not true: at the time of the decision by the AEC to open a second lab, the SAUSAGE had been designed; the Livermore lab opened in September 1952, just a few weeks before the IVY Mike shot.

Of current relations between the labs, Bradbury said:

Livermore laboratory does exist. We are in close contact with them, we are friendly and amiable in our relations with them. We have given them an enormous amount of assistance in the form of consultations, in the form of documents, in the form of experiments. We are actually carrying on certain experimental things for them right now. We have given them every possible assistance in their program. If they are happy to regard themselves as a stimulant or goad to Los Alamos, I am happy to have them.⁴¹

When asked directly if he considered LASL as a competitor to UCRL, Bradbury replied that he didn't think LASL needed competitors in order to carry out its job. Bradbury gave no hint that just two days earlier he had suggested to the AEC that UCRL be placed in an entirely subordinate position to LASL, and be essentially reduced to the status of a consulting subcontractor. When Bradbury said LASL didn't need competition, he meant it in the worst possible way, and he was already doing everything he could to eliminate any competition.

⁴¹Transcript of Press Conference with Norris Bradbury at Los Alamos, New Mexico, September 24, 1954, pp. 8, 9.

Two days later, James Shepley appeared on the NBC television program "Comment" and defended his book while repeating his libelous and highly political charge that LASL had ignored thermonuclear weapons work for four years:

In an accident that was almost miraculous, the U.S. learned just a little over five years ago this month that the Soviets had mastered the secret of the atomic bomb. Thus began the second round of what is now referred to as the hidden struggle over the H-bomb. With my associate Clay Blair, Jr., I wrote a book about this struggle which has attracted some notice this past week.

The hidden struggle over the H-bomb was a remarkable chain of events. Almost as remarkable was the hidden struggle over the H-bomb book. I am going to tell this story now because I think it has an important bearing on the vital subject of how the American people shall be informed on their atomic security.

In my 15 years experience as a Washington and overseas newsman, I have never seen such elaborate attempts, for a mixture of motives, to kill a book or suppress an important news story. When the book was submitted to the Atomic Energy Commission for security clearance, I expected some immediate reaction.

The AEC staff in Washington has been largely untouched by the new (Republican) Administration (sic), which is as it should be. The AEC should be above politics. So when the book was put in for clearance, it was in effect being delivered over to many of the people who would fall under its implied criticism. I expected a reaction, but I was in no way prepared for what actually happened.

A week passed and I received a call from Mr. Lewis Strauss, Chairman of the AEC, whose role in the great H-bomb struggle is treated favorably by the book. He desired that I come to his office immediately. When I arrived, he was seated behind an enormous pile of paper work and at least three United States senators were trying to talk to him on the telephone. Amidst these

interruptions, we had a short, even-voiced conversation.

Some of his associates at the AEC were badgering him almost to distraction to stop the book. "Why stir up the animals?" they demanded. The weapons laboratory at Los Alamos was now working smoothly and the weapons program was progressing satisfactorily. Why should anyone be criticized for past mistakes? Strauss then turned suddenly from his desk and said in a soft voice: "Jim, I want to buy your manuscript, put it in the safe for 25 years, or at least until the animals are no longer alive and kicking, and then publish it as an historical document."

I replied that I assumed Mr. Strauss was kidding. Strauss' motives were, of course, the highest. Certainly he had nothing to conceal. In fact, without him, the Soviets would have had the H-bomb and the U.S. would not.

I could only surmise, therefore, that he was under great pressure and that he wish to shield his scientists from public controversy, even though he must have known that a savage attack on himself, in connection with the H-bomb struggle, was even then in preparation.

I added that Mr. Strauss had already conceded that the book had important information for the American people about their atomic security. Things happened too quickly in the atomic era to wait 25 years for a report on how this most vital of all of the functions of government had been, and was being, handled.

Not long afterward, I learned -- through Mr. Stewart Alsop, among others -- that Mr. Strauss' predecessor, Mr. Gordon Dean, had written (Strauss) an angry letter, blaming the book on Strauss and demanding that he suppress it. Dean began his letter by stating that he had not read the book, and then in several hundred words described its contents, including a statement that it was an attack on him, Dean, which it is not. Needless to say, Strauss in no way inspired the book, which he tried to lock away in

his safe.

This attempt by Dean to suppress the book was, of course, unsuccessful. He had better results in another attempt. A television program which deals with books had scheduled a discussion of "The Hydrogen Bomb." The program director was friendly with Mr. Dean and invited him to appear as a critic of the book. A short while later, we received notice that (our) engagement had been canceled.

Next to be heard from was Dr. Norris Bradbury, director of the Weapons Laboratory (sic) at Los Alamos during the years (when) the (hydrogen) weapon was not built, as well as during the years when it was. He wrote a letter to one of the vice-presidents of Time Incorporated, whom he knew. Essentially, Bradbury argued that because I was not Q-cleared for atomic secrets, I could not know about atomic matters and should not report on them.

In his argument, Bradbury, at one stroke, excluded all newsmen of all media from coverage of atomic matters. By definition, none of them could ever be Q-cleared, since the "clearance" implies a solemn promise to keep atomic secrets secret. But obviously, it is not necessary to know atomic secrets in order to report accurately on how well our atomic affairs are being administered.

The AEC made only minor security points about our manuscript and in the conversation that accompanies the clearance report, had even fewer and less important points on the question of accuracy.

Dr. Bradbury's main line of argument was that if the "secret" facts were all laid out it would be shown that he and Los Alamos made great contributions to the hydrogen bomb program. This is true, and we have said so explicitly in the book. I quote: "Events in time were to prove that initially the greatest use of Livermore would be to energize Los Alamos, as Teller had long predicted.

Once it was actually faced with a competitive situation, the Los Alamos

laboratory threw its full energies and very great talents into the thermonuclear program. In the end, the more mature staff and the more experienced atomic scientists at Los Alamos significantly outpaced the new laboratory at Livermore in providing the hardware that became the hydrogen bomb." Unquote.

The point Dr. Bradbury overlooks is that he, like all of us, are (sic) still bound to the mast of history. For four years, he and his staff made literally no effort to produce thermonuclear weapons, and came within an ace of losing atomic superiority to the Soviet Union. This, only journalists will report. It is unlikely that Q-cleared scientists will report it about themselves.⁴²

⁴²Enclosure to letter dated September 24, 1954 to Lewis L. Strauss, Chairman, USAEC, from James R. Shepley, TIME Incorporated. The cover letter, addressed to "Dear Lewis" and signed with "my warmest regards," suggests that Shepley and Strauss were quite friendly with each other. How much of Strauss' account is true is questionable: Shepley's impassioned defense of his work does not excuse the many errors and apparently deliberate misstatements of fact in it. There remains little doubt that the book was first and foremost an attack on Oppenheimer and LASL, and only incidentally a history of U.S. thermonuclear weaponry. It is likely that Shepley cared little about the latter and much more about the former. Shepley did have a good point about the AEC trying to hold all U.S. nuclear weapons as proprietary information; the editor of this information service would have a similar experience 25 years later with an AEC successor organization. As a fellow writer, this author can sympathize with Shepley's feelings of persecution at the hands of the AEC, but not with his slipshod presentation of the facts and his blatantly political agenda against Dr. J. Robert Oppenheimer. The "savage attack" on Strauss refers to an article by Stewart and Joseph Alsop entitled "We Accuse" which appeared in the October 1954 issue of HARPER'S MAGAZINE and which was a scathing attack on Strauss and the AEC for their treatment of J. Robert Oppenheimer. In an October 12, 1954 letter to Darol Froman, Strauss characterized the Alsop article as "filled with distortions, insinuations, and lies," the same terms used by many other persons to characterize the Shepley-Blair book. Shepley also seems to have misconstrued the AEC's silence on most of the book as evidence of its accuracy; this was probably not the case: it was more likely that the AEC did

On September 28, a group of ten LASL staff members, including Darol Froman, Carson Mark, and William Ogle, wrote to AEC Chairman Strauss and urged him to issue an "uncompromising repudiation" of the Shepley-Blair book, making it "unequivocally clear that it would be folly for the American public to give the (Shepley-Blair book) any credence whatsoever as an account of what actually happened."

The scientists' letter said the book gave an account of work at LASL since 1945 that was "distorted or untrue in all significant respects, as a detailed technical history of that work would show" and warned that public or congressional reaction to the book could damage many AEC programs if the book was not soon either repudiated or discredited.

The scientists stated that "we believe that the current weapons work Los Alamos is of far too great importance to permit anything to be left to chance in a matter which might lead to serious outside interference with that work."⁴³

The latter point might have been a veiled reference to a possible investigation of Los Alamos by the notorious right-wing Republican Senator Joe McCarthy of Wisconsin: in March 1954, McCarthy had charged that "Communist sympathizers" had delayed the U.S. hydrogen bomb project by 18 months.⁴⁴

Both Gordon Dean and Lewis Strauss had made pre-emptive efforts to blunt the effect of the book. Dean sent advance copies of his BULLETIN review to many newspapers and magazines;⁴⁵ Strauss had gone even further: while the book was highly flattering to both him and Teller, his offer to buy the manuscript from its authors later came back to haunt him when critics accused him of attempted book-burning and censorship.⁴⁶

not want to either confirm or deny most of the claims in the book. If there were pre-publication collusion between Strauss and Shepley-Blair, then the meeting in Strauss' office was merely a charade for media and public consumption.

⁴³Letter dated September 28, 1954 to Lewis Strauss from Darol Froman, et. al.

⁴⁴Hewlett and Holl, p. 88. The time span encompassed within the alleged 18 month lag was not clear; McCarthy was prone to making figures up on the spur of the moment.

⁴⁵Advance copy of Dean review dated September 28, 1954 addressed "To the Press."

⁴⁶Letter dated October 12, 1954 to Darol Froman from Lewis Strauss. In spite of his claims of innocence, Strauss had

Strauss had hoped to delay publication of the book "for a period of years during which tempers might cool." Strauss was very concerned at this time about reaction in the national labs to the AEC decision in the Oppenheimer case.⁴⁷

The AEC met on September 30 and decided not to make a public statement on the book, on the grounds that such a statement would only draw more attention to the book and promote sales and because such a public declaration would set a dangerous precedent, calling into question the AEC's silence on previous controversies, and making future comment on other controversial books mandatory.

In response to Froman's letter of September 28, Strauss wrote back two weeks later and stated that the AEC's Office of Public and Technical Information, which had received an advance copy of the Shepley-Blair book, had called his attention to "passages which were embarrassingly over-laudatory of me and other passages which would surely hurt feelings in the laboratories."

Strauss also explained that "there was restricted data in the manuscript which was being called to the attention of the authors in the standard manner."

The AEC chairman noted that he had invited one of the book's authors to his office and offered to buy the manuscript "in good faith" to "prevent its appearance." He explained that his job was already difficult and that the book would make it even more difficult. Strauss closed by pointing out that further AEC comment on the whole matter would only encourage controversy and increase sales.⁴⁸

On October 6, JCAE Chairman Sterling Cole issued a press release which also condemned the Shepley-Blair book.⁴⁹

reviewed at least one early draft of the book in the summer of 1954, a few months before its publication. (Letter dated July 10, 1954 to Lewis Strauss from Pearl Carroll, TIME Incorporated.)

⁴⁷Hewlett and Holl, p. 111.

⁴⁸Letter dated October 12, 1954 to Darol Froman from Lewis Strauss. Strauss' offer to buy the book, if true, was disingenuous at best and cunning at worst, considering that he had probably instigated its creation, and had reviewed the manuscript in advance of publication.

⁴⁹Letter dated October 14, 1954 to Rep. Sterling Cole from M. Stanley Livingston, Federation of American Scientists; letter dated October 25, 1954 to Sterling Cole from Gordon Dean.

There was (and is) little doubt that the book was a thinly-veiled hatchet job on Norris Bradbury, J. Robert Oppenheimer, Gordon Dean, and most of the AEC commissioners. Only Lewis Strauss and Edward Teller emerged as the heroes of the Shepley-Blair account of the development of the hydrogen bomb, and suspicion was rife that Strauss had worked closely with the TIME-LIFE-FORTUNE organization to foster publication, as he had in the spring of 1953 when the notorious FORTUNE article appeared that so strongly libeled Oppenheimer.

The suspicions about Teller's possible participation in the preparation of the book, following closely on the heels of Teller's damning testimony against Oppenheimer earlier in the summer, did not engender good feelings between Teller and many persons at LASL.

Despite their claims of highly-placed sources, Shepley and Blair had little access to accurate classified information about U.S. nuclear weapons development, and it would be another 30 to 40 years before this type of information was declassified in any quantity.

A good part of the blame for the misrepresentations in the Shepley-Blair book can be placed squarely on the AEC's failure to make public more of the history of the U.S. hydrogen weapons program, even in spite of the hysteria of the times and the paranoid U.S. fear of the Soviets. Had declassified details been available, Dean and Bradbury could have made much better refutations of the errors in the book, far beyond the very general denials and vague explanations to which they were forced to limit their criticisms.

To the extent that the book mischaracterized the U.S. nuclear weapons program, the AEC had no one to blame but itself for its own policies, especially in the early 1950s, of overclassifying almost everything, including basic scientific data that was either already in the public domain, or information which could be derived directly from an aggregation of a variety of published sources.

Bradbury, in spite of the damage the TIME-LIFE-FORTUNE organization was doing to his and LASL's reputations, did not call on the AEC to make more information public. He did, however, urge Strauss and the AEC to end their practice, in effect since 1947, of pre-publication reviews of articles and books about the U.S. atomic energy program.

This practice, ostensibly for reasons of "national security," did little except to disclose even more classified information than it protected. In any event, not every author or publisher was bound by either law or conscience to submit every article or manuscript to the AEC for review in advance

of publication. Unfortunately, this practice of pre-publication review was continued by the AEC and its successor organizations until a disastrous and widely-publicized civil lawsuit in 1979.

Norris Bradbury wrote to Strauss on October 1 and suggested that the AEC abandon this review policy:

Since my letter to you of September 27, 1954 I have seen the public statement by Mr. Shepley that his text was "reviewed" by the AEC and that certain portions, "even more damaging" to me, were removed at their request. Regardless of the inaccuracy of the latter statement, it does appear that the AEC suggested, or recommended, or requested, the deletion of certain portions of the text, presumably upon the grounds of classification.

The book as published is, of course, full of matters which, when true, are certainly classified. Most of their statements in the classifiable areas are untrue, but occasional ones such as [deleted] matter, and the [deleted] matter are close to the truth and classified.

....

Ever since the passage of the Atomic Energy Act, I have urged the Commission NOT (sic) to review textual matter by uncleared personnel not connected with the Commission. The Act and its definition of Restricted Data is there for all to read; the Commission is not a judicial body, nor should it be available for free legal service or advice.

Finally, when the Commission tells an uncleared person to omit something, they have told him (in my opinion) Restricted Data. The results of such unnecessary cooperation now stare us in the face, and the overall tacit approval which the authors may now ascribe to the Commission for the remainder of the text is extremely disturbing.

I trust that the Commission has an exact record of the content of every passage of the Shepley-Blair text whose deletion was recommended on the grounds of classification or otherwise so that the absolute necessity

for these deletions may be reviewed in light of the consequences which such a review seems to have brought about.

May I also urge, again, that the Commission consider most seriously a policy of henceforth undertaking no review of material submitted by uncleared personnel and of returning such manuscripts without comment.⁵⁰

Strauss replied three weeks later:

You have made a very important and, to me, a very telling point in your recommendation that we should not review textual matter by uncleared personnel not connected with the Commission. I have no idea how the practice began, but, as you must know, it has now attained the status of "standard operating procedure."

It makes us all kinds of trouble. For instance, one of the Alsops, with the assistance of Mr. Ralph Lapp, did a piece last autumn which I am informed contained an estimate of our weapons stockpile numbers and when the reviewing people in the Commission quite properly declined comment, a perfect hurricane broke over me.

I cannot guess how much grief there will be ahead for us in reversing our policy of seven years standing -- but of one thing I am sure, and that is that your recommendation, if adopted, will be ascribed to me and I will be called some more harsh names.⁵¹

⁵⁰Letter dated October 1, 1954 to AEC Chairman Lewis Strauss from Norris Bradbury, Director, LASL.

⁵¹Letter dated 21 October 1954 to Norris Bradbury, Director, LASL, from Lewis L. Strauss. Strauss' comments about "how much grief there will be ahead for us" was very prescient in light of the PROGRESSIVE case 25 years later. Another letter from Bradbury to Strauss dated April 1, 1955 refers to "steps taken by the Commission in connection with the problems of review of manuscripts of uncleared and non-project connected authors;" Bradbury agreed that "these procedures should materially improve the situation."

Despite Bradbury's suggestion, the AEC continued to review for many years manuscripts which were voluntarily submitted to it before publication.

Meanwhile, by the end of 1954, the U.S. thermonuclear weapons production program had been converted from an "emergency capability" phase to a "quantity production" phase. MK 17 and 24 bombs had been developed to a stage that permitted quantity production.

The MK 15, deliverable by B-47, B-66, and A3D aircraft, was designed and in early production. The MK 21, tailored to the B-47, was nearing design completion. In addition, small lightweight thermonuclear weapon design studies had reached a point that three tests to prove design principles applicable to 3,000 lb. megaton-yield warheads were scheduled for Operation TEAPOT. On a longer time scale, the study of intercontinental ballistic missile warheads had started.⁵²

Toward the end of the year, a classified assessment of relative U.S. and Russian nuclear capabilities was prepared that estimated the capabilities of the two countries to deliver a decisive and devastating attack upon one another. The study concluded that at the moment, neither country could launch such an attack, even if either of them were to strike first.

The major reason for this was that both countries' nuclear arsenals were composed mainly of kiloton-yield bombs; the U.S. was just beginning to build up a stockpile of megaton-yield weapons.

Within a couple of years, the U.S. could destroy the USSR, but only with a pre-emptive attack. The U.S. would have an adequate stockpile of high-yield bombs by then, and the USSR would not.

Starting around 1957-1958 and continuing for a presently undetermined number of years, the U.S. could wipe out the war-making capability of the USSR with and possibly even without a first strike. U.S. war planners also assumed that if Russia had discovered the "secret" of a modern megaton-yield weapon by that time, and unless the U.S. also developed by then distant early warning and control systems, and new interceptor aircraft and anti-aircraft artillery and missile defenses, then the U.S. could be destroyed by a surprise Soviet nuclear attack.

The study assumed that by the late 1950s, both countries

⁵²Atomic Energy Commission, Program Status Report, December 31, 1954, Part III - Weapons Program, pp. 1, 11, 12.

would have adequate stockpiles of high-yield weapons and could inflict devastating damage upon one another. In the years ahead, it was also very likely that both nations would develop intercontinental ballistic missiles with multimegaton warheads.⁵³

Improving Yield-to-Weight Ratios

An obvious "next step" in U.S. thermonuclear weapons development was a reduction in volume and weight for a given yield. The higher-than-predicted yields of almost all of the CASTLE tests were attributable to nuclear reactions in lithium-7, which had previously been assumed to be a much poorer fuel than lithium-6 or liquid deuterium. The higher yields could also be credited partially to better-than-expected uniformity of compression of the device secondaries.⁵⁴

Efforts directed toward reducing thermonuclear weapon size for a given yield began soon after CASTLE ended:

Laboratory efforts in atomic weapons research and development have produced a number of ideas which are highly important to the development of weapons for tactical and air-defense application and of lighter-weight thermonuclear weapons. Verification of some of these ideas within the limits of laboratory techniques will be completed in late 1954.

Further verification will require a series of tests at the Nevada Proving Ground in the Spring of 1955 to maintain progress in this field. Specific details of devices incorporating the new ideas are not now available although it is expected that a detailed test program will be available

⁵³"Time Chart of Relative Gross Capabilities to Deliver a Decisive Nuclear Attack," Dwight D. Eisenhower, Papers as President of the United States, 1953-1961. Ann Whitman File, ACW (Ann C. Whitman) Diary series, Box 3, Folder: ACW Diary - November 1954 (1), Dwight D. Eisenhower Library, Abilene, Kansas. The statement about the Soviets discovering the "secret" of the H-bomb in the future seems to suggest that U.S. policy makers knew by this time that the August 1953 Soviet nuclear explosion was not produced by a true multi-stage thermonuclear device.

⁵⁴Minutes of the Forty-First Meeting of the General Advisory Committee to the USAEC, July 12-15, 1954, p. 15.

late this year.

It should be noted that further developments are likely to add to the numerous devices presently under consideration to such degree as to generate a requirement for a Fall 1955 test series of lesser magnitude at the Nevada Proving Ground.

....

To meet the above requirements a series of atomic tests to be conducted at the Nevada Proving Ground commencing about mid-February 1955 and designated as Operation TEAPOT has been approved by the Commission. As now envisaged the test series will include approximately twelve tests. With the exception of two, possibly three air drops, all tests will be detonated on towers. Four, possibly five of the tests will have yields from 15-30 KT or greater. All others will be less than 5 KT.⁵⁵

Relatively light-weight warheads would also encourage quick development of unmanned intercontinental delivery vehicles (such as ICBMs and cruise missiles).

One result of the "tritium bonus" discovered during CASTLE was that thermonuclear weapons would require much less Li^6D and Li^7H than was loaded into the CASTLE devices for a given yield. Another path to lighter-weight weapons was to design them for lower yields, between several hundred kilotons and a megaton or two, rather than the megaton multiples of most of the CASTLE bombs. This could be accomplished most directly by reducing the amount of uranium-238 in the secondary fusion stage pusher, which would also save weight at a rate of 1,200 lbs. per cubic foot.

Another route to light-weight fusion weapons was to replace their large-diameter unboosted primaries with smaller-diameter (22" to 30") boosted primaries. A substantial volume and weight reduction could be achieved without any loss in primary yield.

⁵⁵Letter dated August 30, 1954, from Lewis L. Strauss, Chairman, U.S. Atomic Energy Commission, to President Dwight D. Eisenhower, Washington, D.C. Emphasis is in original document. The spring 1955 test series became Operation TEAPOT in Nevada; the fall 1955 tests were the DIXIE shots incorporated into Operation REDWING in the spring of 1956.

Planning for Operation TEAPOT

Small boosted fission devices suitable for use as thermonuclear weapon primaries were developed and tested at the Nevada Proving Ground in 1955 during Operation TEAPOT. The advent of sealed-pit boosted primaries essentially tripled the number of weapons that could be made from the available inventory of special nuclear materials. These new small boosted primaries also featured better yield predictability and reproducibility.⁵⁶

TEAPOT had several objectives, especially the determination of the feasibility of new ideas in weapon design including small-size spherical implosion; high-pressure deuterium-tritium gas boosting; linear implosion systems; the use of uranium-233 as a weapon fuel; and mockup tests of Class "D" thermonuclear weapon designs.⁵⁷

The TEAPOT Zucchini, Apple-1, and Apple-2 shots were specifically cited as being important to the development of a Class "D" thermonuclear weapon. The mockups fired in Nevada were designed to limit their yields to a maximum of 50 KT.⁵⁸ These shots, finally fired on May 15, March 29, and May 5, 1955 yielded 28, 14, and 29 kilotons, respectively; the Apple shot was repeated because of the unexpected low yield on the first attempt.

The second Apple shot included some secondary reaction, i.e., energy yield from thermonuclear fuel reactions. In spite of this, the AEC would continue to deny publicly for several years that thermonuclear experiments were occurring in Nevada.

Objectives for LASL during Operation TEAPOT were the firing of a 16-inch diameter two-kiloton experimental system, a 22-inch diameter two-kiloton system to test central initiation, a 22-inch diameter system to test external initiation, a boosting test, and two tests of mockups of small light-weight thermonuclear systems expected to provide casing design and radiation channeling information applicable to the design of Class "D" hydrogen fusion weapons for possible aircraft or guided missile delivery.

⁵⁶MANAGING NUCLEAR OPERATIONS, Ashton B. Carter, John D. Steinbruner, and Charles A. Zraket, eds., The Brookings Institution, Washington, D.C., 1987, p. 28.

⁵⁷Operation TEAPOT, Report of the Test Manager, Joint Test Organization, Nevada Test Site, Spring 1955, p. 75.

⁵⁸Operation TEAPOT, Report of the Test Manager, Joint Test Organization, Nevada Test Site, Spring 1955, pp. 78, 79.

The latter two devices were expected to yield approximately 30 to 50 kilotons and less than 80 kilotons. One of the proposed TEAPOT test devices was called the "Jasper."⁵⁹ All LASL shot yields were expected to be approximately two kilotons, except for the Class "D" case tests that were expected to yield 30 to 50 KT.

The LASL missile warhead tests were to fulfill existing DOD requirements for small-diameter lightweight devices with yields between two and 10 KT. The Wasp, Moth, Bee, and Hornet devices, with diameters between 17 and 23 inches, were the first tests of spherically-imploded devices with diameters less than 22 inches.⁶⁰

LASL shots Wasp and Moth were tests of spherically-imploded, unboosted missile warhead prototypes; the LASL shots Bee and Hornet were tests of spherically-imploded, boosted missile warhead prototypes.

LASL also planned a new test series then called simply "Post-TEAPOT;" the laboratory's objectives for "Post-TEAPOT" were the firing of one or more two-stage devices; one-point safety tests; test of an optimized 30 KT device with a beryllium tamper; a test of a boosted device using lithium-6 deuteride or D-T gases; and a 30" diameter, two kiloton device. In addition, other predetonation and beryllium tamper tests might be conducted.⁶¹

This test series might begin around 1 September 1955. (This proposed series later became known as Operation DIXIE.)

UCRL planned to test during TEAPOT two low-yield devices, probably of less than two kilotons each, of a "basically new design for air defense and tactical purposes." (These were linear implosion tests.) Two different design diameters on the order of 10" were proposed to explore the "region of feasibility" of such devices.

⁵⁹Minutes of AEC Meeting 1020, August 18, 1954, p. 44; Minutes of the Forty-First Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, July 12-15, 1954, p. 26. "Jasper" was also the name of radiation rate meters installed on USAF T-33 and F-84G cloud sampling aircraft. (Letter dated December 24, 1952 to Commanding Officer, 4925th Test Group (Atomic) from H. F. Plank, LASL.)

⁶⁰Operation TEAPOT, Report of the Test Manager, Joint Test Organization, Nevada Test Site, Spring 1955, pp. 76, 77.

⁶¹Minutes of the Forty-First Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, July 12-15, 1954, p. 27.

Specifically, UCRL shots Tesla and Post were tests of linear implosion systems. The UCRL Turk shot, and the LASL Apple and Zucchini shots were tests of Class D thermonuclear weapon prototype mockups.

In addition, a UCRL test of a primary for a Class "D" thermonuclear weapon design was under consideration; this became the 43 kiloton Turk shot on March 7, 1955.⁶²

The LASL and UCRL Class "D" tests were necessary to interpolate CASTLE device principles to a 3,000 to 4,000 lb., megaton-yield thermonuclear system. Auxiliary information of general value to two-stage weapons might also be obtained from such tests.⁶³

Objectives of TEAPOT began to solidify in mid-1954. During the 41st meeting of the GAC at Los Alamos on July 14, LASL director Dr. Norris Bradbury delivered a critique of U.S. nuclear weapon design philosophy since 1945 Bradbury was very introspective about the work of LASL, and he rarely hesitated to expound at length on this subject, either in person or in writing.

From 1947 until 1954, Dr. Bradbury said, the country's thinking had been defined by cores-versus-bomb sizes, in which interchangeability of cores in bombs was a dominant feature; this was appropriate when fissionable material was scarce and stockpile size was limited.

The LASL director believed that this attitude was no longer relevant or appropriate. By now, the U.S. stockpile was in a period of transition from scarcity of materials to relative abundance, from a modest range of yields to an almost unlimited range of very small to very large yields. Interchangeability should no longer dictate design if it restricted the numbers of weapons, their readiness and flexibility of use in time of emergency, or their effectiveness.⁶⁴

⁶²Atomic Energy Commission, Revised Schedule for TEAPOT, AEC 707/38, Note by the Secretary, April 5, 1955; Atomic Energy Commission, Nevada Proving Ground Test Activities For Calendar Year 1955, Report to the General Manager by the Director of Military Application, August 9, 1954, pp. 5, 7; Operation TEAPOT, Report of the Test Manager, Joint Test Organization, Nevada Test Site, Spring 1955, pp. 76, 78.

⁶³Atomic Energy Commission, Nevada Proving Ground Test Activities for Calendar Year 1955, Note by the Secretary, August 9, 1954, pp. 5, 7, 14.

The U.S. nuclear weapons stockpile at this time consisted mainly of MK 5, 6, 7, and 8 fission bombs, MK 9 artillery shells, small numbers of MK 18 high-yield fission bombs, and TX-14, -17, and -24 thermonuclear bombs.⁶⁵

Since early 1954, thermonuclear two-stage weapon Classes "A," "B," "C," and "D" had been established to cover the spectrum of yields and sizes in the new field of megaton-range bombs and warheads. In a number of cases, these new weapon classes appeared to render particular stockpiled standard fission bombs obsolete.

The MKs 6 and 13, whose weights corresponded to Class "C" thermonuclear weapons, were, in Bradbury's words, "dead ducks." Was the Air Force likely to use a B-47 to deliver a kiloton-yield bomb when a three-megaton MK 15 bomb of the same weight was available? Was the 3,000 lb. MK 5 worth carrying when a Class "D" bomb of the same weight and with a yield twenty times as great was available? The "A" to "D" thermonuclear classes seemed to cover the entire strategic spectrum.

Bradbury suggested abandoning the core-versus-bomb size concept, and substituting instead new thermonuclear classes to cover the tactical area. Among these were proposed Class "E" weapons for fighter-bombers, missile warheads, etc.; these might be the size of the 1,600 lb., 30" diameter MK 7. He said that the real point was to fix on a device with characteristics that the military wanted, then to make that weapon the best it could be. Bradbury summarized most military requirements as saying, in effect, "we don't know what we want, but we want to be able to do anything."

The director went on to propose additional bomb classes, in both fission and thermonuclear areas. Classes "F," "G," and "H" embraced existing tactical weapons, such as the MK 7, 15" to 22" diameter air-to-air defense, anti-submarine, and missile warheads, as well as existing gun-type special purpose bombs and artillery shells.

Dr. Bradbury emphasized that he was not proposing what the detailed class descriptions should be, but was suggesting a design philosophy to fix on types for which large numbers were needed, to develop the best possible weapons with the best achievable characteristics of each type, without penalizing the design by requiring that its core be

⁶⁴Letter dated July 16, 1954 to Lewis L. Strauss, Chairman, USAEC, from I. I. Rabi, Chairman, GAC.

⁶⁵Atomic Energy Commission, Status Report for June 30, 1954, Part III - Weapons, July 22, 1954, p. 11.

interchangeable with some other type of weapon, e.g., a strategic weapon.

Tactical weapons would be required in large numbers, widely deployed and always ready for use. Under these circumstances, interchangeability was irrelevant; by abandoning this practice, the existing stockpile could be doubled, even without use of boosting, then a largely-unimplemented technique. If boosting proved out, then any particular weapon could have a variety of yields and the stockpile could be made even more flexible.

On the other hand, if interchangeability was required in the future, thereby minimizing the implementation of boosting, Bradbury believed that further gains in fission weapons were very limited. (Failure to succeed in the design and use of boosted primaries slowed the Soviet thermonuclear weapons program, and led to low yield-to-weight ratios in their warheads, requiring large, heavy devices to obtain high yields.)

Bradbury's comments were followed by an "animated" discussion of their pros and cons; however, there was no strong opposition voiced to the director's proposals.⁶⁶ The GAC recommended that interchangeability be re-appraised as a guiding factor for weapons design in light of the increased availability of fissionable material, and suggested that studies be initiated to provide principles to optimize design of specific weapon classes instead:⁶⁷

(During) the period of scarcity of materials and limited numbers of weapons, interchangeability of nuclear parts was an important criterion of stockpile weapon design. The present is a period of transition -- from scarcity to relative abundance, from a modest range of yields to one almost without limit from very small to very large.

Interchangeability should no longer dominate design if it restricts numbers of weapons, their readiness and flexibility of use in time of emergency, or their effectiveness in accomplishing the specific

⁶⁶Minutes of the Forty-First Meeting of the General Advisory Committee to the USAEC, July 12-15, 1954, pp. 27-29.

⁶⁷"Recommendations, Suggestions, Expressed Interests of GAC - Period, August 1953 - December 1954," December 31, 1954, p. 3.

missions assigned.

We would, therefore, urge a careful reappraisal of the relations between the types of design of nuclear weapons and the missions for which they are appropriate, with the objective of establishing guidance principles for optimizing the design of the several classes of weapons, small and large, which could accomplish these missions. In such a reappraisal, both LASL and Sandia should take part with the Military Departments (sic) in the necessary comprehensive system studies.⁶⁸

Concerns About the UCRL Program

The UCRL weapons program was also reviewed at this GAC meeting. Dr. Edward Teller stated that although the failure of the MORGENSTERN during CASTLE was a bitter disappointment, the laboratory had learned a great deal from the test nonetheless, including the reason for its "fizzle."

Teller also described plans for a 10,000 megaton device. The GAC was surprised by this proposal, and considered it to be "an advertising stunt," not to be taken too seriously. The suggested weapon yield was an order of magnitude greater than that hypothesized by Teller in his September 1947 report, LA-643.

Herbert York then spoke about the UCRL small weapons program. Two lines of research were being pursued, with more progress on one than on the other. Some devices under study or development included a 12" diameter, 42" long, 800 lb. warhead; a 10" diameter, 36" long system; a 7" diameter, 26" long, 240 lb. device (possibly a prototype of the SWALLOW tested during Operation REDWING); a 5" diameter, 100 lb. weapon (possibly a prototype of the SWIFT tested during Operation REDWING); and a 4.2" diameter, 16" long, 50 lb. "gadget" (a possible W-51/54 prototype). UCRL also had plans for a Class "D" two-stage weapon.⁶⁹

⁶⁸Letter dated July 16, 1954 to Lewis L. Strauss, Chairman, USAEC, from I. I. Rabi, Chairman, GAC, p. 3.

⁶⁹Letter dated July 16, 1954 to Lewis L. Strauss, Chairman, USAEC, from I. I. Rabi, Chairman, GAC. According to one source, the UCRL SWALLOW, SWAN, and SWIFT devices were so-named because the "SW" in their names stood for "small weapon." Allegedly, when UCRL classification officers learned of this, they ordered that this particular naming convention be ended immediately.

UCRL was also still studying uranium hydride-fueled weapons; various fabrication and handling methods were being investigated.⁷⁰

The Livermore lab was pursuing the concept of linear implosion at this time; the GAC thought this technique might provide an alternative to the gun or hollow implosion methods and was therefore "interesting to explore."

The GAC had some concerns about the direction of the UCRL small weapons program, believing that it was being planned without sufficient forethought of future delivery vehicles and war-fighting goals (the exact words used by Dr. Rabi was that "there was something amateurish in the way the objectives were defined").

The program objectives seemed to be defined "essentially as to do something more risky than Los Alamos (was doing)." This bold and audacious attitude became a hallmark of the Livermore laboratory; UCRL spent many of its early years trying to establish its own identity and get out from behind the shadow of its more-famous predecessor at Los Alamos.

GAC chairman Dr. Isidor Rabi noted that Livermore had no responsibility for any necessary part of the national nuclear weapons program, and that he would like to see a clear division between LASL and UCRL with respect to defined and different objectives.⁷¹

In other discussions, several members of the GAC expressed concern over Teller's 10,000 MT device; some wondered what fraction of the Livermore lab's effort was being expended on high-yield devices named the GNOMON and the SUNDIAL. Drs. von Neumann and Rabi stated that "the (Livermore) laboratory clearly has very capable people on its staff; it is unfortunate that they are not being effectively utilized up to their abilities."

In addition to Bradbury's suggestions, the short-term direction of thermonuclear weapons development depended to a large extent upon Air Force requirements. The Air Force's main current strategic attack mission was to target Soviet airfields, to catch their aircraft on the ground, which implied an American first strike: if the Soviets attacked

⁷⁰Minutes of the Forty-First Meeting of the General Advisory Committee to the USAEC, July 12-15, 1954, pp. 30-35, 55-57.

⁷¹Minutes of the Forty-First Meeting of the General Advisory Committee to the USAEC, July 12-15, 1954, p. 55-57.

first, it was unlikely that their planes would be grounded.

The destruction of the Soviet long-range bomber force was of paramount importance to lessen the burden on U.S. air defenses, and to minimize destruction of SAC bomber forces and continental U.S. atomic weapons facilities, and urban-industrial centers.⁷² (By February 1955, SAC had targeted no fewer than 645 Soviet airbases.)⁷³

Although destruction of Soviet industry was now secondary, SAC still desired high-yield bombs to drop on urban areas to not only destroy industrial installations, but also to inflict so much damage that the Soviets would be unable to recuperate quickly and continue a war.⁷⁴

The Air Force was still interested in large, high-yield weapons. Existing bombs were generally satisfactory, but current delivery aircraft (the B-36, B-47, and B-50) left much to be desired.

The Air Force also desired contact-fuzed weapons, for surface bursts to crater and contaminate their targets, and was unhappy that this type of fuse was not receiving much attention. SAC war plans at this time called for attacks on even empty Soviet airfields.

While guided ballistic missiles might become very important as future nuclear weapon delivery vehicles, the Air Force believed that at least one more heavy bomber beyond the B-52 was needed (this became the aborted XB-70 project). Nuclear propulsion was desired to the point of being more critical than bomb development. In addition, high-altitude delivery capability was considered more important than high speed delivery.

The Air Force also expected to build many new airbases, to

⁷²STRUGGLE FOR PREDOMINANCE: EVOLUTION AND CONSOLIDATION OF NUCLEAR FORCES IN THE STRATEGIC AIR COMMAND, 1945-1955, PhD thesis by William Stephen Borgiasz, The American University, Washington, D.C., 1991, pp. 238, 239.

⁷³"Toward Armageddon: The Foundations of United States Nuclear Strategy," PhD thesis by David A. Rosenberg, University of Chicago, August 1983, p. 209. How SAC knew the locations of these bases is still a mystery: the U-2s did begin flying over the Soviet Union until July 1956.

⁷⁴STRUGGLE FOR PREDOMINANCE: EVOLUTION AND CONSOLIDATION OF NUCLEAR FORCES IN THE STRATEGIC AIR COMMAND, 1945-1955, PhD thesis by William Stephen Borgiasz, The American University, Washington, D.C., 1991, p. 227.

widely disperse their bombers, keeping no more than five of them at any one site.⁷⁵

TEAPOT Plans, August 1954

In August, UCRL and LASL submitted their test plans and objectives for Operation TEAPOT, then scheduled for early 1955. Tests were pertinent to strategic, tactical, and air defense warhead applications; both the labs and DOD also sought further specific and precise weapons effects data.

Tests of mockups of small, light-weight Class "D" bombs were required to provide information needed to proceed with development of bombs as well as missile warheads. Tests of various low-yield fission devices would enable the laboratories to optimize warheads for air defense and tactical purposes. When weapon systems using such warheads were designed, they would be deployed widely in very large numbers.

As currently envisioned, TEAPOT would comprise approximately six LASL shots, three UCRL shots, and three DOD effects tests. LASL would test a device using uranium-233, perhaps as one of the effects shots. Most of the tests would yield less than five kilotons, with four or five higher-yield shots for thermonuclear weapon casing studies, primarily radiation ducting analyses. The two laboratories also desired a follow-up series of tests in the fall of 1955, after TEAPOT.

All shots would be on 300 to 500 foot towers, except for tests of the LASL "booster" and the two 22-inch diameter systems, which would be air drops.

One aspect of TEAPOT planning was whether or not to publicly announce the purposes of certain tests:

Santa Fe Operations Office and the (AEC) Test Organization (sic) have never recommended public issuance, confirmation, or denial of the specific purpose of individual shots nor of the purpose of the whole or a portion of a full series schedule.

There is a local awareness, it is believed, of some of the national level factors which might determine advisability of announcing the interest and work of the developmental people toward defensive, smaller yield weapons. This is largely a national level determination.

⁷⁵Minutes of the Forty-First Meeting of the General Advisory Committee to the USAEC, July 12-15, 1954, p. 37.

....

It may additionally be decided to suggest that any national level action on test series purpose incorporate a statement clarifying use of NPG in the thermonuclear (weapons) development program.

Test Organization has consistently suggested to the public that thermonuclear devices are not tested at NPG (because of high yield), leaving the implication (not always accepted by news media) that only fission weapons and devices were tested at NPG. The Oppenheimer (hearing) transcript and various official statements have seemed to state that devices involved in thermonuclear development have been tested at (the) NPG.⁷⁶

In fact, by the end of August 1954, thermonuclear devices had been tested at the NPG, both early boosted devices, and thermonuclear weapon prototypes, although no two-stage tests had yet been conducted in Nevada. (They soon would be.)

Theoretical studies of "clean" and "dirty" thermonuclear bombs continued. By late September, the AEC was actively studying contaminating characteristics of thermonuclear weapons, to both increase and decrease fallout; the latter might be accomplished by removing or replacing the U-238 fusion tamper jacketing the secondary. A preliminary study on decreasing fallout had been sent by the AEC to DOD early in September; this study had been prepared by LASL in April 1954.⁷⁷

The Department of Defense was also studying this matter:

Consideration of the effects of nuclear weapons, particularly those with high yields, indicates that radioactive fall-out and contamination are among their most important characteristics, and indeed may be one of the factors governing their use in some situations.

It also appears that it may be possible to control to a considerable extent, by

⁷⁶Memorandum dated August 31, 1954 to Director, LASL, from Director, Office of Information, USAEC, subject: Announcing Purpose of Certain TEAPOT Tests.

⁷⁷Letter dated October 11, 1954 to Donald A. Quarles, Assistant Secretary of Defense, from Lewis L. Strauss, Chairman, USAEC.

different weapon designs, the relative amounts of long-lived radioactive products per megaton of total energy yield. It is understood that the AEC is gathering information as to special materials which could be added or substituted to enhance or diminish the radioactive contamination.

For your information, the Department of Defense is considering the broad aspects of the contamination problem as well as the more detailed operational factors which would influence the degree of contamination. These studies when combined with the results of your investigation should enable the Department of Defense to decide whether or not there is need for high yield weapons with different contaminating characteristics than those for which military characteristics are now specified.⁷⁸

In response to this letter, the AEC wrote to LASL and requested information on work which had been done to date in this area:

We received from Dr. Carson Mark his letter of April 20, 1954 on the enhancement of radioactive contamination by adding special materials to thermonuclear weapons. That letter was made available to several agencies of the Department of Defense. If there are any further thoughts on that aspect at Los Alamos, we would be glad to hear of them.

We would very much appreciate an analysis of the reduction of radioactive contamination and the corresponding reduction of yield that would result from replacing part or all of the natural or depleted uranium in thermonuclear weapons with other materials. (Author's note: such as lead.) Possibilities of more extensive changes in design to reduce the radioactive contamination per megaton (of yield) would also be of interest.

⁷⁸Letter dated 17 September 1954 to Lewis L. Strauss, Chairman, USAEC, from Donald A. Quarles, Assistant Secretary of Defense.

A copy of this letter and enclosure is being sent to Dr. York to elicit any ideas that Livermore may have on the subject.⁷⁹

In early November, TEAPOT was to be followed by a series of "sealed pit" safety test shots in Nevada in the fall of 1955; this safety test series was to be called Operation DIXIE. Since "sealed pit" weapons had nuclear materials permanently installed in them, it was absolutely necessary that they be safe during accidents:

To establish how large an unboosted yield can be obtained in a sealed-pit device that is absolutely or marginally safe, it is probably necessary that the test device employed would have to become critical under one-point detonation, but give no appreciable yield. The design of such a device cannot be determined until test results of similar devices under symmetrical implosion are obtained in TEAPOT.

It is regarded as infeasible, therefore, to perform a one-point detonation shot until DIXIE with any promise of obtaining information that would be generally applicable to the problem of designing safe, sealed-pit type warheads.⁸⁰

Study was underway at this time at both UCRL and LASL to find ways to both increase and decrease, either by weapon design and/or variation of detonation conditions, the local and world-wide contamination resulting from the use of nuclear weapons.

LASL was studying the probable results of varying weapon design in order to increase or decrease radioactive fallout. Dr. J. Carson Mark of LASL had examined the so-called "cobalt" bomb, in which fusion stages were to be jacketed with cobalt in lieu of uranium-238, and had concluded that the addition of cobalt was not a feasible way to greatly increase contamination.

⁷⁹Letter dated September 29, 1954 to Dr. N. E. Bradbury, Director, LASL, from Brig. Gen. K. E. Fields, USA, Director of Military Application, USAEC.

⁸⁰Minutes of the 42nd Meeting of the General Advisory Committee to the USAEC, November 3-4, 1954, p. 11; letter dated November 18, 1954 to Honorable Herbert B. Loper, Chairman, MLC, from K. D. Nichols, General Manager, USAEC.

Other AEC laboratories were measuring neutron capture cross-sections to determine whether any other elements could produce the desired effects. To date, the use of gold, at great expense, appeared to offer possibilities for increasing local contamination.

In general, contamination could be increased or decreased by a factor of two relatively easily, but a factor of 10 or greater was required for any military utility.

A ton of gold, if used in lieu of some of the uranium in thermonuclear weapons, would result in a doubling of local short-lived contamination, but at a cost of a 75% reduction in weapon yield (the example given was a 12 MT bomb reduced to a 3 MT bomb).

Substituting lead, and perhaps iron, for some of the uranium would reduce contamination by an unknown factor, and at cost in total yield almost as great as that engendered by the use of gold.

Other short-lived radioisotopes that might be militarily useful included sodium-24, with a half-life of 15 hours, and silicon-31, with a half-life of nearly three hours. Both are powerful beta and gamma ray emitters. Sodium-24 could be generated either by the radioactive decay of aluminum-28, or from magnesium-24.⁸¹

The use of either of these isotopes offered the military advantage that they would probably decay harmlessly before the fallout containing them drifted out of enemy territory. In addition, when weapons containing these isotopes were used tactically, friendly troops could enter contaminated areas soon after their detonation.

The most desirable alternative to reduce fallout was to decrease the weapon's fission yield and increase its fusion yield. No one knew how to do this at the moment, especially since present weapon design techniques required fission reactions in the primary and the "sparkplug" to initiate fusion reactions. Just trying to reach this goal would, with present designs, probably greatly decrease the yield-to-weight ratio of stockpiled and future weapons.

It might be possible to reduce the fission yield with a corresponding decrease in fusion yield by replacing some of the uranium with tungsten to reflect neutrons and so increase the neutron flux in the secondary.

⁸¹WEAPONS OF TOMORROW, Brian Beckett, Plenum Press, New York, 1983, p. 102.

There really was no such thing as a totally "clean" weapon: even if all fission yield could be replaced with fusion yield, neutron-induced radioactivity would cause major short-term local contamination for bursts in which the fireball contacted the earth or water. Little consideration had been given to this aspect of thermonuclear explosions, and LASL was planning to undertake a theoretical and mathematical analysis to determine the seriousness of this problem.⁸²

The AEC's General Advisory Committee summed up TEAPOT goals at the end of November (the proposed slate of shots had grown slightly):

The Committee reviewed the proposals of the Los Alamos and Livermore laboratories for test explosions of seven and three nuclear devices, respectively, at the Nevada Proving Ground next spring. The four tests of small implosion weapons represent important lines of development in the small weapon field. The prospective efficiency and economy of use of fissionable material are particularly significant in this field where large numbers and wide deployment will be necessary.

The three case tests are directed in different ways toward the development of light Class D thermonuclear weapons, for specific applications in free-falling bombs and in missile delivery. The results of these case tests will be essential to progress in the Class D thermonuclear weapons field, and may also be consequential for future improvements in weapons classes A, B, and C.⁸³

⁸²"Effects of Design Changes," attachment to notes on Coordination Meeting on the Radiological Contamination Problem held November 9, 1954, Room 3E-1060, The Pentagon under the auspices of the Office of Assistant Secretary of Defense, Research and Development, Technical Advisory Panel on Atomic Energy.

⁸³Comments of the General Advisory Committee on TEAPOT, Appendix "F" to Atomic Energy Commission, Proposed Program For Operation TEAPOT, Report to the General Manager by the Director Of Military Application, November 30, 1954; letter dated December 22, 1954 to Lewis L. Strauss, Chairman, USAEC, from I. I. Rabi, Chairman, GAC.

The "case tests" mentioned were full-scale nuclear tests of mocked-up multistage thermonuclear weapon casings, with "live" primaries having yields equivalent to those to be used in the actual multistage device, to determine radiation transmission characteristics within the casing (radiation channel) before insertion of a "live" secondary.

The secondary in a mockup was simulated by a tracer-loaded capsule to ascertain radiation intensity and depth of radiation penetration into the capsule. Even though the capsule and the casing would be vaporized by the explosion of the fission primary, the performance of the system could be determined from high-speed photography, instrumentation, and radiochemical analysis of device debris collected from fallout on the ground around the shot tower or from the mushroom cloud generated by the explosion, especially by-products of irradiated tracer materials which might be strategically placed within the capsule.

At the beginning of December, LASL proposed tests of primaries for possible Class "D" weapons during the forthcoming Operation TEAPOT in Nevada. The lab offered a lengthy justification for these weapons:

By definition, the "Class D Weapon" is an object weighing about 3,000 lbs. or less, with a yield of [deleted] compatible with a number of designated carriers. The carriers contemplated have, however, somewhat inharmonious characteristics and requirements.

Some, like the NAVAHO, REDSTONE, and SNARK missiles, now in an advanced stage of design, were laid out to carry the now extinct XW-13 implosion system, so that they have a built-in warhead compartment and weight-carrying capacity.

Their performance will, consequently, not be approved appreciably by being provided with a smaller and lighter object than they have been designed to carry; and the most favorable warhead for these carriers would presumably be one that makes the best possible use of the space and weight available to them.

Other carriers, notably high-speed fighter-bomber types with plans to carry bombs internally, have an absolute limit on bomb diameter (usually based on MK 7 weapons and smaller than the missiles mentioned above),

obtain little benefit from decreasing the size of bombs below that which they can carry, and are not nearly as sensitive to the weight of the object carried as are the missiles.

The planes which will carry bombs externally are also not particularly sensitive to the weight of the object carried, but will benefit greatly from any possible reduction in its diameter.

Finally, possible or projected carriers, such as the ATLAS missile, are expected to place enormous premiums on reductions both in weight and diameter of the warhead, providing the yield is held up to a sufficiently high level that the problems of required accuracy in guidance can be solved.

Several different approaches to possible Class D weapons are suggested by the considerations outlined above, and the tests proposed for TEAPOT are intended to obtain information which will be useful at each end of the (yield) range of interest indicated.

...
[deleted]

... carriers for Class D weapons (although there appears to be some uncertainty at present as to whether some of the supersonic aircraft which are planned to carry bombs externally can indeed carry an object of this size without serious effects on performance characteristics). Since it is believed that at some small size and weight a two-stage system with a [deleted] will be superior to one with [deleted]⁸⁴

At this time, the AEC was still rather skittish about confirming the behavior or by-products of the devices fired earlier in the year during CASTLE, thus implying the design. AEC Commissioner Willard F. Libby prepared a speech to be given to the Washington Conference of Mayors in D.C. on December 2; he submitted a draft of his speech for classification review.

⁸⁴Letter dated December 1, 1954 to Brig. General K. E. Fields, Director of Military Application, USAEC, from Norris Bradbury, Director, LASL.

Among other objections, the AEC's Office of Classification claimed that the revelation of a particular thermonuclear explosion by-product -- tritium -- revealed weapon constituents, despite the fact that this information was by no means a secret:

We note that in the sentence beginning with "We find, for example," it is stated that a principal radioactive product of the thermonuclear weapon is tritium. To the best of our knowledge, a Commission official has not, to date, stated that tritium is a by-product of a thermonuclear explosion. Stating the fact in this manner implies that the by-product tritium is formed either by the Li-6 neutron reaction, or as the result of a D-D reaction. (Author's note: see "Fusion Physics" in "Weapons Physics" in Volume I for details of these reactions; tritium is also formed by deuterium-neutron reactions and lithium-7 reactions with neutrons, as was demonstrated so sensationally during CASTLE.)

The revelation of this fact will hardly come as a surprise to the scientific community. We feel, however, it will probably have a tremendous public information impact and may result in the generation of a large number of articles on the same subject. ... In any case, we have considered such statements, when made by persons having "access," to comprise Restricted Data. We suggest that the sentence be deleted or revised.⁸⁵

Here was a clear example of the AEC censoring widely-known information (that tritium is a by-product of thermonuclear explosions) merely because it was being reiterated by an AEC official, and because the AEC did not want to have to answer questions about it or see other articles that might analyze or repeat Libby's comment. Libby re-wrote the offending sentence before delivering the speech, removing his statement that tritium was a principal by-product.⁸⁶

⁸⁵Memorandum dated December 1, 1954 to W. F. Libby, Commissioner, from Murray L. Nash, Division of Classification. Emphasis in original.

⁸⁶Comparison of draft of Libby speech with version finally

UCRL made its plans for TEAPOT known early in the month. Dr. Edward Teller proposed a number of tests of systems of widely-varying capabilities:

Our test program for TEAPOT has been discussed quite carefully in the last couple of months. You probably know that we expect to try [deleted]

The shots which we are going to fire will be purely experimental but they should open up the possibility, if successful, toward very small-diameter devices. We are considering at present quite low yields in the neighborhood of a kiloton with the possibility of increasing the yield later.

These devices should become useful, not only as weapons for the Army or for air-to-air missiles, but they might also find some application [deleted]

We are also planning to make an experimental shot (of) a two-stage device, which should eventually lead to a light megaton bomb at REDWING. Some of the ideas in this device have already proved fruitful in that Los Alamos has adopted them for their own plans. Nevertheless, their two-stage shots and our two-stage shots have quite a few differences and I feel certain that the programs complement each other in a very nice way.

I believe that the plans for our TEAPOT shots have been laid in a reasonably conservative manner and I hope that we shall come out of these shots with a lot of additional knowledge and with some progress toward usable weapons.⁸⁷

After its experiences during UPSHOT-KNOTHOLE and CASTLE, UCRL was looking forward to redeeming itself, at least partially, by its performance during TEAPOT.

Late in December, the AEC's Division of Military Application gave its blessings to plans for TEAPOT:

used.

⁸⁷Letter dated December 10, 1954 to Lewis L. Strauss, Chairman, USAEC, from Edward Teller.

By memorandum of April 2, 1954 from the Chairman, Military Liaison Committee, the Department of Defense forwarded the desired military characteristics for a new small-sized implosion warhead for application to missiles, atomic demolition, and other uses in appropriate weapon systems by use of suitable adaption kits.

By memorandum of May 14, 1954 the Acting Secretary of Defense forwarded a memorandum from the Under Secretary of the Army expressing great interest in the work being done by the University of California Radiation Laboratory on new designs of small, very low yield atomic weapons, and emphasizing the importance which the Army places on the rapid acquisition of simple, rugged versions of these small weapons.

By letter dated June 19, 1954, the Chairman, MLC, forwarded the Department of Defense development requirements for [deleted] of 22-inch diameter yielding both [deleted] and of 30-inch diameter yielding [deleted]

By letter (Top Secret) of February 5, 1954, the Department of Defense forwarded a requirement for a Class D warhead of light weight and small diameter having a yield in the megaton range.

The programs submitted by Dr. Bradbury and Dr. York contain the laboratory developments which are most likely to meet the above Department of Defense requirements and which are to be tested during TEAPOT.

... The General Manager advised the Commission that he had instructed the Director, Division of Military Application to make arrangements for including a U-233 test in TEAPOT. Subsequent action by the Division of Military Application resulted in such a test being scheduled during TEAPOT.

The TEAPOT program as submitted by the laboratories includes developments applicable to each of the above requirements established by the Department of Defense. Specifically, the Moth and

Hornet are 22-inch devices designed to meet the requirement for [deleted]; the Bee and Wasp are 15-inch devices designed for [deleted]; Tesla and Post are "linear implosion" devices, the area of small weapons of particular interest to the Army; and Turk, Apple, and Zucchini are devices designed to fulfill the Class D warhead requirements and to further thermonuclear developments in all classes. ... The maximum yield anticipated is about 35 to 40 KT.

The U-233 shot will be conducted in conjunction with the military effects test (MET) in Frenchman's Flat. The estimate of the yield of this device is about 28 KT $\pm 10\%$ which is acceptable to the Department of Defense since their program requires a yield of this magnitude which can be predicted within an accuracy of ten percent.

....

The Bee device originally scheduled to be ready for firing February 15, 1955 has been shifted to a place late in the series. This was necessitated by certain changes just recently found to be necessary in the initiation system of this device. The shift in firing sequence will provide additional time in which to resolve this problem satisfactorily. It is anticipated the device will be ready for firing late in the series although there is a remote possibility it may not be ready at all for TEAPOT.

In conjunction with this proposed program, a letter to President Eisenhower was also prepared to inform him of the purposes of the TEAPOT test series and to ask for his approval:

Following the President's approval of the Atomic Energy Commission's proposal of August 30, 1954 of the conduct of a series of nuclear tests at the Nevada Test Site in the spring of 1955, the Atomic Energy Commission and the Department of Defense have developed the detailed test program considered necessary at this time in the interest of rapid progress in the field of nuclear weapons. The major purposes of this

series of tests are as follows:

- a. The development of small-diameter implosion systems to provide more efficient weapons for air defense, tactical and demolition purposes.
- b. The development of small-size but powerful thermonuclear weapons suitable for delivery by tactical aircraft and guided missiles.
- c. The test of the performance of U-233 as a fission material in comparison with Plutonium (sic) and U-235.
- d. The test of the feasibility of the linear implosion principle as a means of providing very small low-yield weapons of acceptable efficiency for air defense and tactical uses.
- e. The acquisition of additional essential data on the military effects of atomic explosions and on effects for civil defense purposes.⁸⁸

TEAPOT was very important to the future U.S. thermonuclear weapons program. Some light-weight thermonuclear weapons conceived between 1954 and 1956 that used TEAPOT results included the MK 21, MK 27, the MK 28 family, and the MK 39 (a MK 15 follow-on). After the TEAPOT tests in Nevada, another Pacific test series would be required to proof-test these high-yield devices.

By the end of 1954, the U.S. thermonuclear weapons production program had been fully converted from an "emergency capability" phase to quantity production of weapons tested during Operation CASTLE. In addition, specific development and production programs had started on lighter and smaller thermonuclear weapons.⁸⁹

⁸⁸Atomic Energy Commission, Proposed Program for Operation TEAPOT, Report to the General Manager by the Director of Military Application, December 30, 1954; letter dated January 14, 1955 to President Eisenhower from Lewis L. Strauss, Chairman, USAEC.

At the beginning of 1955, the MLC was following thermonuclear weapons developments closely and expressing its desires for future improvements:

Of considerable and continuing importance to the Department of Defense is reduction in the weight and attainment of the specified yields of each class of TN (Author's note: thermonuclear) weapon. The attainment of weights specified for each class should be considered as early objectives; however, any possibility for major reductions of weight in each class should be fully exploited.

The attainment of the specified yield for each class has a direct application to the DOD desire for weight reduction. In the allocation of weapons (to) targets, this will allow such obvious substitutions as a high yield Class B for the MK 17 or MK 24 and of Class C for the MK 26.⁹⁰

The MLC also noted that the attainment of a true contact fuse in all thermonuclear weapons classes at the earliest possible date was a matter of considerable urgency.

Early in January 1955, LASL and Sandia reviewed their heavy future weapons development schedule:

The Sandia Corporation and the Los Alamos Scientific Laboratory, through the mechanism of a joint meeting of the TX-N and TX-Theta Committees have recently had occasion to examine the definition and scheduling of the various new weapon development programs of which the laboratories are presently aware or in which they are currently involved.

The number and variety of these programs are such that some degree of ordering is necessary, both to prevent dissipation of effort on already-available systems and to make sure that essential new items are produced at the earliest possible time.

⁸⁹Atomic Energy Commission, Report to the Joint Committee on Atomic Energy, June-November 1954, Part III: Weapons Program.

⁹⁰Memorandum for the Chairman, USAEC dated January 6, 1955 from Herbert B. Loper, Chairman, MLC, Subject: Thermonuclear Weapons Program.

The purpose of this letter is to inform you of our conclusions in the matter and to suggest that you may wish to inform the various appropriate agencies in the Department of Defense as to the nature of the lines of attack now being given maximum attention. Gun-type weapons, since they do not come to the attention of the Sandia Corporation, are omitted from this study.

In the small weapon field, the XW-25 [deleted] warhead for air-to-air defense has been defined and is the subject of active work. It is further likely that this weapon can, for all practical purposes, be regarded as the optimized version of the [deleted] and the laboratories propose to so regard it.

While it is true that some economies in fissionable material could be effected by making [deleted] it is dubious if the extent of the demand therefor requires the high priority development of a special high explosive system and nuclear geometry.

The [deleted] satisfies any specific requirements for a [deleted] warhead or bomb and this device can remain in the stockpile for as long as any such specific requirement remains. Most delivery systems, of course, will benefit by the reduced size and weight of the XW-25, and will doubtless employ it.

The so-called "optimized" [deleted] weapons which are requested will be met by the development of a somewhat smaller high explosive system to facilitate external carriage by high speed aircraft and to provide some weight benefit for both aircraft and missile delivery.

The precise overall diameter to be chosen will depend upon studies currently under way by the Sandia Corporation, as well as further discussions with AFSWP Field Command. However, it is likely to be in the range of 24" to 26."

[deleted] in this size will probably follow the technique used for the XW-25 and this technique as well as the actual size will be determined after TEAPOT. Needless to say, the nuclear components for this device as well as

those for the XW-25 will be specialized and non-interchangeable.

For [deleted] there is some chance that two-stage systems may be appropriate. This possibility, as well as more conventional techniques, are being explored. If a two-stage system seems to have merit and can be developed in time for DIXIE, a prototype might be tested at that time.

It is to be noted that the "optimized" [deleted] systems are to be met both by improving the nuclear geometries and by improving the delivery characteristics. The reduction in diameter [deleted] to facilitate its external carriage makes its size approach the nominal diameter of [deleted] and consequently also points to the desirability of making the optimized version of the latter weapon of a size significantly different from that of its next larger competitor.

In the area of Class D weapons, primary attention is being given to the development of [deleted] which meets the desired military characteristics for this weapon class and which will be useful for both missile and aircraft delivery systems. Accordingly, development of the proposed [deleted] is being postponed for the present until the need for a warhead for missile applications having a somewhat higher yield than the [deleted] becomes apparent.

If results in TEAPOT indicate the feasibility of significant case weight reductions, the [deleted] will be modified so as to meet the desired military weight characteristics, both for the warhead and bomb versions. It is presently believed that this can be done without loss of yield.

This presents a simpler technical and logistic problem than making a completely new weapon. The [deleted] would not introduce a new military capability, and would also be unlikely to meet the upper limit of the desired military characteristics of yield. Accordingly, development of the [deleted] is being table for the moment to permit the more rapid availability of appropriate modifications in the [deleted]

The Los Alamos Scientific Laboratory and the Sandia Corporation believe that urgency of development is properly assigned at the present time to the various systems described above. Indeed, the development of these specific weapons within the desired time scales will tax the resources of both laboratories and will leave little flexibility for the inclusion of important systems such as the new anti-submarine depth bomb LULU to replace ALIAS BETTY.

Since a military capability now exists in both the Class A and Class B systems, although the extremes of the desired military characteristics are not quite met, it seems appropriate to defer development of replacement systems for [deleted] until some of the more urgent and undeveloped systems have had attention.

A similar statement probably holds true for the [deleted] A military capability already exists in this area with the T-4 munition compared to which [deleted] may actually be inferior in every respect except nuclear economy. Clearly, unless large numbers are required, the latter point may be of minor significance.

The Los Alamos Scientific Laboratory and the Sandia Corporation, through the media of the TX-N and TX-Theta Committees, are proceeding along the above lines and will prepare optimum development schedules accordingly. We would welcome comment on these matters which will assist us in resolving the priority to be given to any of these systems, should competition between them develop to a serious degree.

However, we wish it to be clear that the above programs will tax all our available facilities for the next year or so, and that inclusion of any additional programs (with the possible exception of LULU) will probably have to be undertaken only by deferring one or more of those (projects) described above.⁹¹

⁹¹Letter dated January 11, 1955 to Brig. General K. E. Fields, Director of Military Application, USAEC, from N. E. Bradbury, Director, LASL, and J. W. McRae, President, Sandia

Operations at the NBS-AEC Boulder, Colorado facility had been reduced in lieu of Operation CASTLE. In January, the AEC's Division of Military Application ordered the resumption of deuterium production by the deuterium gas separation plant at Boulder on a 24-hour, 7-day week basis at least until June 30, 1955. The deuterium gas was to supplement deuterium separated at the AEC's Savannah River facility; large amounts were required to be chemically-compounded with lithium-6 at Oak Ridge, Tennessee for the nation's growing thermonuclear weapons stockpile.⁹²

The Killian Committee Report, February 1955

In February 1955, the Technological Capabilities Panel of the Science Advisory Committee to President Eisenhower, also known as the Killian Committee, issued its estimation of Soviet and U.S. warmaking capabilities. The committee concluded that at the moment, the U.S. had an offensive advantage but was also vulnerable to surprise nuclear attack by the Soviets.

Although the strength of the Air Force's Strategic Air Command was great, the U.S. still did not have a large stockpile of multimegaton bombs. There was no reliable distant early warning system, so SAC was vulnerable and the country was open to surprise attack. There was also evidence that the Soviets were developing long-range bombers.

The Soviet nuclear weapons stockpile was also growing. In 1954, both SAC and the CIA had concluded that the Russians had 120 bombs with yields in the 30 to 100 kiloton range, and would by 1957 possess 500 bombs with yields in the 500 kiloton to one megaton range.⁹³

Corporation. The TX-N committee was a joint Sandia-LASL committee overseeing implosion weapons development, and the TX-Theta committee was a joint LASL-Sandia committee which directed thermonuclear weapons development. The LULU used the W-34 warhead; in January 1955, this weapon was still undergoing feasibility studies. The Class A and B weapons referred to were the MK 17/24 and MK 21 bombs; the MK 21 in January 1955 had been successfully tested in prototype form as the SHRIMP device fired during the CASTLE Bravo shot, and was nearing design release. The T-4 was an early gun-assembly atomic demolition munition.

⁹²USAEC, Santa Fe Operations Activity Report, Week Ending Noon, January 18, 1955.

⁹³STRUGGLE FOR PREDOMINANCE: EVOLUTION AND CONSOLIDATION OF NUCLEAR FORCES IN THE STRATEGIC AIR COMMAND, 1945-1955, PhD thesis by William Stephen Borgiasz, The American University,

While neither country could marshal overwhelming nuclear forces for a decisive air strike, the U.S. could conduct a sustained air offensive that would probably inflict massive damage and be conclusive in a general war.

Furthermore, if the Soviets were to become fully aware of present U.S. vulnerability, and if that vulnerability were to last long enough for the Soviets to build up a long-range attack force, then the Russians might be tempted to attack the U.S. before a large stockpile of American thermonuclear weapons could be amassed.

The study also concluded that starting in 1956 or 1957, and lasting through 1958 to 1960, the U.S. would have such a large stockpile of both atomic and thermonuclear weapons that a decisive surprise attack against the USSR would be possible. Both countries were expected to remain vulnerable to sneak attacks.

Even in event of a surprise Soviet attack on the U.S., the U.S. would be severely damaged, but could still destroy the Soviets. This would probably be the period of greatest American military superiority over the Russians.

This could all change if the Soviets demonstrated a capability to produce multimegaton multi-stage thermonuclear weapons. The Killian Committee noted that the Russian test shot of August 1953 was not conclusive evidence of Soviet ability to build modern hydrogen bombs.

If this capability were coupled with the development of high-performance long-range jet-powered bombers that could reach the U.S., then America would be at a highly increased risk of Soviet attack.⁹⁴

The Committee also reviewed the present state of U.S. nuclear weaponry and speculated on future developments:

Following World War II, the fission bomb developments have been primarily aimed at (1) increasing the efficiency of bombs in terms of kilotons per kilogram of fissionable material, and (2) making the bombs more deliverable by means of ordnance engineering and by decreasing

Washington, D.C., 1991, p. 219.

⁹⁴"Meeting the Threat of Surprise Attack," The Report to the President by the Technological Capabilities Panel of the Science Advisory Committee, Volume I, Executive Office of the President, Office of Defense Mobilization, Washington, D.C., February 14, 1955, pp. 10, 11.

the size and weight of the bombs. Under ordnance engineering, applications to specific military problems have been made and will continue to be made.

The small sizes of general purpose atomic bombs now available go down to about a half ton in weight and even smaller sizes are under development. The smaller fission bombs yield about [deleted] whereas the largest fission weapon yields around [deleted]

The advent of thermonuclear bombs brought about three things: (a) the maximum practicable yield of air-delivered bombs was increased by about ten times over that practicable with fission bombs; (b) the comparable cost per megaton was decreased by a factor of about twenty; and (c) the yield per ton of bomb (weight) was increased to about [deleted]

The steps in the future development of thermonuclear bombs are very similar to those in the fission bomb program; namely, the making of thermonuclear compatible with existing or planned delivery systems and in this way, increasing the nuclear capabilities of military forces. This program leads essentially to a variety of general purpose thermonuclear bombs which the military forces can apply to their needs.

....

As noted earlier in this report, the past ten years have seen a significant increase in the explosive yield per ton of bomb. For reasons given below, comparable increases in this characteristic will not occur in the future.

The sources of energy (fission and fusion) now being exploited in thermonuclear bombs will (theoretically) allow ultimately between 15 and 60 times more explosive power per ton of bomb than has now been achieved, depending upon the relative roles of fission or fusion involved in the bomb. This includes no allowance for bomb hardware, and it is not likely to be achieved.

The future can bring only small further increases in the explosive power per ton of nuclear bomb; a factor of two is achievable and a factor of ten might be attained, but certainly not a factor of 60. Planners desiring a thermonuclear bomb or warhead yield of (a

certain number of) megatons will need to think of delivered weights of about the same number of tons. (Author's note: this implies a yield-to-weight ratio of about one-half kiloton per pound of gross bomb weight.)

These limitations, being imposed by nature, will not allow the Soviets to build nuclear bombs with significantly more megatons per ton of bomb (weight) than some already possessed by the U.S. A probable maximum to consider (for both the U.S. and the USSR) is ten megatons per ton of bomb. (Author's note: a yield-to-weight ratio of five kilotons per pound of gross bomb weight.)

It is important to strive for the maximum in megatons per ton of bomb, because delivery is always made easier and cheaper with smaller weight. Only a very small part of U.S. nuclear bombs have [deleted] the other types have ample room for improvement in this characteristic.

With delivery by aircraft, the maximum payload is limited to a bomb yielding tens of megatons, but with delivery by ship or submarine the payload can be a bomb yielding a thousand or so megatons. Such heavy weapons are indeed feasible to make. Comparison of the U.S. and USSR seacoasts show that the U.S. is much more vulnerable to ship-delivered weapons than the USSR.

Clandestine introduction of nuclear bombs, either atomic or thermonuclear, is frequently thought to be limited to small size bombs. Complete atomic weapons do exist, having weights of a few hundred pounds and "suitcase" dimensions, although they are inefficient and have low yield. A perhaps more worrisome possibility is the megaton bomb of large size which could be disassembled into small pieces for clandestine transportation; after introduction, the pieces could be reassembled. It is feasible to make such a weapon.

We conclude that:

(1) U.S. thermonuclear bombs using fission and fusion are already fairly close to their ultimate in terms of megatons per ton of bomb; in other type bombs and in other characteristics there is ample room for improvement.

(2) The ultimate in megatons per ton of bomb, being a fact of nature, applies equally to the U.S. and to the USSR.

(3) Now and for some time, the most efficient thermonuclear bombs will yield no more than about [deleted]

(4) For U.S. defensive planning purposes, the upper limit that should be ascribed to the Soviet bombs is [deleted] a reasonable planning figure is [deleted]

(5) Present U.S. technology could permit the design of a costly weapon yielding a [deleted] but the weight would be many hundreds of tons; we must assume that the Soviets can do the same.

(6) Present U.S. technology would permit the design of megaton (yield) nuclear bombs which could be disassembled into small enough pieces to allow clandestine introduction; we must assume that the Soviets can do the same.

We recommend that:

1. For planning purposes, the maximum yield per weight of nuclear weapons be taken as [deleted]

2. For U.S. defense planning purposes, it be considered technically feasible for the USSR to deliver, by ship or submarine, very large and heavy, though expensive, bombs having yields up to about 1,000 megatons.

3. For U.S. defense planning purposes, it be considered technically feasible for the USSR to construct, for clandestine introduction into the U.S., nuclear bombs having yields as high as megatons.⁹⁵

The committee report also noted that:

...the availability of lightweight

⁹⁵"Meeting the Threat of Surprise Attack," The Report to the President by the Technological Capabilities Panel of the Science Advisory Committee, Volume II, Executive Office of the President, Office of Defense Mobilization, Washington, D.C., February 14, 1955, pp. 53-55.

thermonuclear weapons in quantity will give new significance to the striking power of small aircraft; any light bomber, fighter-bomber, or attack aircraft capable of delivering megaton-yield weapons has a tremendous destructive potential. While small aircraft will be limited in range and will not be able to reach all of the Soviet targets as would the medium and heavy bombers, they nevertheless would be able to reach a significant portion of Soviet targets, even on an unrefueled basis.

The light bombers, fighter-bombers, and attack aircraft planned for deployment by 1958 have the potential of carrying, on a single combined mission, about 10 times the megaton load which could have been carried on a single mission in mid-1954 by the combined heavy and medium bombers of the Strategic Air Command. (The factor of 10 is not an appraisal of the relative military worth of these two hypothetical missions.)

...it must be emphasized that the availability of thermonuclear weapons in quantity will increase by a factor of about 100 the mid-1954 megaton carrying capacity of the medium and heavy bombers. Thus the Strategic Air Command will continue to be a powerful concentration of nuclear striking power.

The short range forces, if equipped with megaton-yield bombs and if deployed around the USSR, would be a force with which the Soviets would need to reckon in their planning for a surprise attack against the U.S. The wide geographical dispersal of the short range forces would compound the Soviet problem of coordinating strikes in Europe, in the Pacific, and in the continental U.S.⁹⁶

Operation REDWING

Operation REDWING, the next set of Pacific tests, was another part of the developmental testing of thermonuclear devices that started in 1951. Operation GREENHOUSE demonstrated the

⁹⁶"Meeting the Threat of Surprise Attack," The Report to the President by the Technological Capabilities Panel of the Science Advisory Committee, Volume II, Executive Office of the President, Office of Defense Mobilization, Washington, D.C., February 14, 1955, p. 69.

feasibility of thermonuclear detonation, IVY detonated a thermonuclear device, and CASTLE tested less awkward solid-fueled devices.

REDWING would test more advanced hydrogen bomb designs and was originally scheduled for the fall of 1954, later rescheduled for the fall of 1955 or the spring of 1956, and finally conducted in the late spring and early summer of 1956,⁹⁷ despite claims by Drs. Edward Teller and Herbert York in early March 1953 that holding the test series as late as the spring of 1956 would "definitely slow down advancement in the development of thermonuclear weapons."⁹⁸

By mid-1953, REDWING was to involve tests of high-yield weapons or devices which might be developed as a result of CASTLE and concurrent weapons development programs. REDWING was also to include the airburst of a high-yield weapon, a shot postponed before CASTLE started.⁹⁹ In June 1954, REDWING was to include five tests: one LASL and two UCRL ground shots, and two LASL barge shots.¹⁰⁰

A month later, these shots were expected to include tests of LASL and UCRL Class "D" devices, a Class "B" weapon proof-test (e.g., a 15,000 lb. shortened and lightened SHRIMP), a Class "C" weapon, and a high-yield (500 KT) boosted weapon.¹⁰¹

While the REDWING tests were not as spectacular or historically significant in some respects as the CASTLE

⁹⁷Atomic Energy Commission, Memorandum for Information, Assignment of Code Names, Note by the Secretary, September 22, 1952; memorandum dated September 7, 1954 to Brig. Gen. K. E. Fields, Director, Military Application, USAEC, from W. B. McCool, Assistant Secretary, USAEC, Subject: Questions Raised in August 30, 1954 Memorandum from Commissioner Murray to the Chairman.

⁹⁸Letter dated March 20, 1953 to Brig. Gen. K. E. Fields, Division of Military Application, USAEC, from Herbert F. York.

⁹⁹Untitled paper reviewing plans for future AEC tests in early 1953, undated (but probably after conclusion of Operation UPGHOT-KNOTHOLE), p. 2.

¹⁰⁰"Schedule of Full-Scale Weapons Tests, FY 1954 - FY 1956," Annex C to "FY 1956 Program Assumptions, Weapons Program, Operations and Construction (3000 and 9300)," Sandia Corporation, March 3, 1954, revised June 14, 1954.

¹⁰¹Minutes of the Forty-First Meeting of the General Advisory Committee to the USAEC, July 12-15, 1954, p. 27.

series had been (there were only two unexpectedly-high yields and solid-fueled weapons were a reality by 1956), they were nonetheless highly significant to the military and the weapons labs.

The devices fired during REDWING were of lower yield and greater variety than those tested in 1954. Many new one-, two-, and three-stage bombs, missile warheads, and test devices, ranging in yield from fractions of kilotons to a few megatons, were fired.

In addition to these proof tests, a number of purely effects tests were conducted to measure radiation, thermal, and blast effects on aircraft, military equipment, structures, animals, and soils from devices too powerful to be fired safely in Nevada. The REDWING tests also featured the first U.S. airdrop of a fully-armed and fuzed thermonuclear gravity bomb.

REDWING and "Clean" Weapons

Another very important objective of the REDWING series was to test "cleaner" warheads that derived most of their explosive power from fusion rather than fission. The bombs detonated during the CASTLE series were big two-stage weapons which generated the majority of their explosive force by the fission of natural uranium jackets around their fusion fuel stages. By comparison, several of the REDWING devices were two-stage weapons deriving up to 95% of their explosive power from fusion alone.

This was an important development toward proving the feasibility of battlefield hydrogen weapons: the less radioactive fallout they generated, the less contamination that would result from their use and the less impedance to friendly ground forces moving around areas where such weapons had been fired.

Reduced contamination would also allow more flexible and "surgical" strikes against both tactical and strategic targets. "Clean" bomb tests might also have the added political effect of muting some of the growing public outcry against continued atmospheric weapons testing.

Studies on how to reduce fallout from thermonuclear weapons began immediately after CASTLE. By early November 1954, LASL had investigated theoretically the behavior of a "clean" version of the TX-21; these early studies indicated that "clean" versions of standard ("dirty") thermonuclear bombs would suffer a significant yield reduction, possibly as much as 66 to 75 percent.

Nonetheless, such a weapon, although still yielding several megatons, might have residual radiation comparable to that

produced by only 200 KT of fission energy.¹⁰² UCRL had two long-range programs underway that might result in "cleaner" weapons.¹⁰³

At the end of the month, AEC Commissioner Thomas Murray stated:

It is my view that important advantages would accrue from early stockpiling of high yield weapons with relatively small amounts of radioactive products per megaton of total energy yield. So important in fact as to require a most vigorous development effort involving the marshaling of our very best talents in the weapon field without impairing the effectiveness of our current weapons program.

It may be that initial surveys will prove discouraging or at best not especially attractive. But this should not deter us from that type of effort which, carried forward in the face of obstacles in the past, has been attended by so much success. Our objective should be to test a weapon of this type at the earliest possible date.¹⁰⁴

Some consideration was given to producing short-lived radioactive contamination for military purposes. If fission reactions were made sufficiently small, then the major cause of radioactive contamination would be isotopes generated by neutron bombardment of the weapon casing. Gold, with a half-life of 2.7 days, was suggested as a case material, but rejected for obvious economic reasons (this would have been real gold-plating).

Dr. Edward Teller suggested in a December 1954 General Advisory Committee meeting that induced radiation caused by high-energy neutrons emanating from a thermonuclear explosion could be minimized by "doping" (impregnating) the dense

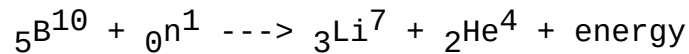
¹⁰²Memorandum dated November 9, 1954 from George I. Bell, Subject: Control of Radioactive Contamination from Large Yield Weapons.

¹⁰³Letter dated November 9, 1954 to Brig. General K. E. Fields, Division of Military Application, USAEC, from Herbert F. York, Director, UCRL-Livermore.

¹⁰⁴Memorandum for Chairman Strauss dated November 30, 1954 from Thomas E. Murray, Subject: Investigations of Different Contaminating Characteristics of High Yield Weapons.

plastic foam in the bomb used to support internal weapon components with boron-10.¹⁰⁵ If enough boron were loaded into the device, neutron emission might be significantly reduced.

The boron-10 isotope has a very high cross section for the following reaction:



Because this reaction generated lithium-7, boron also served as a supplemental source of fusion fuel in the exploding weapon, adding even more power to its yield.

In January 1955, pure boron cost about \$60 per pound, but several minerals containing significant amounts of boron were relatively inexpensive and could be used as "feed" materials to produce high-grade boron enriched in the ${}_5\text{B}^{10}$ isotope.

At the end of 1954, the GAC concluded that there were several methods to reduce the relative fission yield in future U.S. thermonuclear warheads:

In the light of the presentations made to the Committee, we conclude that there are promising possibilities for developing thermonuclear weapons which would generate far less radioactivity than do the present models. Although this would appear to be achievable only at the cost of substantial reduction in yield, there may be military situations in which such a weapon would be preferable to one producing heavy fall-out. We therefore recommend continuing paper studies of this question. Within a few months, such studies should make possible a more detailed and searching consideration of the level at which a development effort

¹⁰⁵Minutes of the Forty-Third Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, December 20, 21, and 22, 1954, Washington, D. C. ${}_5\text{B}^{10}$ has a voracious appetite for neutrons; boron was used as a nuclear "fire-extinguisher" at the late April 1986 Soviet nuclear reactor accident at Chernobyl. Plastic foam contained in a thermonuclear warhead is used as a structural element because it is lightweight, transparent to x-rays, eliminates awkwardly-shaped and radiation-obstructing component supports, and can be fabricated and machined to almost any desired density and shape.

should be pursued.¹⁰⁶

Early in January 1955, both LASL and UCRL submitted memoranda to DOD (via the AEC) concerning the control of radioactive contamination from high-yield thermonuclear weapons. The AEC requested a statement of the possible extent of DOD interest in this subject, noting that "continuing thought will be given by Los Alamos and Livermore to the possibilities for high-yield weapons with both much more and much less associated radioactive contamination."¹⁰⁷

U.S. -- U.K. Cooperation: To Be or Not to Be?

Also at this time, the Joint Chiefs made recommendations to the Secretary of Defense about the type of atomic weapons information that might be shared with the United Kingdom, subsequent to a forthcoming proposed U.S.-U.K. agreement.¹⁰⁸

A letter from Henry Cabot Lodge, U.S. Ambassador to the United Nations, noted that Winston Churchill was unhappy about the lack of cooperation between the U.S. and England; Churchill reportedly said that "the situation was completely unsatisfactory from the British viewpoint, that they had to do (all weapons development) on their own, that they had made a lot of progress, but that the situation between our two countries was not in any way what it should be in this (area)."¹⁰⁹

In March, AEC Chairman Strauss noted that the chief obstacle to better U.S.-U.K. weapons data interchange were

¹⁰⁶Report of the Forty-Third Meeting of the General Advisory Committee, December 20-22, 1954, letter to Lewis L. Strauss, Chairman, USAEC, Washington, D. C., December 22, 1954.

¹⁰⁷Letter dated January 7, 1955 to Donald A. Quarles, Assistant Secretary of Defense (Research and Development), from K. D. Nichols, General Manager, USAEC.

¹⁰⁸Memorandum from the Joint Chiefs of Staff to the Secretary of Defense (Wilson), dated January 26, 1955, document 4 in FOREIGN RELATIONS OF THE UNITED STATES, 1955-1957, Volume 20: Regulation of Armaments; Atomic Energy, John P. Glennon and David S. Patterson (eds.), U.S. Government Printing Office, Washington, D.C, 1990, pp. 10, 11.

¹⁰⁹Letter dated February 23, 1955 to The President from Henry Cabot Lodge, U.S. Ambassador to the United Nations, document 9 in FOREIGN RELATIONS OF THE UNITED STATES, 1955-1957, Volume 20: Regulation of Armaments; Atomic Energy, John P. Glennon and David S. Patterson (eds.), U.S. Government Printing Office, Washington, D.C, 1990, pp. 35, 36.

"deficiencies in the U.K. personnel security system" and that the British were not even adhering to a 1952 U.S.-U.K. agreement on minimum security review requirements for English citizens who would have access to U S.-generated restricted data. Until the British improved their personnel security clearance procedures, the AEC was strongly opposed to sharing weapons data.¹¹⁰

In early March, the Secretary of Defense wrote to AEC Chairman Strauss and stated DOD opinions on the military advantages of, and possible future requirements for, "clean" weapons:

Information that we have received from the (Atomic Energy) Commission and studies that have been made by Defense Agencies (sic) all indicate the great importance that should be attached to the radioactive by-products of high-yield bombs.

Until the CASTLE tests confirmed the feasibility of megaton yields at comparatively small cost, military economy in the atomic weapons field had been largely dominated by blast effects and means of maximizing these (effects) in relation to design and delivery costs.

As important as these blast considerations still are, we are now confronted with perhaps even more important considerations in the radioactive by-products field.

Stated broadly, the problem appears to be that of maximizing the military effect at the desired time and place, and minimizing such effects where they are not desired. While blast effects are essentially instantaneous and local, the radioactive effects may cover very large areas and may persist for very long periods ranging, in fact, from days in the local fallout effects to many years in atmospheric contamination effects.

¹¹⁰Letter dated March 3, 1955 from Lewis Strauss, Chairman, USAEC, to The President regarding negotiations with U.K. about exchange of restricted data, document 11 in FOREIGN RELATIONS OF THE UNITED STATES, 1955-1957, Volume 20: Regulation of Armaments; Atomic Energy, John P. Glennon and David S. Patterson (eds.), U.S. Government Printing Office, Washington, D.C., 1990, pp. 37, 38.

In other words, radioactive effects force us to bring time in as an additional dimension in dealing with this problem. Moreover, the areas subject to lethal radiation are so large, that in planning the use of these weapons we must carefully weigh the damage to friendly as well as enemy installations.

The Secretary thought that methods should be devised to increase the total yield of "clean" weapons:

The Department (of Defense) understands that investigations undertaken to date indicate a fair probability that high-yield thermonuclear weapons could be designed for greatly decreased long-life radioactive effects at the expense of a considerable reduction in blast and thermal effects for the same expenditure of critical materials; and that developments along this line could be pursued on a relatively short time schedule.

It is understood also that there is some possibility that an entirely new approach to the problem, with intensive effort over a longer period, might result in designs which would minimize long-life radioactive yield without much serious compromise of other effects.

The Department believes that the problems associated with the use of high-yield weapons are of such an order as to justify continued vigorous studies on all aspects of the problem, to include further explorations of methods of both increasing and decreasing the radioactive yield of thermonuclear weapons.

These studies, in combination with military studies as to the strategic and tactical implications of the use of such weapons, should provide a firmer basis for evaluating the extent of the requirement for the development of a so-called "clean" bomb. Should our studies reveal the need for development of such a weapon on a priority basis, the Department will wish to furnish further guidance.¹¹¹

On June 15, 1955, the U.S. signed a set of bilateral

¹¹¹Letter dated March 5, 1955 to Lewis Strauss, USAEC, from C. E. Wilson, Secretary of Defense.

Agreements for Cooperation with Canada and the United Kingdom on civil aspects of atomic energy; the portion of the agreement pertinent to the U.K. included additional limitations on the exchange of restricted data.

AEC Commissioner Thomas E. Murray opposed the agreement, because he was 'convinced that the security problems involved have not been resolved in the best interests of the United States." Murray cited a February 1955 AEC memo from the director of the commission's Division of Security which stated that a new U.K. personnel clearance program was not adequate to protect U.S. information classified higher than Confidential, and that personnel clearance investigations were perhaps comparable to the AEC's "L" clearance level, a much lower level than the requisite "Q" clearance for access to weapons information. Murray noted:

If we accept the present proposed U.K. (clearance) standards, we would in effect be admitting that our own security program is unnecessarily stringent. Also, it would be unfair to our own citizens to enter into an agreement with the U.K. establishing a double standard that would in effect be more restrictive of them than of U.K. subjects.¹¹²

Objectives of REDWING

As was done during the CASTLE series in the Pacific in 1954 and the TEAPOT test series in Nevada in 1955, the REDWING series continued the newly-established practice of the two weapons labs at Los Alamos and Livermore testing experimental devices based either on new nuclear technology concepts or on the wishes or requirements of the Department of Defense for warheads of specified sizes and yields for particular delivery vehicles.

Operation REDWING was to include tests of high-yield "sealed pit" weapons, which were bombs and missile warheads with external neutron generators for their primary stages. These weapons could be maintained in combat-ready status aboard bombers or atop missiles for long periods, requiring only periodic refreshment or replacement of the tritium gas reservoirs used to boost their primaries.

Long-term maintenance of these "sealed pit" weapons consisted essentially of pressure checks and continuity tests, and simple component replacement. Complex component replacement

¹¹²Statement of Commissioner Thomas E. Murray concerning Agreement for Cooperation with the U.K., undated but ca. June 9 or later, 1955.

and modifications could be performed either by traveling technician teams, or by return of the weapon to a maintenance depot.¹¹³

In many cases, devices fired at REDWING were not stockpiled weapons but were nuclear test systems that could be weaponized later (this had also been true of most of the devices fired during Operation CASTLE).

As test devices, the devices often had not been assigned TX, XW, or MK numbers, and instead bore cryptic mnemonic names whose significance was known only to their designers. Devices and prototype warheads tested by the two weapons labs during REDWING would, in the near and distant future, influence or confirm the development and operation of the MK 21, 25, 27, 28, 35, 36, 38, 39, 40, 41, 43, 44, and 45 weapons.

The REDWING series was especially important to the scientists at the fledgling University of California Radiation Laboratory: although members of this lab had participated in diagnostic experiments at every test series since Operation IVY, their first three attempts at bomb firings during Operations UPSHOT-KNOTHOLE and CASTLE had ended in embarrassing "fizzles." After successful shots during TEAPOT, REDWING would either prove or destroy the viability of the second weapons lab.

The primary raison d'être of the second laboratory, sponsored largely by the personal efforts of Dr. Edward Teller and the institutional lobbying of the U.S. Air Force, was the early design and stockpiling of large, high-yield bombs for the Air Force.

While Livermore finally tested successful fission devices at Operation TEAPOT in Nevada in 1955, UCRL did not have a successful fusion device shot until 1956, nearly four years after the lab was established. By this time, Los Alamos had already successfully designed, tested, and provided for stockpiling several different high-yield H-bombs.

On November 27, 1953, before the failure of the MORGENSTERN device during Operation CASTLE, UCRL had proposed a pre-REDWING series called "the QAD Operation," for "Quick and Dirty", with a single airdrop test in the winter of 1954-1955 and a second drop four to six months later.

The purpose of these explosions was to confirm the designs and measure the yields of two high-yield multistage thermonuclear warheads which were prototypes of possible

¹¹³Teletype dated 29 May 1958 to USCINCEUR, Paris, France from DA, Washington, D.C., subject: Optimized Sealed-Pit Weapons.

stockpile weapons and follow-ups to the solid-fueled MORGENSTERN (as proposed in the UCRL technical program for calendar year 1954 and FY 1955).¹¹⁴

These devices, which would be ready for testing long before Operation REDWING was conducted, were to be fitted into ballistic casings, equipped with Sandia Laboratory-designed fuzing and firing systems, and airdropped over the western side of Eniwetok atoll. This proposal became irrelevant when the MORGENSTERN device failed during CASTLE, and the other UCRL shot, a test of the RAMROD device, was subsequently canceled.¹¹⁵

In lieu of the canceled "QAD" program, UCRL proposed on April 7, 1954, while CASTLE was still in progress, that the laboratory detonate four low-yield experimental nuclear devices in Nevada during the fall of 1954; this proposed series was to be called "TEAPOT I" and the planned spring 1955 test series would be redesignated "TEAPOT II."

LASL indicated that if the UCRL proposal were approved, then Los Alamos would like to test two small yield devices previously slated for detonation in the spring of 1955 but which would be ready by the fall of 1954. The proposed "TEAPOT I" operation was expected to start in either late October or early November 1954 and conclude by mid-December. After "TEAPOT I" was concluded, four to seven "TEAPOT II" tests would still probably be required in 1955.

UCRL's four shots were predicted to yield from 0.1 to one KT; LASL's tests of proposed 30- and 16-inch diameter spherical implosion air-defense devices would yield between one and two kilotons.

The two LASL systems were "specialized and maximum fissionable material economy two-kiloton 30-inch implosion type devices for potential air defense and tactical purposes;

¹¹⁴Letter dated December 16, 1953 to J. J. Flaherty, Manager, San Francisco Operations Office, USAEC, from Herbert F. York and E. O. Lawrence.

¹¹⁵Memorandum dated November 13, 1953 to Brig. Gen. K. E. Fields, USA, Director of Military Application, USAEC, from Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, Subject: AEC Participation in WIGWAM; letter dated November 27, 1953, from Herbert F. York, Director, University of California Radiation Laboratory, Livermore, California, to Brig. Gen. Kenneth E. Fields, U.S. Atomic Energy Commission, Division of Military Application, Washington, D. C. One of the devices under consideration for test may have been the UCRL TX-22, which was canceled at the end of April 1954.

and a 16-inch implosion system in the one to two KT range with minimized fissionable material to provide precise information on the active material costs of systems for air defense in this size range."

Both LASL experiments were to be airbursts at 1,000 feet with minimal scientific support except for yield and alpha measurements.¹¹⁶

By late July 1954, the AEC was not planning any further Pacific tests until some time in 1956. At least one AEC Commissioner, Thomas E. Murray, felt that this time schedule was not "consistent with optimum progress in weapons development."¹¹⁷

Operation REDWING was designed to achieve four objectives vital to continued progress in research, development, and improvement of atomic and thermonuclear weapons.¹¹⁸ These were:

1. To proof-test certain weapons currently in

¹¹⁶Memorandum dated April 7, 1954 to Brig. Gen. K. E. Fields, USA, Director of Military Application, USAEC, from Donald J. Leehey, Manager, Santa Fe Operations, USAEC, Subject: Full-Scale Test Proposed for the Fall of 1954.

¹¹⁷Memorandum for Chairman Strauss dated August 30, 1954 from Thomas E. Murray.

¹¹⁸The following history of Operation REDWING, except where noted, is drawn from OPERATION REDWING, Report of the Manager, Albuquerque Operations, Pacific Proving Ground, Spring 1956, pp. 20-27, 29; OPERATION REDWING, Operation Order No. 1-56, J3-430 (EX), Task Group 7.1, Headquarters, Task Group 7.1, Joint Task Force Seven, San Francisco, California, 25 January 1956; OPERATION REDWING, Commander Task Group 7.3, Final Report, March-August 1956, Pacific Proving Grounds, pp. 5-7, 58-61, 72; A HISTORY OF OPERATION REDWING: THE ATOMIC WEAPON TESTS IN THE PACIFIC, 1956, Headquarters, Joint Task Force Seven, Washington, D. C., 1 December 1956, pp. iv-vi, 1-21, 23-31, 66-83, 86-90; OPERATION REDWING, WT-1359 (EX), Report of the Commander, Task Group 7.1, May-July 1956, pp. 11-15, 22-34, 38, 62, 63, 66, 72-75, 79-81, 83-85, 88, 89, 91-95, 100-102, 104-106, 113, 138, 143-148, 154-159; OPERATION REDWING, Technical Summary of Military Effects, Programs 1-9, WT-1344 (EX), Pacific Proving Grounds, May-July 1956, Headquarters Field Command, Defense Atomic Support Agency, Sandia Base, Albuquerque, New Mexico, April 25, 1961; and OPERATION REDWING 1956, DNA 6037F, Defense Nuclear Agency, Washington, D. C., 1 August 1982.

the national stockpile or scheduled to be stockpiled in the near future.

2. To continue developmental research on promising weapons.
3. To continue long range weapon research of new techniques, ideas, and designs.
4. To conduct a weapons effects program required by the Department of Defense.

A multiplicity of projects was incorporated in all shots to obtain data for developmental research on promising weapons and for DOD weapons effects programs (all of the REDWING shots except Cherokee tested new weapon developments).

Ten Los Alamos devices were finally scheduled for detonation during Operation REDWING. Some of these shots were proof tests of devices suitable for weaponization and stockpiling; others were experimental devices of a more forward-looking nature which could be weaponized if their performance proved sufficiently attractive.

These included both Class "C" bombs for medium-weight bombers and Class "D" warheads for missiles and for external carriage on fighter planes. By the end of March 1955, plans were being made to test at least one reduced contamination ("clean") weapon during REDWING.¹¹⁹

The LASL-designed MK 15 had entered the U.S. nuclear weapons stockpile in April 1955; the MK 21 in December; and the MK 36 in April 1956. In addition, the TX-25, TX-28, XW-35, MK 39, XW-40, XW-43, XW-44, and XW-45 weapons were all in various stages of study, design or engineering development at Los Alamos in mid-1956 (final development of the W-45 was later awarded to UCRL).

The newly-stockpiled bombs and several of the prototype warheads were all candidates for testing at REDWING. A number of the TX-28 variants were put on the REDWING shot schedule.¹²⁰

¹¹⁹Summary extract of AEC-MLC Meeting No. 96, March 31, 1955.

¹²⁰Letter dated 24 February 1956 from William M. Ogle, LASL Test Director, to Brig. General Alfred D. Starbird, Director of Military Applications, U.S. Atomic Energy Commission, Washington, D.C., Subject: REDWING Quickies. "Quickies" were 16MM black-and-white ten- to-twenty minute long films made after each shot describing and depicting briefly but in great detail the test objective, test device, and preliminary results. (Letter dated February 17, 1956 to RAdm. B. H.

UCRL planned to test and obtain diagnostic measurements on seven devices during Operation REDWING. Among these were prototypes of the MK 27 warhead which UCRL had begun developing in July 1955 (using a Los Alamos-designed primary) and the XW-38 missile warhead.

In addition, two models of a large thermonuclear device (which would lead to the MK 41 aerial bomb) were ready for testing, along with a number of small boosted fission devices suitable for use as air defense warheads.

One of the significant design features of several of the contemporary high-yield two- and three-stage LASL and UCRL devices was the relatively-low fission yield and relatively-high fusion yield. This was quite an advance over the "dirty" bombs tested during Operation CASTLE. These strides in "clean" bomb design were considered at the time of the tests to be "extremely important information which should be closely protected."¹²¹

On each of the LASL and UCRL shots, diagnostic experiments would be made to evaluate its performance, if the device worked properly, or to help determine the reason for failure, if it did not work. In addition, many experiments would be undertaken to obtain more precise understanding of the underlying factors and physical properties of nuclear weapons explosions and their effects.

A wide range of military effects programs, the most extensive yet, were also planned for REDWING. Forty-seven separate projects were conducted under eight programs. Nearly 1,000 men from DOD, AFSWP, and the AEC were involved in these experiments and observations. While the objectives of the programs were numerous, the three major purposes were:

- a. To obtain the special weapons delivery capabilities of late-model Air Force and Navy aircraft, including the Air Force's B-47E, RB-52B, B-57B, B-66B, F-84F and F-101A, and the Navy's new A3D-1.
- b. To document the radioactive fallout from high yield devices, including the initial and final distribution of activity, the time history of accumulation locally and at sea, and the physical and chemical

Hanlon, USN, Commander, JTF 7, from Brig. Gen. Alfred D. Starbird, USA, Director of Military Application, USAEC.)

¹²¹Minutes of AEC Meeting No. 1215, 3:30 PM, Thursday, July 12, 1956.

nature of the radioactive material.

- c. To document the basic blast, thermal, and nuclear radiation effects from a high yield air burst.

A major political objective of REDWING was to demonstrate to the Soviet Union that the newly-deployed B-52 was capable of delivering nuclear weapons. A high-yield airburst was planned; for this shot only, and for the first time in the Pacific since Operation CROSSROADS at Bikini in 1946, American newsmen would be invited to witness the test.

The Secretary of Defense requested (and received) approval from President Dwight Eisenhower to expend a MK 15 ZOMBIE bomb, complete with nuclear components, from the national stockpile.¹²² A 5,000-foot HOB (height of burst) was chosen to allow for an air zero point 1 1/2 times the anticipated maximum fireball radius, corresponding to a one to one-and-a-half megaton explosion (within the yield range of the ZOMBIE).

In addition, it would have been very risky to airdrop an untested bomb over deep water: if it failed to fire, recovery would have been extremely difficult. The objectives of the Cherokee shot were twofold: to demonstrate U.S. capability to carry hydrogen bombs to the Soviet Union, and to determine the effects of a multimegaton airburst on a B-52.

REDWING Planning and Organization

Upon the completion of Operation CASTLE in May 1954, JTF 7 and J-Division at Los Alamos began preparations for the next Nevada operation (which became Operation TEAPOT) and for REDWING (a year later, REDWING was scheduled for the fall of 1955 or the spring of 1956).¹²³ As early as the end of 1952, the name REDWING had been assigned to a Pacific test series.¹²⁴

¹²²The REDWING Cherokee shot was a MK 15 weapon test. The device dropped from a B-52 was a Class "C" stockpiled thermonuclear bomb; the MK 15 was a stockpiled Class "C" weapon. The Cherokee device was 34.5" in diameter, 136" long, and weighed 6,867 lbs. The MK 15 gravity bomb was 34.7" in diameter, 136" to 140" long, and weighed around 7,500 lbs.

¹²³Memorandum for the Chairman, U.S. Atomic Energy Commission, dated December 17, 1953, from Major General James E. Briggs, Acting Chairman, Department of Defense Military Liaison Committee, Subject: Future Tests.

¹²⁴Atomic Energy Commission, Memorandum for Information,

Within a few months of the end of CASTLE, LASL and UCRL knew in general the devices they wanted to test in REDWING. They estimated dates when they would need to verify their theoretical understanding of the new designs with actual detonations.

The major organizational change between CASTLE and REDWING was the establishment of four new task units embodying LASL, UCRL, DOD and Sandia programs, and eight support task units providing timing, firing, radiological safety, photographic, and device assembly services. The cryogenic fuel Production (H. L. Johnston) and Cambridge Corporation task units, TU 2 and TU 3, used at CASTLE were no longer required and were disbanded.

After the completion of CASTLE, planned tests series were TEAPOT, scheduled for the late winter and spring of 1955, and DIXIE in the fall of 1955, both in Nevada. By December 1954 much work had been done on the Nevada shot programs but comparatively little on REDWING. A nebulous list of probable shots had been drawn up by that time, including both LASL and UCRL devices. Since this program appeared similar in scope to CASTLE, preliminary estimates of military support requirements were made accordingly.

The Air Force and AFSWP wanted to include in the test program a megaton-yield weapon dropped from an aircraft, a request first made in 1953. Such a drop would be the first attempted by the U.S. and it would determine how effectively a high yield weapon could be wedded to an operational heavy bomber.

In addition, even as late as September 1954, there had never been a full-scale nuclear test of a standard stockpiled weapon selected at random.¹²⁵ The laboratories agreed to the proposal and the Joint Chiefs of Staff concurred. Only the details remained to be decided: where, when, and at what altitude the burst would take place.

Debate over "clean" low-contamination bombs continued in parallel with REDWING planning. By early January 1955, the MLC had not yet received comments of either Dr. York of UCRL or Dr. Bradbury of LASL regarding this subject (preliminary reports had been furnished by the labs in early November 1954).¹²⁶ It was likely that there would soon be a DOD

Assignment of Code Names, Note by the Secretary, USAEC, Washington, D. C., December 31, 1952.

¹²⁵Minutes of Atomic Energy Commission Meeting No. 1025, 2:15 PM, Wednesday, September 15, 1954.

¹²⁶Letter dated January 7, 1955 to Honorable Donald A. Quarles,

requirement issued for a low-contamination, high-yield bomb.

DOD was primarily interested in the near-term feasibility of developing such a weapon, rather than in a quick improvisation for immediate test. Merely altering present weapons to reduce contamination would result in a considerable decrease in yield, so the desirability of doing this was therefore questionable. Far better would be a program to develop a "clean" weapon with the highest possible yield for its weight class.¹²⁷

Later in the month, the AEC agreed on the urgency of obtaining a reduced contamination thermonuclear weapon and discussed means of pushing development, including the possible establishment of a third (!) U.S. nuclear weapons laboratory. The Air Force was interested in such a weapon, even if it entailed a loss in yield. If devices tested by Livermore during Operation TEAPOT were successful, UCRL would follow up with work on a "clean" thermonuclear bomb.

The AEC hoped that either UCRL or LASL would have a reduced contamination bomb ready for test in the Pacific during Operation REDWING.¹²⁸

On February 15, representatives of LASL, UCRL, and the Santa Fe Operations Office (SF00) of the AEC met to discuss the DIXIE and REDWING programs. Plans were made to conduct DIXIE in the spring of 1956, as the fall of 1955 was too early to allow proper preparations. LASL specifically needed more time to incorporate the results of Operation TEAPOT tests into current weapons then under development.¹²⁹ LASL and UCRL would be in a better position after TEAPOT to estimate DIXIE requirements.

LASL expected to conduct two one-point safety tests and a

Assistant Secretary of Defense (Research and Development), from K. D. Nichols, General Manager, USAEC.

¹²⁷Minutes of 95th AEC-MLC Meeting, January 6, 1955.

¹²⁸Minutes of AEC Meeting 1056, January 25, 1955; memorandum dated January 25, 1955 to K. D. Nichols, General Manager, USAEC, from John H. Stumpf, Recording Secretary, Subject: Recap of Meeting 1056, January 25, 1955.

¹²⁹Letter dated January 28, 1955, from Norris E. Bradbury, Director, Los Alamos Scientific Laboratory, to Brig. Gen. K. E. Fields, Director of Military Application, USAEC, Washington, D. C.; memorandum dated February 1, 1955 to W. B. McCool, Secretary to the Commission, from Col. Vincent G. Huston, USAF, Acting Director of Military Application, USAEC, Subject: Atomic Weapons Test Operations.

test of a primary for a two-stage, small moderate-yield thermonuclear weapon; the overall LASL DIXIE program might be three or four tower shots of relatively-low yield. The laboratory did not expect that any warhead stockpile dates would be unduly delayed whether DIXIE was conducted in either late 1955 or early 1956.

LASL expected to know its DIXIE requirements after TEAPOT and delay of DIXIE until the spring of 1956 would allow the lab time to assimilate TEAPOT results into practical experiments. If DIXIE were conducted in early 1956, REDWING might be delayed until the late fall of 1956 or even to the early spring of 1957. LASL believed that there was great virtue in full-scale nuclear testing of a weapon prototype fairly late in its development cycle in order to have a device which would closely approximate the stockpile model.¹³⁰

UCRL expected to fire during DIXIE two linear implosion shots similar to those in TEAPOT and possibly a third shot similar to the TEAPOT Turk test.¹³¹

On the other hand, if DIXIE were delayed, the spring of 1957 was too late for REDWING because of the growing number of devices that would require full-scale tests by then. The fall of 1956 was unsatisfactory for REDWING because of its proximity to DIXIE. Given these conditions, UCRL and LASL recommended to the AEC that DIXIE and REDWING be combined and scheduled for the PPG in the spring of 1956.

Two days later, the political ramifications of conducting REDWING during an election year were raised by DOD:

The political implications of testing weapons in a presidential election year have been discussed. Mr. Strauss mentioned this matter in a query to General Fields. Col. Huston stated that, in his opinion, the political repercussions from REDWING would be spent before election time in 1956 if the REDWING series were conducted in the Spring (sic) of 1956 as planned.

The political considerations in this field are

¹³⁰Letter dated January 28, 1955 to Brig. Gen. K. E. Fields, Director of Military Application, USAEC, from N. E. Bradbury, Director, LASL.

¹³¹Atomic Energy Commission, Atomic Weapons Test Operations - Dixie and Redwing, Note by the Secretary, AEC 787, February 16, 1955.

real. During Operation IVY, General Clarkson was requested to delay Mike shot until after the November 1952 election. He (chose) to fire as soon as conditions permitted regardless of the election. Mike shot was detonated on 1 November 1952 without apparent political effects.¹³²

By the end of February, DIXIE tests were to be combined with REDWING; Operation JULEP, a continental test series planned for the fall of 1956, would probably be held to test several UCRL devices.¹³³

At about this time, the AEC strongly urged UCRL to proceed with the development of "clean" weapons:

You will recall that during my last trip to Berkeley and Livermore ... we discussed possibilities of relatively low fission content weapons and a Livermore effort in such a direction. Since my return to Washington, the Assistant Secretary of Defense for Research and Development has discussed this subject with the Chairman of the (Atomic Energy) Commission and indicated the desire of the Department of Defense to push the development of such a weapon.

The Commission also considers that this is an urgent project and that there should be instituted at the earliest (moment) a vigorous and extensive effort to disclose fruitful lines of attack and to prosecute ... those that offer promise as rapid(ly) as possible. The Commission would hope that such tests as might be feasible and contributory could be included in REDWING.

In view of the above and of UCRL's informal

¹³²Memorandum for Admiral Momsen dated 17 February 1955 from Col. David O. Byars, Jr., U. S. Army, subject: Delay in REDWING Schedule.

¹³³Minutes of AEC Meeting 1061, February 23, 1955; letter dated 24 February 1955 to RAdm. C. B. Momsen, Commander, Joint Task Force Seven, from Alvin C. Graves, J Division, LASL; Atomic Energy Commission, Test Schedule - REDWING, DIXIE, Report to the General Manager by the Director of Military Application, March 31, 1955; memorandum dated 24 March 1955 to Commander, AFSWC, from Co. H. E. Parsons, USAF, Director, Weapons Effects Tests, AFSWP, Subject: Forecast of Test Requirements.

indication of possible technical approaches of interest, you are requested to submit a proposal as to possible early realignment of your (weapons) program toward a major effort in the field of clean weapons.

We believe the approach should be comprehensive and aimed at radically reducing fission contribution without excessive losses in yields. However, because this is a field in which no present capability exists, we believe the gamble can and should be accepted.¹³⁴

DOD was also expressing its interest in this new technology:

... the Secretary of Defense was prepared to sign a letter on DOD interest in developing thermonuclear weapons with various contaminating characteristics. This letter would state the DOD position that more study on the question of developing both a reduced contamination weapon and an increased contamination weapon is needed before the Department would be in a position to state a development requirement, but that the Department is extremely interested in such weapons.¹³⁵

LASL now planned to fire at least nine devices during REDWING.¹³⁶ Tentative plans were made to detonate all the medium and high-yield REDWING shots at Bikini and the low-yield shots at Eniwetok, with a two-atoll capability to allow small shots to be fired in almost any weather while awaiting good weather for large shots.

With a dual-site capability, Operation REDWING, which finally included almost three times the number of shots as Operation CASTLE, should not take much longer to complete than CASTLE. REDWING might even be shorter than the much-protracted CASTLE, since the number of high-yield shots scheduled was

¹³⁴Letter dated February 9, 1955 to Dr. Herbert York, Director, University of California Radiation Laboratory, from Brig. Gen. K. E. Fields, Director of Military Application, USAEC. This is one of the earliest AEC letters that used the term "clean weapons."

¹³⁵Minutes of the 96th AEC-MLC Meeting, February 24, 1955.

¹³⁶Letter dated 24 February 1955 to RAdm. C. B. Momsen, Commander, Joint Task Force Seven, from Alvin C. Graves, J Division, LASL.

fewer than CASTLE's five.

Toward the end of February, Alvin Graves of LASL mentioned the possibility of a megaton-range weapon airdrop during REDWING. By early May, the airdrop was part of REDWING plans.¹³⁷

On March 5, the Secretary of Defense wrote to the AEC, stating that DOD believed that the problems associated with the use of high-yield weapons were of a magnitude sufficient to justify continued investigation of all aspects of weapons design, including thermonuclear weapons with both increased and decreased radioactive yield.¹³⁸

Spring 1955: A Possible Test Moratorium Looms

Despite the onrushing plans for REDWING, at least one AEC commissioner believed that there was still a possibility of a world-wide nuclear test moratorium. Thomas Murray urged President Eisenhower to pursue a moratorium, especially since the U.S. now had a commanding technological lead over the Soviets:

Dear Mr. President:

In my letter of February 15, 1954, I presented for your consideration thoughts concerning an attempt to negotiate a weapons test moratorium. May I take this opportunity (to expand) those thoughts, particularly in light of the events of the past year.

The most important points to be considered in this connection are:

- (1) The United States is far ahead of the USSR in the field of thermonuclear weapons.
- (2) Weapons testing is essential for rapid development.

¹³⁷FINAL HISTORY, TASK GROUP 7.4, OPERATION REDWING, 1955-1956, M/Sgt. William A. Evans, Historical Section, 4950th Test Group (Nuclear), Air Force Special Weapons Center, Air Research and Development Command, February 1957, p. 48.

¹³⁸Memorandum for the Chairman through the General Manager dated January 22, 1957 from Brig. Gen. Alfred D. Starbird, Director Of Military Application, USAEC, Subject: Clean Weapons.

As you know, the United States has exploded a total of 56 nuclear weapons. Of these, seven have been thermonuclear weapons with yields as high as fifteen megatons. [deleted]

This testing has given us a weapons technology that is highly advanced. So much so, that we could accept a delay of a year or more in testing weapons of yields greater than 100 kilotons without our progress being greatly hampered.

In contrast, with our experience, [deleted] the Soviets have exploded [deleted]. We know that at best only [deleted]

Analyses of the 1945 Soviet test indicates that, contrary to expectations, they did not include a thermonuclear device. On the other hand, our 1954 tests and the current Nevada series did include thermonuclear devices and have enabled us to take tremendous steps forward in this field.

My second point is that testing is essential for the rapid development of nuclear weapons. Our scientists agree that, although some weapons development may continue without testing, the absence of such tests would slow progress to an appreciable degree.

It is my deep conviction that a moratorium on the testing of large thermonuclear weapons would lengthen the time during which the United States would maintain its advantage over the USSR. Our experience is so much more extensive than that of the Soviets that we could use tests of small weapons and components to much greater advantage than they.

Then again, the United States does not plan to test large thermonuclear weapons for (another) year. From what we have been able to detect of the Soviet pattern, we should expect their next test series to take place this Fall (sic). Thus, if a proposal is made to defer tests of large thermonuclear weapons is made soon, the Soviets would be the first to be affected. ...

....

I continue to be a firm advocate of expanding our capabilities in the nuclear weapons field. Under this moratorium proposal, preparations

for (the REDWING) tests next Spring (sic) would be continued with the same vigor as at present. Likewise, we would continue with our extensive program of weapons development.

In conclusion, the information available to me supports the view that, with appropriate safeguards, a moratorium on the testing of large thermonuclear weapons would act to maintain and advance our weapons superiority over the Soviets, and thus would be in the interests of the United States.¹³⁹

The GAC had addressed the question of a test moratorium during its meetings in December 1954 and March 1955. By March, the GAC had concluded:

It is our view that progress can be made in weapon development even without final proof tests. It is possible, by testing scale models and components, together with extensive theoretical calculation, to design weapons which have a very fair assurance of giving the predicated yields, particularly if designs are chosen which, because of their simplicity or symmetry, are susceptible to calculation.

This point is borne out by past experience. A weapons moratorium would slow up progress in weapons development but would not stop it entirely. It is therefore possible that a foreign power could improve its position with respect to nuclear weapons even though it honored an agreement with respect to a test moratorium.

Likewise, we ourselves could make progress in weapon development even with a test moratorium, but at a much slower rate. The political effects of a test moratorium are certainly of importance but beyond the competence of this committee to evaluate.¹⁴⁰

¹³⁹Letter dated March 14, 1955 to President Eisenhower from AEC Commissioner Thomas E. Murray, document 15 in FOREIGN RELATIONS OF THE UNITED STATES, 1955-1957, Volume 20: Regulation of Armaments; Atomic Energy, John P. Glennon and David S. Patterson (eds.), U.S. Government Printing Office, Washington, D.C, 1990, pp. 56, 57. The reference to thermonuclear tests in "the current Nevada series" may be to tests of Class D weapon primaries from LASL and UCRL.

Near the end of the month, REDWING was to embrace a wide range of tests aimed at a number of objectives:

Present plans call for the next test operation to take place at the Pacific Proving Ground in early 1956. This test series will retain the code name REDWING as was previously contemplated.

Whereas previous plans embodied conducting DIXIE at the Nevada Test Site during the latter part of 1955, it is now considered advisable to integrate that test program with REDWING. This will enable the laboratories to make adequate preparations so as to maximize the benefits to be derived from the proposed tests.

It is visualized that the next test series after REDWING will take place in calendar (year) 1957. The decision as to whether it will be conducted as a continental or overseas test will be made as soon as possible but may be no earlier than the latter part of REDWING when the scope and magnitude of the 1957 series will be more accurately known.

Preliminary plans (for REDWING) include three tests of 2 KT or less for one-point detonation (safety tests) and air-to-air rocket (warhead) development; two or three 5-20 KT [deleted] development; one 50-100 KT for fission weapon development; three thermonuclear devices in the low megaton range; and an air-drop of a device in the megaton range.¹⁴¹

At this time, the AEC informed the JCAE that Operation DIXIE had been integrated with REDWING and that the Pacific test series would probably comprise 11 barge, land, airdrop, and tower shots: four by UCRL and seven by LASL. REDWING would include three devices with megaton range yields, one of which might be an operational airdrop of the MK 15 bomb. Combining DIXIE with REDWING would allow more time for the labs to digest results of TEAPOT and incorporate them into REDWING test devices.¹⁴²

¹⁴⁰Letter dated March 5, 1955 to Lewis Strauss, Chairman, USAEC, from I. I. Rabi, Chairman, GAC, p. 2.

¹⁴¹Letter dated March 23, 1955 to Commander, Joint Task Force Seven, from Brigadier General K. E. Fields, USA, Director of Military Application, USAEC.

Later in March, the MLC stated that development of entirely new Class "A" and Class "B" weapons as replacements for the MK 24 and MK 21, respectively, should be deferred until some of the more undeveloped Class "C" and "D" bombs had evolved further. In the meantime, existing MK 17/24 and MK 21/26 weapons should be improved, mainly by increasing their yields.¹⁴³

In mid-April 1955, the 1956 PPG series was to begin in March 1956; at least eleven shots were planned.¹⁴⁴ Instead of six large shots previously contemplated, there would probably be three in the megaton range, one of which might be an airdrop. There would also be at least seven lower-yield (kiloton range) tests.¹⁴⁵

Because of the proposed locations of the shots, requirements for L-20 aircraft, helicopters, boats and vehicles would be greater than for CASTLE, but the construction requirements for scientific test structures would be much smaller. A tentative ready date of March 15, 1956 was set for the first shot at each atoll (Bikini and Eniwetok).

Among aircraft being considered at this time for high-altitude sampling of fission products from mushroom clouds during both TEAPOT and REDWING were the Martin B-57A and B-57B, two-place, twin-jet engine Air Force medium bombers. The B model had a few advantages over the A for this mission. The B-57A, in turn, had succeeded F-84G and T-33A sampler aircraft; these piloted planes replaced unmanned drone B-17s and smaller aircraft used previously during both Nevada and Pacific tests.

Specifically, the B-57B had more effective dive brakes, allowing faster egress from radioactive clouds and quicker return to base, thereby minimizing crew exposure to radiation emanating from material clinging to the skin of the plane, and the design of the B-57B canopy allowed better visibility and more effective defrosting than that of the A model.

¹⁴²Letter dated March 23, 1955 to Senator Clinton P. Anderson, Chairman, JCAE, from Lewis Strauss, Chairman, USAEC.

¹⁴³Memorandum for the Chairman, USAEC dated March 24, 1955 from Herbert B. Loper, Chairman, MLC, Subject: Thermonuclear Weapons program.

¹⁴⁴Unsigned U.S. Navy Bureau of Ships memorandum dated 5 April 1955, Subject: Planning for Atomic Warfare Countermeasures Tests in Operation REDWING.

¹⁴⁵Evans, FINAL HISTORY, TG 7.4, p. 2.

In other respects, performance of the two types of aircraft was similar with the same basic weight configuration. The Air Force believed that the advantages offered by the B-57B might well make the difference between success and failure of high-altitude sampling missions over the Marshall Islands during REDWING.

This was not a minor factor, considering that the collection of satisfactory samples of the debris remaining after the explosion of nuclear or thermonuclear experimental devices or weapons formed an important part of the field operations during tests of these devices. Radiochemical analysis of samples of cloud debris provided key information about the energy release and internal processes occurring during the explosion of some of the more-complex test devices.¹⁴⁶

Operation TEAPOT: Prelude to REDWING

During the spring of 1955, many of the test personnel from LASL, UCRL, AFSWP, and Sandia participated in Operation TEAPOT at the Nevada Proving Grounds. Certain weapon developments started at CASTLE continued at TEAPOT and contributed to the ongoing design of devices for REDWING.¹⁴⁷

The objectives of TEAPOT tests were fourfold:

- a. Strategic applications
- b. Tactical applications
- c. Air defense applications
- d. Effects information

In particular, UCRL planned to test three nuclear devices. One was a mockup of a lightweight high-yield device, and the other two were to investigate the principles of a linear-implosion device.¹⁴⁸

LASL proposed seven TEAPOT tests in three categories:

¹⁴⁶Memorandum dated April 26, 1955 to Assistant for Development Programming, Headquarters, U.S. Air Force, from Col. W. O. Jackson, Jr., Deputy Director of Operations, subject: B-57B Aircraft for Cloud Sampling.

¹⁴⁷Letter dated December 23, 1953, from Lewis L. Strauss, Chairman, USAEC, Washington, D.C., to Honorable Robert LeBaron, Chairman, Military Liaison Committee to the Atomic Energy Commission.

¹⁴⁸Letter dated December 7, 1954 to Brig. Gen. K. E. Fields, Director, Division of Military Application, USAEC, from Herbert F. York, Director, UCRL - Livermore.

The Los Alamos Scientific Laboratory specifically proposes six experimental devices for test in Operation TEAPOT; and a seventh, primarily intended to provide the blast and other conditions required by the Department of Defense to enable them to obtain information on effects, is also of interest to the Laboratory as a nuclear system. These seven devices (include) ... small diameter implosion systems (4 devices); two-stage weapon component tests (2 devices); and the MET shot [deleted] nuclear system for the DOD effects test.¹⁴⁹

Among experiments to be conducted during TEAPOT were two tests of mockups of small, high-yield lightweight thermonuclear systems required for development of a LASL Class "D" weapon design, and a test of the primary for a UCRL Class "D" warhead (which became the 43 kiloton TEAPOT Turk shot fired on March 7).¹⁵⁰

The two LASL tests became the TEAPOT Apple-1, Apple-2, and Zucchini shots: the Apple device was predicted to yield 40 kilotons; the primary of the original Apple shot (Apple-1) on March 29 at the NPG malfunctioned and yielded only 14 kilotons, and a second, modified Apple device was fired on May 5 with a yield of 29 kilotons, still below the predicted value.

The Zucchini shot was a test of a primary for a high-yield, lightweight externally-carried Class "D" thermonuclear weapon; it also detonated with a yield well below its predicted value.¹⁵¹

¹⁴⁹Letter dated December 1, 1954 to Brig. Gen. K. E. Fields, Director, Division of Military Application, USAEC, from N. E. Bradbury, Director, LASL.

¹⁵⁰Atomic Energy Commission, Nevada Proving Ground Test Activities for Calendar Year 1955, Report to the General Manager by the Director of Military Application, USAEC, August 9, 1954.

¹⁵¹Atomic Energy Commission, Proposed Program for Operation TEAPOT, Report to the General Manager by the Director of Military Application, USAEC, December 30, 1954; Atomic Energy Commission, Revised Schedule for TEAPOT, Report by the Director of Military Application, USAEC, April 5, 1955; letter dated April 7, 1955 to The President, The White House, from Lewis L. Strauss, Chairman, USAEC; letter dated April 11, 1955 to Senator Clinton P. Anderson, Chairman, JCAE, from Lewis L. Strauss, Chairman, USAEC.

Despite these problems, TEAPOT results were stunning: the test series established the feasibility of 16" diameter implosion systems and proved effectiveness of tritium gas boosting in hollow-pit systems, thus establishing design principles for the XW-25, XW-30, XW-31, XW-34, and XW-37 warheads; and boosted primaries for the TX-15-X4, TX-21-X2, TX-27, TX-28, and TX-39-X1.

TEAPOT also established the feasibility of lightweight cases for TX-27 and TX-28 type weapons; revealed limitations on the degree of secondary stage implosion in thermonuclear weapons; and tested linear implosion fission systems.

The general purpose of TEAPOT was to obtain information needed for design of small-diameter tactical and air-defense weapons and small missile- and fighter aircraft-delivered thermonuclear warheads.

TEAPOT tests could be grouped in accordance with these principal purposes. One group included three 22" diameter devices (Wasp, Wasp Prime, and Hornet) and two 16" diameter devices (Moth and Bee). These detonations provided information related to the development of optimized spherical implosion devices simultaneously making more efficient use of fissionable materials while being safer against accidental one-point detonation.

In addition, the Bee and Hornet devices were boosted with tritium gas; their tests successfully demonstrated the feasibility of using gas boosting to increase energy yield of small-size, low-yield weapons.

A second group of four devices was tested to obtain information related to the development of Class "D" thermonuclear weapons, i.e., thermonuclear warheads weighing less than 3,000 lbs. These shots included the Turk, Apple, Apple-2, and Zucchini tests, and they confirmed the feasibility of Class "D" weapons. LASL air defense warhead tests were named after flying insects; Class "D" bomb primary tests were named after fruits and vegetables.

A third group of tests included the Tesla and Post devices, fired to determine the feasibility of linear implosion. Unlike earlier UCRL shots, these tests met or exceeded expectations.

The MET shot, or Military Effects Test, had two purposes. MET and the Ess and HA shots were used by DOD to gather more information on atomic explosion effects. MET also tested a uranium-233 bomb core.¹⁵²

Foremost among TEAPOT results was the successful proof-testing of boosted fission devices suitable for use as primary stages of lightweight, small-diameter thermonuclear bombs:

The "fission triggers," or primaries, of the early TN (thermonuclear) weapons did not differ appreciably from the pure fission weapons already deployed, and they suffered in this new role from safety and reliability problems. They were also relatively inefficient in their use of fissile material, and the early TN weapons were very large and heavy, severely taxing the bomber aircraft of that time and scarcely suitable for delivery by guided or ballistic missiles.

Many of these problems soon were addressed by the second new concept: the boosted fission primary, which was first tested in 1955...This concept contributed to reliability, efficiency, and significant reduction in size and weight as well. The 1955 tests proved the feasibility of this approach to primary design. Developmental weaponization followed quickly in a 1956 test series (Author's note: Operation REDWING) directed toward the B-47 and B-52 bombers that were to replace the ponderous, propeller-driven B-36, and toward future missile systems.¹⁵³

A remarkable set of improvements occurred from 1952 to 1956 in the physics, design, and engineering of the primary stage of a nuclear weapon. The diameter of the primary stage decreased by more than a factor of three and the weight decreased by more than a factor of 30.

Collectively, these improvements allowed the (U.S.) to rapidly expand the flexibility and utility of its nuclear stockpile. The improved stockpile made

¹⁵²Atomic Energy Commission, Program Status Report, Part III - Weapons Program, July 27, 1955, pp. 11, 12.

¹⁵³HYDRONUCLEAR EXPERIMENTS, Robert N. Thorn and Donald R. Westervelt, LAMS-10902-MS/UC-2, Los Alamos National Laboratory, February 1987, pp. 1, 2.

possible multiple delivery platforms, including ballistic missiles and tactical applications, and facilitated the shift of national policy from massive retaliation and targeting of cities and populous areas to a flexible response strategy designed to deter and counter Soviet war-fighting capabilities.¹⁵⁴

One rather unusual event during TEAPOT was the disclosure by the AEC in early March 1955 of the yield of possible tactical weapon prototypes tested in the series. The AEC released this data in an attempt to allay public fears about fallout engendered by the CASTLE Bravo test a year earlier:

The nuclear test explosion at Bikini on March 1 of last year was indeed colossal, and the explosion was on the surface of a coral island, and because of these two facts, it did produce heavy fallout over a very large area. But, the device tested was not a tactical weapon such as our Army Navy, and Air Force would use on the battlefield. Our tactical atomic weapons are very much smaller in every way, ranging down to possibly 1/10,000ths of the explosive power released in our test detonation at Bikini a year ago. (Author's note: this "tactical" yield would then be 1.5 KT; TEAPOT tests Wasp and Moth were in this yield range.) Some are nearer in explosive power to the "block busters" of World War II. ...

As a result of the close cooperation between the Department of Defense and Atomic Energy Commission, our Army, Navy, and Air Force now have small tactical atomic weapons that can be used in the warheads of guided missiles, or in shells fired from guns, or in bombs that can be employed accurately against military targets on land, in the air, or at sea.¹⁵⁵

¹⁵⁴"Los Alamos Timeline: Significant Events and 50 Years of Research Highlights," DATELINE: LOS ALAMOS, Los Alamos National Laboratory, Special Issue 1995, p. 30.

¹⁵⁵Draft of Chairman's Supplemental Statement on Fallout, undated, but probably ca. February or March 1955. Emphasis in original. The final paragraph was rather premature in its claims, considering that TEAPOT test devices had not yet been weaponized.

An earlier suggestion for this statement was even more specific regarding the availability, nature, and probable targets for tactical atomic weapons:

... as a result of the joint efforts of the Department of Defense and the Atomic Energy Commission, there are now tactical atomic weapons, in sizes equivalent to or smaller than those tested in Nevada, available to our Army, Navy, and Air Force for use against such small military targets as heavy gun emplacements, formations of armored tanks, railway bridges, or against planes, hangars, and fuel tanks on airfields, or against landing craft and small boats attempting an amphibious assault.¹⁵⁶

Planning for REDWING Continues

Planning for REDWING was discussed during TEAPOT and a detailed general concept of Operation REDWING was published by JTF 7 on April 12. The 13-shot schedule at that time is shown in Table 3-8.

Table 3-8 Preliminary REDWING Shot List,
April 1955¹⁵⁷

<u>Shot Name/Sponsor</u>	<u>Site</u>
Hopi/LASL	Eniwetok (Runit) 100-foot tower
Shawnee/LASL	Eniwetok (Runit) 100-foot tower
Yuma/UCRL	Eniwetok (Aomon) 200-foot tower
Blackfoot/LASL	Eniwetok (Runit) 200-foot tower

¹⁵⁶Memorandum for the Chairman dated March 11, 1955 from Paul F. Foster, Subject: Proposed New Statement on Fallout.

¹⁵⁷Commander Task Group 7.1, Joint Task Force 7, General Concept No. 1-55, Operation REDWING, 12 April 1955; Headquarters Task Group 7.4 (Provisional), 30 June 1955, subject: Concept of Air Task Group Operations, Operation REDWING. The shots are not listed in order of firing at this time.

Osage/LASL	Eniwetok (Runit) 200-foot tower
Kickapoo/UCRL	Eniwetok (Aomon) 200-foot tower
Pueblo/LASL	Eniwetok (Bogairikk) Ground
Mohawk/UCRL	Eniwetok (Eberiru) 300-foot tower
Zuni/UCRL	Bikini (Eninman) Ground if small yield; Bikini (Yurochi) barge if large yield
Flathead/LASL	Bikini (Yurochi) Barge
Navajo/LASL	Bikini (Yurochi) Barge
Apache/LASL	Bikini (Yurochi) Barge
Cherokee	Bikini (lagoon) Air-drop

Of these shots, there was only a 50% chance that Pueblo would be fired. Expected yield for Mohawk was 20 KT. Only the Hopi, Shawnee, Osage, Apache, and Cherokee shots had specific devices assigned to them.

The first REDWING shot was tentatively slated for April 15, 1956. There was a remote possibility that a fourth UCRL shot at Eniwetok with a maximum yield of 5 KT would be fired. If detonated, it would be from atop a tower erected on Aomon after the Mohawk test.¹⁵⁸

Tower, barge, and surface shots would be fired at Eniwetok from the timing and firing station on Parry island, and at Bikini from Station 70 at Enyu. Shots would be fired by radio command from the USS CURTISS if Bikini had to be evacuated (as it eventually was for the 3.5 MT UCRL BASSOON device shot on Eninman). **Figure V-1** is an aerial view of Enyu,

¹⁵⁸"Task Group 7.4, Concept of Operations," Air Task Group 7.4 (Provisional), Kirtland AFB, New Mexico, 30 June 1955.

which, at 1.7 miles long and 0.36 miles wide, was the second largest island at Bikini.

All personnel at Bikini other than the firing party would be evacuated before each shot; personnel at Eniwetok would be relocated to Parry island on shot days. As was done during CASTLE, plans were made to conduct test operations from afloat, if necessary.

Temporary camps were set up at Bikini on Enyu, Eninman, and Romurikku islands; at Eniwetok, the main camp was located at Parry with temporary work camps on Rojoa, Runit, and Teiteiripucchi islands. Principal laboratory, machine shop, photographic, warehouse and stock room facilities were also located at Parry.

A meeting was held at the NTS on April 14 to review REDWING plans:

We propose, in this meeting, to cover those parts of REDWING which are now reasonably firm, stressing the problems as we now know them, and to see where we are at the moment and where we have to go.

The number and type of shots for REDWING is not at all firm at this time. There will be a number of changes. However, an overall picture can be given. It is known that Operation DIXIE, which was to have been held at the NTS this fall, will be combined with REDWING at Eniwetok in the Spring (sic).

I will list the shot proposals in two groups, those which have been included in REDWING and those which might have been DIXIE shots. Hence, the listing bears no relationship to a firing sequence.

In the REDWING portion of the series there will be [deleted] a Class D weapon in the megaton region weighing less than 3,000 pounds, to be fired off Yurochi on a barge. The yield will be on the order of [deleted]

A second device will be either an attempt at [deleted] Either gadget would be detonated off Yurochi on a barge. In the first case, we would have a yield (not well specified) of [deleted] In the second case, it might yield [deleted]

The third shot will be a device which might

be fired for the DOD or LASL, or a combination of the two. It would be an air drop in the Bikini lagoon and might be a [deleted] In the first case, the yield would be [deleted] and in the second, about [deleted] The former appears to be more likely.

The fourth (test) will be a UCRL device. It is a [deleted] so-called clean weapon which is designed to reduce fallout problems to a minimum. This [deleted] weapon will be fired as a surface shot at Eninman and is expected to have a yield of about [deleted]

Finally, in the REDWING series, there is a possibility of a [deleted] This is an attempt to find out whether two-stage weapons can be made to compete with ordinary fission weapons. This would be fired at Bogairikk at Eniwetok atoll as a surface shot, and might have a yield of [deleted] There is a possibility that it may be necessary to test the [deleted] primary before testing the two-stage weapon.

There is a remote possibility of attempting to get ready and test a warhead for an ATLAS missile. If done, it would be fired off Yurochi on a barge. Its yield, at the moment, is completely unspecified.

The following are the sort of shots which might have been considered for DIXIE.

First, we will proof-test [deleted] warhead which is to be used for air-to-air or ground-to-air missiles. This will be detonated on Runit on a 200 foot tower, and the yield might be [deleted]

Next, we will have two one-point detonations on Runit on 100 foot towers. These might have yields as high as 0.1 KT or as low as a few pounds, or zero. The Los Alamos Laboratory (sic) will probably design three gadgets as the prediction of yield of one-point detonations is most uncertain. On the second of these shots, we would select a higher or lower yield device depending on the results of the first experiment.

We will fire an [deleted] device on a 200 foot tower on Aomon. The yield will be [deleted]

A [deleted] will also be fired on Aomon on a 300 foot tower. The yield can be [deleted] depending on whether [deleted]

An [deleted] two-stage weapon is being considered and would be detonated on a 500 foot tower on Aomon. The yield, including secondary fuel, would be [deleted] depending on whether the [deleted] was selected.

This is as conservative an estimate as can be made at the moment. The feeling is that many of these shots are simple and can be done with very little worry about the weather. We would plan to schedule the little ones to be fired on Eniwetok while awaiting suitable weather for bigger shots on Bikini or Eniwetok. ...

The Air Force has indicated a desire to do the air drop over land in order to obtain additional information on fallout problems. Until it is determined whether this shot is for weapons development, or to show a TAC capability or a SAC capability, the zero point cannot be decided. There is still some worry about a tsunami probability.

The tentative target date for the first shot of the operation is 15 April. This date is not firm since the Division of Military Application has not, as yet, formally approved it. However, both weapons laboratories are convinced that an earlier date is impossible because of the time required to design, construct, and ship the test devices themselves. As in Operation CASTLE, we should, in general, try to be ready with all shots before shooting the first.

At this meeting, LASL scientist William E. Ogle noted that for thermonuclear shots, the yield was usually determined primarily by fireball growth rate photography, since only a relatively small portion of the yield came from fission and could be measured by radiochemistry. This was especially true for the proposed "clean" weapons; the CASTLE devices, by comparison, gained most of their yield from fission, not fusion.

Ogle also said that two other diagnostic measurements, primary alpha and time between primary and secondary reactions, were of paramount importance. Alpha was required since the yield of the primary could not be measured directly; time interval between stages indicated whether the primary and the radiation channel between stages worked as intended. UCRL planned to measure radiation flow times by means of "hot spot" techniques, as was done with the IVY Mike device.

As regarded the air drop, plans at this time were for the use of a B-36 to release the weapon at an altitude of 40,000 feet, with visual bombing of an air zero point at a "practical" height-of-burst, with a CEP of no more than 1,500 feet.¹⁵⁹

REDWING and "Clean" Weapon Tests

On April 15, LASL submitted its first thoughts on "clean" bomb design to the AEC; the lab -- choosing its words carefully -- thought it could have such a device ready for test during REDWING:

Your letter of April 8, 1955 inquires as to the status of our present understanding and prospects with respect to so-called "clean" weapons.

The studies we have available at present seem to us to establish the point [deleted]

... Depending on the mode of use and the nature of environment, the fission debris would, of course, be accompanied by other material made radioactive by neutron capture, but unless the materials surrounding the bomb were rather carefully chosen for the purpose, the fission debris would still constitute the major source of residual radioactivity.

The calculations we have made with respect to such models are, in general, very similar to those which, when applied to the models tested at Operation CASTLE, give results [deleted]

Studies which have been made here to date have been concerned only with models which

¹⁵⁹Meeting on Operation REDWING Held at the Nevada Test Site on 14 April 1955, pp. 1-5, 7.

are geometrically similar to models presently being considered for stockpile [deleted]

With respect to the matter of a possible test in REDWING of such a device, we feel, as you know, that the list already available of objects which (must) be tested merely to meet demands specifically related to rather immediate military requirements is impressively long.

We are consequently reluctant to consider adding to this if that can be avoided. One of the items which we believe to be of great importance to establish at REDWING is [deleted]

... we would expect that the evidence obtained in this way could be used to make estimates of the sort referred to above much more certain and precise.

Although this proposal has received a preliminary consideration, we are not yet in a position to lay down a design of such a device and it will require a rather favorable rate of progress to be able to prepare this test in time for REDWING. At the moment, however, we are rather optimistic about being able to do so.¹⁶⁰

UCRL was also continuing to work on "clean" weapon concepts; Dr. York summed it up in a letter in late April:

I. Requirement and Meaning

A clean weapon is one in which the radioactive aftereffects are reduced to a minimum. The need for such a weapon has been outlined and discussed in a number of places; for instance, in a letter of General Fields to Dr. York (2/9/55); a letter of Secretary (of Defense) Wilson to Admiral Strauss; and in discussion between the writer and (AEC) Commissioners Libby, Murray, and von Neumann.

¹⁶⁰Letter dated April 15, 1955 from N. E. Bradbury, Director, LASL, to Brig. Gen. K. E. Fields, Director of Military Application, USAEC.

A purely thermonuclear bomb, exploded on the surface, would still give some local radioactive hazard due to induced activities produced in the ground (or water) by the neutrons which would inevitably escape. We have calculated that for the case of "average" dirt, the local fallout intensities would be reduced [deleted]

At this time, the Livermore lab was considering two general methods for reducing residual fallout. The first method was simple substitution of materials, primarily the replacement of the fissionable fusion tamper on the secondary with a heavy non-fissionable material, such as lead. This technique incurred a significant loss in yield for no change in weapon weight.

The second method was far more radical and daring. One way to maintain high yield without increasing residual radiation was to gain a higher compression of the fusion stage. This might be achieved by inserting an intermediate stage between the primary and the original secondary, extending the original Teller-Ulam principles and creating a three-stage device.

In this case, the new secondary would explode with a force of several hundred kilotons, and act as an x-ray emitting, highly-compressive driver for the main fusion fuel charge in the tertiary. Both the secondary and tertiary would be jacketed with a non-fissionable heavy metal. The tertiary would explode with a force of several megatons.

UCRL noted that since the currently-stockpiled Class "A" and "B" weapons were under LASL cognizance, UCRL would limit its "clean" weapons research to mainly experimental devices which could not be immediately and easily "weaponized."¹⁶¹

At the end of April, the surface zero and height of burst for the airdrop test were still somewhat uncertain. In order to make fallout reasonably typical of continental conditions, it would be necessary to detonate the bomb over a land surface. However, only a single small island, Namu at Bikini, appeared to be available for the test. The island was so small, in fact, that its size placed severe restrictions on the accuracy of the drop.

There was somewhat less uncertainty over the height of burst.

¹⁶¹Letter dated 25 April 1955 to Col. Vincent B. Huston, Acting Chief, Research Branch, Division of Military Application, USAEC, from Herbert F. York, Director, UCRL - Livermore.

It seemed most likely that it would be in the range of one to one-and-a-half fireball radii to make sure that the fireball did not touch the ground.¹⁶² (Fireball radius is directly proportional to yield.)

The shot schedule at this time included eight at Eniwetok: four at Runit, three at Eberiru, and one at Bogairikk, and five at Bikini: three barge shots in the lagoon off Yurochi island, one ground shot at Eninman, and one air drop, possibly over Namu.¹⁶³ The Air Force recommended that the air drop be from a B-36, using a drogue chute on the bomb to prolong time-of-fall to 100 seconds in order to permit the bomber sufficient time to escape the blast.¹⁶⁴

In early May, AEC Commissioner Willard F. Libby, in his temporary capacity as acting chairman, advised the Secretary of Defense that the AEC was in agreement with DOD's desire for variable-contamination weapons, and that methods to both the increase and decrease radioactive yields would be investigated but that priority would be given to the latter.

Increasing radioactive yield would be confined to only theoretical studies at this time; the AEC hoped to test a decreased radioactive yield device at an early date, possibly during REDWING.¹⁶⁵

Also at this time, in response to a telegram from the State Department which forwarded a request from the United Nations that the U.S. release certain information about the amounts of fission products put into the earth's atmosphere by nuclear testing since 1945, Dr. John von Neumann, in a memorandum to AEC Commissioners Willard F. Libby and Thomas E. Murray, warned against disclosure of this data, even in the face of evidence which suggested that much of it was already in the hands of the Soviets:

I take it that this proposal, if made and

¹⁶²Trip Report dated 25 April 1955 from Maj. S. E. Singer, USAF, Research Directorate, AFSWC, Subject: HQ AFSWP Meeting on REDWING Fall-Out Programs, 11 April 1955.

¹⁶³Memorandum for Record dated 29 April 1955 from Col. David O. Byars, Jr., USA, Subject: Resume of Conference on Military Support Requirements.

¹⁶⁴Memorandum to Commander, ARDC, Baltimore, Maryland from Commander, AFSWC, 5 May 1955.

¹⁶⁵Letter dated May 4, 1955 to Honorable Charles E. Wilson, Secretary of Defense, from W. F. Libby, Acting Chairman, USAEC.

accepted, means that we undertake to give the UN committee certain information, including the following:

(a) The total quantity of fission products released by nuclear (bomb) experiments into the atmosphere.

This is restricted data, and goes beyond the type of information which can be communicated to friendly nations even under the more relaxed provisions of the (Atomic Energy Act) of 1954. In the case of thermonuclear weapons, which represent the bulk of the fission product release, this involves aspects of the inner structure of the bomb -- the partition of the total yield into fission and fusion yields.

After REDWING, where "pure" (fusion) bombs will become a significant part of the picture, this will be even more fundamental information on detailed bomb design.

The actual fallout data (I mean fallout up to the present) are clearly not in this sensitive category. However, it will require a very difficult and delicate decision, to give the UN committee the latter (fallout) data without referring to the former (total bomb fission yields) as well.

Indeed, if we did, so, we might mislead the (UN commission) since we know the former amount exceeds the latter, as far as we can determine the latter, by the factor of about 10. In other words, about 90% of the bomb fission products are, at this moment, unaccounted for.

Furthermore, the steadily continuing increase of strontium-90 deposition on the ground indicates that at least part of this "unaccounted" quantity is not removed forever from the biosphere. This latter information is also restricted data, and of a highly sensitive nature.

The Russians have made a thermonuclear experiment, and the British may make one in the not too distant future. Hence, they too may now be or may soon become aware of such circumstances. It would, therefore, be unwise in the extreme for us to assert that we have given complete information on (a), unless we also face the question that we may have to disclose at some point, not only the present (cumulative) fallout level, but also the considerably higher total bomb fission product output.

Since (the latter) is very sensitive information, the problems of possibly having to declassify it should be faced and evaluated now -- i.e., before any undertaking of the type (a) above is made.

(b) The total quantity of radiokrypton released into the atmosphere.

Statements on this subject would disclose our total plutonium production to date, i.e., our present plutonium stockpile. This is restricted data of high sensitivity, and it is at present clearly illegal to communicate it to anyone under any conditions.

It is true that the British very probably possess a good estimate of this, and it is not at all unlikely that the Russians also have some estimate. Nevertheless, an explicit announcement of ours, with our official guaranty of exactness, would go significantly farther. Clearly, a change in the (Atomic Energy Act) of 1954 would be required (to release this information), and it doesn't seem likely that we could get such a change even if we wanted it.¹⁶⁶

¹⁶⁶Memorandum dated May 5, 1955 for W. F. Libby and Thomas E. Murray from John von Neumann. Radiokrypton is a by-product of both nuclear detonations and the chemical separation of plutonium from irradiated uranium; Dr. von Neumann was correct in his assertion that the U.S. plutonium stockpile might be

Near the middle of the month, LASL expected REDWING to include a total of thirteen plus or minus two tests. Shots Hopi, Shawnee, Osage, and Yuma were certain; Blackfoot was probable, but no decision to test had been made. Kickapoo was reasonably firm; Mohawk and Zuni were quite definite; Pueblo was not yet certain. Other LASL and UCRL shots were still only in a planning stage. One UCRL device scheduled for testing was the BASSOON.¹⁶⁷

Predicted yields for Yuma, Kickapoo, Mohawk, and Zuni were revised; Kickapoo and Mohawk were now to be fired from 300 and 500 foot towers, respectively, reflecting increases in expected yields. Zuni was now to be fired on the ground at Eninman island at Bikini atoll.¹⁶⁸

By the beginning of May, the ready date for the first REDWING shot had been changed to April 15, 1956; by June, due to delays in scientific station construction, it had slipped to May 1.

REDWING was now expected to comprise 14 shots, with the smaller yield tests to be detonated at Eniwetok and the high yield shots at Bikini. There might be eight or nine Eniwetok firings, all on islands; Bikini might be the site of five detonations, including three barge shots in the lagoon off Yurochi island, one air drop over Namu, and one ground shot at Eninman island.¹⁶⁹

At the beginning of June, yield ranges for shots Pueblo and Zuni were estimated to be between 50 and 100 kilotons, and from one to three megatons, respectively.

divined from an official authoritative statement on the quantity of radiokrypton in the atmosphere. His comments about the continuing deposition of Sr-90 being "of a highly sensitive character" referred to the subject's political sensitivity, and not to any particular technical sensitivity.

¹⁶⁷Minutes of the Forty-Fifth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, May 2-4, 1955, p. 6.

¹⁶⁸Memorandum dated 25 May 1955 to Distribution from W. T. Kerwin, J-3, Plans and Operations, Headquarters Joint Task Force 7, Subject: Minutes of Meeting Held at Los Alamos on 18 May 1955 to Discuss Proposed DOD Participation in Operation REDWING.

¹⁶⁹SWOTP Programming Plan 3-55 dated 9 May 1955, Headquarters Air Force Special Weapons Center, Air Research and Development Command, Kirtland AFB, New Mexico, p. 3.

While shot Cherokee, the proposed airdrop, was of primary interest to DOD for its effects, it would also provide LASL with a check of new developmental principles into high-yield weapons and would constitute a proof-test of a stockpiled weapon. The proposed height-of-burst was to be between one and one-and-a-half fireball radii, i.e., at an altitude from 5,000 to 8,00 feet.¹⁷⁰

By the latter part of June, REDWING included a potential total of 10 LASL and six UCRL shots, including three LASL one-point safety tests:

Table 3-9 Preliminary REDWING Shot List, June 1955¹⁷¹

<u>Shot Name</u>	<u>Ready date</u>	<u>Site/Sponsor</u>
Cherokee	1 May	Bikini (Namu) Airdrop (LASL)
Pueblo	1 May	Eniwetok (Bogairikk) Ground (UCRL)
Blackfoot	1 May	Eniwetok (Runit) 200-foot tower (LASL)
Yuma	1 May	Eniwetok (Rojoa) 200-foot tower (UCRL)
Shawnee I	1 May	Eniwetok (1-point safety test) (LASL)
Osage	4 May	Eniwetok (Runit) 200-foot tower (LASL)

¹⁷⁰Letter dated 13 May 1955 to Chief of Staff, United States Army, from Major General A. R. Luedecke, USAF, Chief, AFSWP, subject: Military Weapons Effects Program for Operation REDWING, Spring 1956, pp. 2, 3, 4.

¹⁷¹Classified Notebook No. 681(L) issued by Information Division, UCRL Livermore; issued originally to Paul Harrold, L Division Operations, Livermore; reissued to John Flynn, L-3, L Division - Operations, Personal Notebook; HQ, FCDASA memorandum dated 5 July 1955 to Director, Ballistic Research Laboratories, U.S. Army, Aberdeen Proving Ground, Maryland, from Col. E. W. Niles, Deputy Director, Weapons Effects Tests, Subject: Transmittal of Ground Zero Locations.

Shawnee II	4 May	Eniwetok (1-point safety test) (Engebi) (LASL)
Shawnee III	7 May	Eniwetok (1-point safety test) (Engebi) (LASL)
Kickapoo	8 May	Eniwetok (Rojoa) 300-foot tower (UCRL)
Huron	11 May	Bikini (Yurochi) Barge (LASL)
Zuni	18 May	Bikini (Eninman) Ground (UCRL) (est. yield was 3 MT)
Lacrosse	25 May	Eniwetok (Runit) Ground (LASL) (est. yield was 40 KT)
Flathead	25 May	Bikini (Yurochi) Barge (LASL)
Mohawk	29 May	Eniwetok (Rojoa) 300-foot tower (UCRL)
Navajo	1 June	Bikini (Yurochi) Barge (LASL)
Apache	8 June	Bikini (Yurochi) Barge or airdrop (UCRL)

In July, several "one-point safe" weapon safety tests -- the Shawnee series -- originally scheduled for REDWING were rescheduled for Nevada in the late fall and winter of 1955. These became the four "Project 56" plutonium-dispersal shots (including at least one XW-25 firing) at Nevada between November 1955 and January 1956. (The many acres of contaminated ground resulting from these tests is now known as "Plutonium Valley" at the eastern edge of the Nevada Test Site.)

Among weapons to be tested during REDWING were two new Class "D" lightweight thermonuclear weapons, LASL's TX-28 and UCRL's XW-27. The latter, using a weaponized ZITHER secondary, was given high priority at Livermore, although the new lab's weapons development program also embraced several

other projects as well:

Assuming (UCRL does) XW-27, our program is affected as follows:

a) XW-27 will receive first priority and [deleted]

b) [deleted] will have next priority and will proceed as before. This system is potentially of interest in connection with normal thermonuclear weapon development also. [deleted] device is planned for REDWING.

c) [deleted] Work on this will not begin for at least three to six months, but we still hope tentatively to have a device of this sort for test in REDWING.

d) [deleted] Theoretical work on this series of devices will be dropped. Some experimental work with materials on hand may be continued for a short time. (A) report describing our present views on the feasibility of [deleted] will be prepared and submitted within the next few weeks.

e) SMALL WEAPONS. We plan at present to proceed as before. However, this program uses approximately one-half of our computational facilities, and should unexpected difficulties arise in connection with [deleted] some decrease might possibly be necessary.¹⁷²

Also at this time, serious consideration was being given by both labs to a test of a "clean" device during REDWING, and to a half-month postponement of the series start date:

¹⁷²Dr. Herbert York quoted in memorandum dated June 13, 1955 to K. E. Fields, General Manager, USAEC, from Col. Vincent G. Huston, USAF, Acting Director of Military Application, USAEC. Item (d) in this list may have been UCRL's uranium hydride weapons program; item (e) is probably a reference to linear implosion devices. A zither is a plucked stringed instrument; UCRL was naming its large thermonuclear devices after musical instruments.

We have also explored with Livermore the extent of their interest in that part of the clean weapon program which would involve the test, for example, of [deleted] They would apparently prefer to work upon [deleted] systems of general applicability rather than undertake a straightforward experiment.

Accordingly, restricted solely to the question of the behavior of a clean but otherwise familiar type of system, the LASL will (determine) with the Commission at the time of the July 13 meeting the extent of the AEC interest in this field and whether or not the Commission wishes a clean [deleted] experiment carried out. If so, the LASL will probably do it.

Finally, the two laboratories have agreed that May 1, 1956 is a preferable starting time for the REDWING operation compared to the April 15 date currently stated.¹⁷³

During the spring and summer of 1955, as changes in weapon design and configuration occurred, the REDWING shot schedule was revised with additions, substitutions, and deletions of devices and weapons, much as the pre-CASTLE schedule had been extensively revised two years earlier.

By mid-1955, SAC's appetite for nuclear weapons seemed to be insatiable:

After studying the nuclear weapons allocation for U.S. forces for Fiscal Year 1956, I am convinced that political considerations and interservice disputes have resulted in a decision which does not best serve national interests.

The simple fact is that while no nuclear weapons are provided for more than 200 combat-ready SAC bombers and the entire SAC fighter force, there is evidence that some theater commanders have been allocated more than one weapon for every aircraft which may become atomic-capable during the period.

¹⁷³N. E. Bradbury quoted in memorandum dated June 13, 1955 to K. E. Fields, General Manager, USAEC, from Col. Vincent G. Huston, USAF, Acting Director of Military Application, USAEC.

This command is the only force with a proven capability to deliver an atomic weapon regardless of weather or time of day. It has the only force capable of striking the enemy if our advanced bases are destroyed by surprise attack and/or sabotage, or are denied us by political considerations.

It has the only control structure which will permit launching a fully-coordinated strike. And yet one-third of the combat-ready SAC force is left without weapons in order to satisfy the desires of theater commanders.

I cannot see how this decision can be justified. In effect, it indicates that more weight is being given to holding a defensive line in Europe than to the prevention of atomic attacks on U.S. cities. It implies that there is as great a threat involved in retreat in any theater as loss of the air battle. It means that the American people have invested millions of dollars in aircraft and air crews for which collateral plans have not provided ample weapons.

I feel certain that the FY 56 allocation is a result of compromise. And when the survival of the nation is at stake, no compromise can be justified.

As you are aware, I have never committed atomic weapons to my strategic fighter wings. This was not because these wings lack the capability to deliver weapons, but rather because careful analysis discloses that the B-47 can match every capability of the fighters, is not restricted by weather and range considerations, and provides at least 100% more (explosive) yield for the critical material invested.

So long as (the stockpile of) nuclear material is critical, good judgment dictates that the bomber be given priority in weapon assignment ahead of any other delivery vehicle. If the task of winning the air battle is to enjoy the priority it deserves, those strategic fighter wings which can be maintained on rotation in forward areas should be furnished weapons before the support of ground combat is considered.

Through the SAC control structure, attacks by

SAC fighters can be coordinated so as to enhance the penetration by the bomber force. Targets (the fighters) can attack include communications centers, air defense forces, and enemy radars.

But not one weapon should be allocated to these fighters, or to the theater (of operations), until every combat-ready SAC bomber is furnished with a high-yield weapon to perform its initial mission.

I strongly urge that the following action be taken on an emergency priority basis:

a. All Charlie and Dog capsules and all TN (thermonuclear) weapons in the JCS reserve be specifically allocated to this command through 30 June 1956.

b. An additional 70 Baker capsules in the JCS reserve be allocated to this command through 31 December 1955.

Only through this action can we be assured that each SAC bomber and each SAC fighter on rotation have at least one atomic weapon allocated to accomplish the first assigned mission and are supported by realistic deployment and pickup schedules.¹⁷⁴

Many of LeMay's points harkened back to the basic question of the use of air power during World War II: strategic bombing versus ground combat support. LeMay's sympathies were obviously with the strategic mission. His comments also reflected the eternal struggles between the Air Force, the Army, and the Navy over their respective modern nuclear war roles, allocation of resources, and funding.

In mid-summer 1955, fourteen REDWING shots were scheduled, at least five of which would be UCRL shots.¹⁷⁵ Eight would be at

¹⁷⁴Letter dated 6 June 1955 to General Nathan F. Twining, USAF, Chief of Staff, United States Air Force, from General Curtis E. LeMay, USAF, Commander in Chief, Strategic Air Command, USAF. The references to the Baker, Charlie, and Dog capsules are unknown at this time.

¹⁷⁵Letter dated 7 July 1955 to Col. Alfred D. Starbird, Director, Division of Military Application, USAEC, from Mark Mills, Acting Director, UCRL-Livermore.

Eniwetok atoll: three on the Eberiru complex plus one on Rujoru; three on Runit and one in the air above Runit. Six were slated for barges in Bikini lagoon near Yurochi island.

When weather for large shots proved more favorable at Eniwetok than at Bikini and completion of the Bikini program lagged, two of the high-yield barge shots, Huron and Apache, were rescheduled and relocated to the IVY Mike crater at Eniwetok. Towers for Blackfoot and Yuma were not built on Runit until after the Lacrosse detonation.

A high-yield megaton-range airburst was scheduled; a second megaton burst was to be a high-yield "clean" land (surface) burst at Eninman island at Bikini (this became the Zuni shot).¹⁷⁶

By mid-July, LASL planned to fire four groups of shots. The first group consisted of two devices. The objective of the second group was to develop a Class "D" thermonuclear weapon. The objective of the third group was to determine the efficiency of small boosted primaries.

The fourth group of tests was aimed toward the development of a small two-stage weapon which would be suitable for application to a rocket. This group would start with a test of the primary; if it were successful, a second shot would test the complete weapon.

During an AEC meeting, Norris Bradbury noted that although no military requirement currently existed for this type of weapon, DOD was extremely interested in its development. (This test became the late-July 1956 Huron shot of the EGG device; it may have been an early prototype of the W-50 warhead for the Army's PERSHING missile.)

Bradbury noted that current REDWING plans did not include any one-point safety tests of new air-defense and anti-submarine warfare warheads since he assumed that such tests would, in all probability, be conducted in Nevada before REDWING. He also thought that a worst case one-point test might result in a yield of two kilotons (this was considerably higher than the 100 ton upper-limit yield then being used as a planning value).

Bradbury also described an experiment that might be conducted to develop a reduced contamination ("clean") thermonuclear weapon. He said that this would be a difficult addition to

¹⁷⁶Memorandum for the Record dated 5 July 1955 by Capt. R. H. Maynard, USN, Acting Director, Weapons Test Division, AFSWP, Subject: Factors Bearing on Scheduling of Events - Operation REDWING.

LASL's REDWING program, but that if the AEC believed such a test was necessary, then LASL would conduct it and consider it technically useful. The AEC quickly decided that the shot should be conducted.

During the course of his comments, Dr. Bradbury observed that LASL was especially interested in using REDWING to confirm the reproducibility and predictability of boosted weapon yields, as a follow-up to TEAPOT testing.

Dr. Mark Mills of UCRL reviewed his laboratory's proposed REDWING experiments. Among them was a reduced contamination thermonuclear device; the ZITHER thermonuclear weapon for guided missiles, which, if successful, would be weaponized as the XW-27; and an experiment to develop small two-stage thermonuclear weapons.

With respect to the latter, Mills believed that although a single shot might suffice to test both primary and secondary, they might have to be tested separately. UCRL also planned to test two small gun-type devices.

Alvin Graves of LASL stated that REDWING could be completed within three months of the first shot, and would be the most ambitious nuclear test program yet undertaken at the Pacific Proving Ground.¹⁷⁷

By this time in the U.S. nuclear weapons program, many of the suggestions made by Dr. Bradbury a year earlier were in effect, either by choice or by the course of "natural" design evolution. The TX-28, W-30, W-31, and W-34 "sealed pit" warheads, and generally all weapons that were foreseen in the future, were gas-boosted when feasible and applicable to provide greater yield than would otherwise be possible from these small-diameter devices.

To enhance inherent safety, the objectives of the weapons labs in New Mexico and California were to produce designs that would give no nuclear yield if the high explosives surrounding the nuclear material were to detonate at only one point, as might happen in an aircraft or missile accident. This feature permitted weapons to be stored and transported in a ready condition with nuclear components in place; i.e., no manual or automatic in-flight insertion of the core was required to arm the warhead.

The TX-28, and W-30, -31, and -34 also exemplified the current trend in weapon design away from interchangeability

¹⁷⁷Minutes of AEC Meeting 1011, July 14, 1955. Dr. Mills was later killed in a helicopter crash during Operation HARDTACK in 1958.

of nuclear components among several types of weapons, and toward single-purpose "dedicated" designs among which nuclear components were not interchangeable.

The shift toward non-interchangeability resulted in higher nuclear efficiency and better use of fissionable material that was inherent in dedicated designs, since fewer design compromises had to be made to reconcile competing or conflicting yield, weight, or adaptability requirements.¹⁷⁸

Construction of new facilities for REDWING at Eniwetok and Bikini started in mid-1955. By the beginning of the test series a year later, 735 scientific stations would be completed.

Revising the Test Schedule

Following TEAPOT, the schedule of future atomic tests was changed. Operation DIXIE, originally slated for the fall of 1955 at the NTS, was canceled and tests planned for DIXIE were incorporated into REDWING. A DOD proposal to conduct certain "nuclear safety tests" in Nevada before REDWING was being considered.

These tests were planned to verify calculations that indicated no nuclear yield would result from accidental one-point detonation of implosion weapons containing permanently-installed "sealed pit" nuclear components. While the elimination of in-flight arming was a great technological improvement, it also introduced the possibility of disaster if the new "sealed pit" weapons were not at least as safe as the older interchangeable-core warheads.

Operation JULEP, originally scheduled for the fall of 1956, was canceled and tests from JULEP were postponed to Operation HARDTACK now slated for the PPG in the fall of 1957. Another Nevada series, Operation PILGRIM, was planned for the spring of 1958.

REDWING was now expected to include 14 shots (exclusive of one-point safety tests), and HARDTACK and PILGRIM would each include 10 shots.

The AEC now had 11 weapons storage sites in operation and one under construction; two new sites were planned for

¹⁷⁸Atomic Energy Commission, Program Status Report, Part III - Weapons Program, July 27, 1955, pp. 8. 9. The more things change, the more they stay the same: during the 1990s, in preparation for a possible comprehensive nuclear test ban treaty, the U.S. turned once again toward interchangeable warheads for a number of different delivery systems.

construction by June 1956.¹⁷⁹

In early July, the issue of DOD's high-yield weapon requirements had arisen again, this time in connection with the impending release by the AEC of a new edition of its nuclear weapons effects handbook.

At an AEC-MLC conference, there was considerable debate over just how high a yield should be included on a set of yield-versus-effects curves. AEC Chairman Lewis Strauss was somewhat alarmed at a 100 megaton upper limit, and was afraid that both the U.S. public and the Soviets might conclude from it that the U.S. either had stockpiled or was planning to build bombs with that very high yield. The Soviets might be sufficiently alarmed to try to match the U.S. weapons with their own high-yield bombs and warheads.

Speaking for the DOD, Major General Herbert Thatcher, acting chairman of the MLC, stated that the 100 megaton figure was merely a hypothetical limit that was a compromise between other values of 70 and 125 MT. DOD did not want the effects curve to stop at present bomb yields, or to approximate the yields of stockpiled thermonuclear warheads.

Strauss stressed the danger of unrealistic yield values on one hand, while noting the possible interpretation by the Soviets that 100 MT was a U.S. goal.

Dr. Herbert Scoville claimed that the use of only 20 MT as an upper limit in the handbook might tend to limit the views of both weapons designers and civil defense planners to only 20 MT effects. There was no guarantee that Soviet bomb designers would limit their weapons to an upper yield limit of 20 MT; indeed, the U.S. was already seeking higher yields for its own warheads.

AEC Commissioner Thomas Murray commented on the moral responsibility of the AEC for the release of high weapon yield values, and the use or misuse of that information by the U.S. public and other nations. Murray stated that if the AEC sanctioned the possibility of 100 MT weapons, then it would be inferred that the U.S. was planning to make 100 MT bombs, and the AEC would bear the responsibility for the inference.

Murray suggested that if DOD wanted 100 MT warheads, then it should make them a formal requirement to the AEC. Like Strauss, Murray was also concerned that by just mentioning such high yields, the Soviets would be spurred to achieve

¹⁷⁹Atomic Energy Commission, Program Status Report, Part III - Weapons Program, July 27, 1955, pp. 13, 14.

them.

The meeting finally closed with a recommendation that the AEC and the AFSWP evaluate the probable psychological impact of including high-yield curves in the effects handbook.¹⁸⁰

Following a series of hearings that culminated in July, the Joint Committee on Atomic Energy issued a report which, among other things, called for increased tritium production and more emphasis on both "clean" and small-diameter weapons:

Moreover, the evolution of weapons technology now suggests that, even without a marked step-up in our total requirements for weapons, we will need larger quantities of tritium and lithium-6 than will be forthcoming under our existing production plans.

This conclusion is based upon the trend toward utilizing thermonuclear reactions in weapons going well down toward the lower end of the yield spectrum, and on the apparent advantages of "boosting" in a wide variety of weapon models.

We find that both the Los Alamos and Livermore laboratories have development programs of outstanding excellence under way. We find, nonetheless, that there is a need for increased effort in two specific areas -- the design of "clean" thermonuclear weapons minimizing radioactive fallout, and the design of very small weapons in the [deleted] diameter range. We also find that there is need for more intensive study within the military establishment concerning how such weapons might be most effectively utilized in combat.¹⁸¹

By the end of July, the REDWING shot schedule had been revised as shown in Table 3-10.

¹⁸⁰Minutes of Ninety-Ninth AEC-MLC Conference, July 5, 1955, pp. 4-6.

¹⁸¹Report of the Military Applications Subcommittee of the Joint Committee on Atomic Energy, ca. July 29, 1955, p. 4.

Table 3-10 REDWING Shot List, July 1955¹⁸²

<u>Shot Name</u>	<u>Ready Date</u>	<u>Site</u>
Cherokee	1 May 1956	Bikini (Namu) Airdrop
Erie	1 May	Eniwetok (Runit) 300-foot tower
Yuma	1 May	Eniwetok (Aomon)
Flathead	3 May	Bikini (Yurochi) Barge
Osage	7 May	Eniwetok Airdrop (originally 300 ft. tower shot on Runit)
Zuni	15 May	Bikini (Eninman) Ground
Blackfoot	15 May	Eniwetok (Runit) 300-foot tower
Kickapoo	15 May	Eniwetok (Aomon) 300-foot tower
Huron	19 May	Bikini (Yurochi) Barge
Navajo	28 May	Bikini (Yurochi) Barge
Mohawk	5 June	Eniwetok (Eberiru) 300-foot tower
Lacrosse	12 June	Eniwetok (Runit) Ground (may be repeated if necessary)

¹⁸²Project Officers Meeting, 27-28 July 1955, Operation REDWING, Headquarters, Task Group 7.1, Joint Task Force Seven, p. 13; Planning Directive for Operation REDWING, CJTF Seven No. 1-55, Headquarters, Joint Task Force Seven, 10 August 1955, p. 2; Evans, FINAL HISTORY, TG 7.4, p. 50.

Apache

1 July

Bikini (Yurochi)
Barge or airdrop

Cherokee, a TX-15-X1 airdrop, was to be both a proof-test and an effects shot; burst height of 5,000 feet was to be slightly higher than the expected fireball radius.¹⁸³ Although a B-36 was first considered as a bomb drop aircraft, it was too slow to escape the thermal effects of the free-falling test device, so a faster B-47 was substituted.¹⁸⁴ Some thought was also given to using a 100-second delay drogue parachute drop technique.¹⁸⁵

The Osage shot, a proof-test of the XW-25 air defense warhead, was also to be an air drop, although from a much lower altitude.¹⁸⁶

The large number of LASL tests scheduled sharply reduced the amounts of time and manpower that LASL could devote to helping UCRL with its extensive REDWING plans. In particular, the LASL GMX division was forced to cut back its fabrication of high explosive components for UCRL tests.

Since UCRL did not at this time have its own equivalent to LASL's "S" site, and since the Livermore high explosive fabrication and test facilities -- later known as Site 300 -- was still under development, the California lab turned to DOD for assistance. In particular, UCRL was leaning toward the use of the Navy's Naval Ordnance Testing Station (NOTS) at Inyokern, California, and the Navy Ordnance Laboratory (NOL). As an alternate HE fabrication site, the Army's Picatinny Arsenal in New Jersey might be useful.¹⁸⁷

At the beginning of August, UCRL had made contingency plans to conduct a test of the primary of the Mohawk device before the full-scale two-stage shot; if this were done, the shot would be detonated on a 200-foot tower on Rujoru island at

¹⁸³Memorandum dated 12 August 1955 to Distribution from Commander Task Group 7.1, Subject: Minutes of the Project Officers' Meeting Held in Los Alamos July 27-28, 1955, p. 10.

¹⁸⁴Monthly History - Operations Directorate, 8 September 1955.

¹⁸⁵Evans, FINAL HISTORY, TG 7.4, p. 27.

¹⁸⁶Memorandum to Commander, 4925th Test Group (Atomic), from Headquarters, Task Group 7.4 (Provisional), 25 July 1955, Subject: Air Drop Requirements for REDWING.

¹⁸⁷Letter dated July 25, 1955 to Lewis L. Strauss, Chairman, USAEC, from Edward Teller.

Eniwetok atoll.

If necessary, UCRL also planned to conduct a "Lacrosse II" shot at Eniwetok atop a 300-foot tower; this would be the contingent Pawnee shot.¹⁸⁸

By late August, a prototype "clean" weapon was also slated for test during REDWING.¹⁸⁹

In late September, during the 46th meeting of the GAC, one question addressed was whether some information on the configuration of contemporary U.S. hydrogen bombs might safely be passed on to allies. This discussion echoed a similar debate in December 1953 over the possibility of sharing fission weapons data with the British:

One question addressed to us was whether information on the external characteristics of a thermonuclear weapon (including size, weight, shape, and center of gravity, but specifically excluding any details about the interior) and certain information on yields, effects, and delivery systems would reveal to another country "important information concerning the design or fabrication of the nuclear components" of the thermonuclear weapon.

We arrived at the following opinion after devoting a great deal of discussion to this difficult question. We believe that the information cited would not directly reveal important information concerning the design and fabrication of the nuclear components of the weapon.

Nevertheless, we feel that the information, taken in combination, may give valuable clues and stimulus and direction to the development of similar weapons. We are particularly concerned in this regard with the thought that revelation of the very high specific energy release attained in our weapons (on the order of two megatons

¹⁸⁸Classified Notebook No. 681(L) issued by Information Division, UCRL Livermore; issued originally to Paul Harrold, L Division Operations, Livermore; reissued to John Flynn, L-3, L Division - Operations, Personal Notebook.

¹⁸⁹Minutes of Executive Session of AEC Meeting 1118, August 24, 1955.

per ton weight) would call attention to the fact that a radical new technique has been developed and is being exploited.¹⁹⁰

At the end of September, REDWING had been expanded to 14 shots, six at Bikini and eight at Eniwetok. Of these, nine were LASL tests and five were UCRL experiments. It might be necessary for each lab to conduct at least one additional shot to attain the desired objectives of the test series.

Although REDWING was expected to produce far less fission yield than CASTLE, an enlarged danger area would be declared around Bikini and Eniwetok atolls. The half-million square statute mile semicircular danger area finally established for CASTLE was replaced by a nearly 516,000 square statute mile rectangular area with Bikini and Eniwetok atolls near its southern edge; most of the danger area lay to the north and east of the two atolls.

This danger area was to be closed to shipping and aircraft on April 20, 1956, approximately ten days before the first scheduled REDWING shot. The proposed warning notice stated that "grave hazards, as a consequence of tests of military weapons, will exist in this area."¹⁹¹

In addition, all six CASTLE shots were surface bursts that pulled very large quantities of pulverized coral and other radioactive debris into the atmosphere. As now planned, of the three high-yield "dirty" or "unclean" devices to be fired during REDWING, the one with the highest yield, the Cherokee device, was to be fired as a true airburst, at an altitude sufficiently high to preclude its fireball touching the ground, thereby eliminating most of the radioactive soil, mud, and water from the mushroom cloud.

REDWING was to commence on May 1, 1956 and extend through the early part of the following July. The series was to test weapons for both strategic and tactical employment, and to continue research and development of small designs suitable for use in air-to-air and surface-to-air missiles.

Another concept to be tested was the so-called "clean" low-fallout thermonuclear weapon design. The total yield expected

¹⁹⁰Letter dated September 23, 1955 to Lewis L. Strauss, Chairman, USAEC, from I. I. Rabi, Chairman, GAC. The numbers listed translate to a yield-to-weight ratio of one kiloton per pound of gross weapon weight.

¹⁹¹Letter dated 29 December 1955 to Lewis L. Strauss, Chairman, USAEC, from RADm. B. Hall Hanlon, USN, Commander, JTF 7.

for all 14 REDWING shots was predicted to be less than half the total 48.2 MT yield of the six CASTLE shots in 1954. The shot sequence at this time is shown in Table 3-11.

Table 3-11 REDWING Shot List, October 1955¹⁹²

<u>Shot Name</u>	<u>Sponsor</u>	<u>Site</u>
Cherokee	LASL	Bikini (Namu) Airdrop
Yuma	UCRL	Eniwetok (Aomon) 200-foot tower
Lacrosse	LASL	Eniwetok (Runit) Ground
Osage	LASL	Eniwetok Airdrop (1,000 ft. HOB)
Kickapoo	UCRL	Eniwetok (Aomon) 300-foot tower
Zuni	UCRL	Bikini (Eninman) Ground
Erie	LASL	Eniwetok (Runit) 300-foot tower
Seminole	LASL	Eniwetok (Bogairikk) Ground (formerly Pueblo shot)
Mohawk	UCRL	Eniwetok (Eberiru) 300-foot tower
Flathead	LASL	Bikini (Yurochi) Barge
Blackfoot	LASL	Eniwetok (Runit) 200-foot tower
Huron	LASL	Bikini (Yurochi) Barge
Navajo	LASL	Bikini (Yurochi)

¹⁹²Atomic Energy Commission, Pacific Proving Ground Test Activities for Calendar Year 1956, Report to the General Manager by the Director of Military Application, September 30, 1955.

		Barge
Apache	UCRL	Bikini (Yurochi) Barge
Pawnee	LASL	Eniwetok (Aomon) Tower
(Unnamed)	UCRL	Unspecified

The first fourteen shots were required for several reasons. Cherokee, the LASL-sponsored airburst over Namu island at Bikini, was to satisfy both military effects information needs and supply data for future weapons.

Lacrosse was a test of a new gas-boosted, externally-initiated pit for the TX-39; expected yield was 30 to 55 kilotons. Zuni, a test of a "clean" UCRL device, was expected to yield two to three megatons. Seminole, a test of a TX-28 primary, was expected to yield approximately 10 kilotons.

Two provisional shots, Pawnee and an as-yet-unnamed test, were extra LASL and UCRL experiments that would be conducted only if results of earlier tests were unsatisfactory.¹⁹³

Following re-evaluations in mid-October by the AFSWC, a B-52 was finally chosen as the Cherokee shot drop aircraft. The B-36 was considered marginal in regard to safety for free-fall delivery of a MK 15; the B-47 could safely deliver a drogued (parachute-retarded) weapon at maximum speed from certain altitudes, but it was also less than ideal for the Cherokee test.

The marginal aspects of free-fall drop by either the B-36 or the B-47 and the current uncertainty of a drogue parachute program for the MK 15 made use of a B-52 desirable. Since a drogued MK 15 appeared unlikely in the near future, then free-fall delivery ultimately dictated use of a B-52.¹⁹⁴

¹⁹³Atomic Energy Commission, Pacific Proving Ground Test Activities for Calendar Year 1956, Report to the General Manager by the Director of Military Application, September 30, 1955.

¹⁹⁴History of Task Group 7.4, Provisional, October 1955, Joint Task Force Seven, pp. 15 - 19. For a while, F-84F and F-86F or F-86H aircraft using LABS (Low-Altitude Bombing System) were considered as delivery aircraft for the Osage shot; however, their likely bombing errors were too great for the scientific measurements that were required for the shot. (Monthly History - Operations Directorate, TG 7.4, JTF 7, 7 October 1955.)

The U.S. nuclear weapons stockpile at this time comprised the MKs 5, 6, 7, 8, 9, 12, 15, 17, 18, and 24 bombs, artillery shells, and missile warheads. Among other weapons under development were the TX-19, -21, -23, and -33.¹⁹⁵ Results of REDWING would soon expand this arsenal.

Also in mid-October, the AEC advised the JCAE that "clean" weapon prototypes were to be tested during REDWING:

Two clean TN weapon devices are being fabricated now for test at the Pacific Proving Ground during REDWING, planned for the Spring of 1956. The research connected with this effort is of great interest to the AEC and has been underway for some time. Performance under test will have a major effect in establishing the avenues of further research and development.

REDWING was also to include tests of small-diameter, low-yield fission devices for different applications and delivery systems, including 105MM howitzers:

Both Los Alamos Scientific Laboratory (LASL) and University of California Radiation Laboratory (UCRL) at Livermore have been giving great attention for some time to the practicability of low yield devices of very small diameter. Such devices require relatively large amounts of fissionable material for the yield obtained.

Tests of such devices are planned for REDWING. LASL will test an 11" diameter spherical implosion system and UCRL will test three [deleted] diameter with yields in the range of 0.1 to 10 KT. Both 11" devices could lend themselves to serving as primaries for small diameter two-stage systems. We are now studying jointly with the Department of Defense possible applications of the 11" devices and the application of [deleted]

The EGG device is being tested with a view toward applying its nuclear system to such

¹⁹⁵Letter dated 14 October 1955 to Director, Division of Research, USAEC, from Lt. Col. H. O. Beeth, Adjutant General, AFSWP.

weapons as the [deleted] It may be, however, that the ultimate efficiency of such devices, and consequently their ultimate cost (either in terms of dollar expenditures or in terms of use of fissionable material other weapon types) will not make desirable their employment in all of the delivery systems which would otherwise offer attractive use possibilities. If REDWING results indicate possibility of extending the [deleted] to small diameter, its applicability to the 105MM gun will be investigated.

We believe the REDWING tests, together with studies underway with the Department of Defense, will provide for early exploitation of low-yield weapon possibilities.¹⁹⁶

One of the factors affecting both REDWING and the one-point safety tests planned to occur before REDWING was the Joint Chiefs of Staff's current expressed priority for future weapon stockpiling. In the early fall of 1955, the most desirable sequence of programs, from the JCS point of view, was:

- a. XW-25, approximately 17" in diameter, approximately 230 lbs., [deleted] unboosted acceptable in initial version.
- b. TX-28, not more than 20" in diameter, approximately 1,500 lbs., for the bomb version, [deleted] boosted primary.
- c. XW-34, approximately 18" in diameter, not more than 300 lbs., [deleted] boosted version.
- d. XW-30, 22" in diameter, approximately 350 lbs., [deleted] boosted version.

¹⁹⁶Letter dated October 12, 1955 to Senator Clinton P. Anderson, Chairman, JCAE, from Lewis L. Strauss, Chairman, USAEC. During REDWING, LASL ultimately tested an 11.5" diameter device in the Blackfoot shot; it yielded 8.5 KT. UCRL tested 5", 8", and 11" diameter devices -- the SWIFT, SWALLOW, and SWAN, respectively -- during the Yuma, Kickapoo, and Inca shots, respectively. The Inca shot yielded 16 KT and the SWAN was used as a primary with a FLUTE secondary in the 350 KT Mohawk shot. Some of the small diameter UCRL REDWING test devices may have used linear implosion, possibly with boosting.

e. XW-27, 30" in diameter, not more than 2,800 lbs., [deleted] boosted primary.

f. XW-31, 30" in diameter, not more than 1,000 lbs., [deleted] boosted version.

The MLC noted that this sequence was not in accord with an earlier sequence submitted by the AEC to the MLC in August 1955; however, the MLC also noted that the AEC had stated it could develop one new weapon per quarter and requested that new operational availability dates be provided by the AEC to the MLC as soon as possible. ¹⁹⁷

As far as potential REDWING tests addressed the Joint Chiefs' desires, the ZITHER, a possible UCRL candidate for the XW-27 secondary, was to be tested with a LASL primary in the Apache shot, as was a LASL candidate for the XW-25 air-to-air rocket warhead in the Osage shot.

Several TX-28 tests were to be conducted by LASL during the Erie, Seminole, Flathead, and Dakota shots. Full-scale tests of the XW-30, -31, and -34 would not be conducted until the next Nevada series.

A preliminary report on the advisability of on-site inspection as part of a possible international nuclear test ban prepared at this time by a task force chaired by Ernest O. Lawrence noted that a test moratorium would be dangerous because the momentum of technology development would be lost, and stated that "guarding of TN secret (was) imperative."¹⁹⁸

On October 25, 1955, the purposes of the REDWING tests and their expected magnitude were summed up in a letter from the Chairman of the AEC to Secretary of State John Foster Dulles:

The proposed tests, designated Operation REDWING, are designed to achieve objectives vital to our continued progress in the research and development of atomic weapons and in the improvement of our atomic defenses. In addition to testing weapons for strategic and tactical employment, this

¹⁹⁷Memorandum for the Chairman, USAEC, dated October 5, 1955 from Herbert B. Loper, Chairman, MLC, Subject: Atomic Weapon Development Guidance. The deleted information presumably specifies yields.

¹⁹⁸Memorandum for Mr. Harold E. Stassen dated October 13, 1955 from McKay Donkin (Donkin was an aide to Lewis Strauss). The "TN secret" presumably refers to the "three concepts."

program will proof-test a prototype warhead for air-to-air rockets and will continue research and development of small designs suitable for use in air-to-air and surface-to-air missiles.

Although there will be a greater number of shots in REDWING than there were in the previous Operation CASTLE, the total cumulative yield expected for REDWING is considerably less than one-half the cumulative yield obtained during CASTLE.

Even more important to considerations of the possible effects of radioactive fall-out, the total cumulative fission yield expected from REDWING is less than one-quarter of that obtained in CASTLE. Furthermore, the maximum expected yield of the largest single shot during REDWING is about one-third the maximum obtained on the largest single shot in CASTLE.¹⁹⁹

A "two-shot" concept meeting was held at Los Alamos on October 26; a result of this meeting was a plan to fire shots at either atoll on the same day (this was done on two occasions; identical shot times were used in one case). This required that task units have the capability of operating at both Eniwetok and Bikini atolls simultaneously.

In the same way that the Soviet nuclear test of August 1953 stimulated the American "emergency capability" program, an airburst test of a true two-stage 1.6 MT Soviet thermonuclear device on November 22 spurred REDWING. Adding further urgency was the fact that the Soviet device was dropped from a bomber, thus making it the first operational parachute-retarded airdropped hydrogen weapon.²⁰⁰ The device used U-235, U-233, U-238, and lithium-6 deuteride.²⁰¹

¹⁹⁹Letter dated October 25, 1955, from Lewis L. Strauss, Chairman, USAEC, to the Honorable John Foster Dulles, Secretary of State. The air-to-air rocket warhead proof test was the Osage shot. An earlier draft of this Strauss letter appears as a memorandum dated October 21, 1955 to Elmer B. Staats, Executive Officer, Operations Coordinating Board, from Paul F. Foster, Special Assistant to the General Manager (Liaison), USAEC, Subject: Atomic Tests at the Pacific Proving Ground, commencing about May 1, 1956.

²⁰⁰Zaloga, p. 105, 106.

²⁰¹NUCLEAR WEAPONS DATABOOK, Vol. IV, p. 349.

Just the day before the Soviet test, LASL director Norris Bradbury had written to an AEC officer and contemplated the future role of the laboratory, especially if weapons work were to cease; Bradbury bemoaned the apparent lack of a near-term technological challenge comparable to the odyssey in search of the Super:

Replying to your letter of October 6, 1955, ten pages will not be required to describe for the General Advisory Committee what appears to me to be the fundamental long-range Los Alamos problem since it can be very simply stated as follows: What should be the eventual technical role of Los Alamos as atomic weapons gradually cease to have a dominant role in international relations?

Although the developmental demands upon the Laboratory (sic) are very heavy today and five or ten years of very hard work can be foreseen, the future beyond that point looks somewhat unrewarding. Fissionable material will go on and on being made until the efficiency of atomic weapons will become of academic interest. Everyone will ultimately have all the weapons in all the variety wanted, and the number will probably be more than the world can safely tolerate being used.

Inspection, control, (and) atomic disarmament are words continually in the public eye and, in one form or another, (are) an announced objective of our Government. (sic) The steady advance of our potential enemy -- along with our own advance -- in the atomic weapons field points to the eventual decline of this technique as a unique source of international dominance.

Moreover, there does not appear to be any unscaled peak on the horizon in 1955 comparable to the challenge of a thermonuclear weapon a decade ago. In all frankness, the things the weapons laboratories are now doing and foresee doing are modifications, variants, and extensions of basic ideas going back, in some cases, many years.

Because of this prospect, and because the

laboratory must have an intelligible, realistic, and exciting future for its staff if they are to remain and work hard on weapons now, we have increased our interest and effort in the reactor field, the nuclear propulsion fields, the Sherwood field, and in basic research. However, in most of those fields, we are unclear as to the eventual relationship the (Atomic Energy) Commission desires between its laboratories and industry.

In short, if atomic weapons are abandoned, what should Los Alamos then be in a position to do? How should we plan today?²⁰²

Although the Livermore lab might have argued that LASL was being short-sighted and there were still many nuclear weapons design breakthroughs to be made at this time, the questions Bradbury raised in 1955 were the same issues that both labs would face nearly 40 years later at the end of the Cold War.

At the beginning of December, the REDWING shot sequence had changed yet again, as shown in Table 3-12.

Table 3-12 REDWING Shot List, December 1955²⁰³

<u>Shot Name</u>	<u>Ready Date</u>	<u>Sponsor</u>	<u>Site</u>
Lacrosse	1 May	LASL	Eniwetok (Runit) Ground
Cherokee	1 May	LASL	Bikini (Namu) Airdrop, 5,000 ft. HOB
Zuni	15 May	UCRL	Bikini (Eninman) Ground

²⁰²Letter dated November 21, 1955 to T. E. Johnson, Director, Division of Research, USAEC, from Norris E. Bradbury, Director, LASL. Project Sherwood was an early and long-lived program of research into controlled thermonuclear fusion for commercial power generation.

²⁰³Classified Notebook No. 681(L) issued by Information Division, UCRL Livermore; issued originally to Paul Harrold, L Division Operations, Livermore; reissued to John Flynn, L-3, L Division - Operations, Personal Notebook.

Erie	23 May	LASL	Eniwetok (Runit) 300-ft tower
Seminole	28 May	LASL	Eniwetok (Bogairikk) Ground
Yuma	1 June	UCRL	Eniwetok (Aomon) 200-ft tower
Flathead	2 June	LASL	Bikini (Yurochi) Barge
Blackfoot	7 June	LASL	Eniwetok (Runit) 200-ft tower
Inca	8 June	UCRL	Eniwetok (Rujoru) 200-ft tower
Huron	12 June	LASL	Bikini (Yurochi) Barge
Osage	14 June	LASL	Eniwetok Airdrop (700 ft. HOB)
Kickapoo	18 June	UCRL	Eniwetok (Aomon) 300-ft tower
Navajo	18 June	LASL	Bikini (Yurochi) Barge
Mohawk	1 July	UCRL	Eniwetok (Eberiru) 300-ft tower
Apache	1 July	UCRL	Bikini (Yurochi) Barge
Tewa	7 July	UCRL	Bikini (Yurochi) Barge

Dates were still pending for the Dakota, Pawnee, and Wyandotte shots. There was also the possibility that delays might protract the test series until mid-August 1956.²⁰⁴

In mid-December, UCRL's post-REDWING weapons development plans included very-small diameter devices; cleaner "clean" weapons; smaller bombs; and a boostable all-oralloy gadget. There were over a dozen programs being considered at Livermore; only four to six would be pursued, depending upon REDWING results.²⁰⁵

By the end of December, the total number of REDWING shots was up to fifteen, with total fission and fusion yield expected to be one-half the cumulative CASTLE yield. Total REDWING fission yield was predicted to be approximately one-third that of CASTLE.²⁰⁶ Both of the total yield figures represented slight increases over the predictions made two months earlier.

In its regular quarterly progress report to the congressional Joint Committee on Atomic Energy, the AEC stated that a test of a "clean" device was a high-priority objective of REDWING:

Also to be tested is a "clean" design concept for thermonuclear weapons. The objective is to obtain the greater portion of the energy yield from fusion rather than from fission, so that the radioactive contamination, which results from the

²⁰⁴Joint Task Force Seven, Task Group 7.3, Concept of Operation, Annex A to CGT 7.3 Operation Plan No. 1-56, 24 January 1956, p. A-2.

²⁰⁵Minutes of the Forty-Seventh Meeting of the General Advisory Committee to the USAEC, December 13-15, 1955, pp. 12, 13.

²⁰⁶Letter dated December 29, 1955 to Senator Clinton P. Anderson, Chairman, JCAE, from W. F. Libby, Acting Chairman, USAEC. Total CASTLE fission and fusion yield was 48.2 MT. Of this, nearly 30 MT was fission yield. According to the NRDC's NWD 86-2 (p. 18), the total fission yield for REDWING was 9 MT, only 30% that of CASTLE. The largest REDWING shot was the 5 MT Tewa test, which was indeed a third the size of the 15 MT CASTLE Bravo (SHRIMP device) shot. The total REDWING fission and fusion yield was just under 20.6 MT; although REDWING included nearly three times as many shots as CASTLE, the total fission and fusion yield for the entire test series was indeed finally less than half as large as the 48.2 MT fission and fusion yield of the six CASTLE shots two years earlier.

fission products, can be much less than from current weapons designs.²⁰⁷

Lewis Strauss had told President Eisenhower earlier in December that "our tests scheduled for the next two years are extremely important to our attainment of warheads for ballistic missiles, high-yield weapons for delivery by tactical-type aircraft, and warheads for air defense weapons."

Strauss also noted that a press release announcing REDWING had cleared the State Department, but was being delayed in its publication because of fears of repercussions in the UN General Assembly and in Japan. General Fields was proposing that the release be delayed until just after the beginning of 1956 when both the UN General Assembly and the Japanese Diet (the parliament of Japan) would be in adjournment.²⁰⁸

In early January 1956, REDWING included 15 scheduled shots and two potential additional tests contingent upon the results of previous tests. Among UCRL test devices was a new three-stage thermonuclear system, the BASSOON, proposed as a method of satisfying revised specifications for Class "A" and "B" weapons.²⁰⁹

UCRL further proposed that an early operational capability for the BASSOON could be achieved if a second version of the device, named BASSOON PRIME and identical to the BASSOON except for the use of a U-238 fusion tamper on its secondary and tertiary stages, were also built and tested during REDWING. The first BASSOON was to be "clean" and used lead tampers on its fusion stages in lieu of fissionable tampers.

²⁰⁷Program Status Report, December 31, 1955, Part III Weapons, quoted in memorandum for the Chairman dated July 23, 1956 from Brig. Gen. A. D. Starbird, Director of Military Application, USAEC, Subject: Notifying Joint Committee Relative to Test Shots.

²⁰⁸Memorandum entitled "Possible Subjects to be Discussed with the President on 19 December 1955," dated 17 December, attachment to note by Lewis Strauss dated 12-19-55; the note states: "These were the items discussed by me with the President at his home in Gettysburg this morning."

²⁰⁹COPB 4547, OPERATION REDWING - POST OPERATIONAL REPORT, PROGRAM 22 - REACTION HISTORY MEASUREMENTS, by L. F. Wouters, University of California Radiation Laboratory, Livermore, California, July 1960, states (on pp. 16 and 17) that time intervals were measured between secondary and tertiary stages on the Zuni and Tewa shots, which were tests of the BASSOON and BASSOON PRIME devices, respectively.

If the BASSOON PRIME were successful, it would provide a new Class "A" or "B" bomb approximately one year earlier than would be possible if the test were conducted at the next overseas test series, Operation HARDTACK, then slated for the spring of 1958. (The BASSOON may have been a prototype for the three-stage UCRL MK 41 gravity bomb which began entering the stockpile in September 1960).

In addition, for a while there was a possibility that REDWING might also include a one-point safety test of a LASL XW-34 prototype whose design had not been firmed up in time for the Project 56 one-point safety test series in Nevada in early November 1955 (this safety test was deleted from REDWING when the experiment was conducted at the Nevada Test Site on February 29, 1956.)²¹⁰

On January 12, the AEC announced publicly that "the forthcoming series of nuclear tests at the Eniwetok Proving Grounds ... will involve weapons generally smaller in yield than those tested during the 1954 test series. It is anticipated that the energy release of the largest test will be substantially below that of the maximum 1954 test."²¹¹ A subsequent release four days later noted that "one of the important purposes of this series will be the further development of methods of defense against nuclear attack," an oblique reference to the planned testing of small-yield air defense warhead prototypes.²¹²

Before these official announcements, press speculation had begun. A January 7 Associated Press wire service story had stated:

A new super hydrogen blast, dwarfing both this country's and Russia's previous best efforts, reportedly will

²¹⁰Atomic Energy Commission, Pacific Proving Ground Test Activities for Calendar Year 1956, Report to the General Manager by the Director of Military Application, January 10, 1956; "Timetable of MK 34 Events," September 22, 1986. If the W-34 was fired in a one-point safety test on February 29, 1956, this test has never been announced by DOE.

²¹¹Statement by Lewis L. Strauss, Chairman of the Atomic Energy Commission dated January 12, 1956.

²¹²Memorandum dated January 25, 1956 to All Members of Task Group 7.1 from Commander, Task Group 7.1, subject: Official AEC-DOD Announcement Regarding Operations at the Pacific Proving Ground.

be set off by the United States this year. Congressional sources said the test would be part of a series the Atomic Energy Commission is planning in the Pacific Ocean testing area. The AEC itself declined to comment.

Speculation put the size of the major test between two and three times the force of the hydrogen device the United States detonated at Bikini on March 1, 1954, which caused a small island to disappear completely. That would put the strength of the new blast at the equivalent of between 30 and 50 million tons of TNT.

Russia reported last November that it had set off its most powerful hydrogen bomb. AEC chairman Lewis L. Strauss said the Russian explosion was "in the range of megatons" and later published reports set it at between one and five megatons. A megaton is the equivalent of one million tons of TNT.

The United States' 1954 Bikini blast is generally thought have had a force of 15 to 17 megatons. Widespread radioactive fallout from that blast reportedly caused U.S. officials to cancel plans for a second explosion with perhaps as much as 30 megatons of force.

The AEC has made no announcement of any kind concerning forthcoming tests, even to the Senate House Committee (sic) on Atomic Energy.

...

In theory, a hydrogen or fusion bomb, unlike a conventional atomic fission (bomb), could be developed to almost any size desired. Practically, such size would be limited by the need for fitting it into a transportable weapon, such as a bomb casing for air delivery.

Under present plans, more than one island site (will) be used, it was said, in order to conduct a series of

tests (rapidly).²¹³

The final concept of Operation REDWING was completed on January 20, 1956, including the final shot schedule of January 16. The tentative ready date for the first shot was scheduled for May 1 and the test series was estimated to end by August 15. Sixteen definite shots were now scheduled, with three contingent tests, as shown in Table 3-13; shot sponsors and predicted yields are also listed. This schedule was virtually identical to the November 1955 schedule listed above, except for relocation of the Seminole shot.

Table 3-13 REDWING Shot Schedule, 25 January 1956²¹⁴

<u>Shot Name</u>	<u>Ready Date</u>	<u>Shot Site</u>
Lacrosse (LASL)	1 May	Eniwetok (off Runit) Ground (25-50 KT)
Cherokee (LASL)	1 May	Bikini (Namu) Air drop, 5,000 foot HOB (MT range)
Zuni (UCRL)	15 May	Bikini (Eninman) Ground (2-3 MT)
Erie (LASL)	23 May	Eniwetok (Runit) 300-foot tower
Seminole (LASL)	28 May	Eniwetok (Bogon) Ground (10 KT)
Yuma (UCRL)	1 June	Eniwetok (Aomon) 200-foot tower (boosted device)

²¹³Associated Press release dated January 7, 1956. The misidentified "Senate House Committee on Atomic Energy" is the JCAE.

²¹⁴OPERATION REDWING, Operation Order No. 1-56, Task Group 7.1, Headquarters, Task Group 7.1, Joint Task Force SEVEN, San Francisco, California, 25 January 1956, pp. C-4, C-5; Atomic Energy Commission, Pacific Proving Ground Test Activities for Calendar Year 1956, Report to the General Manager by the Director of Military Application, January 10, 1956; OPERATION REDWING, Operation Plan No. 1-56, CJTF 7, 20 January 1956, pp. A-1, A-2. Predicted yields are in parentheses. "Ready date" was the date was when device was to be available for test.

Flathead (LASL)	2 June	Bikini (Yurochi) Barge over 125 ft. of water
Blackfoot (LASL)	7 June	Eniwetok (Runit) 200-foot tower
Inca (UCRL)	8 June	Eniwetok (Rujoru) 200-foot tower
Huron (LASL)	12 June	Bikini (Yurochi) Barge
Osage (LASL)	14 June	Eniwetok (Runit) Air drop, 700 foot HOB
Kickapoo (UCRL)	18 June	Eniwetok (Aomon) 300-foot tower
Navajo (LASL)	18 June	Bikini (Yurochi) Barge over 180 ft. of water
Mohawk (UCRL)	1 July	Eniwetok (Eberiru) 300-foot tower
Apache (UCRL)	1 July	Bikini (Yurochi) Barge over 140 ft. of water
Tewa (UCRL)	7 July	Bikini (Yurochi) Barge (6-8 MT)

Dakota -----

Pawnee	- Contingent LASL shots; to be announced later
Wyandotte --	

Some of the shots on this list were rescheduled at least once again before the series began, with Lacrosse slated for May 5, Cherokee on May 8, Erie on May 27, Seminole on June 1, Flathead on June 5, Navajo on June 8, Huron on June 15, Dakota (an added TX-28 test) on June 16, and Apache on June 20. Dakota was slated to be a barge shot near Yurochi island at Bikini atoll.²¹⁵

If the Lacrosse shot were unsuccessful, a second attempt to fire the same type of device would be made in a shot named

²¹⁵Atomic Energy Commission, Program Status Report, December 31, 1955, Part III - Weapons, February 15, 1956, p. 15.

Pawnee, to be detonated on a 300-foot tower on Engebi island at Eniwetok.²¹⁶

The bomb drop for shot Cherokee was originally assigned to a B-36 from the Air Force's 4925th Test Group (Atomic), based at Kirtland AFB, New Mexico; by the end of January, plans for using a B-52B (serial 52-013) instead were progressing satisfactorily, and much information had been gained from practice drop missions during the month.²¹⁷

Faced with mounting international criticism of atmospheric testing, the AEC justified Operation REDWING at this time with the following statement:

Testing is a vital part of the development of any complicated mechanism, whether it be automobile, aircraft, nuclear reactor, or nuclear weapon. To build a better device, new principles, ideas and designs must first be tested. Little progress in the development of nuclear weapons could be made on the basis of laboratory experiments and calculations alone. Without tests, our weapons development would soon come to a virtual standstill. To be more specific, tests of nuclear weapons and devices are undertaken for the following reasons:

- a. To proof-test a weapon before it enters the stockpile, as a check on the engineering, fabrication, and integration of the components into an effective whole.

²¹⁶History of Task Group 7.4, Provisional, March 1956, Joint Task Force Seven, Joint Chiefs of Staff, p. 14.

²¹⁷Memorandum dated 30 June 1955 to Distribution from Headquarters, Task Group 7.4 (Provisional), Kirtland AFB, New Mexico, Subject: Concept of Air Task Group Operations, Operation REDWING; History of Task Group 7.4, Provisional, January 1956, Lt. Col. Worth E. Ober, Jr., USAF, Joint Task Force Seven, pp. 8-10. This B-52B is now displayed at the National Atomic Museum in Albuquerque, New Mexico; besides being the REDWING Cherokee drop aircraft, it also participated in device and instrumentation drops during Pacific Proving Ground Operations HARDTACK Phase I and DOMINIC in 1958 and 1962, respectively.

- b. To test new designs of weapons and components before undertaking their fabrication.
- c. To test new principles and ideas which show promise.
- d. To provide information required to confirm or correct theoretical calculations.
- e. To obtain information on the effects of weapons needed by civilian and military authorities for defensive purposes.²¹⁸

Other statements suggested to the AEC at this time concerning REDWING were less than honest: a January 3, 1956 memorandum for Lewis Strauss from Theodore C. Streibert, Director, U.S. Information Agency, stated that the AEC should stress the defensive nature of REDWING, i.e., that defensive weapons were to be tested.

While this was true, it was far from the whole truth: besides small-diameter prototype air-defense warhead candidates, REDWING was also to test multikiloton and multimegaton "clean" and "dirty" aerial bomb and missile warheads which were possible prototypes for strategic offensive warheads.

In fact, of the 17 REDWING shots finally fired, only five of them could be accurately characterized as purely "defensive" devices. The USIA also suggested that public announcement of REDWING by the AEC be timed so as to be buried by news of President Eisenhower's State of the Union speech on January 5, 1956. The test series was finally announced a week later, on January 12.

The January 16 shot schedule was revised slightly on February 15 and February 24. Lacrosse was rescheduled for April 29; Cherokee for May 8; Zuni for May 15; Navajo for June 8; and Apache for June 20. A new LASL barge shot, Dakota, was scheduled for Bikini lagoon on June 13; this addition brought the shot total to 17.

²¹⁸Memorandum dated February 1, 1956, to Clark C. Vogel, Assistant Director/Plans, Division of International Affairs, USAEC, from Colonel Jack L. Armstrong, USAF, Assistant Director, Operations, Division of Military Application, USAEC, Subject: Request by State Department for Technical Information Concerning Pacific Tests.

In addition, the name Wyandotte was assigned to the heretofore unnamed LASL contingency shot, if the shot were to be added later to the test series.²¹⁹

By the middle of February 1956, no fewer than 10 different nuclear bombs and warheads were in the U.S. weapons stockpile, including the MKs 5, 6, 7, 8, 9, 12, 15, 17, 18, and 24. At the end of 1955, the U.S. possessed a total of 2,280 nuclear cores, including 1,499 in the custody of the AEC and 781 in the custody of DOD.

Many factors influenced the number of nuclear assemblies added to the stockpile in any particular period, including the types of weapons produced, the quantities of plutonium and uranium-235 produced, and the amounts of fissionable material allocated to other uses, including civilian applications, such as power reactors.

New weapons under development were designed specifically for particular tasks, could be stored in a ready state with their nuclear components permanently installed, and made more efficient use of fissionable materials. Older weapons, whose nuclear cores were removable and interchangeable, were being slowly phased out of the stockpile.

The new Rocky Flats plant was under construction; when it was finished, it would fabricate composite plutonium-uranium cores. The plant was expected to open early in 1957. An additional U-235 fabrication facility was scheduled to open at Oak Ridge in the summer of 1957.²²⁰

The DOD had by this time also formally requested that the following additional weapons be developed in the indicated order of priority:

1. Laydown Weapons - This is a family of weapons nominally 18" in diameter weighing between 1,000 and 2,000 pounds, yielding respectively [deleted] which (can be) dropped from low altitude (down to 50 feet) by high-speed fighter-type aircraft and

²¹⁹Test Bulletin No. 20 dated 23 February 1956 to Distribution, from Roger Preston, L-3, Operations Division, UCRL, Subject: Revised Shot Schedule for Operation REDWING; Test Bulletin No. 23 dated 13 March 1956 to Test Distribution, from Vern Denton, Operations Division, UCRL, Subject: Revised Shot Schedule for Operation REDWING.

²²⁰Atomic Energy Commission, Program Status Report, December 31, 1955, Part III - Weapons, February 15, 1956, pp. 1, 5, 6.

incorporating a delay fuse which will permit the delivery aircraft to escape (before the bomb explodes).

2. Class B TN Bomb - The new Class B weapon for the bomb version is not to be more than 60" in diameter, weigh 10,000 lbs. and with a yield of [deleted] If the yield is not attainable in this weight, it may be reduced to the extent necessary. Although a maximum bomb weight of 10,000 lbs. is specified, a maximum bomb weight of 8,500 lbs. is desired, provided a yield of [deleted] can be attained. (Author's note: this became the MK 41 bomb.)

3. Class A TN Bomb - The new Class A weapon for the bomb version should be compatible with the B-52 and P6M and weigh 25,000 lbs. with a yield of [deleted] If the yield is not attainable in this weight, it may be reduced to the extent necessary.

4. 10-12" Diameter Implosion Warhead - Small implosion warhead, 10-12" in diameter, 100-150 lbs., [deleted] yield (Author's note: became W-45)

5. Class C TN Bomb - New Class C, not more than 50" in diameter, 6,400 lbs., [deleted] If the XW-27 is unable to achieve [deleted], the new Class C should have a yield of [deleted] (Author's note: became TX-46)

6. Optimized warheads, 30" in diameter, 800-850 lbs., [deleted] yield (Author's note: became XW-37)

7. 8" Diameter Atomic Shell - 200-240 lbs., [deleted] KT employing advanced nuclear techniques (Author's note: became TX-33)

REDWING test results were to make significant contributions to the advancement of these programs.²²¹

²²¹Minutes of the Fiftieth Meeting of the General Advisory Committee to the USAEC, July 16-18, 1956, p. 28. The P6M was the long-range Martin Seamaster jet-propelled flying boat flown by the U.S. Navy; capable of carrying up to 30,000 lbs. of bombs, only a dozen Seamasters were built before the Navy canceled the contract in August 1959.

In addition to these stated DOD desires, the following areas of military interest might grow into definite requirements:

1. Penetrating high-yield warhead conceived of as a gun-type thermonuclear weapon [deleted]
2. Low-fission yield thermonuclear weapons which might be used as "clean weapons" or might be designed to produce controlled-duration heavy fall-out contamination (referred to as "salted weapons")
3. [deleted] 155MM, and 175MM atomic shells
4. Nuclear warheads for such systems as torpedoes, anti-ballistic missile missiles, atomic hand grenades, warheads which act as nuclear power generators for missile propulsion prior to detonation, grenade-type or grape-shot type warheads for covering large areas with multiple nuclear explosions, etc.²²²

If REDWING results were reasonably successful, then it would be possible to stockpile a number of weapons without additional tests, including the XW-25, TX/XW-27, TX/XW-28, XW-34, XW-37, TX-39, TX-15-X4/TX-39-X1, and the XW-40. Of these, the TX/XW-28, XW-34, XW-37, and XW-40 might require additional one-point safety testing.

In addition, some REDWING experiments might add immeasurably to knowledge of multistage thermonuclear system behavior and directly affect plans for later tests:

In large measure, the question of what military necessity will attach to post-

²²²"Weapon Program Background Information for Study Relative [to] Effects of a Test Moratorium," U.S. Atomic Energy Commission, February 14, 1956. Item (4) is an early suggestion of what is now generally-known as "third generation" nuclear weapons, including directed-energy ("particle beam") weapons, "blast bombs," reduced residual radiation weapons, etc. It is interesting to note that a concept as seemingly outlandish as "atomic hand grenades" was being considered, even if only in passing. The reference to nuclear-powered missiles using their engines as warheads is reminiscent of George Gamow's "New Mexico Jumping Bean" concept of October 1948.

REDWING tests will depend upon the confidence gained from the [deleted] tests in REDWING. With great confidence resulting, it is conceivable that ICBM and IRBM warhead secondary burn could be considered predictable and that as a result it would be considered possible to stockpile such warheads after only mock-up tests of secondary implosion wherein the test device yield was less than [deleted]

On the other hand, if these tests leave a number of questions unanswered, it would be doubtful whether confidence in weaponized designs of these devices oriented toward the exacting requirements of applications to anti-missile warheads and laydown weapons would permit their being stockpiled without proof test, and almost surely, the ICBM and IRBM warheads would have to be tested full-scale to establish that they functioned as desired.²²³

In terms of extremely-high yield warheads, DOD had at this time a standing requirement for a 100 MT weapon within prescribed volume and weight limits. Even larger yields were believed possible: a report prepared by UCRL discussed the feasibility of a 1,000 MT warhead.²²⁴ (This was reminiscent of Dr. Edward Teller's July 1954 proposal for a 10,000 MT weapon.)

In a meeting with the Joint Chiefs of Staff on February 10, President Eisenhower noted that "production (of fissionable material) is now so vast that it may be possible to have everything that is needed." Recent developments were reducing stockpile flexibility, with the gradual elimination of interchangeable cores and their replacement by warheads specifically tailored to, and optimized for, particular delivery systems.²²⁵

The President met with the Joint Chiefs again on February 27

²²³"Weapon Program Background Information for Study Relative [to] Effects of a Test Moratorium," U.S. Atomic Energy Commission, February 14, 1956.

²²⁴U.S. Atomic Energy Commission, Statement of Commissioner Thomas E. Murray at Hearings before the Joint Committee on Atomic Energy on February 23, 1956, p. 6.

²²⁵Memorandum for Record, Conference of Joint Chiefs of Staff with the President, 8:30 AM, 10 February 1956.

to review future U.S. nuclear weapons stockpiling requirements and current strategic war plans.

During this meeting, Eisenhower favored a buildup of the total number of high-yield weapons through fiscal year 1959. An alternate proposal he did not like involved sacrificing quite a large number of high-yield weapons to gain a relatively small increase in the number of lower-yield weapons.

The President also thought that there was, in some cases, too much "overlapping," i.e., overkill, of bombs on certain targets. He was inclined to question whether thermonuclear weapons were not being assigned indiscriminately to targets, especially to industrial targets and population centers of relatively little importance with regard to initial counterattacks, which should focus on Soviet military targets.

There was no shortage of justifiable targets: a 1955 JCS targeting study had noted that more than 77% of all targets were within three miles of several Soviet city limits. Three cities produced 50% of all major-caliber artillery. Four cities contained 50% of all Soviet aircraft engine manufacturing. Three cities manufactured 97% of the Red Army's tanks and self-propelled guns. One city built 90% of the Soviet Navy's combat ships, and another city made all of its submarines.

Production of motor vehicles, torpedoes, sea mines, and chemical and biological warfare agents were all within urban areas. Two cities contained one-third of the raw materials and petroleum refineries; four others held 50% of all Soviet industrial resources; and of 185 government centers, 70 were also military targets.²²⁶

Eisenhower was also inclined to think that weapons with yields in excess of two megatons should be used for only very special purposes, and only after the Joint Chiefs justified their use. Airbursts, rather than surface bursts, should also be used whenever possible, especially against airfields, in order to reduce radioactive fallout.²²⁷

²²⁶STRUGGLE FOR PREDOMINANCE: EVOLUTION AND CONSOLIDATION OF NUCLEAR FORCES IN THE STRATEGIC AIR COMMAND, 1945-1955, PhD thesis by William Stephen Borgiasz, The American University, Washington, D.C., 1991, p. 227.

²²⁷Memorandum dated February 29, 1956 for Admiral Arthur W. Radford, Chairman, Joint Chiefs of Staff, from Col. A. J. Goodpaster, U.S. Army, Defense Liaison Officer to The President.

The latter point was contrary to some Air Force thinking, which held that the only good Soviet airfield was one that had been turned into a smoking, radioactive crater. If this were the case, then a surface or subsurface burst would be required, with much resultant soil vaporization and fallout.

At the end of February, UCRL REDWING plans were altered by a change from a boosted to an unboosted device in the Kickapoo shot (this decision was later reversed and a boosted device was fired during Kickapoo) and by an increase in size and yield of the Mohawk shot. If the high-yield Apache and Tewa shot devices could be prepared before their respective scheduled ready dates, these two shots might be advanced to relieve weather problems for REDWING as a whole.²²⁸

Early in March, the AEC outlined REDWING goals for President Eisenhower; the development of "clean" weapons was perhaps the most important of these objectives:

Certain aspects of this test series, we believe, will be of particular interest to you. The program will proof-test a stockpiled fusion weapon and a small fission warhead for air-to-air rockets which will be stockpiled in early 1957. In addition, Operation REDWING incorporates development tests of devices to be adapted as strategic and tactical weapons, and includes tests designed to continue the research and development of small designs suitable for use in air-to-air and surface-to-air missiles.

These latter, of course, have application to the air defense problem. Another concept to be tested is the so-called "clean" design for thermonuclear weapons whereby the major portion of the weapon yield will result from fusion rather than from fission, and a reduction in contamination is achieved.

As was the case with all U.S. nuclear tests, this letter to the President also sought approval to expend particular amounts of special nuclear materials. For REDWING, these included plutonium, oralloy, uranium-233, tritium, and lithium deuteride enriched to 95% in the Li-6 isotope.²²⁹

²²⁸Minutes of the 99th Meeting of the Special Weapons Development Board, Part Two, SWDB-56-99-2, 29 February 1956, p. 23.

A large danger area in the Pacific around the Marshall Islands was announced by the AEC on March 1. This area was generally rectangular in shape and encompassed roughly 375,000 square nautical miles. The region was many times larger than the initial danger area used in 1954 for Operation CASTLE, and all ships, aircraft, and persons were warned to remain clear of the area during REDWING.²³⁰

At the same time that Eisenhower was being informed that REDWING was to test low-contamination "clean" bombs, the Air Force was expressing its interest in "salted" or "dirty" high-contamination weapons:

The DOD has asked the AEC to study salted weapons. This letter is written in the belief that it will be helpful to AEC scientists working on this project to know a little more about AF requirements contemplated for this sort of weapon.

The AF has made some studies of the military effects of salted weapons. A summary of some results is attached for your information.

Initial emphasis should be placed on large weapons. The AF is interested in a "salted version" of the new Class A (25,000 lb.) and of the MK 21, if feasible. The ideal isotope should emit an energetic gamma ray and have a half-life of the order of a week or so. Yield losses should be no more than 25%.

These remarks as to the most desirable ranges of parameters -- size, half-life, tolerable yield loss -- are not intended to preclude your consideration of wider ranges.

SUMMARY OF SALTED WEAPON STUDIES

²²⁹Letter dated March 5, 1956 to The President, The White House, from Lewis Strauss, Chairman, USAEC; letter dated April 9, 1956 to Senator Clinton P. Anderson, JCAE, from K. E. Fields, General Manager, USAEC. The "stockpiled fusion weapon" mentioned was the TX-15-X1/XW-39 detonated during the Cherokee shot; the "small fission warhead for air-to-air rockets" was the XW-25 fired during the Osage shot.

²³⁰"Public Health and Safety Precautions for Eniwetok Tests Announced," undated draft press release ca. early March 1956.

General

The main effect sought by salting is the paralysis over large areas of practically all above-surface activity for a period of weeks. Surface bursts are contemplated since this has the desired effects of increased cratering and increased fraction of radioactive products falling out in the general area.

Size

The AF has greatest interest in salted versions of large salted weapons. Among the reasons for choice of large versus small weapons are:

- a. Their intended use on strategic targets distant from friendly forces.
- b. The high level radiation intensities already achieved with "normal versions" of multimegaton bombs.
- c. Better technical prospects for salting techniques in large weapons.
- d. The huge area covered by fallout.

Isotope

The most desirable half-life should be of the order of a week or so. A much greater half-life would result in intensities that are too low. The intense radiation from a much shorter-lived isotope would cause a high casualty rate over wide areas among exposed people (and animals), but would allow early resumption of activities by initially-exposed personnel. A more desirable goal is the extension of the time period during which above-surface activities are essentially paralyzed. Emission of energetic gamma rays is desired. If there are techniques that make decontamination more difficult, these would be of interest.

Yield Loss

It is generally presumed that a salting

technique will involve some loss in yield. Should this be unavoidably true, the following considerations should apply:

- a. The first objective is prevention of enemy attacks with thermonuclear weapons on the U.S.
- b. An airbase or missile base will certainly not operate if it has been replaced by a crater.

Therefore, blast effect is still of primary importance, and initial studies of salted weapon design should be aimed at no more than 25% yield losses. This situation will change if it should become possible to actually rely on radiation to preclude emergency operations from enemy bases.²³¹

By the end of March, ten LASL devices were to be tested during REDWING. Efforts were directed to weapons in the very small and medium size ranges, with one test of an 11-inch diameter spherical implosion system (the Blackfoot shot device). Smaller-diameter implosion devices were expected to be tested during Operation PILGRIM, now scheduled for the spring of 1957 in Nevada.

UCRL was just starting weaponization activities, having recently added a weapons engineering division: Sandia's Livermore office began operations in early 1956 to support fusion weapon research at UCRL. Heavy emphasis would be placed at UCRL on the development of small weapons and a high-yield warhead for the ATLAS ICBM. Livermore expected to weaponize two new devices per year.²³²

At the beginning of April, the first REDWING shot was still scheduled for May 1, with five additional large-yield shots and various low-yield shots to follow, with the large-yield tests separated from each other by approximately two weeks.²³³

²³¹Letter and attachment dated March 7, 1956 to Brig. Gen. Alfred D. Starbird, Division of Military Application, USAEC, from Swofford, Headquarters, USAF (signature illegible).

²³²Minutes of the Forty-Ninth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, March 28-30, 1956, p. 39.

²³³Letter dated April 6, 1956 to Edwin L. Pauley, Chairman, Board of Regents, University of California, from Lewis L.

As happened during CASTLE, these plans changed almost as soon as the series started.

In preparation for "open shots" that would be observed by print and broadcast media reporters, the AEC issued a briefing that described REDWING experiments (later editorial changes are shown as italicized text in parentheses):

Tests of nuclear and thermonuclear devices have been conducted at approximately two year intervals at the Pacific Proving Ground since Operation CROSSROADS of 1946. The last test series, Operation CASTLE, in 1954, established the practicability of some thermonuclear principles, and opened the way for further development in the thermonuclear field.

The two AEC weapons laboratories, Los Alamos Scientific Laboratory and University of California Radiation Laboratory at Livermore, have exploited those developments theoretically since that time. They have now, however, reached the stage where the state of understanding must be checked by the actual detonation of devices based on that knowledge.

This (together with the need for additional information related to the development of defensive weapons) is the primary reason for the present test series, Operation REDWING. ... The acceleration in the United States' small weapon program afforded by (successful results of low yield tests at) the TEAPOT series of tests in Nevada last spring has led to a desire on the part of the Commission weapons laboratories to conduct further tests of small (low yield) devices at about this time.

Thus it seemed appropriate to combine the requirements for large (high) and small (low yield) device tests into one operation, in hopes of taking advantage of the lulls between the firing of large (high yield) devices to detonate the less hazardous small (low yield) devices.

....

... the large (high yield) devices are fired at Bikini Atoll, where there is no

Strauss, Chairman, USAEC.

large permanent installation, and where our personnel can be conveniently evacuated for each shot. Eniwetok Atoll, on the other hand, furnishes a base from which to work that we cannot afford to lose.

Small (low yield) devices can then of course be fired at either atoll ... However, ... most small (low yield) devices are planned to be fired at Eniwetok.

....

The particular shot you are to witness is one in which the Los Alamos Laboratory, who developed it, are (sic) interested in the yield of the device. ... It will be dropped from a B-52 with the intention that it detonate at 5,000 (an altitude of several thousand) feet above sea level over the center of Namu Island at Bikini Atoll. The size of the shot will be of the order of several megatons.²³⁴

With respect to the assertion that REDWING would advance thermonuclear development, LASL also assumed at this time that there was "a fairly wide-spread belief that the emphasis in warhead development is shifting from a situation holding promise of orders of magnitude gain through application of new principles to one of application of tested principles to a wide variety of specific uses."²³⁵ However widespread Froman might have believed this idea to be, it is unlikely that it was shared by the mavericks at UCRL

A similar briefing was prepared for Canadian and British military officers who were scheduled to attend REDWING:

Operation CASTLE at the Pacific Proving

²³⁴"The Scientific Mission of Joint Task Force Seven," paper by Dr. William E. Ogle, AEC, Acting Deputy Commander for Scientific Matters, Joint Task Force Seven, undated (but probably ca. April 1956). The later editorial changes generally expunged references to relative weapon sizes, replacing such references with yield sizes instead, and re-emphasized REDWING's importance to defensive weapons development (a major justification for the test series). In addition, the exact height of burst of the Cherokee shot was blurred to a more general range..

²³⁵Memorandum dated May 4, 1956 to Technical Division Leaders, LASL, from Darol Froman, Subject: LASL Technical Program in Nonweapon Fields.

ground in 1954 tested several concepts with respect to large weapons that have led to the design of engineered devices which may, upon proof testing, be added to the U.S. national stockpile.

Since that time, other ideas have been generated and appear theoretically sufficiently interesting to warrant testing on the field. It was to accomplish these purposes that Operation REDWING was initially set up over a year ago and is now being conducted.

In the spring of 1955, Operation TEAPOT at the Nevada Test Site made similar advances in the small weapons field, and generated concepts that again warranted testing as soon as possible.

As has been mentioned by others, the probability of firing a large shot, that is, in the megaton range, on any given day in the Pacific Proving Ground is small, perhaps one in ten. Thus it seemed at that time simple and cheap to plan to fire small devices while waiting on the weather for large shots, and thus better to do the small shots here than to set up another simultaneous operation in Nevada. Therefore, last summer, the plans for REDWING were changed to include the firing of several small devices.²³⁶

²³⁶Rough draft (as sent to Division of Military Application, USAEC) of Proposed Talk to UK - Canadian Observers to be given by William E. Ogle; undated, but later than May 5, 1956 and before May 8, 1956. The first paragraph in this briefing was later edited extensively to be more general:

Since our last Pacific test operation two years ago, the last U.S. continental test series a year ago, we have made important design advances in our laboratories. It is now necessary to test these new concepts or to test new devices which have entered or will enter stockpile. To accomplish these purposes, Operation REDWING was scheduled over a year ago and is now being conducted. (Proposed Talk to UK - Canadian Observers to be given by Dr. William E. Ogle, Acting Deputy Director for Scientific Matters,

Although the largest REDWING test was to be only one-third the size of the biggest CASTLE test, the 15 MT SHRIMP device fired during the Bravo shot, this did not mean that the DOD was interested in reducing thermonuclear bomb yields. Quite to the contrary: even before REDWING started, DOD had expressed a desire for a weapon with a yield several times that of the SHRIMP, partly to offset expected ICBM inaccuracy, and the AEC had been conducting studies at both LASL and UCRL of very-high yield weapons on the order of 60 MT:

The DOD has indicated a developmental requirement for a [deleted] megaton bomb. The DOD has stated a modest requirement for this large yield weapon commencing in FY 1959.

We have no formal statement from the DOD as to why bomb yields larger than [deleted] megatons are needed. It is believed that this figure came about through DOD desires to have the largest yield for a certain weight and size, and an indication by our laboratories that this was about the maximum yield that could be expected in the relatively near future for the specified conditions.

In informal discussions as to why such large yields are necessary, DOD staff members have pointed out that there will be targets where bombing accuracy or size of target is such as to warrant the larger burst. SAC representatives, to include General LeMay, have emphasized that the expected size of our operational bomber force makes mandatory use of the largest yield possible. ...

The first ICBM will have a relatively small payload. Later models, particularly if nuclear propulsion is possible, should have much larger payloads and yields. Considering the dispersion to be expected in any ICBM, a very large yield may prove necessary. Something in the neighborhood of [deleted] megatons might be useful, though we know of no systems analysis to estimate what might be the optimum.

The President, after CASTLE, made a comment

Joint Task Force Seven, undated, but ca. 1 June 1956.)

to the press to the effect that there was no military requirement which could lead us into the production of a larger bomb than was tested at CASTLE [deleted] He went on to say that he knew of no reason for building a bigger bomb than was found to represent as great an efficiency as was needed. However, the impression generally made was that we would not need bombs with a higher yield than that achieved at CASTLE.

As concerns the President's comments after CASTLE, the DOD has stated a military requirement for a much larger weapon. It now appears that weapons with much greater yield are feasible and can be carried in the aircraft available now and in the future. Ballistic missiles following ATLAS will require, and may be able to carry, a multimegaton weapon.

We have been proceeding with research, at DOD request, for a bomb much larger than the President publicly indicated was the largest required.

....

The AEC is not in a position to determine whether a bomb of greater than [deleted] megatons is required by the military. However, the Joint Chiefs of Staff should be asked to inform the President of their stated requirement and should ask Presidential sanction before we proceed with production.

Presentation to the President should include information as to the destruction (sic) effects and strategic benefits to be expected from use of these larger weapons, and in general terms, the price (if significant) (of the development and production of high-yield weapons in terms of) the (loss) of other (lower-yield) weapons. The views of the (Atomic Energy) Commission should be part of the (presentation) to the President.²³⁷

²³⁷Atomic Energy Commission, Statement of Commissioner Thomas E. Murray at Hearings before the JCAE on February 23, 1956, Report to the General Manager by the Director of Military Application.

President Eisenhower was not alone in his belief in limiting the yield of the largest U.S. weapons. During an April 12 meeting of a congressional subcommittee on disarmament, AEC Commissioner Thomas E. Murray was quite outspoken about what he saw as a moral imperative to control bomb yields:

Before developing my proposals I should like to comment on one central aspect of the past history of our nuclear weapons program. The fact is that from the outset we were primarily seeking the answer to one question: What can be done with nuclear energy in the form of bombs?

The problems we faced were scientific ones. Questions concerning the development of a national policy with regard to the use of nuclear arms were not in the foreground. We were simply looking for results, always with some antecedent doubt about what the results might be.

Ever since the first nuclear explosion at Alamogordo, the specific and continuing objective was to develop ever larger ones. This was true of the H-bomb program, which achieved in 1952 a spectacular success.

Throughout my six years on the Commission, I have strongly supported this mode of procedure. However, I long ago recognized the need to develop a national nuclear policy which could keep pace with scientific and technological progress and tried to help in developing it.

The "Mike" thermonuclear explosion on November 1, 1952, and our CASTLE tests in the Spring (sic) of 1954 presented the nuclear weapons problem in an altogether different light. These tests re-emphasized in a most dramatic way the urgent need for a national nuclear policy interrelated with the types of bombs to be produced. The major question no longer is what can we do with nuclear energy in bombs. We know what we can do. We know that there is no upper limit to the size of bombs we can build.

A new question therefore assumes the primacy. Should our future program proceed on the same basis as in the past, and with the same major objective in view -- to keep

on making bigger and bigger bombs? Would this, in our present circumstances, which are different from those of the past, be a rational course of action?

I think that policies will have to be adopted that will set our nuclear weapons program on a basis more in conformity with the realities of the present day international life, more in conformity with certain principles of reason, and better calculated to fit into a new national policy on the use of nuclear weapons.

The new direction of our policies as I see them could be summed up in the phrase "rational nuclear armament." This phrase means that decisions with regard to the size and type of nuclear weapons to be placed in stockpile should be made in light of two principles.

There is the military principle that the weapons we manufacture should be demonstrably useful for the purposes of actual war. And there is the higher principle that the use of force in warfare is subject to the dictates of the moral conscience.

The moral norm is the more important. There is something in the atmosphere of the age of violence through which we have lived, in which we are still living, that tends to dull the moral sensibilities. In a strange way, the sheer brilliance of our technical achievements with nuclear weapons also tends to dull the moral sense. It is therefore necessary constantly to remind ourselves that a concern for justice in the use of force is an obligation upon a civilized nation.

....

... our stockpile should include only weapons that are actually useful in war. Moreover, it should include only weapons that we can legitimately intend to use. I am altogether opposed to any school of thought that would move on toward weapons of ever-increasing magnitude, while at the same time disclaiming the intention of using them.

To say the least, this position is unrealistic. Under the stress of a threat to its survival, a nation will use any weapon it has in hand, however recklessly, if there is any seeming advantage in its use.

....

Furthermore, a special problem presents itself in judging the military usefulness of large thermonuclear weapons. We know a great deal more about the effects of nuclear explosions than we did a few years ago. For instance, we now know considerably more about the contamination of the atmosphere and soil by radioactive strontium. ...

When the military usefulness of a weapon is being determined, all the knowledge that we have must be brought to bear. Serious account must also be taken of the gaps in our knowledge; these gaps make it imperative to proceed with caution in evaluating the military usefulness of large nuclear weapons.

Here the chief question is this: how does the factor of contamination resulting from large thermonuclear explosions affect the military usefulness of these weapons? Surely it must limit their military usefulness. Indeed, in certain circumstances the factor of contamination might well cancel all apparent military usefulness. This is particularly true in view of the further fact that this contamination lingers for years and would affect life on earth long after hostilities had ended.

Murray suggested limitations on the yields and numbers of high-yield stockpiled weapons, and called for the increasing development of low-yield warheads. He also noted the increased cost of smaller weapons in terms of fissionable materials and called for a moratorium on tests of large weapons:

We come now to the question, what concrete conclusions result when these criteria of rational nuclear armament are applied. I have reached the following conclusions. Taken together, they constitute the program of rational armament that I propose as part

of the substance of disarmament.

My first proposal is that an upper limit should be set on the size of thermonuclear bombs to be stockpiled. From the scientific and technological point of view, there is, as I said, no upper limit to the size of bombs that can be produced. It is for us, as rational human beings, to set limits to their size, on the basis of moral principles and military usefulness.

Concretely, I know of no reason why we should develop bombs more powerful than those we now have. I would add that, when a full military theory with regard to moderate and discriminating use of nuclear weapons evolves in light of moral principles, the conclusion might be that the bombs we presently have are already too large.

This proposal to limit the size of thermonuclear bombs seems to have been supported by the President at a press conference on March 21. He is reported to have said:

While I have been told many times that there is no theoretical limit to the size of these instruments which can be made, there is, I think, a practical limit, as long as you are thinking of using it against someone else, and there is size of targets.

There is an old saying: "You do not drive a tack with a sledgehammer." So when you have gotten something of a, let's say, of a maximum tactical size, useful size, why I would think you would go no further.

My second proposal is that a limit should be set on the number of large thermonuclear bombs stockpiled. I have had the need for this in mind since I witnessed the Mike thermonuclear detonation at Eniwetok. That event impressed me with a lasting sense of the menace to humanity inherent in an unlimited program of nuclear armament.

....

My third proposal is that increasing concentration should be placed on the stockpiling of a wide range of very small nuclear weapons.

The reasons for this proposal are both military and moral. I have said that limited nuclear war is a possibility that the moral conscience can face and accept. It is also a possibility that is inherent in the present state of international tensions.

We have, of course, been preparing for this possibility with our program of small weapons. However, this program should be intensified, to the end that it will put us in possession as soon as possible of larger numbers of smaller weapons.

When I speak of small weapons, I mean very small weapons. When I speak of larger numbers, I mean tens of thousands of weapons in this range. These tens of thousands of weapons could form a broad and solid base upon which could be superimposed a stockpile in the image of a pyramid of intermediate and large size weapons.

A stockpile of this composition will support a theory of moderate and discriminating use of force against all aggression, including aggression that may be minor in degree. With this diversified nuclear armament, possessed in abundance, American and allied military forces of reasonable size would be equipped to handle all the various wartime contingencies that might arise.

We shall have to face the fact that an intensified small weapons program will be more costly in terms of fissionable material. This increased cost would result not only from the larger numbers, but also from the inherently less efficient use of material in smaller weapons. ...

....

My fourth proposal concerns the testing of atomic weapons. The testing of small weapons must of course go on, and even be

greatly accelerated to keep pace with the intensified program that is necessary in this field.

However, I also propose that the tests of multi-megaton thermonuclear weapons be stopped. This proposal follows from the position that I have already taken, namely that the weapons that we now have in hand are large enough. I am not recommending cessation of these large tests on the grounds that they are dangerous to health and safety. The evidence now available does not warrant stopping tests for this reason.²³⁸

Research into very high-yield thermonuclear weapons, and differences of opinion between the AEC, DOD, and the President over them and their uses, would continue in the coming years. Of more immediate benefit from REDWING was the possibility of developing and stockpiling a wide array of small nuclear weapons:

Both of our nuclear laboratories are now placing great emphasis on achieving the smallest practical devices, both nuclear and thermonuclear. If designs being tested in REDWING prove successful, LASL intends to attempt as a next step an [deleted] device, and Livermore intends to try a [deleted]

Small weapons of, say, 30 inches and less can be carried or delivered by a variety of planes or missiles. The DOD is placing very heavy emphasis on the development of these missiles. However, such missiles, in general, have limited range and flexibility. They are tied for employment to relatively limited areas and generally are of shorter range.

Warheads will have to be dispersed to the near-vicinity of all potential users if they are to be available when needed. With the greater dispersion incident to smaller size, a tremendous stockpile will be

²³⁸Statement of Thomas E. Murray, Commissioner, United States Atomic Energy Commission, before the Subcommittee on Disarmament of the Committee on Foreign Relations on April 12, 1956.

necessary if such weapons are to be available at any place where they may be needed.

....

All practical emphasis is now being placed on the research and development of small weapons. DOD is placing adequate emphasis on the development of (delivery vehicles).

The AEC also felt that emphasis on small-weapon development at the cost of possible delays in large-weapon development was not advisable. This dilemma recalled a recurring argument in the early days of hydrogen bomb development, namely, whether fission weapon development should have priority over the Super program:

Complete elimination now of all work in the laboratories on multimegaton weapons might advance slightly completion of stockpiling of certain weapons in the [deleted] range. However, if work is to continue on ICBM-type (thermonuclear) warheads and clean larger weapons, it is doubtful that significant improvement in small weapons schedules would be achieved by eliminating other high yield weapons work.

REDWING was also to advance the state-of-the-art of "clean" low-fallout weapons; however, such devices were currently very costly in terms of loss of yield due to improved "cleanliness:"

We are extremely interested in developing clean multimegaton weapons (bombs and perhaps warheads). Designs now being looked at are relatively sensitive. It is doubtful that we would be willing to stockpile and place any reliance on such weapons without full and adequate development and proof tests.

Both the AEC and DOD were also still very interested in improving yield-to-weight ratio:

Deliverability, as well as availability, govern the effectiveness of our stockpile. We are at work developing improved versions of stockpiled (thermonuclear) weapons -- lighter and with higher yield per pound of weight. Improved versions can be carried in lighter (and hence additional) aircraft. They will permit thereby improved

deliverability.²³⁹

By early spring 1956, President Eisenhower -- as he did for every U.S. nuclear test series conducted during his terms of office -- had authorized the AEC to expend specific amounts of fissionable and fusionable materials. For REDWING, these comprised plutonium, uranium enriched to 93.5% in the U-235 isotope, lithium deuteride enriched to 95% in the Li-6 isotope, tritium, and uranium-233.²⁴⁰

REDWING Test Operations

As had been done during previous Pacific test series, a number of task units within JTF 7 were again created for REDWING. TU 4, Sandia Corporation Programs, was responsible for the assembly of the LASL and UCRL Cherokee, Osage, and Apache devices.

TU 5, Timing, had two principal objectives: to provide a remote control system that would arm and fire a surface-detonated device with a high degree of reliability, and to supply experimenters with an accurate sequence of timing signals before or after zero time to start or stop their equipment during airdrop or surface shots.

The timing task unit worked as part of the firing team. TU 5 also provided voice-time announcements and established a ship-borne control station for remote operation of the Bikini timing system.

A complete timing system was assembled and tested at EG&G's Nevada laboratory; 55 tons of timing equipment was shipped to the forward area in the Marshall Islands between April 1955 and May 1956. Dry runs and tests of timing equipment were conducted before and during the shot series.

An exact zero time was determined for each shot by means of the initial flash of detonation. This burst of light triggered a fiducial marker atop the control points, Station 70 on Enyu island at Bikini and Station 71 on Parry island at Eniwetok, which in turn triggered a flash tube that took a photograph of a clock in the command bunker.

This clock was synchronized with a World Time clock at short-

²³⁹Atomic Energy Commission, Statement of Commissioner Thomas E. Murray at Hearings before the JCAE on February 23, 1956, Report to the General Manager by the Director of Military Application.

²⁴⁰Letter dated April 9, 1956 to RAdm. B. Hall Hanlon, USN, Commander, JTF 7, from K. E. Fields, General Manager, USAEC.

wave radio station WWVH at the National Bureau of Standards Radio Propagation Laboratory in Honolulu, Hawaii Shot time could be determined to the nearest millisecond, a requirement for certain experiments.

Figure V-2 shows the 300 foot high camera and communications antenna on Enyu. **Figure V-3** shows part of the interior of the firing bunker on Enyu (this was the bunker in which Bernard O'Keefe and his colleagues were trapped during the CASTLE Bravo shot).

Magnetic tape recording and playback equipment provided pre-recorded audio voice-time announcements synchronized with the timing system. The recorder was started automatically at 15 minutes before detonation and continued to give time hacks down through zero hour. The voice countdown originated at the control stations and was distributed to ships and stations throughout the atolls by radio.

Another radio link was used to transmit tone signals from the USS CURTISS to the master control station on Enyu. These signals were used to remotely control the master timing system. Timing signals were also radioed to various scientific stations to activate equipment at sites inaccessible by hard-wire.

TU 6, Firing, armed and fired nuclear devices at REDWING. Two teams were provided so that arming and firing capabilities could be maintained at both atolls. These teams worked in close liaison with TU 5 and the weapon assembly teams, Task Units 10, 11, and 12. The TU 6 work was divided into four parts: pre-arming tests, arming, firing, and disarming (in case of failure to fire). Checklists were used to monitor steps in weapon preparation.

The pre-arming tests included visual inspection, functional checks, and adjustments with live equipment while connected in a simulated final arrangement. The interlocks in the firing circuit were checked, along with monitor lamps, meters, and recorders.

The arming process started anywhere from six to 14 hours before zero hour and included final connections and preparations. During final preparations, a safe-separation timer, whose contacts completed the arming circuit, was set to close two hours before zero time and open four hours later. This timer provided a safe time interval after the weapon or device had been armed for complete evacuation of the atoll when necessary and allowed the firing team time to retire to the control bunker.

If a shot were canceled within two hours of zero time, the disarming team entered the zero area after the timer opened the arming circuit. On some of the smaller shots at Eniwetok,

where the diagnostic measurements were not limited to a certain early time period, the safe-separation timer was set to open the arming circuit up to eight hours after zero time.

This allowed several hours in which zero time could be rescheduled when the weather was unsettled. When possible, arming operations were conducted during daylight hours, but on several occasions it was necessary to arm after dark. Transportation for the arming team was provided by airplane, helicopter, or boat.

Shots were fired from the control bunkers at Bikini and Eniwetok by starting a motor-driven timer 15 minutes before zero hour. This timer automatically provided timing and firing signals. Firing equipment in the control bunker at Enyu could be remotely controlled by radio from the CURTISS. Shot Zuni, the UCRL BASSOON device test, was the only occasion where the remote control station aboard the CURTISS was used.²⁴¹

Several disarming operations were conducted at both atolls because of postponements due to unfavorable weather conditions. Only one disarming was conducted under emergency conditions. This was during the Huron shot after the detonation was canceled within three minutes of zero time when the pit in the primary of the test device failed to maintain boost gas pressure.

TU 6 was not responsible for the two airdrops, Cherokee and Osage, conducted during REDWING. The fuses and firing circuits of the airdropped weapons operated flawlessly.

All weapons and devices were detonated on time, after several reschedulings in a few cases, and no misfires occurred during the operation, although the original Huron shot was aborted within three minutes of zero time.

There were three separate weapon assembly teams at REDWING. Task Unit 10, LASL Assembly, was formed in July 1955 to assemble and make ready for firing all LASL weapons. TU 10 was also responsible for drawing up schedules of operations at zero cabs to avoid conflicts between weaponeers and diagnostic program engineers. TU 10 was composed of members of LASL's W Division, including groups W-1 and W-7, and GMX Division, including groups GMX-3 and GMX-7, and technicians from the Sandia Corporation.

TU 10 activities in the forward area began in April 1956 with shipment of 11 REDWING test devices from LASL. These included several TX-28 prototypes, a TX-21C weapon, and a complete MK

²⁴¹WT-1359 (EX), p. 148.

15 for the Cherokee airdrop test. In addition, early models of prototype devices not yet assigned XW or TX numbers were brought to the Pacific from New Mexico. Two others, possibly the Blackfoot and Huron devices, were not ready for shipment aboard the CURTISS at the end of March, and were airlifted by Air Force C-124s after REDWING began.

All but two of the test devices aboard the CURTISS were shipped almost completely assembled. At this time, REDWING was expected to encompass at least 15 shots and possibly as many as 19 between May and late August 1956; REDWING would be nearly three times the size of any previous Pacific test series.

If all the test devices performed according to theoretical computations, then minimum results expected from REDWING just before the series started included early stockpile entry of smaller-sized weapons with greater nuclear efficiency; a useful nuclear missile warhead; and information on which to better formulate plans for defense against enemy nuclear weapons.²⁴²

A wide range of zero stations was planned for LASL tests, including two ground sites, two airdrops, three towers, and four barges. Of the eleven devices, an alternate, the Pawnee device scheduled for a surface burst, was canceled after the Lacrosse shot succeeded.

TU 11, UCRL Assembly Team A, was formed from a group of engineers and physicists to assemble and detonate low-yield fission devices in the PPG. They were members of the UCRL small weapons program who were also responsible for the original design and fabrication of three UCRL devices fired during the Yuma, Kickapoo, and Inca shots.²⁴³ A design and assembly team accompanied each device to the PPG.

An Air Force C-124 was used to transport each bomb to Eniwetok, allowing maximum time before shot day for the design and fabrication of the devices at Livermore and assembly at the PPG. TU 11 also cooperated with TU 12 on the assembly of the Mohawk shot device.

²⁴²"Operation REDWING: Briefing for UK - Canada - PJBD," attachment to Atomic Energy Commission, Foreign Observer Briefing at REDWING, Note by the Acting Secretary, May 7, 1956.

²⁴³WT-1359 (EX), p. 158. Shot dates for the three devices are listed as May 28, June 14, and June 22; these correspond to dates announced for the Yuma, Kickapoo, and Inca shots.

More nuclear test devices were moved by air during REDWING than during previous Pacific operations, mainly because late design changes made by UCRL and LASL delayed final fabrication. Quick delivery by air gave the labs more flexibility in operations and also got the devices to the PPG on schedule. In a number of cases, the time between device fabrication and testing was only a few days.

TU 12, UCRL Assembly Team B, was made up of UCRL A Division (Large Weapons) and Device Engineering Division personnel who had followed the high-yield Zuni, Mohawk, Apache, and Tewa shot devices through their design and fabrication phases. They were thoroughly familiar with these devices by the time REDWING began. The task unit was further subdivided into device assembly teams.

Full trial assembly of each device was made in the continental U.S. This assembly including matching support stand and local diagnostic equipment to the device. On some devices, jigs that connected external equipment to the device were flown out to the PPG from Livermore to speed early completion of field diagnostic work.

Another TU 12 procedure was to ship devices in nearly-assembled condition; weapons were made nearly complete in the U.S. before being loaded aboard a C-124 for the long flight to the PPG. This permitted rapid final assembly at Eniwetok. UCRL devices were flown by C-124 to the PPG to arrive 11 to 19 days before a particular shot.

More than 13,700 men, military and civilian, were in JTF 7 for the REDWING tests.

By the time of Operation REDWING, the military forces on both Eniwetok and Bikini were heavily equipped with various small arms, in addition to the aircraft and ships that could be called in to defend the Pacific Proving Ground against a Soviet attack.

Among the weapons available to ground forces to repel air attacks and amphibious raids were .30-, .38-, .45-, and .50-caliber pistols, rifles, submachine and machine guns; 57MM and 75MM recoilless rifles; and 60MM mortars. A total of approximately 191,000 rounds of ammunition were available for these weapons. The Air Force detachment was best-armed, with 48,000 rounds of .30-caliber rifle ammunition and nearly 2,000 rounds of .50-caliber machine gun ammunition.

In the event that either or both Bikini and Eniwetok had to be abandoned in the face of an overwhelming attack, plans had been made to destroy or remove classified documents and weapon or test device components and critical test

equipment.²⁴⁴

Preparations for REDWING

Preparations for REDWING were very similar to those for CASTLE. The MSTs provided sealift of personnel and cargo. The Air Force furnished aircraft for weapon drop, cloud sampling, weapons effects programs, photographic flights, and other scientific projects; helicopters at Bikini and Eniwetok; meteorological aircraft; liaison planes; inter-atoll aircraft; and off-atoll support. MATS provided airlift of the majority of personnel and many tons of urgent freight, including a number of test devices flown from Travis AFB in California to Eniwetok.

Equipment for REDWING began arriving at the Naval Supply Center in Oakland, California in early December 1955. Approximately 3,100 tons of materiel were moved by MSTs ships to Eniwetok. All ships were unloaded at a new deep water pier at Parry island. Cargo vessels were scheduled to sail from Oakland at three-week intervals between January and April 1956. About 2,000 tons of cargo destined for Bikini was unloaded at Eniwetok and shipped to the other atoll via Navy LSTs.

Once again, as she had done many times before, the USS CURTISS transported a number of devices to the forward area, carrying 85 tons of weapons, including the Lacrosse, Cherokee, Osage and Navajo devices, and other critical materials from the Port Chicago naval weapons center on March 26, 1956. She also doubled as the seaborne command ship for REDWING, replacing the ESTES. **Figure V-4** shows the ESTES during REDWING, with the CURTISS in the background of the photo.

The CURTISS was showing her age by this time: Shop 18 aboard the ship had been modified before Operation SANDSTONE in 1948 to transport and assemble the weapons used during that test series. Between 1948 and 1956, the sizes and configurations of nuclear weapons had changed drastically and fittings in Shop 18 had become obsolete and unusable. Hold space had also become cramped and inefficient to use.²⁴⁵

During the next U.S. Pacific test series, Operation HARDTACK Phase I in 1958, all test devices would be partially assembled in California and New Mexico before being flown out to the Marshall Islands for final assembly and preparation

²⁴⁴Annex E (Hostile Action Alert Plan) to CGT 7.2 Operation Plan 1-56, Headquarters Task Group 7.2, Joint Task Force Seven, 15 April 1956.

²⁴⁵WT-1359 (EX), p. 138.

for firing. It was likely that REDWING would be the CURTISS' last nuclear test series. (The ship was decommissioned in September 1957 and scrapped sometime after July 1963.)

The USS CATAMOUNT (LSD-17) carried shot barges with test devices loaded aboard between the assembly area on Parry island at Eniwetok and lagoon shot sites at Bikini atoll.

The actual device firing program began early in May; scientific station construction delays had moved the start date back a few days. One of the many new features of REDWING was the so-called "two-atoll concept," whereby JTF 7 had the capability of firing either a large-yield device on Bikini or a small-yield device at Eniwetok on any one day.

Although weather conditions at any particular time might not be suitable for a shot at Bikini, they might be entirely satisfactory at the same time at Eniwetok. This "two-atoll concept" required LASL and UCRL to have at least two devices ready for test simultaneously, one at each atoll.²⁴⁶

The LASL Lacrosse device, an externally-initiated boosted primary for the TX-39 and the UCRL XW-27,²⁴⁷ arrived at Eniwetok aboard the CURTISS on April 10; two weeks later, a dummy device was moved to a 17-foot high platform on an artificial island at the end of a causeway about 450 feet off the north end of Runit.

Detonators were installed to check out the firing circuit the following day. The live unit arrived at Runit the same day (April 25) and detonation was scheduled for April 29. Bad weather delayed the shot for a full week, and it was not fired until 0625 hours on May 5 (all shot dates are local times; each date was one day earlier in the continental U.S.).

Lacrosse yielded 40 kilotons and was very close to the predicted value of 39 kilotons. The performance of Lacrosse was so satisfactory that the back-up shot Pawnee, which was scheduled on a contingency basis, was removed from the schedule.

Lacrosse dug a crater 44 feet deep and 404 feet in diameter.

²⁴⁶"Operation REDWING: Briefing for UK - Canada - PJBD," attachment to Atomic Energy Commission, Foreign Observer Briefing at REDWING, Note by the Acting Secretary, May 7, 1956.

²⁴⁷Operation REDWING, A Preliminary Report of [deleted] (Lacrosse), 50005JFE, 5 July 1956, p. 4; OPERATION REDWING - POST OPERATIONAL REPORT, PROGRAM 22 - REACTION HISTORY MEASUREMENTS, COPB 4547, L. F. Waters, July 1960, p. 67.

The shot vaporized and lifted into the stratosphere about 725 tons of steel, iron, aluminum, copper, and wood that comprised the shot cab, along with 112,000 cubic yards of coral sand and rock.²⁴⁸

The mushroom cloud reached only 35,000 feet, about 10,000 to 15,000 feet less than average for similar weapons tested in Nevada. Atmospheric conditions in the humid Pacific tropics were quite different from the cool, dry air over the high Nevada desert.

The LASL TX-15-X1 bomb²⁴⁹ for the Cherokee proof test shot also arrived aboard the CURTISS on April 10. The bomb was checked out on May 6 and first scheduled for detonation on May 8. The test was delayed due to bad weather; on May 10, the weapon was taken from Parry to the air base at Eniwetok island and loaded aboard a B-52B (serial 52-013).²⁵⁰ The drop was delayed again, and the bomb was unloaded and returned to Parry on May 12. It was taken again to Eniwetok four days later and loaded aboard the drop aircraft.

On May 21, after more delays, the free-falling bomb was air-dropped from 40,000 feet and detonated at 0550 hours at an altitude of 4,350 feet northeast of Namu island at Bikini, operating as expected and yielding 3.8 megatons.²⁵¹ Ten

²⁴⁸Letter dated June 29, 1959 to Joe B. Sanders, Director, Pacific Operations, USAEC, from S. P. Howell, Manager, Engineering and Construction, Holmes & Narver, subject: Estimated Weight of Construction Materials, Operations CASTLE, REDWING, and HARDTACK I.

²⁴⁹Operation REDWING, Preliminary Report of [deleted] (Zuni), JO-269, 15 October 1956, p. 46.

²⁵⁰Operations Directorate History (April 1956), 15 May 1956; Operation REDWING, A Preliminary Report of [deleted] (Cherokee), 5379JFE, 17 July 1956, p. 5. This particular B-52B also participated in airdrop tests during Operation DOMINIC in the Pacific in 1962.

²⁵¹WT-1344 (EX), pp. 177, 180; OPERATION REDWING - PROJECT 5.2, INFLIGHT PARTICIPATION OF A B-52, May-July 1956, WT-1328 (EX), F. L. Williams, Capt., USAF, Project Officer, and Staff of the Boeing Airplane Company, Wright Air Development Center, Wright-Patterson Air Force Base, Ohio, 23 October 1959, p. 31. There are many reports in the open literature that the Cherokee yield was 10 megatons, e.g., BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 11, 1956, p. 274 (on p. 230 in the same volume, the BULLETIN initially reported Cherokee yield as 15 to 18 megatons, with a detonation altitude of 15,000 feet); and IRON EAGLE, Thomas M. Coffey, Crown

minutes after detonation, the resultant mushroom cloud had a stem diameter of 15,000 feet, with a lower cloud elevation of 40,000 feet and its top at 70,000 feet; the clouds had spread out to a diameter of 50 miles.²⁵²

The cloud created its own weather:

From personal observation, shot Cherokee occurred precisely at zero time as judged from the voice countdown aboard the AINSWORTH. The fireball was partially obscured by low-hanging clouds, and the shot cloud rose rapidly without the formation of "skirts." The blast reached us as a sharp bang without any physical sense of pressure.

As the cloud rose and cooled, a very

Publishers, Inc., New York, 1986, p. 342. These statements are incorrect: all of the REDWING shots were considerably lower in yield than the CASTLE tests two years earlier. In addition, the 5,000 foot burst height for the MK 15 was set at a distance 1 1/2 times the expected maximum fireball radius to minimize local fallout (GAC Report dated June 10, 1954, from Col. Vincent G. Huston, USAF, Deputy Director of Military Application, USAEC; History of Task Group 7.4, Provisional, March 1956, Joint Task Force Seven, Joint Chiefs of Staff, p. 13.). For a 10-megaton explosion (according to the Nuclear Bomb Effects Computer from the 1964 edition of the AEC's EFFECTS OF NUCLEAR WEAPONS) the maximum fireball radius is about 1.48 miles, or 7,809 feet (for a 3.5 MT blast, the maximum fireball radius is about a mile; for 1.7 MT explosion, about 3,800 feet). The Cherokee fireball did not touch the surface. On May 5, 1956, more than two weeks before the Cherokee shot, at least one newspaper account estimated that Cherokee would yield between three and five megatons. ("Atom Blast Opens Tests at Eniwetok," New York Times, May 5, 1956.) By May 9, the same NY Times writer was guessing Cherokee yield at five to 10 megatons. ("Weather Delays H-Bomb Blast for Third Time," New York Times, May 9, 1956.)

The Coffey statement is especially misleading: according to him, "...one of SAC's B-52s...missed its battleship target in the Bikini atoll...The bomb exploded ten thousand to fifteen thousand feet above the target...with a force of ten million tons of TNT." The actual detonation altitude was 4,350 feet $\pm$ 150 feet; there were no battleships -- or any other ships, for that matter) moored in Bikini lagoon during the Cherokee shot -- and yield was only 3.8 MT, not 10.

²⁵²"General Notes - Cherokee, 21 May 1956, 0551."

intense purple with Szchrinkoff (sic) radiation. Rain started in the area at about H+3 minutes after the burst, and at about H+10 minutes, a thunderstorm developed within the stem. Mr. Tanner and I counted 21 flashes appearing exactly like lightning flashes within a cloud.²⁵³

Seven minutes after the bomb detonated, the radiation intensity within the mushroom cloud was measured at 20,000 roentgens per hour (a cumulative dose of just 500 roentgens kills approximately half the persons exposed to it). A group of newsmen aboard the USS MOUNT MCKINLEY (AGC-7) 34 miles from Namu, characterized the heat from the blast as feeling "as though the door of a nearby blast furnace had been suddenly opened."²⁵⁴

The bomb missed its target, a lighted asphalt-paved pad, by 19,700 to 20,000 feet due to human (bombardier) error (shades of the CROSSROADS Able shot in 1946): the weapon was dropped 21 seconds too soon when the bombardier mistook another lighted island for the ground zero site.²⁵⁵

This error was quite costly in terms of lost scientific data:

CHEROKEE results extremely meager because of

²⁵³Cherokee Field Report Bikini Operations, p. 10.

²⁵⁴"U.S. Triggers Enormous H-Bomb Explosion," International News Service wire story by Bob Considine, ca. May 21, 1956.

²⁵⁵Letter dated 30 May 1956 to Dr. E. F. Cox, Sandia Corporation, from M. L. Merritt, TU-4, TG7.1, JTF-7; Memorandum to Director, U.S. Naval Radiological Defense Laboratory, from Eugene P. Cooper, Associate Scientific Director, Subject: Preliminary Information on Zuni and Cherokee; letter dated 31 May 1956 to James T. Ramey, Executive Director, JCAE, from Brig. Gen. Alfred D. Starbird, USA, Director of Military Application, USAEC; memorandum dated June 17, 1956 to James T. Ramey from George Norris, Jr., subject: Test at Bikini. The latter document quotes a June 16, 1956 public statement by Air Force Secretary Donald Quarles which claimed either erroneously or falsely that "while the (bombing) error was considerably more than normally expected, it was not great enough to have a serious effect on the (success of) the experiment." The Report of the Cherokee Board Proceedings, 23 May - 8 June 1956, p. 2, states that the bombardier aboard ZIGZAG (the code name for the drop aircraft) repeatedly mistook a light two miles from true ground zero for the target both during practice runs and the final bomb drop run.

bombing error. Burst point three to four miles northeast of expected burst point, hence, few ball of fire cameras saw detonation and cameras to determine zero point accurately were pointed in wrong direction. Small number of films and unknown distance from cameras to actual burst point both contribute to uncertainty of fireball yield.

Fact that detonation occurred over deep water far from land and that cloud consequently had little dirt in it may have helped cause us to get small radiochemical samples and hence cause more than usual uncertainty in radiochemical yield. All alpha data lost because of distance between detectors and detonation. Sandia's transit time and alpha lost. DOD fallout program lost. Most of air blast and structures program lost. Some airplane effects obtained.²⁵⁶

Cherokee was the first air drop by the U.S. of a thermonuclear weapon (the Soviets had airdropped a 1.6 MT two-stage thermonuclear bomb on November 22, 1955), and the first drop of a "live" thermonuclear device from a USAF aircraft. The radioactive cloud from Cherokee finally reached a height of 70,000 to 80,000 feet with a diameter of 25 miles; no militarily-significant fallout was detected.

The Lacrosse and Cherokee shots were observed from aboard the USS MOUNT MCKINLEY by 15 American press, radio, and television reporters and 17 civil defense officials. There had been considerable discussion and debate about the desirability of inviting foreign observers to witness REDWING tests.²⁵⁷

²⁵⁶Teletype message dated June 13, 1956 from Dr. Alvin C. Graves, Director, Joint Task Group 7.1, to Brigadier General Alfred D. Starbird, Director, Division of Military Application, USAEC. This document directly contradicted the Air Force's dubious public claim that the Cherokee bombing error did not have a serious effect on the success of the test.

²⁵⁷FOREIGN OBSERVERS AT U.S. NUCLEAR TESTS, DOE/DP/50040--T1 (DE86004389), Ruth A. Dudgeon and Ruth R. Harris, History Associates Incorporated, Rockville, Maryland, November 18, 1985, pp. 12-14. A letter dated April 2, 1956 to Senator Clinton P. Anderson from Herbert B. Loper, Assistant to the Secretary of Defense (Atomic Energy) notes that as of March

These were the first uncleared U.S. civilians in ten years to observe an American nuclear test in the Pacific. The AEC believed that an uncleared observer program designed around these shots "should promote favorable reporting of the entire series."²⁵⁸ Cherokee was also one of the largest explosions (3.4 to 3.8 MT) of the entire test series.²⁵⁹

At a joint AEC-MLC conference on May 24, General A. D. Starbird, USAEC Director of Military Application, reported on the progress of REDWING. He said that at the beginning of the operation, firm plans had been made for 17 shots and tentative plans had been made for two others.

One of the firm shots had now been canceled since the test device would not be ready in time and one of the tentative shots (Pawnee) would definitely not be fired. However, he stated that there was a possibility that a test of a TX-28C, the other tentative shot, might be added to the test series.

Discussing other matters, Starbird said that besides the Cherokee shot, two other high-yield devices would be tested. The general also commented that news media coverage of the Cherokee shot had been satisfactorily handled and that attendance of news media representatives at REDWING had been extremely worthwhile.²⁶⁰

The BASSOON device for UCRL's first high-yield REDWING shot, Zuni, arrived via USAF C-124 in April; the bomb was stored and checked in Building 341 on Parry island at Eniwetok. It was loaded aboard a trailer and driven onto a Navy LCU (landing craft utility) which was taken aboard the CATAMOUNT like a shot barge for transport to Bikini atoll on May 23.²⁶¹

At Bikini, the LCU left the CATAMOUNT and proceeded to a ramp on Eninman island where the device, still aboard the trailer, was driven to the zero point. The final assembly was done in a shot cab on a concrete slab less than 100 feet from the eastern edge of the water-filled CASTLE Koon crater.

30, 1956, the Deputy Secretary of Defense had issued invitations to five British and 10 Canadian observers to attend REDWING.

²⁵⁸Appendix "A", Background and Discussion, Public Information Program, Operation REDWING, undated (ca. March 1956).

²⁵⁹AEC press announcement dated April 20, 1956.

²⁶⁰Minutes of AEC-MLC Meeting 104, May 24, 1956.

²⁶¹OPERATION REDWING, Commander Task Group 7.3 Final Report, March-August 1956, J. H. Wellings, p. 79.

The BASSOON device was detonated just before dawn at 0556 hours on May 28, and gave the predicted yield of about 3.5 MT.²⁶² For this shot, the firing party was evacuated from Enyu and the device was fired by radio command from the USS CURTISS at sea outside Bikini lagoon. Zuni was a very successful "investigation" of a three-stage high-yield experimental device.

The BASSOON was the first three-stage American nuclear device ever tested.²⁶³ The shot was also UCRL's first completely-successful thermonuclear detonation; its purpose was to determine whether a high-efficiency burn could be obtained from a device whose size and weight were chosen to approximate Class "C" weapon requirements (a weight of 10,000 lbs. and a 40" diameter).²⁶⁴ The BASSOON weighed just over 12,000 lbs. and was 39" in diameter.

The Zuni shot blasted a crater 103 feet deep and 2,310 feet in diameter at Eninman, vaporizing 62 tons of steel, iron,

²⁶²WT-1359(EX), pp. 105, 148. Apparently UCRL was naming large thermonuclear devices after musical woodwinds at this time; small fission devices were named after birds. See "Revealing Report on H-Bomb Was Declassified by Mistake," San Francisco Examiner, May 19, 1979, p. 11; and "U.S. Goof Put H-Bomb Secret on the Shelf of a Library," San Francisco Chronicle, May 19, 1979, p. 8. These are reports on the discovery of UCRL-4725 at Los Alamos by Dimitri Rotow during the PROGRESSIVE case; Rotow and an assistant were forcibly ejected from the Los Alamos Lab library for their efforts to uncover DOE declassification incompetence. These articles name specific devices (SWAN, SWIFT, and SWALLOW) tested during REDWING by UCRL in June and July 1956. See also NUCLEAR WEAPONS DATABOOK, Vol. II, p. 23, for reference to later UCRL devices named DOVE and STARLING (other Livermore "birds" include the OWL and QUAIL). During this same period, LASL was naming devices after snakes, and insects, e.g., COBRA, VIPER, MOCASSIN, RACER, BEE, HORNET, SCORPION, etc. (Letter to author dated December 17, 1988 from M/Sgt. Don Brandt, USAF (Ret.); ATOMIC ENERGY COMMISSION, SUMMARY OF OPERATION HARDTACK PHASE II, AEC 987/15, Note by Secretary, November 24, 1958, p. 5; McPhee, p. 108.)

²⁶³OPERATION REDWING - POST OPERATIONAL REPORT, PROGRAM 22 - REACTION HISTORY MEASUREMENTS, COPB 4547, L. F. Waters, July 1960, p. 17, contains reference to time intervals between secondary and tertiary stages.

²⁶⁴Classified Notebook No. 839-L, issued by Information Division, UCRL, Livermore, to Roger Preston, L-Division, UCRL, p. 19.

copper, and wood comprising the shot cab and its associated equipment. More than three million cubic yards of coral sand and rock also disappeared into the shot cloud.²⁶⁵

Among the materials lifted 85,000 feet into the stratosphere by the explosion was most of the western end of Eninman island, including residues from UCRL's MORGENSTERN device, which had scattered plutonium and uranium on the island after its "inefficient detonation" two years earlier.²⁶⁶ The cloud diameter grew to 75,000 feet.

The MORGENSTERN debris notwithstanding, BASSOON was a clean device: 85% of its total yield came from fusion.²⁶⁷ (The tertiary stage was jacketed with lead.) Nonetheless, 48% of Zuni's fission products was returned as radioactive fallout over more than 17,000 square miles of the Pacific Ocean.

Although the Zuni shot was not announced at the time it was detonated (only the Cherokee and Lacrosse shots, the tests observed by the news media, had been announced by the end of May),²⁶⁸ the firing was detected by a 5,000 ton Japanese freighter which, like the unfortunate FORTUNATE DRAGON more than two years earlier, was dusted with fallout from the test

²⁶⁵Letter dated June 29, 1959 to Joe B. Sanders, Director, Pacific Operations, USAEC, from S. P. Howell, Manager, Engineering and Construction, Holmes & Narver, subject: Estimated Weight of Construction Materials, Operations CASTLE, REDWING, and HARDTACK I.

²⁶⁶DNA-6037F, p. 183.

²⁶⁷"(Item) e. The high yield clean land burst tentatively scheduled at Eninman island to be conducted as the second of the megaton events." (Memorandum for the Record dated 5 July 1955, by Capt. R. H. Maynard, USN, Acting Chief, Weapons Test Division, Department of Defense, Armed Forces Special Weapons Project, Washington, D. C., Subject: Factors Bearing on Scheduling of Events - Operation REDWING.) In addition, an internal AEC memorandum dated August 7, 1956, from Brig. Gen. A. D. Starbird, U. S. Army, Director of Military Application, to AEC Commissioners Strauss, Murray, Libby, Vance, and Fields, Subject: REDWING Fallout Program, states (p. 3) that of the REDWING Cherokee, Zuni, Navajo, Flathead, and Tewa shots, three are standard and two are "clean," and that the total yields, fission yields, and efficiencies vary considerably among them.

²⁶⁸Memorandum to Those Listed Below dated June 5, 1956 from Shelby Thompson, Acting Director, Division of Information Services, USAEC, Subject: Draft Statement on Operation REDWING Progress.

(this time at a much greater distance). The incident was reported in Japan on June 5:

A Japanese sea captain said the U.S. appeared to have exploded a hydrogen bomb May 28 without warning to ships in the Bikini atoll area.

The captain of the MIZUHO MARU said his ship was drenched with radioactive rain. Two of his crew were reported suffering an acute shortage of white corpuscles.

"We were navigating 150 miles southeast of an island, 1,660 miles from Bikini atoll on May 28 when a hydrogen device probably was exploded without warning," he said.

"It was raining that day and the entire crew wore rain-gear. Radioactivity then was only 45 Geiger counts (per minute) but it was 1,250 (per minute) on May 30. We washed the ship with white soap and seawater." Japanese weather observers have claimed the U.S. employed a hydrogen device at Bikini on May 28. They base their claim on recordings of earth and atmospheric shock waves. The United States has neither confirmed nor denied the claim.²⁶⁹

AEC Commissioner Willard F. Libby later wrote: "We have learned from the Japanese that they can follow every bomb we fired in REDWING."²⁷⁰

²⁶⁹Teletype dated 5 June 1956 to Admiral Hanlon, JTF 7, Eniwetok, from Gen. A. D. Starbird, USAEC, Subject: Possible Contamination of Jap (sic) Vessel; memorandum dated June 20, 1956 to H. C. Brown, Executive Officer, Division of Biology and Medicine, USAEC, from Gordon M. Dunning, Division of Biology and Medicine, USAEC, subject: Report of Injury to Japanese Fishermen (sic). The teletype quotes a press release from Tokyo. A memorandum to file dated 5 June 1956 from Wayne Brobeck of the JCAE staff notes that upon arrival in Tokyo, the ship's entire crew was given physical examinations, and two crewmembers had white blood cell counts about half normal. Dunning, in the memo cited, claimed that fallout was not the source of the Japanese seamen's low white blood cell counts.

²⁷⁰"Study Proposal by Commissioner Libby Concerning Limitation on Weapons Testing," AEC 859/3, August 6, 1956. While this may have been true, the Japanese were not always correct in

Shot Yuma was the second UCRL REDWING test device, and the lab's first intentional low-yield Pacific shot. The Yuma device (the SWIFT) was detonated on a 200-foot tower on Aomon island at Eniwetok atoll at 0756 hours on May 28, two hours after the Zuni shot at Bikini. It yielded less than a quarter kiloton and provided experimental information on small boosted fission devices. The SWIFT was a small boosted asymmetric bomb.²⁷¹

The radioactive Yuma mushroom cloud reached only 10,000 feet. The device did not boost as planned; predicted yield had been much higher than actual test yield.²⁷²

The Erie device, a low-yield boosted LASL design, was airlifted to the PPG and arrived at Eniwetok on May 12. The device was installed atop a 300-foot high tower near the center of Runit island on May 28 and was successfully fired at 0615 hours three days later, yielding about 17 kilotons. The cloud reached 32,000 feet in about ten minutes. Erie produced high radioactivity near the zero site on Runit. **Figure V-4A** shows the Erie fireball surrounded by a condensation cloud as the shock disc move outward from the surface zero point.

Following Erie, the test schedule was revised. Seminole was now slated for June 5; Flathead and Blackfoot for June 9; Apache, June 12; Kickapoo, June 13; Huron, June 16; Dakota, June 21; Inca, June 22; Osage, June 25; Navaho, June 30; Mohawk, July 4; and Tewa, July 7. Weather delays, primarily at Bikini, could still prolong the test series until as late as August 1 to August 15.²⁷³

their assumptions: they reported that the July 3 (Mohawk) shot was detonated at an altitude of 22 miles after being lofted by a rocket. ("Twenty-Two Miles High," Science, TIME magazine, July 16, 1956.) There was no rocket-borne warhead test during REDWING.

²⁷¹OPERATION REDWING, Project 2.51, Neutron-Flux Measurements, WT-1313 (EX), Pacific Proving Grounds, May-July 1956, Headquarters Field Command, Defense Atomic Support Agency, Sandia Base, Albuquerque, New Mexico, September 30, 1959, p. 7.

²⁷²OPERATION REDWING - POST OPERATIONAL REPORT, PROGRAM 22 - REACTION HISTORY MEASUREMENTS, COPB 4547, L. F. Waters, July 1960, p. 82.

²⁷³Teletype dated June 1, 1956 to COFS, USAF, Washington, D.C. from CJTF Seven Eniwetok. (Teletype is mistakenly time-stamped as May 1 rather than June 1.)

By the end of May, after the firing of the 85% fusion UCRL BASSOON device, the AEC realized that it could take immediate political advantage of the fact that "clean" weapons were being tested during REDWING. Alternate draft statements for President Eisenhower and AEC Chairman Lewis Strauss were prepared for dissemination.

These statements were intended primarily to respond to rising international outcry over continued atmospheric nuclear testing by the U.S.; by claiming great gains in relative weapon "cleanliness," the United States might be able to mute some of the outcry over testing. It would also help President Eisenhower's re-election bid against Democratic presidential candidate Senator Adlai Stevenson; Stevenson favored a halt to nuclear weapons testing.

Also of great concern to the AEC was any useful intelligence the Soviets might derive from such an authoritative statement: during REDWING, Dr. Edward Teller noted in response to a proposed AEC statement on the development of "clean" weapons that:

There is considerable pressure to stop A-bomb testing and even more to stop H-bomb testing. I feel strongly that all types of tests must continue. Therefore, it is important to publish valid arguments for the test program.

The statement that we have made progress on defensive weapons is such a valid argument. Furthermore, it does not appear to give away valuable information. I feel, therefore, that this statement should be made. (I understand that this is a repetition of an older statement.)

The statement that we are trying to minimize fallout is equally important. There is, however, a possibility that such a statement will give away valuable information. I shall, therefore, discuss this possibility.

There is no simple and direct way to determine the ratio of fission yield to fusion yield from radiochemistry. The Russians, therefore, might think that Zuni (shot) produced most of its energy by fission. Our announcement might help them recognize such an (erroneous interpretation

of fallout from Zuni).²⁷⁴

To address and allay the concerns of Dr. Teller and others, the AEC drafted statements that were as general as possible, while nonetheless laying claims to great advancements in testing "cleanliness:"

1. In accordance with your instructions a statement has been prepared on the development of Clean Weapons (sic) as one result of the current REDWING test series. Two versions are attached, one which would be appropriate for use by the President, and the other for use in the event you made the public announcement.

2. The (AEC's) Division of Classification after review of the proposed statement has advised us that under current security regulations it is classified Secret-(Restricted Data). In view of this, declassification action by the Commission would be required prior to release. Also, it is believed that the statement should be coordinated with the Department of Defense since they requested us to study the possibilities of "Clean Weapons." Relative to declassification, see (paragraph) 3c below.

3. With Dr. Fine and Dr. English, consideration has been given to the benefits that the Russians could secure from this statement.

a. In this connection, it is believed that we must assume that the Russians will get samples on two clean devices, the BASSOON and the TX-21C, and that they will make analyses comparable to the manner and with the success that we would, thereby determining that these shots are different in the ratio of fission to fusion yield from our other high yield shots.

From this information they could

²⁷⁴Letter dated June 5, 1956 to General A. D. Starbird from Dr. Edward Teller.

be expected to (suspect) strongly the same thing as indicated in the statement, that we are studying weapons producing debris with less radioactivity.

b. Assuming they know how to build a two-stage weapon of our design principles, the debris will tell them that we can get burning with lower fission to fusion ratio. This might focus their attention with less effort and, hence, lead to their getting a "Clean Bomb" earlier. However, from the point of view of their stockpile we would like their weapons to be entirely "clean" since for the same supply of source and nuclear materials it is to be expected that the total yield of their stockpile of high yield weapons would be substantially less if composed of "Clean Weapons" than if composed of two-stage weapons using our conventional design.

c. Assuming that they do not know how to build an efficient two-stage weapon, it is believed that the proposed statement would not point them toward our two-stage design any more than the information that they are able to get by analysis of our (bomb test) debris.

d. On balance, we cannot say the Russians could not profit from the statement; however, they probably would not profit materially and it is believed this information could be published "without undue risk to the common defense and security."

4. Though not a part of the problem as assigned me, there appear to be some matters of policy which should be considered in connection with a possible statement on the development of "Clean

Weapons."

a. It is possible that pressure would come from the United Nations or other sources to limit tests in the future to "Clean Weapons." Such a limitation would seriously handicap our development effort, especially for small two-stage designs such as are required for the ICBM. In fact, at this time we do not have a good idea on how small a "Clean Weapon" can be made.

b. It is possible that there would be agitation for converting our present stockpile of high-yield weapons to a "clean" design, thereby reducing our military capability since more nuclear materials are required for "Clean Weapons" than for conventional weapons of the same yield.²⁷⁵

Attached to this memorandum were draft statements for President Eisenhower and AEC Chairman Strauss:

Statement by the President

(on assumption statement will be made at opening of a news conference)

In remarks to newsmen recently in connection with suggestions that the United States should halt its testing of hydrogen bombs, I stated that our research is not aimed just at making a thermonuclear weapon go off with a bigger bang and cause more destruction. Rather, I said we are seeking ways to limit the weapon, to make it useful for defensive purposes; that we want to make it more of a military weapon and less one of mass destruction. And I also said we want to reduce the amount of fallout.

²⁷⁵Memorandum for the Chairman dated 31 May 1956 from General Alfred D. Starbird, USA, Director of Military Application, USAEC, Subject: Statement on the Development of Clean Weapons. Emphasis in original.

Reports from the present series of nuclear tests at the Eniwetok Proving Ground indicate progress with investigations of high yield devices which produce substantially less radioactive material than present weapons of comparable yield. Results to date indicate that new weapons with such a desirable lower potential radioactive fallout are possible. Additional tests are necessary for further scientific advancement in this direction.

Statement by Lewis L. Strauss, Chairman,
U.S. Atomic Energy Commission:

President Eisenhower has authorized me to make the following statement in connection with his recent remarks to newsmen with regard to suggestions that the United States should halt its testing of hydrogen bombs:

The President stated at that time that our research is not aimed just at making a thermonuclear weapon having greater destructive capability. He said that we are seeking ways to limit the weapon and to make it more useful for defensive purposes; that we want to make it more of a military weapon and less one of mass destruction. And, he also said that we want to reduce the amount of fallout.

Reports from the present series of nuclear tests at the Eniwetok Proving Ground indicate progress with investigations of high yield devices which produce substantially less radioactive material than present weapons of comparable yield. Results to date indicate that new weapons with such a desirable lower potential radioactive fallout are possible. Additional tests are necessary for further scientific advancement in this direction.

Most noteworthy from these statements -- from a technological point of view -- was the admission that "clean" thermonuclear weapons required more nuclear material than conventional

designs of the same yield. With the removal of the U-238 fusion tamper around the secondary, and its replacement by a heavy metal non-fissionable jacket, more lithium deuteride was required to achieve the same yield with the same confinement time. If this were not done, yield would be reduced, thus diminishing the weapon's military utility.

In addition, in order to gain a better pre-ignition compression of the thermonuclear fuel, a vital necessity if yield were to be maintained, a larger, more powerful primary was required, leading to the burning of more or alloy and plutonium than would be required for the primary used to compress a conventional, i.e., U-238-jacketed, secondary stage.

There was a limit to how big these more-powerful primaries could be: they should certainly not generate more fallout than the fusion stages, or they would defeat the entire purpose of a 'clean' weapon.

Jacketing the secondary with an inert heavy metal such as lead or tungsten also decreased total weapon yield-to-weight ratio, since the yield of the secondary was reduced substantially while total weapon weight remained the same or increased.

The requirement for a large, powerful primary and the likely reduction of the yield-to-weight ratios of "clean" bombs were the basis of Starbird's admission that in May 1956 the AEC did "not have a good idea on how small" a "clean" bomb might be made in the future.

As the yield of the primary rose, its own contaminating characteristics began to negate the reduction in fallout from the secondary. To increase confinement time, a heavier, thicker tamper around the secondary was required. Obviously there was a point below which "clean" weapon size could not go, and a limit to the potential "cleanliness" that could be obtained from a given total weapon yield.

Meanwhile, back in the Pacific, REDWING continued. The next scheduled test, LASL's Seminole shot, was one of the more bizarre nuclear experiments undertaken in the PPG in the 1950s. The purpose of the test was to determine cratering effects of a high-yield device under carefully-controlled conditions. (See Volume VI for a discussion of cratering weapons and their uses.) The bomb could not be buried underwater on the atoll reef, so a means of simulating an underground burst had to be devised.

The test device was positioned in a 15' diameter, 15' high circular chamber located about 10 feet inside a 50' diameter, 25' high water tank holding nearly three million pounds of sea water (**Figure V-5**). The center of gravity of the nuclear

explosive was located at a point 3' off the center of the larger tank, and 4 1/2' above the ground. Entry to the dry inner chamber was by means of a 7'-high, 30"-wide passageway through the surrounding water tank.

The tank was large enough for the yield of the Seminole device to go from radiative transport to shock before the fireball reached the outside of the tank. This provided better coupling of the energy to the soil and produced a non-symmetrical crater shape around ground zero. The latter effect was caused by the burst point not being in the geometric center of the water tank.

The tamping effect of the water also created conditions equivalent to a 12-foot deep subsurface positioning of the explosive. Each cubic foot of water was equivalent to about 0.6 cubic feet of earth.

The Seminole device, a TX-28, arrived at the PPG on May 12; it was moved to the shot chamber within the water tank on Bogon island at Eniwetok a week later and fired at 1255 hours on June 6 with a yield of 13.7 KT. It demonstrated that the design of the device was satisfactory and gouged a crater 47 feet deep and 660 feet in diameter from the island.

The shot vaporized tons of steel, iron, aluminum, copper, lead, paraffin, and wood, and more than 265,000 cubic yards of coral sand and rock.²⁷⁶

The Seminole shot cloud rose in a uniform cylinder to 16,000 feet with no mushrooming and considerable initial fallout of water carried aloft from the tank. Since Seminole was in effect a subsurface cratering burst, it scattered a large amount of radioactive sand and coral near the zero site.

Figure V-6 shows the Seminole shot just after detonation, with the shock disc spreading out just beyond the debris cloud encircling the rising fireball (the IVY Mike crater is visible to the right of the burst point).

Figure V-7 depicts the condensation cloud caused by the rapid compression and decompression of the humid tropical air as the shock wave passes. The shock disc has now spread out considerably, engulfing Teiteripucchi island.

Figure V-8 shows the Seminole debris throwout cloud which

²⁷⁶Letter dated June 29, 1959 to Joe B. Sanders, Director, Pacific Operations, USAEC, from S. P. Howell, Manager, Engineering and Construction, Holmes & Narver, subject: Estimated Weight of Construction Materials, Operations CASTLE, REDWING, and HARDTACK I.

severely contaminated Bogon and the surrounding area.

LASL's Flathead device, a TX-28 with a uranium-tamped secondary, arrived at Eniwetok on May 22. It was loaded aboard a barge which was moved to Bikini by the USS CATAMOUNT on June 5. The barge was anchored in Bikini lagoon about 2,000 feet south of Yurochi island the following day. Detonation was originally scheduled for three days later (June 9).

After a short delay, Flathead was fired at 0626 hours on June 12. It yielded 370 KT and proved the correctness of various TX-28 warhead design parameters, including the yield reproducibility of the primary.²⁷⁷ Flathead dumped heavy contamination into Bikini lagoon and on the northern islands of the atoll. Its radioactive mushroom cloud rose 61,500 feet and spread out to a diameter of 40,500 feet.²⁷⁸

The Flathead device was a relatively dirty weapon: 73% of its total yield came from fission of uranium isotopes in its components.

By early June, the AEC was still concerned about what the Soviets might learn from tests of "clean" devices; the recent Japanese press release on the May 28 Zuni shot was cited as evidence of the impossibility of hiding a high-yield atmospheric detonation:

On June 4, 1956, we discussed with personnel of AFOAT-1 (Author's note: the Air Force's nuclear test detection and intelligence-gathering organization) the question of whether long-range detection methods could identify that the BASSOON test and the (TX-21C) test (when it is conducted) were in fact tests of "clean" devices. The following is our understanding of the situation, based on expectations and not on any experimental evidence from the BASSOON detonation.

Long-range detection methods include analyses of radioactive debris, from which can be obtained various ratios of fission products, isotopes present in the original fissionable material, neutron-capture

²⁷⁷Teletype dated June 4, 1956 to Gen. A. D. Starbird, USAEC, from W. Ogle, LASL.

²⁷⁸Letter dated 17 July 1956 to Lt. Col. Jack James, Program 9, Task Group 7.1, JTF 7, from D. F. Seacord, Jr., Edgerton, Germeshausen, & Grier, Inc.

products, products of (n,2n) reactions, etc. Long-range detection methods also include measurements of acoustic, seismic, and electromagnetic signals, from which can be obtained estimates of yield.

These yield estimates are not precise, often ranging for one standard error from one-half to twice the best value. If the yield is in the megaton region, this is readily available. (Perhaps the recent Japanese announcement on the BASSOON shot is a good indication of this.)

For yields in the megatons, the possible sources of the energy released are:

- (1) Fission of special nuclear material (plutonium, U-233, and uranium enriched in U-235).
- (2) Fission of U-238 (or thorium).
- (3) Fusion of thermonuclear materials.

With regard to possibility (1) above, if this were the source of a large part of the yield of multimegaton devices, hundreds of kilograms of special nuclear material would have to be consumed and even more would have to be present initially. This would be a very expensive device and would not appear reasonable.

With regard to possibility (2) above, analyses of the debris of "clean" devices would indicate low values of the following ratios compared with the values that could have been obtained from the debris of other multimegaton shots. [Remainder of memo deleted]²⁷⁹

²⁷⁹Memorandum dated 7 June 1956 to Brig. Gen. A. D. Starbird, Director, Division of Military Application, USAEC, from S. G. English and P. C. Fine, Subject: Long-Range Indication of Clean Weapon Tests. During REDWING, AFOAT participated in the collection and analysis of electromagnetic pulse signals and radiochemical samples of gaseous and solid particulates from a number of the shots. (Operation Plan 2-56, Headquarters United States Air Force (AFOAT-1) Participation in Operation REDWING,

By this time, Dr. Norris Bradbury of LASL had seen the drafts of the proposed presidential and AEC statements on the development of "clean" weapons, and he was strongly opposed to their release:

I have seen the proposed statements with respect to the development by the United States of weapons of reduced fall-out and less mass destruction and have the following quick comments:

(1) I would personally prefer very strongly to withhold any such statement until the debris from BASSOON has been examined and compared with (debris from) previous conventional (unboosted), boosted, and two-stage weapons. It is probably not possible to infer from the debris the ratio of fission to fusion yield, and this statement will thus be the first indication to the Russians that we are developing "clean" bombs.

The relative amount of Pb-203 (Author's note: lead) in the samples they collect compared to previous samples of large yield experiments may, however, suggest to them that the amount or disposition of Pb in this device may be both relatively greater or disposed differently with respect to the neutron flux. In any event, I would regard a detailed comparison of [deleted] as an important prelude to any release.

(2) I would regard the release as confirming press speculation that present U.S. weapons are of the so-called "fission-fusion-fission" type and that the U.S. is endeavoring to decrease the latter process and increase the central one. This, of course, is true, and it will probably suggest to an informed reader

16 March 1956.)

that we are endeavoring to reduce the amount of fissionable material (including ordinary uranium) in our hydrogen bombs.

If increased Pb is detected by the Russians, they will probably suspect correctly that Pb is replacing the U in regions of high neutron flux. I doubt if they can infer from this alone much about the two-stage principle or extensions of it.²⁸⁰

As noted earlier, Dr. Edward Teller also opposed the release of such a declaration of U.S. "clean" weapons development. A few weeks after his earlier statement on this matter, Teller wrote:

It is highly probable that the Russians have recognized our first [deleted] as a peculiar shot. A statement on our part that we are working on clean weapons is not likely to aid them substantially. The same is true about the Los Alamos two-stage clean weapon. [deleted]²⁸¹

Back in the Pacific, the LASL Blackfoot device had arrived aboard a C-124 on May 22. It was assembled at Parry island on June 6, and moved to a 200-foot shot tower on Runit island three days later. First scheduled for detonation on June 10, Blackfoot was detonated successfully at 0626 hours on June 12, simultaneous with the Flathead detonation at Bikini. This was more by accident than design: Flathead was delayed and almost canceled by unfavorable winds.²⁸²

²⁸⁰"Comments by Dr. N. E. Bradbury on Proposed Public Statement on Development of Clean Weapons," attachment to Memorandum for the Commissioners, from W. B. McCool, Secretary, Subject: Proposed Public Statement on Development of Clean Weapons. The references to lead (Pb) concern the tamper around the third stage of the BASSOON device.

²⁸¹Memorandum for the Chairman through the General Manager dated 11 July 1956 from Brig. Gen. Alfred D. Starbird, Director of Military Application, USAEC, Subject: Opinions of Dr. Teller and Dr. Bradbury relative to the Clean Weapons Matter.

²⁸²The shots occurred within approximately 0.126 seconds of each other. (Teletype dated June 14, 1956 to CTG 7.1, Eniwetok, Marshall Islands, from Commander, Field Command

Blackfoot was a low-yield (8.5 KT) boosted device; its radioactive mushroom cloud rose to 32,000 feet, about 10,000 feet higher than expected.

UCRL's Kickapoo device, the SWALLOW, was detonated atop a 300-foot tower on Aomon island at Eniwetok at 1126 hours on June 14 and yielded 1.4 kilotons. It provided design information required for the development of a small diameter air-to-air missile warhead. The Kickapoo device, like the Yuma device, was a small asymmetric boosted weapon.²⁸³ Its mushroom cloud rose to 18,000 feet.

The low-yield XW-25 for LASL's Osage shot arrived at Eniwetok on April 10 aboard the CURTISS. The warhead, contained in a heavily-instrumented MK 5 ballistic drop case, was loaded aboard a B-36 on June 15.²⁸⁴ The following day, it was airdropped with "excellent bombing accuracy" and detonated successfully at 1314 hours 680 feet over Runit island at Eniwetok.²⁸⁵

The burst was within a 180-foot horizontal position of ground zero and yielded 1.9 kilotons. The detonation cloud rose to 21,000 feet, about 8,000 feet higher than predicted.

By June 21, as a result of earlier TX-28 tests, JTF 7 decided to fire the Dakota shot, which had heretofore been a contingency test. In addition, a one-point safety test of the LASL XW-31 was dropped from current REDWING plans.²⁸⁶ The Dakota device had arrived at the PPG on June 15.²⁸⁷

AFSWP, Sandia Base, Albuquerque, New Mexico.)

²⁸³WT-1313 (EX), p. 7; Minutes of the 99th Meeting, SWDB-56-99-2, 29 February 1956, Part Two, Special Weapons Development Board, pp. 23, 24.

²⁸⁴WT-1313 (EX), p. 7. Osage was a "low yield prototype stockpile warhead unit in a bomb case." HISTORY OF TASK GROUP 7.4, Provisional, for June 1956 states on p. 10 that T-62 shapes were "mockups of the Osage device." The T-62 was a training shape for the MK 5 bomb. See W-25 history in Volume VII.

²⁸⁵Teletype dated June 17, 1956.

²⁸⁶Letter dated June 21, 1956 to James T. Ramey, Executive Director, JCAE, from Brig. Gen. Alfred D. Starbird, Director of Military Application, USAEC.

²⁸⁷OPERATION REDWING, Commander Task Group 7.3 Final Report, March-August 1956, J. H. Wellings, p. 79.

Shot Inca, a test of a UCRL SWAN device, was detonated atop a 200-foot tower on Rujuro island at Eniwetok on at 0956 hours June 22. It yielded about 15 KT and provided scientific information on a device designed to fulfill several purposes, including serving as a primary for a three-stage device. The shot cloud reached 42,000 feet. One description of shot damage stated that "the (shot) tower is completely gone and the shot island is cleaned off down to bare sand."²⁸⁸

Early in June, Dr. William E. Ogle, leader of LASL's J Division and Acting Deputy Commander for Scientific Matters in JTF 7, prepared a briefing for Canadian and British military observers at REDWING. In this talk, Ogle outlined the reasons for the sequence and placement of many of the current tests:

... it appeared clear that the high-yield shots not requiring closely-aligned or stable platforms should be barge shots off Yurochi, using the structures built for CASTLE. The Zuni shot used instrumentation requiring land so it inherited the Eninman region. The DOD requirement for fall-out studies on the Cherokee air-drop required a fairly large land mass, so its ground zero was placed in the center of Namu island.

The Seminole (test) was originally conceived to be a fairly high-yield shot requiring detailed instrumentation, so with some reluctance it was put on Teiteiripucchi island in order to use the IVY recording structure on Bogon. Later, after the rest of the program was fairly well settled, the design yield was reduced and the zero point was moved to Bogon.

In order to use the same recording station for several shots and to reduce both the experimental effort and construction effort, all of the Livermore small shots (the Yuma, Inca, Kickapoo and Mohawk) were put on the Aomon-Bijjiri (island) group as tower shots, and by ordering the shots so as to have the largest yield last, the towers could all be built before the operation began.

Similarly, the LASL small shots, Blackfoot, Erie, Lacrosse, and the Osage air-drop,

²⁸⁸Teletype dated June 25, 1956 to Gen. A. D. Starbird, USAEC, from Alvin Graves, LASL.

were put on the only remaining piece of land (exclusive of) Engebi; that is, Runit island. However, since Lacrosse was a highly-instrumented shot, it had to come first, and since it was also fairly large, the towers for Erie and Blackfoot could not be built ahead of time.

Engebi was not used initially, since fall-out from shots there would certainly land on the Bogon installation. However, we have the capability of putting up a tower there later in the series and making the appropriate measurements if it should prove necessary.

Now as to time. The Lacrosse shot had to be fired first of the Runit shots because of the construction problems. The Erie, Seminole, Flathead, and Dakota (shots) were (or are to be) fired at Runit, each depending upon the results from the shot before it. (Author's note: these four tests were of various TX-28 configurations; see MK 28 history in Volume VI.)

Since we were anxious to get started, the Erie was the second shot on Runit, some three weeks after Lacrosse, (the time required) to build a new tower and instrumentation. The Seminole (shot) on Bogon was fired then as soon as possible after Erie. As I mentioned, the Mohawk test on Eberiru must be the last shot; Yuma, Inca, and Kickapoo were detonated as (the devices became) available.

On Bikini, the Cherokee shot was essentially an effects shot, concerned with fall-out and structures and, hence, had to be fired first, since the instrumentation would be adversely affected by an earlier shot. The Zuni installation on Eninman would not suffer seriously from the Cherokee blast but would be seriously damaged by a barge shot near Yurochi; therefore, the Zuni shot was the second at Bikini.

From then on, the shot order depended essentially on the readiness of the device; for instance, Flathead and Dakota must wait until after the Seminole (test) on Bogon, the Apache (device) will not be in

existence until late in June, etc.

....

As a final subject, I would like to mention some of the measurements we make on a typical high-yield shot. Pictures will be taken from photo stations which will allow a measurement of the fireball diameter as a function of time and, hence, permit a determination of the yield of the device.

Other cameras in the same stations are used for other purposes; for instance, a fan of rocket (smoke trails) is photographed to study blast effects. Canisters dropped by parachute from aircraft just before zero time telemeter free air blast information to receiving stations. The locations of the canisters are determined by cameras.

Fall-out stations all around the atoll will measure fallout activity as a function of time, and will also collect the actual fallout material as it comes down, for further study in the (United) States. Similar (collection) stations are on rafts in the (atoll) lagoon, on anchored skiffs outside the lagoon, and on manned ships farther away in the fallout pattern.

The intensity and color of the device (explosion) light will be measured. The characteristics of the water wave produced in the lagoon will be studied.

Many different aircraft will be passing overhead to study the effects of blast and thermal radiation. These airplanes are accurately positioned by radar.

The system on the ground and in the air is tied together by the timing station on Enyu (island at Bikini) which sends out time signals by wire and radio to all stations needing time signals.²⁸⁹

²⁸⁹"Proposed Talk to U.K.-Canadian Observers," To be Given by Dr. William E. Ogle, Acting Deputy Commander for Scientific Matters, Joint Task Force 7, Appendix "B" to Atomic Energy Commission, Release of Atomic Information to U.K.-Canadian Observers (Proposed Talk by Dr. Ogle), Report to the General Manager by the Director of Military Application, USAEC, June

LASL's Dakota device (a TX-28C)²⁹⁰ arrived at the PPG on June 8 and was taken aboard a shot barge to Bikini by the CATAMOUNT six days later. The barge was anchored in the CASTLE Union crater in Bikini lagoon south of Yurochi island and detonation scheduled for June 18. After a number of weather-caused delays, the shot was fired successfully at 0606 hours on June 26. It yielded a megaton and provided additional design information required for two-stage high-yield fusion devices. Dakota's actual yield exceeded its maximum predicted yield by 25%.²⁹¹

The radioactive Dakota shot cloud ascended to 81,000 feet with a diameter of 93,000 feet.²⁹² The upper umbrella of the cloud extended almost from Bikini to Eniwetok, a distance of about 200 statute miles.

Dakota was the only shot which caused extensive damage to Air Force weapons effects aircraft, due mostly to the higher-than-expected yield (many aircraft had been positioned on the assumption of a lower yield). A B-47E suffered moderate overpressure damage, torsional wrinkling, to its bomb bay doors and forward landing gear wheel well doors and thermal damage to its control surfaces.

The bomb bay doors, flaps, landing gear brakes, and braking chute on an RB-52B (AF serial 52-004) were damaged by thermal effects and overpressure. Both aircraft were about five miles from surface zero at detonation time; the B-47E was exposed to a thermal burst of 547⁰.

The B-52 was diverted to Hickam AFB on the island of Oahu in Hawaii where it landed safely on a 13,000 foot runway without using its flaps, brakes, or braking chute (the runway at Eniwetok island was too short for the B-52 to land without brakes or flaps).²⁹³ As it turned out, the B-52 suffered only

25, 1956. Another draft of this briefing is labeled "To British -- As Given." Among subjects not to be discussed with the British and Canadians were boosting, external initiation, or the "three concepts" of thermonuclear explosive device design.

²⁹⁰Operation REDWING, A Preliminary Report of [deleted] (Mohawk), JO-265, 15 October 1956, p. 42.

²⁹¹Draft Final Report by Commander, Joint Task Force SEVEN on Operation REDWING, 27 August 1956, p. 5.

²⁹²Memorandum dated 4 July 1956 to Distribution from D. J. Barnes, Edgerton, Germeshausen, & Grier, Inc., Subject: Cloud Dimensions - Dakota.

minor isolated paint blistering and one dent.²⁹⁴

UCRL's Mohawk device was stored and checked in the CMR building, Building 341, on Parry island, then shipped by boat and truck to a 300-foot tower at Eberiru island. Final assembly was completed July 2 by a joint TU 11-TU 12 assembly team.

The bomb was fired the following day at 0606 hours. It yielded 350 kilotons and was a satisfactory test of a high-yield two-stage thermonuclear device with a boosted SWAN primary and a FLUTE secondary. The shot cloud rose to 65,000 feet. The explosion carved a crater eight feet deep and 1,340 feet in diameter, obliterating most of Eberiru island.²⁹⁵

In order to expedite gamma ray measurements for one diagnostic experiment, a massive lead and paraffin shield had been erected near the Mohawk device in the tower cab. The shield consisted of 6" of lead and 12" of paraffin on the device side to "hold back" neutrons: paraffin, with its high hydrogen content, is an excellent neutron absorber.

The Mohawk shot shield caused a "pimple" to form in the fireball, as shown in **Figure V-10** and **Figure V-11**.

Shields of this and other types were to be used quite extensively during Operation HARDTACK in 1958 to either collimate or block various kinds of radiation and particles from exploding test devices.

Results of Mohawk showed that assembly time, i.e., the time to critical mass, for the two-stage device primary was somewhat longer than anticipated and that the compressed geometry was neutronically not as favorable as expected.

Nonetheless, within these limitations, the device functioned at least as well as designed. The positive results of the shot indicated that even more radical and closely-coupled radiation implosion geometries could be developed.²⁹⁶

²⁹³Teletype dated June 29, 1956 from Commander, JTF 7 to Distribution. Part of the Eniwetok airstrip is shown in **Figure V-9**; among the aircraft visible are a B-52, B-47s, B-50s, a B-66, a C-124, F-84s, and a number of other types of transports, light planes, and helicopters.

²⁹⁴Evans, FINAL HISTORY, TG 7.4, p. 115.

²⁹⁵OPERATION HARDTACK I 1958, DNA 6038F, Defense Nuclear Agency, Washington, D.C., December 1982, p. 38.

²⁹⁶OPERATION REDWING - POST OPERATIONAL REPORT, PROGRAM 22 - REACTION HISTORY MEASUREMENTS, COPB 4547, L. F. Waters, July

The day after the Mohawk shot, AEC Commissioner Thomas Murray wrote to President Eisenhower and questioned a current DOD requirement for the development of a very high-yield weapon:

Two circumstance in combination have led me to write this letter. First, the Atomic Energy Commission now has a requirement from the Department of Defense to develop a [deleted] megaton weapon. Second, a recent test at the Pacific Proving Ground, which I witnessed, indicated that we can manufacture weapons with this yield in a form which can be delivered by our presently-available aircraft. Even larger weapons are technologically feasible.

My basic question is whether the stockpiling of a weapon of this size is in the national interest. The particular questions are two:

(1) Is a weapon of this size necessary or useful for military purposes?

(2) Would its use be consistent with the dictates of the moral law with regard to the moderate and discriminating use of force in warfare?

... bombs that we now have in stockpile are already large enough and may possibly be too large. I stated this criticism in testimony ... on April 2, 1956.

I have made every effort to have the Commission as a body bring to your attention this vital question of setting an upper limit on the size of nuclear weapons. Since these efforts were unsuccessful, I felt it my duty personally to present the question to you.²⁹⁷

1960, pp. 113, 115, 116, 118, 119.

²⁹⁷Letter dated July 3, 1956 to The President from AEC Commissioner Thomas E. Murray, document 144 in FOREIGN RELATIONS OF THE UNITED STATES, 1955-1957, Volume 20: Regulation of Armaments; Atomic Energy, John P. Glennon and David S. Patterson (eds.), U.S. Government Printing Office, Washington, D.C, 1990, pp. 408, 409. Eisenhower referred

As REDWING neared completion, the AEC re-emphasized the sensitivity of "clean" weapons information:

The fact that we have succeeded in producing a clean thermonuclear explosion is highly sensitive weapons design information because it discloses a basic principle. During Operation CASTLE, the unfortunate deposition of fallout on the islands revealed highly sensitive weapons design information, that our so-called H-bomb involved an enormous amount of fission energy.

This led to the popular concept of the fission-fusion-fission bomb, which, of course, was essentially correct. [deleted] This is the information which is revealed by a statement that we have made an H-bomb which has very little fallout, and which would be most revealing to the Russians who never have shot such a device ...²⁹⁸

Assembly of the UCRL Apache device was completed in the CMR building on Parry and the complete test package -- warhead, REGULUS I missile nose cone, and support stand -- was transferred to the shot barge. The barge was towed to the firing position in the IVY Mike crater. Apache had originally been scheduled for Bikini; the shot site was moved to take advantage of favorable weather at Eniwetok and to shorten the test series (the Mohawk shot had not significantly contaminated the northern Eniwetok atoll islands).²⁹⁹

Apache was detonated at 0606 hours on July 9. A test of a prototype XW-27 UCRL missile warhead design for the REGULUS, the shot yielded 1.9 MT. The mushroom cloud rose to 85,000 feet. The Apache detonation produced exceptionally heavy fallout throughout the northern islands of Eniwetok atoll. Water in the northern end of the lagoon was also heavily contaminated by radioactive fallout.

The XW-27 was fired in a REGULUS nose cone to offer the most

Murray's letter to the National Security Council.

²⁹⁸Memorandum for the Chairman dated July 6, 1956 from W. F. Libby, Subject: Sensitivity of Clean Weapons Information.

²⁹⁹Teletype dated July 3, 1956 to Chief of Naval Operations, Washington, D.C. and Chairman, USAEC, from CJTF 7, Eniwetok, Marshall Islands.

stringent mass discontinuities, which might degrade yield.³⁰⁰ Very few experimental warheads were ever fired within their actual delivery vehicles, and even fewer were ever tested operationally.

The test device used the same type of boosted LASL primary fired earlier during the Lacrosse shot. In order to provide the best possible comparison of the performance of the XW-27 primary with that of the Lacrosse device, a special collimator was set up to help measure alpha by narrowing the initial gamma radiation to a small beam. The collimator, placed about six feet from the warhead, was a lead shield about five feet square and several inches thick with a central 4" diameter hole opposite the center of the primary.³⁰¹ **Figure V-12** shows the REGULUS nose cone and the lead collimating shield inside the cab on the shot barge.

The Navajo device, LASL's TX-21C bomb,³⁰² a development of the CASTLE Bravo shot SHRIMP devices, arrived aboard the CURTISS on April 10 and was stored at Eniwetok until July 5 when it was loaded aboard a barge and taken to the Bikini lagoon anchorage in the CASTLE Union crater south of Yurochi island, where the Dakota shot had been fired nine days earlier.

After a two-day delay, Navajo was detonated successfully at 0556 hours on July 11. The TX-21C yielded 4.5 to 4.7 MT; it was a very "clean" bomb: 95% of its total yield came from fusion of hydrogen isotopes in its thermonuclear fuel charge.³⁰³ Navajo's yield was within its predicted range limits.³⁰⁴

The Navajo shot cloud stabilized at an altitude of 103,000 feet with a diameter of 140,000 feet;³⁰⁵ 50% of Navajo's

³⁰⁰Operation REDWING, A Preliminary Report of [deleted] (Apache), JO-266, 15 October 1956, p. 4.

³⁰¹OPERATION REDWING - POST OPERATIONAL REPORT, PROGRAM 22 - REACTION HISTORY MEASUREMENTS, COPB 4547, L. F. Waters, July 1960, p. 67.

³⁰²"Timetables of Weapons Events," DOE Albuquerque Operations Office, September 22, 1986.

³⁰³Shelton, p. 8-15.

³⁰⁴Teletype dated July 10, 1956 from CJTF 7, Eniwetok, Marshall Islands, to Chief of Naval Operations, Washington, D.C. and Chairman, USAEC.

³⁰⁵Inter-Office Memorandum dated July 18, 1956 to Distribution from D. Berkowitz, Edgerton, Germeshausen, & Grier, Inc.,

fission yield products was returned as fallout over nearly 14,000 square miles of the Pacific. The high fusion yield of the TX-21C would become a politically-important and often-cited figure in debates about the future of U.S. nuclear testing.

On July 13, JTF 7 planned to fire Tewa as the next shot; the predicted shot date was July 18. Huron would be fired about a week later; it could not be fired much sooner than July 25 because a new pit was required for the test device.³⁰⁶ An attempted firing of Huron on July 2 had been unsuccessful.

The BASSOON PRIME device for Tewa, the last UCRL shot during REDWING, was stored and checked in the CMR building on Parry island, then transported to the barge slip and loaded aboard a shot barge on July 15.³⁰⁷ The barge was shipped aboard the CATAMOUNT to the zero location over 20 feet of water above the Bikini atoll reef halfway between Yurochi and Namu islands.³⁰⁸

Figure V-13, Figure V-14, and Figure V-15 show the Tewa shot barge being unloaded from the CATAMOUNT (the significance of the palm frond atop the the pyramid of corner radar reflectors is unknown). **Figure V-16** shows the barge anchored in Bikini lagoon between Namu and Yurochi islands. **Figure V-17** shows the BASSOON PRIME device inside the cab aboard the barge; the device was 39" in diameter, 135.5" long, and weighed 15,735 lbs.

The BASSOON PRIME was similar to the BASSOON fired during the Zuni shot; however, it was nearly 3,600 lbs. heavier and also featured two "light pipes," diagnostic tubes welded to the casing to measure transit time of the radiation flow from the primary to the end of the device case. In addition, the tertiary stage was wrapped with U-238 instead of non-fissionable lead. A test of a second BASSOON had been planned as early as December 1955.³⁰⁹

Subject: Cloud Dimensions - Navajo.

³⁰⁶Teletype dated July 13, 1956 to AEC, Washington, D.C. from CJTF 7, Eniwetok, Marshall Islands.

³⁰⁷OPERATION REDWING, Commander Task Group 7.3 Final Report, March-August 1956, J. H. Wellings, p. 79.

³⁰⁸Memorandum dated August 2, 1956 dictated by Senator Clinton P. Anderson.

³⁰⁹Classified Notebook No. 839-L, issued by Information Division, UCRL, Livermore, to Roger Preston, L-Division, UCRL, p. 28. The Tewa shot is referred to here as "Zuni Prime."

Final assembly was completed aboard the barge and the device was detonated at 0546 hours local time on July 21. The observed yield was 5 MT, based on fireball expansion rate. The test provided vital design information for a three-stage high-yield thermonuclear bomb. The explosion blasted a crater 133 feet deep and 4,000 feet in diameter in the reef below its anchorage.

The Tewa shot cloud rose to 99,000 feet; fallout heavily contaminated the northern islands of Bikini atoll and reached Eniwetok, where it dusted several of the task force ships and Parry and Eniwetok islands.³¹⁰ Hazardous fallout from Tewa covered more than 2,000 square miles; 28% of Tewa's fission products covered an area of more than 57,000 square miles of ocean.

Like the Flathead device, the Tewa device was a very "dirty" bomb: 87% (4.35 MT) of its total yield came from fission reactions in its primary and two thermonuclear stages.³¹¹

The LASL Huron device (the EGG) had arrived at Eniwetok on May 22. It was taken by barge to Bikini on June 27. An attempt was made to detonate Huron on July 2, but the shot was aborted within three minutes of zero time by a failure of the fissile primary pit to maintain D-T boost gas pressure.³¹²

The EGG device was secured on the shot barge for safe transport and returned aboard the CATAMOUNT to Eniwetok, where the bomb was unloaded and disassembly started on July 5.³¹³ Five days later a new primary pit arrived at Eniwetok and was installed.³¹⁴

³¹⁰Letter dated August 2, 1956 to James T. Ramey, Deputy Director, JCAE, from Brig. Gen. Alfred D. Starbird, Director of Military Application, USAEC.

³¹¹Another source states that 70% (3.5 MT) of the Tewa yield was from fission. ("U.S. Shots 1945 - 1957 Inclusive, Pacific Tests - EPG," undated AEC document probably prepared late in November 1958.)

³¹²Teletype dated July 1, 1956 to Brig. Gen. Griffith, USAF, from Chief of Staff, USAF, Washington, D.C. Dates in the continental U.S. were one day earlier than dates in the Marshalls, which were located west of the international date line in the Pacific.

³¹³Teletype dated July 10, 1956 to A. D. Starbird from Alvin Graves.

³¹⁴Memorandum dated 4 July 1956 to Commander, Joint Task Force

On July 17, the reassembled device was loaded aboard a shot barge and moved to the IVY Mike crater near Engebi island. The bomb was detonated successfully at 0616 hours on July 22 following a two-day delay caused by unfavorable winds aloft. The final REDWING shot, the EGG device functioned properly and yielded 270 kilotons. (The EGG may have been an early progenitor of the LASL W-50.)

During the course of the Bikini shots, all the islands except Enyu were contaminated by radioactive fallout. This was very fortunate for the firing party at Station 70 on Enyu.

Although there were no confirmed submarine incidents during REDWING, on July 18 a Navy patrol aircraft investigated a reported sighting of "a possible underseas craft" off Darritt Island in Majuro atoll. Search results were negative.³¹⁵

Table 3-14 lists final ready and detonation dates for the 17 REDWING tests.

Table 3-14 REDWING Detonations

<u>Shot Name</u>	<u>Ready Date</u>	<u>Detonation Date</u>	<u>Yield</u>	<u>Location & Shot Type</u>
Lacrosse	1 May	5 May	40 KT	Eniwetok-ground
Cherokee	1 May	21 May	3.4 MT	Bikini-air drop
Zuni	8 May	28 May	3.5 MT	Bikini-ground
Yuma	1 June	28 May	0.19 KT	Eniwetok-tower
Erie	23 May	31 May	16.8 KT	Eniwetok-tower
Seminole	28 May	6 June	13.7 KT	Eniwetok-ground
Flathead	2 June	12 June	370 KT	Bikini-barge
Blackfoot	7 June	12 June	8.5 KT	Eniwetok-tower
Kickapoo	18 June	14 June	1.4 KT	Eniwetok-tower
Osage	14 June	16 June	1.9 KT	Eniwetok-air drop

Seven from R. J. Van Gemert, subject: Special Air Mission - Huron Device Parts.

³¹⁵Joint Task Force Seven, Commander Task Group 7.3, History of Operation REDWING, Period ending 31 July 1956, September 4, 1956, p. 10.

Inca	8 June	16 June	16	KT	Eniwetok-tower
Dakota	13 June	26 June	1	MT	Bikini-barge
Mohawk	1 July	3 July	350	KT	Eniwetok-tower
Apache	27 June	9 July	1.9	MT	Eniwetok-barge
Navajo	8 June	11 July	4.7	MT	Bikini-barge
Tewa	7 July	21 July	5	MT	Bikini-barge
Huron	12 June	22 July	270	KT	Eniwetok-barge

Pawnee -- canceled after success of Lacrosse shot

By May 1957, only three REDWING shots had been announced publicly by the AEC.³¹⁶ Even as late as February 1964, the Yuma, Kickapoo, Inca, and Mohawk shots had not been made public.³¹⁷

Results of REDWING

The officially-announced purpose of REDWING was to develop a variety of atomic weapons to meet a wide range of military uses, to provide defense against air attack, and to perfect devices with reduced fallout hazards. Even before the last REDWING detonation, the AEC announced on July 19 that real progress had been made toward meeting these goals:

Following my return from our Eniwetok Proving Grounds in the Marshall Islands, I have reported to the President on the results of the current weapons tests which are now nearing completion.

As a result of this series we have gained confirmation of important objectives stated by the President on April 25 just prior to the beginning of the tests. At his news conference on that date the President discussed our nuclear testing program,

³¹⁶"Background Information on Nevada Nuclear Tests," Office of Test Information, Nevada Test Organization, Nevada Test Site, Mercury, Nevada, May 1, 1957, pp. x, 4.

³¹⁷"Announced Tests - Pacific & South Atlantic," undated AEC document ca. early October 1962; THE EFFECTS OF NUCLEAR WEAPONS, Samuel Glasstone, ed., U.S. DOD and USAEC, Washington, D.C., April 1962, reprinted February 1964, p. 674.

aimed at the development of a variety of weapons calculated to meet a wide range of military usage including defense against air attack. He noted that there was also the objective of making weapons with reduced fallout.

Our tests to date have demonstrated that real progress has been made with respect to these objectives, including achievement of maximum effect in the area of a target with minimum widespread fall-out hazard.

It has been confirmed that there are many factors, including operational ones, which do make it possible to localize to an extent not heretofore appreciated, the fall-out of nuclear explosions.

Thus the current series of tests has produced much of importance not only from a military point of view but from a humanitarian aspect.

We are convinced that mass hazard from fall-out is not a necessary complement to the use of large nuclear weapons.³¹⁸

The suggestion by Strauss of a "humanitarian" H-bomb was later subjected to much press and congressional criticism and ridicule. On July 25, Congressman Chet Holifield of California, while questioning Admiral Strauss during hearings of the congressional Joint Committee on Atomic Energy, stated:

I am concerned over the repeated pattern of applying advertising agency techniques to alleviate the natural fears of hydrogen destruction which trouble our people and the people of the world. Huckstering is no substitute for policy as we grope for solutions to the atomic and hydrogen

³¹⁸July 19, 1956 statement to the press by Lewis Strauss, Chairman, U.S. Atomic Energy Commission. Dr. Edward Teller, fearful of revealing a U.S. nuclear weapons design breakthrough, opposed release of any information about "clean" bombs, in spite of Eisenhower's statement at his April 25, 1956 press conference. Fearful of alerting the Soviets to U.S. nuclear weapon technology breakthroughs, and of providing significant design insights, Teller was not in favor of any reference to "clean" weapons. (Hewlett and Holl, p. 346.)

dilemma.

....

How can anyone suggest that there are humanitarian aspects to hydrogen bombs? They are designed for destruction and contamination. They are the most inhumane device ever conceived by man.

....

No matter how much you change an atomic or hydrogen weapon there is still a great deal of radioactivity. An atom bomb or hydrogen bomb burst on a military target without dangerous fall-out would be a miracle. Such a miracle has not been achieved.³¹⁹

Aside from their technological challenge and their value as a public relations tool in the AEC's campaign to continue nuclear testing, the value of "clean" warheads was questionable. If they could be made small and light, they might have some use as battlefield tactical weapons because troops could quickly occupy territory over or on which "clean" devices had been detonated without having to worry about hazardous lingering radiation aside from that induced by neutrons from the blast.

The only strategic value "clean" weapons might have would be limits on personnel casualties and fallout both around the immediate area of the explosions and from stratospheric distribution. In this context, "clean" bombs might in fact be more "humane" than "dirty" conventional thermonuclear explosives.

However, even if the U.S. were to refrain from using "dirty" warheads in a nuclear attack, there was no guarantee that the Soviets would reciprocate and be equally "compassionate." All in all, the chief immediate value of "clean" thermonuclear devices was their usefulness to the AEC in its attempts to mute public outcry over radioactive fallout resulting from continued atmospheric nuclear testing. Aside from being an interesting technological challenge, "clean" weapons were primarily just an expensive public relations gimmick.

In mid-July 1956, the GAC characterized REDWING results as "spectacular."³²⁰ Not only had additional devices been proven

³¹⁹BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 11, 1956, p. 263; speech of Representative Chet Holifield (D., Cal.) on the Floor of the House of Representatives, July 25, 1956.

³²⁰Minutes of the Fiftieth Meeting of the General Advisory

as weapons suitable for stockpiling, but certain "clean" weapons had achieved predicted effects with minimum fallout.³²¹ Most noteworthy among these were the LASL TX-21C and the "clean" version of the UCRL BASSOON devices; the total fusion yield of the two averaged over 90%.

The top-secret technical results of REDWING, kept hidden from the American public and the Soviets, were dramatic and extensive. REDWING proof-tested a stockpiled MK 15 gravity bomb under field conditions; proof-tested TX/XW-28 variants and the boosted primary for the TX-15-X1 (MK 39) and W-27; and proof-tested early XW-45 antecedents to establish feasibility of 12" diameter implosion systems.

The test series also obtained design information on implosion systems in 8" diameters and on 13" to 15" diameter two-stage systems; explored three-stage weapon design principles; proof-tested TX-21C, XW-25 and XW-27 prototypes; and explored feasibility of both "clean" thermonuclear weapons that produced relatively small amounts of fission products and "salted" weapons that produced relatively large amounts of fission products.

REDWING led to stockpiling of MK 21, W-25, MK 27, MK 28, and MK 39 weapons and advanced design of small warheads for air-to-air and surface-to-air missiles. REDWING also showed the way to an early deployment of high-yield ICBM warheads.³²²

The Commander of JTF 7 summed up REDWING results in a report to the Chairman of the AEC and the Joint Chiefs of Staff:

Certain aspects of the test series were of particular interest. The program incorporated proof tests of a stockpile fusion weapon, a TX-15-X1, and a small

Committee to the USAEC, July 16-18, 1956, p. 48.

³²¹According to the NRDC's NWD 86-2 (p. 18), the total fission yield for all REDWING tests was 9 MT; the total fission yield for tests over 1 MT was 8 MT, which indicates that almost all the REDWING fission yield came from shots in excess of 1 MT, and that only one megaton of fission yield was distributed among the many lower-yield shots which made up the majority of REDWING tests. This 8 MT total fission yield is reiterated in an AEC press release dated May 5, 1959. Most of the tropospheric fallout from REDWING went over Mexico. (Minutes of AEC Meeting No. 1238, 2:30 PM, Monday, October 22, 1956.)

³²²Minutes of the Fiftieth Meeting of the General Advisory Committee to the USAEC, July 16-18, 1956, p. 48; Orientation Guide, 1 June 1957, J-3 Division, Joint Task Force Seven, p. 18.

fission warhead (the XW-25) for air-to-air rockets. The quest for a Class D thermonuclear weapon, that is, one weighing not more than 3,000 lbs. and producing a yield of one megaton occupied a considerable portion of the REDWING test schedule. There were in fact five shots devoted to this program. (Author's note: shots Erie, Seminole, Flathead, Dakota and Apache.)

Four shots were made investigating the TX-28 and one the XW-27. A feature associated with two of the larger devices, the BASSOON and the TX-21C, involved a "clean" design of thermonuclear weapons. In the "clean" weapon the major portion of its yield came from fusion and a reduction in fallout contamination was achieved. In addition, Operation REDWING incorporated development tests of device and continued the research of small devices suitable for use in air-to-air, surface-to-air, and surface-to-surface missiles.³²³

The 17 weapons and devices tested during REDWING performed essentially as predicted. The broad scientific and military achievements of the test series included further progress toward fulfilling requirements for new atomic warheads; in addition, progress was made toward development of new Class "A" and Class "B" weapons. Small-diameter two-stage devices were tested successfully, and further progress was made toward fulfilling requirements for IRBM and ICBM warheads. Three single-stage devices with comparatively cheap atomic warheads for small-diameter guided missiles and rockets were now available.

Five new weapons were now available for stockpiling, including one Class "C" weapon (the TX-15-X1), two Class "D" weapons (the MKs 27 and 28), and two boosted primaries which might also be used in small bombs or missile warheads (the SWIFT and the SWAN).

Design principles for three-stage systems were confirmed, and the "clean" weapon principle was proven.³²⁴ The high yields

³²³Draft Final Report by Commander, Joint Task Force SEVEN on Operation REDWING, 27 August 1956, p. 7.

³²⁴Memorandum for Mr. Strauss, Mr. Murray, Dr. Libby, Mr. Vance, and Mr. Fields dated October 23, 1956, from Brig. Gen. Alfred D. Starbird, USA, Director of Military Application,

obtained from weapons of greatly decreased size and weight, compared to the CASTLE devices, were significant in terms of projected stockpile weapons.

One of the major results of Operation REDWING was a near-reversal in the yield mechanisms of U.S. thermonuclear weapons: during Operation CASTLE in 1954, an average of 63% of the total yield of all six devices fired came from fission of plutonium and uranium. During REDWING, 59% of the total yield of shots over 1 MT came from fusion and only 41% from fission.³²⁵

This was a significant advance in weapons technology. It could mean that relatively-low yield tactical battlefield hydrogen bombs were militarily feasible, although none of the "clean" devices fired during REDWING could be characterized as a tactical weapon.

Thermonuclear weapons could now be tailored to be as "dirty" or "clean" as desired. "Clean" bombs used either very thin uranium-238 pushers, or pushers made of a nonfissionable heavy metal, around their secondary stages.

USAEC, Subject: Final Report by Commander, Joint Task Force Seven, on Operation REDWING; OPERATION REDWING, REPORT OF THE MANAGER ALBUQUERQUE OPERATIONS, PACIFIC PROVING GROUND, SPRING 1956, p. 23; ORIENTATION GUIDE, J-3 Division, Joint Task Force Seven, 1 June 1957, p. 18.

³²⁵"U.S. Shots 1945 - 1957 Inclusive, Pacific Tests - EPG," undated AEC document probably prepared late in November 1958. The REDWING tests that yielded over 1 MT were Cherokee (3.8 MT), Zuni (3.5 MT), Apache (1.9 MT), Dakota (1.1 MT), Navajo (4.5 MT), and Tewa (5 MT). This totals 19.8 MT; if only 8 MT of this came from fission (as NWD 86-2 claims), then 11.8 MT, or 59% of the total yield of these six shots came from fusion of hydrogen isotopes. More than 61% of this fusion yield (7.25 MT) came from the Zuni and Navajo shots alone ($7.25 = 0.85 \times 3.5 + 0.95 \times 4.5$). More than half (4.35 MT) of the 8 MT fission yield came from the Tewa shot alone.

This leaves 3.06 MT of fusion yield spread among the Cherokee, Apache, and Tewa shots; the total yield (fusion + fission) for these three devices was 10.3 MT. Tewa's fusion yield was only 0.65 MT (13% of its total yield); the combined fusion yield for Cherokee and Apache was then 2.41 MT. This means that the average fusion yield between the latter two shots was about 45% ($= 2.41/5.3$) of total yield.

In terms of total test series fission yield percentages, REDWING was comparable to CASTLE: 58% (12.1 MT) of REDWING's total 20.8 MT yield came from fission, compared to CASTLE's 63%.

The Cherokee airdrop proved conclusively that the B-52 was entirely suitable for the free-fall delivery of Class "C" thermonuclear weapons. An Associated Press wire story dated May 21 noted that "a standard Boeing B-52 bomber" with "no special changes made in the plane's bomb-bay design or the bomb-dropping device" made the Cherokee drop.³²⁶

In addition, besides demonstrating the capability of dropping an atomic weapon of megaton yield from a deployed bomber, REDWING better defined aircraft delivery capabilities and optimum airspeeds and release altitudes for bomb drops.

In general, the objectives of the military weapons effects program were attained. Data was obtained that better defined the atomic weapon delivery capabilities and limitations of late model Air Force and Navy aircraft for weapons in the yield ranges tested in REDWING, including safe delivery envelopes for B-47, B-52, B-57, B-66, F-84F, F-101, and A3D-1 type aircraft.³²⁷

Experiments to investigate fallout under a great variety of conditions were successfully carried out; and limited blast, thermal, and nuclear radiation effects data was obtained from the high-yield Cherokee air burst, due to the nearly four-mile miss distance.

Structural responses of military equipment to blast gust and thermal radiation and tests of naval washdown decontamination systems, protective aircraft coatings, and radiation shielding were successfully determined or accomplished. All the military services benefited from tests of radiation and fallout, blast, and heat with respect to ground installations, equipment, and personnel.

The capability of the task force to maintain simultaneous two-shot, two-atoll operation was proven conclusively. This capability would figure prominently in the next series of Pacific tests two years hence.

REDWING had also put Livermore "on the charts" with its first successful set of thermonuclear tests since the lab's establishment in September 1952. Because of a promise that Edward Teller had made to the Navy while REDWING was still in

³²⁶"No Special Changes Made In B-52 for H-Bomb Drop," Seattle Times, May 21, 1956. Boeing built B-52s in Seattle and in Wichita, Kansas.

³²⁷Minutes of the 102nd Meeting of the Special Weapons Development Board, SWDB-56-102-2, Sandia Corporation, Albuquerque, New Mexico, 29 August 1956, p. 12.

progress, UCRL was also about to embark upon its most ambitious undertaking to date: the creation of a lightweight, high-yield and compact warhead for the POLARIS submarine-launched ballistic missile. (See W-47 history in Volume VI.)

The total cost of the REDWING operation was nearly \$170,000,000, about \$10,000,000 per shot, exclusive of the cost of the nuclear materials expended and the expense of the design and production of test devices. Of the total figure, about \$120,000,000 covered the participation of civilian and military test organizations; the remainder represented experimental expenses, including about \$24,000,000 for expendable scientific construction.

One additional result of REDWING was consideration of the use of UCRL's three-stage BASSOON device to fulfill the DOD's new Class "A" thermonuclear bomb requirement. On June 24, 1955, the AEC had received a DOD request for a feasibility study of a new Class "A" weapon with a maximum weight of 25,000 lbs. and a 60 MT yield.³²⁸

About five months later, the Assistant Secretary of Defense for Research and Development indicated that the Joint Chiefs of Staff had promulgated a requirement for the development of a Class "A" weapon, which was accorded fourth priority among several projects then under study. Inauguration of a specific weapon development was not expected until after REDWING ended.

By late June 1956, DOD had issued a stockpile requirement memorandum for a number of the new Class "A" bombs by July 1, 1959. If the BASSOON concept were pursued with sufficient priority, it might be entirely possible to have a number of these weapons ready by that time. A major delay in the program might also be occasioned by an Air Force request that "laydown" (delayed surface burst) capability be included as a feature of the bomb.³²⁹

REDWING also made material contributions toward the weapons programs desired by the DOD in February 1956, including weapons in a 10" to 12" diameter range for missile applications, including the NIKE I, TERRIER, LITTLE JOHN, Follow-On DING DONG, LACROSSE, and ASROC (this small-diameter weapon program eventually became the W-45 project).

REDWING also assisted the development of a new Class "C" bomb

³²⁸Minutes of AEC Meeting No. 1221, August 2, 1956.

³²⁹Memorandum dated June 26, 1956 to K. E. Fields, General Manager, USAEC, from Brig. Gen. Alfred D. Starbird, USA, Director of Military Application, USAEC, Subject: New Class "A" Weapon.

to replace the TX-15, with a maximum weight of 6,400 lbs. and a diameter less than 50 inches.³³⁰ The new Class "B" bomb was expected to yield up to 10 megatons; potential warhead-version carriers included the intercontinental NAVAHO cruise missile.³³¹

As REDWING ended, the current status of the AEC's weapon development program was summarized in a report to the National Security Council:

Emphasis in weapon development will continue to be directed toward smaller, lighter weight, more immediately-ready weapons, with increased economy of utilization of nuclear materials. In general, the trend is away from interchangeability of nuclear components and toward weapons optimized for a single specific purpose, with the fissionable material incorporated as an inseparable part of the weapon.

Safety from accidental detonation has become an important design objective, particularly in view of the widespread deployment anticipated for such weapons as those devoted to air defense.

Weapons under development include improved bombs for strategic and tactical use, missile warheads, and artillery-fired atomic projectiles. Small-diameter weapons are being developed for air defense and tactical purposes.

Primary efforts in the thermonuclear field are toward reducing weight and increasing yield. Also included are programs directed toward the development of so-called "clean" weapons that produce less radioactive fallout. Such weapons would have a greatly reduced fission yield in proportion to their over-all yield.

Radioactive fallout from a weapon is a

³³⁰Minutes of the Fiftieth Meeting of the General Advisory Committee to the USAEC, July 16-18, 1956, p. 29.

³³¹"B. United States Weapon Program," USAEC, undated, but ca. June 18, 1956; Minutes of the Fiftieth Meeting of the General Advisory Committee to the USAEC, July 16-18, 1956, pp. 28, 43.

function of the fission yield. By the above-mentioned reduction in fission yield, a weapon of greatly reduced fallout can be secured.

Operation REDWING. The series of test detonations conducted at the Pacific Proving Grounds from May 4, 1956 to July 21, 1956, Washington time, has provided significant information relating to achievement of the foregoing design objectives.

Although test results have not yet been completely analyzed, preliminary conclusions indicate that encouraging progress has been made, particularly with respect to (1) proving the possibility of developing large-yield (clean) weapons having a greatly reduced fallout for the yield concerned (discussed in a preceding paragraph), (2) the use of tritium gas to increase the efficiency of fissionable material in a weapon, (3) proving the possibility of small-diameter implosion warheads, which have important applications in large multistage as well as small single-stage weapons, and (4) examining the feasibility of increasing the yield of weapons to be carried by our family of bombers to include that of securing a yield as high as 60 megatons, if attainable, in a weapon which could be carried by our heaviest bombers.³³²

Specific systems now under development at UCRL and LASL included warheads for an air-to-air rocket (the XW-25), and warheads for the TALOS, NIKE-B, and BOMARC anti-aircraft missiles. A small nuclear depth bomb, the LULU, was also under development by LASL and the Navy; the weapons labs and the Air Force were working on nuclear warheads for both intercontinental and intermediate-range ballistic missiles.³³³

³³²"Annual Report of the Atomic Energy Commission to the National Security Council, Fiscal Year 1956," USAEC, Washington, D.C., July 1956, pp. 6, 7, and a later edition of the same document dated August 13, 1956.

³³³"Report of the Atomic Energy Commission to the National Security Council, An Estimated Three-Year Projection (through Fiscal Year 1959) of the Status of the Atomic Energy Program," USAEC, Washington, D.C., July 1956, pp. 6, 7.

During fiscal year 1956, six new weapons -- the MKs 11, 19, 21, 23, 33, and 36 -- had entered the stockpile. Also during this period, stockpiles of "special nuclear materials" skyrocketed: production of plutonium, tritium, uranium-235, and lithium-6 were all at record levels.

Production of plutonium was 82% greater during FY 1956 than FY 1955; tritium production was approximately 50% greater than the amounts produced in either FY 1952 or FY 1954, the two earlier years in which production was at all significant. Separation of U-235 was 62% greater in FY 1956 than in FY 1955, and production of Li-6 for thermonuclear weapons was 546% greater in FY 1956 than in FY 1955.³³⁴

Late in July, AEC Commissioner Thomas E. Murray called for the National Security Council to develop a policy directive oriented toward the earliest possible integration of "clean" weapons into the U.S. stockpile and defense planning.³³⁵

Early in August, AEC Chairman Strauss, in the face of disparaging criticism of REDWING results -- especially regarding "clean" weapons -- and in an attempt to justify future nuclear testing, felt compelled to clarify his statement of July 19 regarding "clean" weapons:

On July 19, on behalf of the Atomic Energy Commission, I reported on the success of the 1956 nuclear test series then nearing completion at our Pacific Proving Grounds in the Marshall Islands.

The tests have since been concluded and more recent reports to the Commission have confirmed the results which were described in the July 19 statement.

It appears, however, that in some quarters there has been misunderstanding of the statement, particularly in regard to the progress made in producing nuclear weapons of maximum effect in the immediate area of the target with a minimum amount of

³³⁴"Annual Report of the Atomic Energy Commission to the National Security Council, Fiscal Year 1956," USAEC, Washington, D.C., July 1956, pp. 7, 8, and a later edition of the same document dated August 13, 1956.

³³⁵Memorandum for Lewis L. Strauss, Chairman, USAEC, from Thomas E. Murray, Commissioner, Subject: National Policy on Clean Weapons.

widespread fallout.

It was stated that the 1956 test series had confirmed that there are many factors including operational ones which do make it possible to localize, to an extent not heretofore appreciated, the effect of nuclear explosions, and that as a result of this achievement we are convinced that mass hazard from fallout is not a necessary complement to the use of large nuclear weapons.

The mention of "operational" factors led some to speculate that our success in minimizing heavy fallout has been due primarily to exploding the weapons under particular conditions rather than to any fundamental development in the weapons themselves. Others assumed that the area and amount of fallout was reduced simply by testing weapons of smaller size.

It is true that firing conditions such as the altitude of the burst and whether the fireball touches the surface of land or water contribute importantly to the reduction of widespread fallout. It is likewise true that smaller weapons, including those designed for tactical and defensive use, produce more limited fallout. These, however, are secondary factors in reducing the mass hazard from fallout associated with the use of large nuclear weapons.

The scientists in the Commission's weapons laboratories at Los Alamos and Livermore have for some time been actively engaged in pursuing the objective of reducing the fallout which would result from weapons of a given yield.

They have achieved real progress in this direction, as was demonstrated during the recent tests.

Weapons were tested which, because of the nature of their design and composition, have a yield of radioactive fission products which is only a small fraction of that produced in previously-tested weapons having about the same total energy release.

The Commission feels strongly that the people of the United States and other countries have a right to know the facts about fallout hazards to the fullest extent that such information can be prudently disclosed within the limitations of security. Such limitations forbid disclosure of further details of this important achievement in reducing fallout.

In the absence of an agreement on effective international safeguards which would enable us to cease the production and testing of our nuclear weapons of defense, the United States has taken the initiative in reducing to some extent the basis for worldwide fears associated with the fallout effect of large nuclear weapons beyond the military targets against which they would be used.³³⁶

General Starbird reviewed Strauss' statement, and offered the following comments:

As I mentioned at lunch Friday, I would see no security danger in the new (press) release. I still worry somewhat over the public's placing us in a position in the future where we can test no weapons except clean ones. As you know, certain critical designs for the (near) future cannot be clean and still give the desired yield.

They should be tested. On balance and after further thought, however, I believe the clarifying statement probably would not build up substantially more pressure in this respect. In other words, I am changing my opinion to that of seeing no serious objection to the (press) release.³³⁷

As REDWING ended, the AEC outlined to the MLC its plans for future weapons development:

³³⁶Draft Statement dated August 3, 1956 by Lewis L. Strauss, Chairman, U.S. Atomic Energy Commission.

³³⁷Memorandum for Mr. Strauss dated 6 August 1956 from Brig. Gen. Alfred D. Starbird, Director of Military Application, USAEC, Subject: Draft Press Release in Clarification of the Matter of Clean Weapons.

Based on our analysis of results of REDWING tests conducted jointly with our field agencies, we have established a possible nuclear weapons development program for weapons with operational availability dates (OAD's) subsequent to October 1958. This development program is predicated upon the current status of the nuclear weapons art and the tests scheduled for the Nevada Test Site (NTS) in 1957 and at the Eniwetok Proving ground (EPG) in 1958.

In tabular form this program may be outlined as follows:

<u>Weapon</u>	<u>Estimated OAD*</u>	<u>Responsible Nuclear Laboratory</u>
ICBM/IRBM warhead (XW-35)	Dec 1958	LASL
XW-40	Jan 1959	LASL
New Class B weapon	Feb 1959	UCRL
10-12" warhead/ASROC	Apr 1959	LASL
New Class A weapon	Jun 1959	UCRL
New Class C weapon	Aug 1959	LASL
10-12" warhead/NIKE I, Air-to-Air missile, TERRIER, LITTLE JOHN, Project C ADM	Oct 1959	UCRL
Laydown weapon	Dec 1959 to Feb 1960	LASL, UCRL
Improved ICBM/IRBM warhead (XW-38)	Apr 1960	UCRL
8" Artillery Fired Atomic Projectile	Jun 1960	UCRL

*Estimated OAD represents the best available estimates as of September 1956 based on current status of research and development and on (the extent of) full-scale testing required. Some adjustments in programs may be possible based on further DOD views regarding program priorities.

We believe that the foregoing tabulation represents the best utilization of our laboratories in fulfilling the requirements of the DOD. It must be emphasized that the OAD's shown are those which appear to be achievable at this time. Naturally, these OAD's will be subject to change should possible breakthroughs occur in the state of the art, should unforeseeable difficulties be encountered in development, or if changes in design criteria are needed.

We would expect the nuclear tests contemplated for 1957 and 1958 to assist in crystallizing or removing these uncertainties.³³⁸

The Operational Availability Date (OAD) of a nuclear weapon was the date by which a given number of weapons had been produced, any deficiencies corrected, and the weapon was being routinely stockpiled at a given rate.

These assignments represented an equitable distribution of work between the two national weapons laboratories, and also showed the increased scope of post-REDWING projects allocated to UCRL.

With the current post-REDWING program, it was likely that at least nine major AEC weapon programs would reach their first production unit stage (Phase 5) in 1957 and 1958.³³⁹

The new Class "A" weapon, assigned to UCRL and to be stockpiled by mid-1959, bore a close resemblance to the new Class "B" weapon program also assigned to UCRL (which became the TX-41 project). The development of the Class "A" warhead

³³⁸Memorandum for Chairman, Military Liaison Committee dated September 11, 1956 from K. E. Fields, General Manager, USAEC, Subject: AEC Post-REDWING Weapon Development Program.

³³⁹Letter dated September 12, 1956 to Frank C. DiLuzio, Albuquerque Operations Office, USAEC, from Sandia Division 2521, Re: Final Weapon Assembly at ALO Prime Contractor Plants in 1957 and Beyond.

was at least partially dependent upon a test shot currently planned for Operation HARDTACK in the Pacific during the spring of 1958.

The objective of the program was a 25,000 lb. total weight bomb; while the highest attainable yield would not be known until after HARDTACK, it was believed that 60 MT could be achieved. For the next four to six months, UCRL planned to study optimum designs for weapons in this weight class.

Among specific factors to be examined would be both two- and three-stage thermonuclear systems and possible characteristics of a "clean" version. It appeared that firm military requirements, including guidance on whether a conventional or "clean" version was desired, would not be necessary until January 1957.³⁴⁰

The development of this extremely-high yield bomb seemed to be at odds with a statement made by President Eisenhower two years earlier on April 1, 1954 after the ill-fated CASTLE Bravo shot:

President Eisenhower said today that the United States has no intention of building a bigger hydrogen bomb than those now in production. At his press conference this morning, the President declared that he knew of no military reason for building bigger ones.³⁴¹

In the late summer of 1956, the AEC believed it was possible to develop "clean" weapons in all present classes of thermonuclear weaponry, i.e., Classes "A" (25,000 lbs. gross weight), "B" (10,000 lbs.), and "C" (6,400 lbs.). If the gross weight of the "clean" Class "B" weapon were allowed to rise to 17,000 lbs., the weight of the "conventional" currently-stockpiled MK 21 bomb, then a higher "clean" yield could be obtained.

In any case, "clean" Class "A," "B", and "C" weapons would not be just simple modifications of the conventional versions and would instead require extensive redesign to achieve

³⁴⁰Memorandum for Chairman, Military Liaison Committee dated September 11, 1956 from K. E. Fields, General Manager, USAEC, Subject: AEC Post-REDWING Weapon Development Program.

³⁴¹Quotation from New York Times newspaper of April 7, 1954 cited in letter dated November 3, 1956 to Hunter H. Morrison, Robson Construction Company, from Senator Clinton P. Anderson, Chairman, JCAE.

reasonably-high yields.³⁴²

By mid-August, at least three different concepts of Operation WILLOW -- the next Pacific test series following Operation HARDTACK -- had been prepared. Concept number two covered Johnston Island operations; concept number three covered open sea testing; and concept number four included an EPG "Quick Proof" test series.³⁴³

An AEC Operations and Analysis Office study in September 1956 examined the advantages and disadvantages of "clean" weapons when used in theoretical attacks against Soviet targets:

There are several reasons why the United States may wish to include clean weapons in its thermonuclear stockpile:

1. To reduce contamination hazards to our own or friendly troops located near enemy targets during bombing or moving into such target areas after bombing;
2. To reduce the effects of close-in fallout on civilians of friendly or neutral countries occupied by the enemy or located near the borders and targets of enemy countries;
3. To reduce the effects of close-in fallout on enemy civilians;
4. To alleviate postwar occupational difficulties in an enemy country resulting from long-lived radioactive products; and
5. To reduce the effect on the world of long-range fallout

³⁴²Memorandum for Chairman, Military Liaison Committee dated September 11, 1956 from K. E. Fields, General Manager, USAEC, Subject: AEC Post-REDWING Weapon Development Program.

³⁴³Administrative Plan for Operation WILLOW Concepts #2, 3, and 4, Headquarters, Joint Task Force Seven, Arlington Hall Station, Arlington, Virginia, undated, but all after August 18, 1956.

resulting from detonation of our own thermonuclear stockpile on enemy countries.

Items 1 through 4 above could be accomplished to a large extent by detonating current types of weapons at an altitude sufficiently high to keep the ball of fire from touching the ground. However, numerous targets are more effectively damaged by cratering or increased blast pressure resulting from ground bursts.

Furthermore, airbursts contribute more to the global fallout of long-lived fission products than do ground bursts and therefore do not accord with item 5 above.

It may be noted in this regard that the long-term contamination of the United States would be more than 50 times greater for each megaton of fission from ground-burst weapons used against the United States than from such weapons used against the Soviet Union. (Author's note: probably because of the inherently "dirtier" nature of Soviet thermonuclear weapons compared to U.S. weapons.)

The clean TX-21C device tested in Operation REDWING yielded only about 1/2 that of the standard, 60% fission 10 MT weapon. (Author's note: a reference to the stockpiled MK 36 bomb.) Estimates have been made that, for future designs of weapons weighing from 10,000 to 25,000 pounds, the yield from a clean weapon may be about one-third to one-half the yield of a dirty weapon of the same size and weight.

The magnitude of this loss in yield, together with the loss of radiological contamination in the target area, (represents) an important (loss of) military effectiveness. Whether the possible advantages of clean weapons in the situations cited above outweigh the decreased effectiveness depends on the type and location of the target, the size and weight of the weapon, and our national objectives and constraints.

Consideration of all these factors will not

be attempted here, but a quantitative comparison of the effectiveness of clean and dirty weapons against certain important targets will be presented.

[Table deleted]

The table compares the effectiveness of dirty and clean weapons in single-weapon attacks, using the criteria:

1. Probability of damaging a strategic airfield beyond repair;
2. Percentage of Moscow industry destroyed; and
3. Areas of radiological contamination.

The denial to the enemy of the use of airbases from which strategic bombing attacks can be launched is a military objective of highest priority. In the view of the military, the only satisfactory way of assuring that the enemy will not be able to use an airfield is to crater the runways to the extent that the maximum length of any uncratered section is less than the minimum takeoff distance for medium and heavy bombers (about 5,000 feet).

The probability of airfield destruction given in the table are based on a runway length of 8,000 feet and a bombing accuracy such that there is a 50 percent chance of a bomb falling within one-half nautical mile of the center of the runway. (Author's note: one nautical mile = 2,025 yards, or 1.15 statute miles.) Such accuracy is considered to be achievable in radar bombing in combat by the best United States bombardiers. Even with this accuracy, only the [deleted]

The study also noted, very presciently, that yield losses would be even greater as "clean" weapons decreased in size:

For thermonuclear weapons of lower weight than those considered in the table, the losses in yield for clean versions may be even more significant; in some (sizes), clean weapons may not be possible at all.

This situation may well apply to weapons for delivery by tactical aircraft.³⁴⁴

Operation HARDTACK (Phase I)

The next Pacific nuclear test series occurred two years after REDWING, following the biannual cycle established by IVY, CASTLE, and REDWING.

Objectives and Planning

Planning for Operation HARDTACK began as early as 1952. HARDTACK was to be one of a number of Pacific test series after IVY to develop thermonuclear weapons. It was to follow REDWING, and was originally scheduled for the fall of 1955.³⁴⁵

By June 1954, HARDTACK was estimated to include five tests, three by LASL, including one ground shot plus two barge shots, and two UCRL ground shots in the Pacific in the spring of 1957.³⁴⁶ A year and a half later, in late December 1955, HARDTACK was slated for the Pacific in the spring of 1958, following Operation PILGRIM, a Nevada test series then scheduled for the spring of 1957.³⁴⁷

Development of "clean" thermonuclear warheads, specifically, "clean" high-yield warheads, continued after REDWING. In addition, the AEC wanted to conduct several special weapon effects tests that had been proposed before REDWING but postponed for a number of reasons. HARDTACK would afford more full-scale testing opportunities.

³⁴⁴"Operations Analysis and Planning - August 1956," FPRS #1711, September 20, 1956, attachment to letter to Dr. James W. McRae, President, Sandia Corporation, from Paul C. Fine, Acting Director, Office of Operations Analysis and Planning, USAEC. Emphasis in original.

³⁴⁵Atomic Energy Commission, Memorandum for Information, Assignment of Code Names, Note by the Secretary, USAEC, Washington, D. C., December 31, 1952.

³⁴⁶"Schedule of Full-Scale Weapons Tests, FY 1954 - FY 1956," Annex C to "FY 1956 Program Assumptions, Weapons Program, Operations and Construction (3000 and 9300)," Sandia Corporation, March 3, 1954, revised June 14, 1954.

³⁴⁷Memorandum dated 30 December 1955 to Scientific Department Division and Branch Heads from Eugene P. Cooper, Associate Scientific Director, U.S. Naval Radiological Defense Laboratory, San Francisco, Subject: Information and Guidance on Planning for Field Operations 20 and 22.

Among these tests was a very-high altitude weapons effects shot; such a test, using a CORPORAL missile and a W-7 warhead detonated at 100,000 feet, was first planned for REDWING but deferred because of its complexity and the short time then available for its preparation.³⁴⁸

Plans for such a test had been underway for several months before REDWING started. As early as October 7, 1955, Professor E. O. Lawrence of UCRL expressed concern over the possibility that the Russians might be able to detonate nuclear devices at very high altitudes, thereby defeating U.S. atmospheric test detection systems, and gaining a great advantage in their knowledge of the effects and military usefulness of such explosions as compared to what little mostly-theoretical data the U.S. then possessed on these types of blasts.

Dr. Lawrence urged the AEC to undertake a direct experimental investigation of high-altitude bursts by detonating one or more high yield weapons at very high altitudes. In November, the AEC queried the DOD as to which currently-deployed missiles might be appropriate for the tests.

By mid-January 1956, Lawrence was proposing that a 50 KT burst at about 100,000 feet would prove that megaton-range shots could be fired at even higher altitudes. The Air Force supported this proposal, believing it useful to their program for detection of high-altitude clandestine nuclear tests.³⁴⁹

DOD stated on April 16, 1956 that the CORPORAL II short-range surface-to-surface rocket could carry a W-7 warhead to an altitude of 100,000 feet and detonate it there within a reasonable "window" of arrival times and locations.

By 1958, the time of HARDTACK, both the REDSTONE intermediate range ballistic missile and the NIKE-B anti-aircraft missile would be available; the REDSTONE could carry a 6,500 lb. warhead as high as 600,000 feet while the NIKE-B could carry only a much smaller warhead to a maximum altitude of 100,000 feet.

In early May 1956, Sandia believed that it would be possible to adapt a W-7 to the CORPORAL II and to detonate it with a reasonable certainty at 100,000 feet. Among other modifications, the automatic in-flight insertion equipment would be set to operate only after the rocket had passed

³⁴⁸Furman, pp. 502, 503; Minutes of Executive Session of AEC Meeting No. 1203, May 29, 1956.

³⁴⁹Memorandum for the Chairman, USAEC dated January 19, 1956 from McKay Donkin.

through an altitude of 35,000 feet on the way up, and to extract the nuclear capsule if the rocket passed through this altitude on its way down after warhead detonation failure.

However, these arrangements could not be completed in time for REDWING, particularly in regard to preparation of appropriate aerial instrumentation to measure and record effects.³⁵⁰

In order to prepare for a very-high altitude test during the next scheduled Pacific test series, Operation HARDTACK, the Air Force's Cambridge Research Center was designated in June 1956 to provide administrative, technical, and logistical support to the test.³⁵¹

A high-altitude test was necessary for three reasons. First, the AEC was very interested in determining whether or not the Soviets could fire a large-yield shot at such an altitude that detection of the blast at great distances from air zero would be very difficult or impossible.³⁵² In this context, the AEC thought that a high altitude test would also show the practicability or impracticability of this method of nuclear testing.

Second, a high altitude test, or a series of high altitude shots, would provide data that would materially assist the design of an anti-ICBM warhead and help to devise protective features for U.S. ICBM and IRBM warheads to make them less vulnerable to hostile countermeasures.

Third, the phenomena associated with a very-high altitude test shot might be quite different from predictions based on extrapolation of phenomena observed at lower altitudes.

An alternate delivery system for a high-altitude shot might be a very large balloon, such as those used for weather data collection and upper atmosphere studies. In early August 1956, DOD requested Sandia Corporation to submit a project proposal for the use of a 600 lb. payload for a balloon to be

³⁵⁰Atomic Energy Commission, Proposed High Altitude Test, Report to the General Manager by the Director of Military Application, June 1, 1956; Minutes of Executive Session of AEC Meeting No. 1205, June 1, 1956; teletype dated June 6, 1956 to Adm. Hanlon, et. al. from Gen. Starbird.

³⁵¹Memorandum dated 20 June 1956 to Chief of Staff, USAF, from Maj. Gen. A. R. Luedecke, USAF, subject: Very High Altitude (VHA) DOD Effects Test (Program VII, HARDTACK).

³⁵²Minutes of the Fiftieth Meeting of the General Advisory Committee to the USAEC, July 16-18, 1956, p. 45.

used in a 100,000 foot nuclear detonation. The payload would include 250 lbs. of nuclear device and its container, fuzing, firing, and arm/safe systems.³⁵³

Another purpose of HARDTACK was to conduct nuclear explosions either too powerful or too "dirty" (causing extensive fallout) for Operation PILGRIM at the Nevada Test Site in the middle of 1957. PILGRIM would complement HARDTACK by testing several of the primaries, or first-stage fission triggers, for the thermonuclear devices and weapons to be fired during HARDTACK in 1958.

A number of underwater detonations planned for HARDTACK were to help develop radiological information on water bursts at varying depths. Up to this time, only the WIGWAM and CROSSROADS Baker shots had been fired underwater, and much remained to be learned about nuclear explosion phenomenology in this media.³⁵⁴

In addition, both shallow and deep underwater bursts could help to determine the minimum safe stand-off distances of surface ships for delivery of nuclear anti-submarine weapons; obtain information to design ships and submarines to withstand nuclear explosions; and determine the kill and damage ranges for surface ships and submarines.³⁵⁵

By mid-September 1956, plans for PILGRIM comprised 17 LASL and UCRL tests; with an operational starting date of April 1, 1957, twelve of the experiments could be concluded by June 30 with the remainder being fired by August 15.

A new "twist" in PILGRIM plans, adapted from British nuclear testing methods, was the use of 42-to-60 foot diameter tethered helium-filled balloons to position devices instead of the normal 200, 300, or 500 foot shot towers. Balloons, while susceptible to winds, offered a chance to detonate high-yield devices at higher-than-normal air zeros, perhaps as high as 1,500 or 2,000 feet.³⁵⁶ For a while, robot "drone"

³⁵³Letter dated 9 August 1956 to Sandia Corporation from Lt. Col. H. O. Beeth, Adjutant General, AFSWP.

³⁵⁴Memorandum dated 6 September 1956 to Chief, Bureau of Ships, USN, from Commanding Officer and Director, U.S. Naval Radiological Defense Laboratory.

³⁵⁵Memorandum dated 31 July 1956 to Chief of Naval Operations from Chief, Bureau of Ships, U.S. Navy, subject: Request for establishment of ad-hoc committee to determine the seamanship requirements for Operation HARDTACK.

³⁵⁶Letter dated November 14, 1955 to James W. McRae,

helicopters that could lift 100 KT test devices to an altitude of 1,500 feet were also considered as test device platforms.³⁵⁷

At an AEC meeting on September 25, Commissioner Willard F. Libby expressed his opinion that "clean" weapons could be produced at less than the 70% cost in yield (compared to standard weapons) then believed necessary for reasonable "cleanliness." He indicated that future development of these weapons would increase their yield without increasing fallout and said that he believed it necessary to continue weapons tests in order to develop more powerful "clean" warheads.

In response to a question from Chairman Lewis Strauss during the meeting as to whether or not two "clean" weapons could be used together to approximate the total yield and damage from one "normal" weapon, Dr. Paul Fine stated that the combined fallout resulting from the detonation of even a pair of "clean" bombs would still be less than that resulting from the explosion of a single high-yield "normal" thermonuclear bomb. However, Fine also noted that more nuclear material and more delivery vehicles would be required for two bombs as opposed to one.³⁵⁸

Also at this time, the code names DOMINO, DIXIE, and JULEP, which were assigned to test operations that never got beyond

President, Sandia Corporation, from K. F. Hertford, General Manager, USAEC, Subject: Study of Balloon Suspensions for Test Devices; letter dated December 12, 1955 to K. F. Hertford, General Manager, USAEC, from R. C. Bice, Director of Field Testing, Sandia Corporation, Subject: Feasibility Study of Balloon Suspensions for Test Devices; letter dated December 16, 1955 to Dr. Alvin C. Graves, Leader, J-Division, LASL, from James E. Reeves, Director, Test Division, USAEC, Subject: Study of Balloon Suspensions for Test Devices; teletype dated September 18, 1956 to Brig. Gen. A. D. Starbird, USA, Director of Military Application, USAEC, from K. F. Hertford, USAEC.

³⁵⁷Memorandum dated August 9, 1956 to James W. McRae, President, Sandia Corporation, from K. F. Hertford, Manager, Albuquerque Operations, USAEC, Subject: Feasibility Study of Robot Helicopters for Test Devices; memorandum dated October 11, 1956 to K. F. Hertford, Manager, Albuquerque Operations, USAEC, from James W. McRae, President, Sandia Corporation, Subject: Feasibility Study of Robot Helicopters for Test Devices; Memorandum dated March 20, 1957 to Members of the Planning Board for Nevada Weapons Tests from Alvin Graves, Subject: Minutes of Meeting of the Nevada Planning Board held on 24 January 1957.

³⁵⁸Minutes of AEC Meeting No. 1228, September 25, 1956.

the planning stage, were withdrawn pending future re-assignment. PILGRIM was the name now assigned to the forthcoming spring 1957 Nevada test series, while HARDTACK was to be the next Pacific test series a year after PILGRIM.

For long-range planning purposes, the names TRUMPET and ROUNDUP, in that order, were assigned as the next full-scale nuclear test series in Nevada after PILGRIM, and the code word WILLOW was assigned to the next Pacific series after HARDTACK. No dates were presently designated for either WILLOW, ROUNDUP, or TRUMPET; these three code words had been reserved for nuclear tests in 1951 and 1952.³⁵⁹

Late in September, the AEC believed that "clean" thermonuclear bombs, yielding five to six megatons, could be produced within 18 months. Even in the absence of any stated DOD requirements for such weapons, and although the AEC had not completed post-REDWING data reduction (analysis), the AEC was nonetheless interested in stockpiling "clean" weapons as soon as possible. The Commission also believed that since REDWING had proven the feasibility of "clean" bombs, then future reduced-fallout weapons could be designed with less yield loss.³⁶⁰

By the end of the month, the AEC was contemplating the possibility of developing high-yield "clean" weapons that would meet the size and weight requirements of the new Class "A," "B," and "C" weapons. The AEC had requested specific guidance from DOD on whether conventional or "clean" versions were required.

However, because "clean" warheads would be more than simple modifications of existing "standard" or "dirty" versions, both versions would not be ready for stockpiling at the same date. The AEC planned to take four to six months to study the question of optimum designs for new Class "A," "B," and "C" weapons.³⁶¹

Two weeks later, early in October, LASL had been directed to proceed with the development of a "clean" version of the TX-21 with an operational availability date of April 1958. Studies of data obtained during REDWING also indicated that

³⁵⁹Letter dated September 25, 1956 to Honorable Herbert B. Loper, Chairman, MLC, from K. E. Fields, General Manager, USAEC, Subject: Assignment of Code Names.

³⁶⁰Minutes of the 106th AEC-MLC Conference, September 27, 1956.

³⁶¹Notes for the 106th AEC-MLC Conference, Memorandum by the Director of Military Application, September 26, 1956.

"clean" Class "A," "B," and "C" weapons were feasible:

This letter replies to your letter of March 5, 1955 wherein you indicated that the Department of Defense believes that the problems associated with the use of high-yield weapons are of such an order as to justify continued vigorous studies on all aspects of these problems, to include further exploration of methods of both increasing and decreasing the radioactive yield of thermonuclear weapons.

This letter will deal only with the problem of decreasing the radioactive yield; increasing the radioactive yield will be treated by separate correspondence.

During the recently completed Operation REDWING, we conducted field experiments which developed some information on the feasibility of thermonuclear weapons in which the fission yield constitutes a relatively minor percentage of the total yield and which, consequently, possess the virtue of producing greatly less fission product fallout than would a conventional thermonuclear weapon of comparable yield detonated under similar conditions. Such a weapon has been called a "clean" weapon. Possible programs to produce "clean" weapons for stockpile are discussed below.

The currently stockpiled TX-21 (SHRIMP) which weighs 17,500 pounds and has a yield [deleted] can be modified to produce a "clean" bomb, unchanged in weight and external shape, which would produce a yield [deleted] A comparison of the yields of these two versions of the TX-21 bomb is indicated in the following table:

<u>Version</u>	<u>Total Yield</u>	<u>Fission Yield</u>
Conventional	[deleted]	[deleted]
Clean	[deleted]	[deleted]

In view of various expressions of interest in clean weapons and pending receipt of a Department of Defense requirement for development of a "clean" weapon, the Los Alamos Scientific Laboratory has been

requested to proceed with the development of a "clean" version of the [deleted] with an operational availability estimated for April 1958.

Study of data obtained during the REDWING operation also indicates that we could develop "clean" bombs with total weights corresponding to those prescribed for the new Class A, B, and C weapons. A tabular comparison of our estimates for "clean" and conventional versions of the new Class A, B, and C weapons follows:

(The table listed a new Class "A" weapon at 25,000 lbs.; a new Class "B" weapon at 100,000 lbs.; and a new Class "C" weapon at 6,400 lbs. The total yield and total fission yield for each device, in both "clean" and conventional configurations were all deleted.)

It also appeared possible to build a 17,000 lb. high-yield "clean" bomb, the weight of the currently-stockpiled standard Class "B" bomb, the MK 36. "Clean" Class "B," "A," and "C" bombs might be operationally available in February, June, and August 1959, respectively, but only if DOD gave priority to the development and stockpiling of "clean" bombs over standard weapons. Both "clean" and conventional versions of these could not be developed concurrently within this schedule.

The AEC advised DOD of its estimated capabilities in "clean" weapons; specifically, the latest estimates of the total yields and fission yields of "clean" versions of the new Class "A," "B," and "C" bombs. AEC also requested guidance from DOD as to the numbers of different types of "clean" bombs required for the stockpile. A decision as to which type of weapon to develop in each class would have to be made by the end of 1956.³⁶²

Plans for "Clean" Weapon and High-Altitude Tests

Plans for a possible ultra-high altitude test during HARDTACK

³⁶²Letter dated October 10, 1956 to Honorable Charles E. Wilson, Secretary of Defense, from W. F. Libby, Acting Chairman, USAEC; Memorandum for the Chairman through the General Manager dated January 22, 1957 from Brig. Gen. Alfred D. Starbird, Director Of Military Application, USAEC, Subject: Clean Weapons.

were further advanced in October when the high-altitude payload capabilities of a number of potential delivery vehicles were examined.

Among candidate systems were the Lockheed X-17 re-entry test vehicle (eventually to be used during Operation ARGUS in 1958), and the Army's CORPORAL II, SERGEANT, NIKE-B, and REDSTONE missiles. The Army believed that modifications of the X-17 required to make it compatible with likely test warheads would involve design complications and costs which could be avoided by use of an operational Army missile system. Use of such a missile would also provide an opportunity to train troops.

By 1958, the Army would have the following payload-versus-altitude capabilities:

<u>System</u>	<u>Payload & Altitude Capacity</u>
CORPORAL II	750 lbs. up to 340,000 feet 1,500 " " " 254,000 " 2,800 " " " 218,000 "
SERGEANT	Comparable to CORPORAL II
NIKE-B	1,120 lbs. up to 300,000+ feet
REDSTONE	500 lbs. up to 1,740,000 feet 1,500 " " " 1,480,000 " 2,800 " " " 1,240,000 " 6,400 " " " 820,000 "

For the CORPORAL, the 1,500 and 2,800 lb. weights corresponded to those for the W-7 and W-27 warheads, respectively. For the REDSTONE, the 1,500 lb. weight corresponded to that of the projected W-35 warhead; the 2,800 lb. weight corresponded to the W-27; and the 6,400 lb. weight to the W-15, and W-39 warheads.

The Army's JUPITER-C missile was also a possible candidate; the JUPITER-C, then being used to test composite ablative re-entry vehicles, could reach an altitude of 500 miles.

Based on these comparative capabilities, the Army recommended that the REDSTONE be selected as the high-altitude delivery vehicle for HARDTACK tests.³⁶³

³⁶³Letter dated October 15, 1956 to Chief, Armed Forces Special Weapons Project, from Major General John P. Daley, USA, Director of Special Weapons, subject: High Altitude Test.

In the fall of 1956, the AEC summarized the military and long-term postwar advantages of "clean" weapons, in particular, the "clean" 17,000 lb. MK 21 tested during the REDWING Navaho shot:

The recent tests of clean versions of thermonuclear weapons have shown very large reduction in fission contribution and in resultant contamination effects. Situations are possible where political considerations could bring about restraint on the use of conventional (nuclear) weapons to achieve important military objectives but where the availability of clean versions could make operations feasible.

These weapons may be useful in individual situations where a large destructive capacity is to be coupled with minimum contamination of a local population; or, in a larger sense, they may provide a method of achieving a large total yield from many weapons without world-wide contamination affecting the originating country as well as other distant populations.

It seems likely that study of the differing effects will lead eventually to a requirement for clean weapons in some quantity, and it is clear that the earliest possible establishment of such requirements is desirable to make possible provision of the weapons within a reasonable time.

The sacrifice in blast or crater-producing effectiveness would be much less in the clean weapons than the decrease in contamination. The tabulation and chart show a comparison of the destructive and contamination possibilities of the two weapon types for simple conversion of the MK 21 17,000 pound weapon:

	<u>Dirty</u>	<u>Clean</u>
Total Yield (Megatons)	[deleted]	
Fission Yield (Megatons)	[deleted]	
Crater diameter	[deleted]	
Area contaminated to a level of 500	4000	55*

Roentgens/hr one
hour after blast
(square miles)

Downwind extent of 500 Roentgens/hr one hour after blast (mi.)	200	22*
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Crosswind extent of 500 Roentgens/hr one hour after blast (mi.)	30	5*
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*Estimates made by extrapolation of available measurements on weapons of the same yield; areas for clean weapons may be substantially smaller than those estimated here.

Attack with a dirty weapon anywhere near an area populated by friendly or neutral civilians would obviously entail a probability of fatalities among them in great numbers. The clean weapon maintains a substantial capacity for blast and crater production, but at the same time, the contamination is very drastically reduced. The utility of such a weapon in special and limited situations is worthy of most serious consideration.

In addition to development of the clean weapon for use where contamination is to be kept very low, a greatly reduced fission contribution opens the possibility of [deleted] The potential effects of these developments have not been thoroughly evaluated but seem to merit serious study in planning for maximum utilization of nuclear weapons.

A perhaps more serious limit on the use of large numbers of conventional high-yield nuclear weapons can arise from the possibility of world-wide fallout continuing for many years at levels that can affect the health of the entire world population. To the degree that weapons with reduced fission contributions can be used, this contamination may be markedly reduced, and a substantial increase may be realized in the total yield available within tolerable limits of fallout.

...

An approximate comparison of long-range strontium-90 contamination from clean and dirty MK 21 weapons shows the following differences:

[deleted]

...

The comparison has been made for levels of contamination that will avoid widespread distress, assuming that effects over many years on children and the general population do not present a much less favorable picture than that for industrial workers.

This comparison at levels of quite small world-wide risk shows clearly the contrast between the effects of present weapons, where any substantial increase over the totals indicated would move them into the area of technical uncertainty and political controversy, and the effects of clean weapons, where the limits of freedom from long-range strontium-90 concern are so high that world-wide effects from this material would come to represent a rather minor result of nuclear war.³⁶⁴

One of the chief objectives of HARDTACK was to further perfect "clean" weapons.

Late in October, Operation PILGRIM was to encompass two DOD effects tests of 40 and 1.8 KT; four to seven LASL tests, ranging in yield from two to 35 KT; seven UCRL tests between 0.5 and 80 kilotons; and a special Sandia Corporation test on a 500-foot column. Towers ranging from 300 to 500 feet in height and two sizes of balloons were being considered as test platforms: 67-foot diameter balloons that could lift 2,000 lbs. to 1,500 feet, and 75-foot diameter envelopes capable of lifting 4,000 to 5,000 lbs.

The next Nevada series after PILGRIM might be held in the fall of 1957, as the labs prepared for HARDTACK. In this proposed Nevada series, LASL planned four balloon shots between one and 10 KT, while UCRL expected to conduct five tower or balloon shots and one underground test; yields would range from 10 to 40 KT.³⁶⁵

At the 51st meeting of the GAC in Washington, D. C. on October 31, Dr. Norris Bradbury presented a list of the next ten U.S. nuclear warhead types expected to enter stockpile between January 1957 and January 1959. Of these, UCRL had responsibility for only the TX-27 gravity bomb and warhead.

³⁶⁴Undated paper entitled "Clean Weapon Effects."

³⁶⁵Minutes of the first meeting of the Planning Board for Nevada Weapons Tests, October 26, 1956.

At the moment, there were also stated or written development requirements for seven or eight other systems in different sizes and yields; LASL and UCRL had divided these projects evenly between them. Those weapons slated for stockpiling would require tests in Nevada during PILGRIM.

REDWING tests of the TX-28 series had established the feasibility of a megaton-yield weapon with a diameter on the order of 20" and a total weight between 1,800 and 2,000 lbs. The first warhead for an American ICBM, the W-49, would be a modification of the TX-28C tested during REDWING. Development of an ICBM warhead would also be a prime objective of PILGRIM.³⁶⁶

Among UCRL's longer-range development projects were the XW-38, a heavy, high-yield follow-on ICBM warhead slated for stockpiling in 1960, and an improved 8" artillery-fired atomic projectile based on the linear-implosion principles proven at Livermore. By 1958, UCRL expected to be involved in no fewer than 55 weapons projects, including major modifications to existing weapons, applications of warheads to various guided missiles, new development programs, and related materials and high explosives studies.

UCRL planned a number of PILGRIM shots, including three or four tests of 10" to 12" diameter systems, and two or three case test studies (weapon mockup tests) for a small-diameter two-stage system. PILGRIM would also be used by the Livermore scientists to more fully explore a possible design principle for the new Class "A" and "B" thermonuclear bombs. In addition, a deep underground shot would be conducted to investigate this test method.

By the fall of 1956, UCRL employed about 2,000 people at the Livermore lab; this number was expected to rise to 3,000 by January 1959 as the expected weapons workload increased. Sandia had also established a laboratory in Livermore; by the end of 1956, it would employ 120 people; this figure was expected to rise to 1,000 or 1,500 by the spring of 1958.³⁶⁷

In the middle of October, the AEC was circulating a draft statement on reasons for continued nuclear testing; a major argument was the necessity for testing of "clean" weapons:

Considerable progress has been made already
on a program to develop high yield weapons

³⁶⁶Minutes of the Fifty-First Meeting of the General Advisory Committee to the USAEC, October 29-31, 1956, pp. 21-22.

³⁶⁷Minutes of the Fifty-First Meeting of the General Advisory Committee to the USAEC, October 29-31, 1956, pp. 23-24.

which minimize fallout so that their use in atomic war would not constitute a world-wide fallout hazard. During the spring and summer at the Eniwetok Proving Grounds in the Pacific, the initial models of a family of megaton (yield) bombs essentially free of fallout have been proof-tested with favorable results.

We must continue this development so that our defensive armament can be completed and weapons testing in the megaton range is vitally necessary to this task. For this reason alone it is completely clear that an H-bomb testing ban short of a general disarmament agreement is not in the interest of the defense of the Western World (sic).³⁶⁸

This statement was somewhat less than accurate in its characterization of the multimegaton TX-21C and the BASSOON devices as "defensive" weapons, and even less truthful in its claim that these devices were "essentially free of fallout," considering the large amounts of radioactive material their tests dumped into the Pacific during REDWING.

A few days later, Senator Clinton P. Anderson, then chairman of the JCAE, disputed this AEC statement and noted that testing should continue only to help develop small nuclear weapons:

The real strength of our stockpile is based on H-bombs already built and flawlessly maintained. ... We have large stocks of H-bombs. We need not test further to learn how to make them bigger or to improve their yield.

We know how to make them much larger. What we need to learn is how to make them smaller and still retain a reasonable amount of intensity. I favor research and testing to achieve that, but I believe that to fire off more multimegaton hydrogen bombs teaches us nothing, loses us friends, and might fill the atmosphere with dangerous radioactivity which will imperil the world for generations to come.

....

³⁶⁸"Draft Proposed Statement on H-Bomb Testing," October 15, 1956.

To stop testing of multimegaton hydrogen bombs need not and surely would not deter research into, and development of, improved nuclear weapons. The recent Pacific tests prove that. We learned much from the testing of small devices in the range of a few kilotons, but the big multimegaton bomb was showpiece only and developed little, if any, we did not know before. The work on ground-to-air, air-to-ground, and air-to-air missiles could go right along, as could the program for a nuclear-propelled rocket.³⁶⁹

At the end of October, UCRL submitted to the AEC some optimistic projections for future "clean" weapons:

Our most recent estimates for clean or standard 1960 versions of the new A and B (bombs) are as follows:

[deleted]

We have also estimated that in the period 1962-1963 the following yields might be obtained:

[deleted]

These 1962-1963 versions are extreme extrapolations of present design ideas. Of course, we really don't know that those will work, because if we did, we would be designing them for 1960. However, they do represent our best guess regarding the progress that can be expected in another generation.

The only other clean bomb we have discussed in connection with the 1962-1963 period is one weighing about 300 lbs. and having a yield in the 10 KT range. This device involves new principles now being discussed but not yet subjected to any sort of test.
[deleted]

³⁶⁹"Answers to Questions Proposed by Time Life Inc. under date of October 17, 1956." These questions followed the TV broadcast on October 15, 1956 of an interview of Democratic presidential candidate Senator Adlai Stevenson; Stevenson claimed he had the support of Anderson for several of his disarmament and nuclear test-suspension proposals.

Intermediate sizes have not been studied to the same extent, but we believe that down to 6,000 lbs. or perhaps less, clean bombs can be made which have at least half the yield of standard bombs. This also applies to the 1962-1963 period.

From the above it can be seen that we do not believe that clean and standard versions will give the same yield, but that we do believe that with increasing yield and/or increasing development time, the gap will narrow substantially and the difference will not be very important for large warheads. In addition, we believe that probably the standard yield is not sensitive to whether we optimize the design for clean or standard yield.³⁷⁰

The GAC had also pledged that future Pacific tests would not spread radiation far beyond the limits of the PPG:

In the last tests, carried out in the Pacific a few months ago, encouraging progress was made toward nuclear bombs which for the same explosive power produce much less radioactive byproducts.

We do not foresee that our future military requirements will call for such number or such kind of tests which would cause a harmful contamination outside of the immediate test area.³⁷¹

Also at this time, DOD replied to the early October AEC request for guidance on "clean" weapons, noting that there was still insufficient data on which to base a realistic assessment of the military requirements for such a weapon.

DOD also asked (1) whether full enrichment of fuel for the "clean" MK 21 might interfere with MK 36 production (the MK 36 was then the stockpiled "conventional" 60% fission Class "B" gravity bomb), (2) for the total yield of the "clean" MK

³⁷⁰Teletype dated October 24, 1956 to Director of Military Application, USAEC, from Director, University of California Radiation Laboratory, Livermore, California.

³⁷¹Statement by General Advisory Committee, attachment to Minutes of the Fifty-First Meeting of the General Advisory Committee to the USAEC, October 29-31, 1956.

21, assuming the use of fully (95% Li-6) enriched fuel, and (3) for predicted operational availability dates for the new Class "A," "B," and "C" bombs in both "clean" and conventional ("dirty") versions.

The AEC replied to these questions late in November. The TX-21C was expected to become operational by April 1958; its production was not expected to interfere with the MK 36 program. The AEC also expected to be able to predict operational dates for "clean" and "dirty" Class "A," "B," and "C" weapons within a month.

The AEC was greatly interested in "clean" versions of the new Class "A" and "B" weapons and was pushing the investigation of these systems, which were expected to be very useful. It might also be desirable to produce these weapons in both "clean" and conventional version.³⁷²

Planning for Operation PILGRIM

In the meantime, planning continued for the 1957 Nevada series. By early November 1956, PILGRIM objectives included:

- a. Proof tests of air defense and anti-submarine implosion-type warheads which are scheduled to be stockpiled in the near future;
- b. Development tests of primaries and mockups which provide design information for high yield devices, including the ICBM warhead, that are scheduled for test at the Eniwetok Proving Ground in spring 1958;
- c. Other developmental tests required for advanced type fission weapons for which an established requirement exists;
- d. Exploratory tests directed toward more efficient use of active materials, improved (weapon) case structures, and warheads of

³⁷²Memorandum for the Chairman through the General Manager dated January 22, 1957 from Brig. Gen. Alfred D. Starbird, Director Of Military Application, USAEC, Subject: Clean Weapons; Atomic Energy Commission, Notes for the 108th AEC-MLC Conference, Memorandum by the Director of Military Application, 23 January 1957; letter dated October 29, 1956 to Lewis Strauss, Chairman, USAEC, from Deputy Secretary of Defense Reuben B. Robertson, Jr.; letter dated November 27, 1956 to Deputy Secretary of Defense Reuben B. Robertson, Jr. from Lewis L. Strauss, Chairman, USAEC.

smaller size and weight; and

e. A deep underground test designed to explore the technique of testing in which the nuclear explosion is completely contained underground thereby canceling the fallout hazard and freeing the detonation of the device from weather constraints; and

f. A separate military and civil effects shot.

DOD also planned a military effects test of a 38-40 KT "assured yield" device to be detonated from a balloon at an altitude of 500 feet. In addition, the Air Force planned a full systems test of an XW-25 in an MB-1 GENIE missile, to be airburst following its launch from an interceptor. This was a more realistic test of the XW-25 than the REDWING Osage shot which used an XW-25 in a MK 7 bomb case.

During calendar year 1957, LASL and UCRL wished to fire 26 devices; tests would be conducted on an "as needed" basis throughout the year, including possibly during a two-phase spring and fall series. A number of one-point safety tests would also be conducted. PILGRIM might be completed in two phases: Phase I, 16 tests, between May and July, and Phase II, 10 shots, in October and November.³⁷³

The 26-shot PILGRIM program would yield an estimated total of nearly 510 KT as compared to the 167 KT total yield of Operation TEAPOT: two air drops for 62 KT; 12 balloon shots totaling 170.8 KT; 11 tower shots for 275 KT; and a deep underground test yielding two kilotons.

Several of the shots could be moved to the HARDTACK program without delaying the stockpile entry of weapons for which requirements had been established. However, moving tests from PILGRIM to HARDTACK would incur higher costs per shot.³⁷⁴

The proposed spring series included eight LASL tests, split equally between balloon shots and tower shots; yields were anticipated to be between 0.1 and 40 KT. UCRL planned two balloon shots, two balloon or tower shots, and two tower

³⁷³Atomic Energy Commission, Test Activities for Calendar Year 1957, Report to the General Manager by the Director of Military Application, November 9, 1956.

³⁷⁴"Background and Discussion," Appendix "A" to Atomic Energy Commission, Test Activities for Calendar Year 1957, Report to the General Manager by the Director of Military Application, November 9, 1956.

shots, with yields from 0.2 to 80 KT. DOD also wanted a 500-foot high balloon shot of 38 to 40 KT and a 2 KT airdrop.

The shorter fall phase of PILGRIM was to include, for LASL, two balloon shots and an airdrop, with yields between two and 60 KT. UCRL planned three balloon shots, three tower shots, and a deep underground shot, with yields from two to 40 KT.

If an experiment slated for January 1957 showed that tethered balloons were not practical test platforms, then all but two of the balloon shots would be moved to towers; the other two would become additional airdrops.

Among exploratory shots proposed by LASL were two to increase knowledge and understanding of small weapons and to obtain very small (fractional kiloton) yields with a minimum of fissionable material. Three LASL tower shots were to develop new or improved primaries for existing or future multistage weapons.

The high priority assigned to the U.S. ICBM program might require repeat testing of an ICBM warhead primary if results of a preceding test were unsatisfactory.³⁷⁵ Two of the small LASL devices were the FLEA and the GNAT.³⁷⁶ These new small weapons would require more "boost" tritium gas, thereby increasing the AEC's requirements for future tritium production.³⁷⁷

A later estimate of PILGRIM Phase I shots included 16 tests ranging in yield between 0.02 and 80 KT, detonated atop 200, 300, 400, and 500-foot towers, and from balloons at 1,500 feet. Fourteen shots had been assigned names and ready dates: Wilson, Galileo, and Newton, on May 1, 1957; Boltzmann and Diablo, May 8; Laplace, May 22; Shasta, May 23; Lassen, May 27; Everest, June 6; Owens, June 18; Whitney, June 22; Hood, June 27; and Tycho and Huygens with no ready dates.³⁷⁸

By November 1956, two high altitude shots were planned for HARDTACK: a one-to-four megaton burst at about 250,000 feet,

³⁷⁵"Outline of CY 1957 Full-Scale Test Program," Appendix "B" to Atomic Energy Commission, Test Activities for Calendar Year 1957, Report to the General Manager by the Director of Military Application, November 9, 1956.

³⁷⁶Minutes of AEC Meeting No. 1246, November 14, 1956.

³⁷⁷Minutes of the 106th AEC-MLC Conference, September 27, 1956.

³⁷⁸Minutes of the Second Meeting of the Planning Board for Nevada Weapons Tests, November 13, 1956.

and a burst of at least 40 KT at about 100,000 feet.³⁷⁹ The nuclear explosives (warheads) would be lofted by missiles or balloons to burst altitude. During the shots, an array of sounding rockets would position suitable instrumentation at appropriate points in space.

Other instruments would be carried aloft by manned aircraft and still more instruments would be deployed on the ground. All this instrumentation would be used to measure and record neutron spectra, neutron flux, gamma spectra, gamma flux, thermal flux, time between detonation of the weapon stages, electromagnetic phenomena, peak overpressure, and overpressure as a function of time. Dozens of cameras would photograph fireball development and aftereffects.³⁸⁰

These tests were to be conducted to determine the basic effects of nuclear explosions at altitudes above 100,000 feet. The data gathered would be used in the development of several things: ICBM "kill" mechanisms; defenses against manned bombers, the so-called "Sunlamp" effect, which was intended to kill aircrews by nuclear radiation above their aircraft;³⁸¹ methods of detecting high-altitude nuclear blasts; and to determine whether or not any unexpected phenomena might result from such explosions.

The "Sunlamp" principle was a theoretical idea that involved bombarding enemy aircraft and aircrews with incapacitating doses of fast 14 MeV neutrons from an exploding thermonuclear device, presumably at ranges beyond which blast and thermal effects would be negligible.

As early as 1953, after neutron measurements during the unboosted and boosted TX-7 tests of the Easy and Dog shots of Operations BUSTER and TUMBLER, respectively, indicated that certain alterations in weapon design -- especially boosting -- led to sizable increases in neutron yield, it became theoretically possible to incapacitate bomber crewmen flying at altitudes around 40,000 feet at longer ranges, perhaps as much as 13 times the distance, than those at which aircraft could be destroyed by blast, gust loading, or thermal effects.

³⁷⁹Memorandum for the Chairman, U.S. Atomic Energy Commission, 19 December 1956, from Herbert B. Loper, Chairman, JCAE, Subject: High Altitude Test for Operation HARDTACK.

³⁸⁰Letter dated November 26, 1956, from Kenneth E. Fields, General Manager, USAEC, to the Honorable Herbert B. Loper, Chairman, Military Liaison Committee to the AEC, Subject: Inclusion of High Altitude Test in HARDTACK.

³⁸¹Shelton, p. 9-25.

Radiation doses of 2,000 to 5,000 rem, the equivalent in biological damage to that caused by 2,000 to 5,000 roentgens, would be required to produce immediately-lethal physiological effects on aircrews.

At altitudes above 20,000 feet, nuclear radiation became a significant "kill" mechanism.³⁸² By mid-1954, neutron flux per kiloton yield for tactical weapons was 10 to 30 times higher than normal.³⁸³

Possible ICBM kill mechanisms being considered during the fall of 1956 included neutron heating of the warhead and x-ray pulse on the missile skin. The latter effect was believed to be lethal at distances up to two miles from a kiloton-yield warhead.³⁸⁴

An Army requirement for a high-altitude HARDTACK test noted:

Army surface-to-air missile systems of the future will engage targets at altitudes above 300,000 feet; yields may vary from 2 KT to several MT. In order to design warheads of optimum destructive effect and minimum vulnerability for these missiles, reliable information on high altitude effects must be obtained at an early date. Such information will also provide guidance in the design of intermediate range and intercontinental ballistic missile warheads.³⁸⁵

Also by late November, LASL was able to state that "our theoretical knowledge of the conditions obtaining inside a

³⁸²THE RELATIVE IMPORTANCE OF NUCLEAR RADIATION FROM ATOMIC WEAPONS DETONATED AT HIGH ALTITUDES, AFSWP Technical Analysis Report No. 500, Paul W. Ifland, Weapons Effects Division, Headquarters, Armed Forces Special Weapons Project, April 23, 1953, pp. 1, 3, 4, 12, 13.

³⁸³Memorandum dated 11 May 1954 to Chief, Armed Forces Special Weapons Project, from J. P. Mitchell, Chief, Radiological Division, U.S. Army Chemical Corps Chemical and Radiological Laboratories, subject: Coordination of Project 2.7, p. 4.

³⁸⁴According to Shelton (p. 9-45), the idea of using x-rays to kill ICBMs dated (in the U.S.) back to September 1955.

³⁸⁵AFSWP report dated 3 November 1956, subject: Ultra High Altitude (UHA) Effects Test, pp. 1, 2-1, 4-1, 4-2, 4-3, App-1.

two-stage bomb has increased appreciably in the last three years so that many questions can be answered now by calculation that previously (required) experimental measurements."³⁸⁶

In December, DOD concurred with AEC desires for high-altitude testing and promised to furnish warheads for the shots.³⁸⁷

In early December, the AEC studied the ramifications of conducting all of Operation PILGRIM in the Pacific:

a. It would eliminate the problem of further contaminating the Nevada area. It would, however, bring forth some world-wide opposition which would not develop from a Nevada series.

b. The first shot probably could not be fired until late fall of 1957. The development of weapons scheduled for stockpile entry from now through 1960 will tax the capabilities of the laboratories and will be difficult to meet even with tests conducted as presently planned. Specific implications of this delay on AEC weapons readiness schedules include:

(a) For the first ICBM warheads, the primary [deleted] would not be tested until a year or less ahead of stockpile entry. Delay in test of the XW-35 mock-up would delay the firming-up of the preferred [deleted] warhead design and would result in meeting the October 1958 stockpile date with a W-49 warhead based on the TX-28C device tested in Operation REDWING as part of the Class D bomb program.

(2) Proof testing of warhead for LULU, the anti-submarine weapon, would be delayed from early fall

³⁸⁶Memo for the Record dated November 23, 1956 from William Ogle, LASL.

³⁸⁷Memorandum for the Chairman, U. S. Atomic Energy Commission dated December 19, 1956, from Herbert B. Loper, Chairman, JCAE, Subject: High Altitude Test for Operation HARDTACK.

of 1957 to late fall which might affect entry of a tested weapon into stockpile by the scheduled date, April 1958.

(3) Test of [deleted], the primary for the [deleted] weapon, would be delayed six or more months with probably some effect on stockpile entry of the [deleted] with the desired primary. The [deleted] is a shot aimed at reducing the Pu content of the earlier [deleted] device thereby resulting in a better balance between the production and utilization of Pu and Oy (Author's note: or alloy) and making a larger number of weapons available to the DOD.

(4) Test of the [deleted] new primary for [deleted] would be delayed about 3 months, with possibly some effect on stockpile entry.

(5) Test of the XW-30 warhead for the TALOS-W air defense missile would be delayed six months. Information from the test is required before design of the warhead can be completed. This would result in delay of stockpile entry scheduled for July 1958.

(6) Certain experimental or developmental shots in the 11-inch range [deleted] would be delayed six months. These are designed for a large number of applications to include new air-to-air, ground-to-air, and ship-to-sub missiles. (Author's note: possible XW-44, -45, and -51 prototypes.) It is most likely that this would result in delay of stockpile entry.

(7) The [deleted] studies directed toward the new Classes A and B (bombs) would be delayed six months. The latest information

from UCRL indicates that it will be difficult even with no delay in tests to meet the planned stockpile entry dates for the new Classes A and B.

c. It is considered mandatory that Operation HARDTACK, now scheduled for the Eniwetok Proving Ground in May, June and July of 1958, be conducted as scheduled. PILGRIM, if conducted in the Pacific, could be expected to start no earlier than November 1957 and continue for two to three months. If the information resulting from PILGRIM is to be digested properly, and new devices fabricated, it is most doubtful HARDTACK could start May 1, 1958 and be completed in a reasonable period.

It probably would prove necessary to delay its start and extend materially the completion date. HARDTACK will be made up of proof-tests of weapons that will be stockpiled more than 9 to 18 months later, and of development tests of devices that will enter stockpile more than 30-36 months later. The following comments indicate the impact of delaying HARDTACK.

(1) The XW-35 warhead for the ICBM/IRBM will be proof-tested in HARDTACK prior to stockpile entry
[deleted]

(2) HARDTACK will proof-test weapons with a normal development sequence which enter stockpile after February 1959. By present commitments these are:

Class B February 1959
Class A June 1959
Class C August 1959
Laydown Bomb February 1960

(3) If HARDTACK is not completed until after December 1958, then the OADs of the Class A, B, and C would be delayed until between October 1959 and June 1960, and the laydown bomb family would interfere with these programs in this time period, suggesting possibly even greater slippage for

the Class A, B, and C.

(4) The effect of delaying HARDTACK, upon programs relating to devices that will be subjected to development testing in 1958, would be to delay weapons with OADs after June 1960. It is not possible to predict more specific effects than a general 6-month slippage of such programs, which would include advanced ballistic missile warheads, high-yield anti-aircraft warheads, and clean weapons of advanced design.

....

Movement of (shots from Nevada to the Pacific) would have the following implications:

....

c. Implications on our weapons readiness schedules would be as follows:

(1) The [deleted] Mockup test probably could not be fired in sufficient time to permit its results to be evaluated and incorporated in the design of the XW-38 proof test. Therefore, the ICBM warhead to be stockpiled in October 1958 would be a 1 MT device based on the TX-28C instead of the 3 to 4 MT warhead which the new design would permit, if successful.

(2) Delays to HARDTACK of the [deleted] case studies and the Class B Mockup tests would prevent the effective use of their results in developing the

Class A and Class B designs to be proof-tested in HARDTACK. This would require the redesign of these development programs, which would undoubtedly delay stockpile entry dates (1959) for these important weapons.

(3) Delay in the entire long-range UCRL large-weapons development program would result since their five test devices involved include extensive case research tests. Since the results of these tests are required before the full-scale devices can be fired, scheduling of events are more difficult, and additional delays may be anticipated.

At the same time greater risks are assumed because failure of the mockups would lead either to extension of the operation or delay of test to the next operation.³⁸⁸

Soon after this study was completed, the AEC decided to conduct PILGRIM as a continuous operation in Nevada between May and October 1957.

The Class "A" weapon referenced in the study would be either a "clean" or normal ("dirty") weapon. The specifications for

³⁸⁸"Background and Discussion," Appendix "A" to Atomic Energy Commission, Means of Reducing Off-Site Fallout which May Result from Operation PILGRIM, Report to the General Manager by the Director of Military Application, USAEC, December 3, 1956.

the test device were to be received from DOD about January 1, 1957; the bomb was still just a feasibility study. If a "clean" weapon were required, it was expected to yield less than 60 MT (the AEC had a requirement for a study of the feasibility of 60 MT weapons, either "clean" or normal).³⁸⁹

At this time, PILGRIM was to be conducted in two phases. The first phase, comprising 16 shots, would begin about May 1, 1957 and continue through July. Phase II, consisting of an additional 10 shots, would begin about October 1, 1957 and last for about two months.

The proposed tests included:

- a. Air defense, anti-submarine and universal warheads to be proof-tested for near-future stockpiling.
- b. Primary, case, and mockup tests for high-yield tests to be conducted at the Eniwetok Proving Ground in the spring of 1958.
- c. Development tests required for advanced-type fission weapons for which an established requirement existed.
- d. Exploratory tests directed toward more efficient use of active material, improved case structure, and generally smaller and lighter warheads.
- e. A deep underground test designed to explore the technique of complete containment of the nuclear detonation and thereby independence from weather constraints.³⁹⁰

AEC Commissioner Thomas Murray spoke out on December 20 about his advocacy of an accelerated "clean" weapons program:

It seems to me that we must take steps to accelerate the development and stockpiling of "Clean Weapons." From the information available to me I feel that the Department

³⁸⁹Minutes of AEC Meeting No. 1252, December 5, 1956.

³⁹⁰Memorandum dated 4 December 1956 to Commander, Field Command, Armed Forces Special Weapons Project, from Capt. R. G. Shull, USA, Acting Assistant Adjutant General, subject: Outline of Calendar Year 1957 Full-Scale Test Program.

of Defense is not giving sufficient emphasis to these weapons in its planning.

Dr. Libby's remarks yesterday reinforce the opinion I have expressed for some time that our present stockpile plans include quantities of "Dirty Weapons," only a portion of which could be used without disastrous effects. Under these circumstances, I recommend that the Commission take action either through the Department of Defense or by requesting a Presidential Directive to give the development and stockpiling of "Clean Weapons" the additional emphasis required.³⁹¹

Commissioner Murray's desire for "clean weapons" was not altogether altruistic or "humanitarian:" the AEC could make a better argument for the continuation of atmospheric nuclear testing if most of the shots involved "clean" devices with limited fallout.

By this time, PILGRIM goals were becoming firm. A letter dated December 21, 1956, to President Eisenhower from AEC Chairman Lewis Strauss included these remarks about the objectives of Operation PILGRIM:

A number of nuclear devices for early entry into our defense stockpile, or important to the most rapid advancement of later weapons designs, will require testing in calendar year 1957. Such tests would include:

- a. The proof-firing of certain air defense and anti-submarine implosion-type warheads scheduled for early production. Included this category are a warhead for the TALOS-W missile, two warheads for the NIKE-B missile and an atomic depth charge.
- b. Development tests of primaries and mockups which provide design information for thermonuclear devices, which we plan to test in the Eniwetok Proving Ground in 1958. These include an ICBM warhead and devices directed

³⁹¹Memorandum for Chairman Lewis L. Strauss dated December 20, 1956 from Thomas K. Murray, Commissioner, Subject: Acceleration of Clean Weapons Development and Stockpiling.

toward a higher yield-to-weight ratio.

- c. Exploratory and development tests directed toward achieving more efficient use of active material, improved (weapon) case structures, and warheads of smaller size and weight.
- d. A deep underground test designed to explore this manner of testing. This testing technique would guarantee no off-site fallout and would furnish information of importance to several new possible applications of atomic weapons.³⁹²

Plans for HARDTACK Underwater Tests

On the same day, the Navy released a revised draft of its operations plan for HARDTACK. Among the contemplated tests were two named "Percolator" and "Little Wigwam." Percolator was to be a LULU-type weapon (simulating a W-34 warhead) detonated on the bottom of Eniwetok lagoon at a depth of 150 to 200 feet. Little Wigwam was to be a detonation of an ASROC-yield weapon (W-44 warhead simulation) at a depth of 500 feet in 6,000 feet of water. (These shots were later renamed Umbrella and Wahoo, respectively.)

The shots were to use W-7 warheads; suitable MK 90 pressure vessels, the same ballistic casings used for the Navy's BETTY depth bombs, would be made available for test use.

The two tests were recommended because of the importance of the Navy's antisubmarine mission and the extreme urgency of providing naval operating forces with complete and accurate weapons effects data associated with both current nuclear depth charges and those now under development.

The first objective of the tests was to thoroughly document the basic effects data about initial and residual radiation, air overpressure, underwater shock pressures, blast crater measurements, mechanics of base surges, and radiological contaminants.

³⁹²Draft letter to the President, attachment to Memorandum for Mr. Strauss, Mr. Murray, Dr. Libby, and Mr. Vance dated approximately December 14, 1956, Subject: Proposed Test Plans for Calendar Year 1957 (when finally sent to Eisenhower, the phrase "improved case structures" in item "c" was deleted); Atomic Energy Commission, Program Status Report to the Joint Committee on Atomic Energy, December 31, 1956, Part III - Weapons, February 11, 1957, p. 8.

The second test objective was to document the mechanical responses of selected targets to underwater shock pressures.

Successful tests were expected to provide information necessary to determine:

1. Safe minimum stand-off distances for delivery of nuclear anti-submarine warfare weapons by existing airborne, surface, and underwater delivery vehicles.
2. The lethal ranges of nuclear ASW weapons against both submarines and surface ships in both shallow and deep water.
3. The minefield-clearance capability of subsurface-bursting atomic weapons.

In particular, one program was designed to study target responses to shock waves from atomic explosions. Probable targets included ship hulls; shipboard machinery, personnel, and equipment; and sonar equipment near an underwater atomic burst. The information obtained was to be used to devise improved delivery techniques for the ASROC and ASTOR nuclear weapon delivery systems currently under development for U.S. Navy destroyers and submarines.³⁹³ (ASROC was a rocket-launched nuclear depth charge; ASTOR was a nuclear-tipped torpedo).

Additional purposes of Percolator and Little Wigwam included the determination of radiological hazards and cratering effects resulting from shallow-water burst weapons; the attainment of data on which to base the optimum yield and burst depth of atomic depth bombs or charges being designed and developed; and to determine the relative effectiveness of atomic depth bombs against surface ship convoys and naval task forces as compared with the effects of air- or surface-burst nuclear weapons.

An early functional test of the Navy's MK 101 LULU depth bomb under operationally realistic conditions was desirable. An underwater shot at a depth of 150 to 200 feet would correspond closely to the important tactical situation wherein LULU would be detonated in shallow water. The Little Wigwam test would not correspond to the operational use of the Navy's ASROC, ASTOR, or LULU weapons unless the

³⁹³SPECIAL WEAPONS EFFECTS TEST PLANNING GROUP, OPERATION HARDTACK, REVISED NAVY PLANS, SWET-4, Office of Naval Research, Department of the Navy, 21 December 1956, pp. 2, 3, 110.

detonation occurred at a depth between 750 and 1,000 feet.

The proposed Percolator shot covered tactical situations that might arise in operational LULU use for important areas off the east coast of the U.S., the North Sea, coastal waters of the Far East, and many other locations where the weapon might be used. The test would not replicate LULU detonation at depths between 750 and 1,000 feet.³⁹⁴

Eisenhower approved Operation PILGRIM a week later; by January 25, 1957, PILGRIM had been redesignated Operation PLUMBBOB.³⁹⁵

At this time, the Nevada series was to consist of 24 shots, including two DOD-sponsored effects tests; 13 weapon-development shots: six for LASL and seven for UCRL; two LASL proof tests of the XW-25 and XW-30; six exploratory tests: four for UCRL and two for LASL; and one testing technique shot, the deep-underground UCRL test.

At least one PLUMBBOB shot was of critical importance to HARDTACK:

The Class [deleted] mockup...is the most difficult shot of the PLUMBBOB schedule. ... The shot is directed toward development of a full-scale [deleted] to be tested in Operation HARDTACK. If the [deleted] mockup were not fired in PLUMBBOB, design of the full-scale [deleted] for HARDTACK would have to be much more conservative, since no test will have been conducted prior to design freeze, which is scheduled for October 1957, and complete design release, April 1958.

This would probably result in decreased

³⁹⁴Page Committee Recommendations for Navy Tests on Operation HARDTACK, NAVORD Report 4457, U.S. Naval Ordnance Laboratory, White Oak, Maryland, 1 February 1957, pp. 1, 5, 10.

³⁹⁵Atomic Energy Commission, AEC 944/6, Note by Secretary, Change in Code Word for Test Operation, January 25, 1957; Semiannual Historical Report, Headquarters, Field Command, The Armed Services Special Weapons Command, Sandia Base, Albuquerque, New Mexico, Activities for the Period 1 January 1957 - 30 June 1957, p. 236. Shelton, on p. 8-18, claims that the test series name was changed after public objections. He also claims that the shots named Priscilla and John were related to the original PILGRIM designation.

yield for the [deleted] weapon (possibly as much as one-third decrease). It is estimated that the deletion of the [deleted] mockup from PLUMBBOB could delay the stockpile entry date by as much as one year (presuming a 1959 Pacific series) if trouble should be experienced with the HARDTACK test. The new [deleted] would be affected similarly, if the [deleted] were deleted from PLUMBBOB.³⁹⁶

In mid-December 1956, the DOD agreed to the AEC request to conduct very-high altitude testing during Operation HARDTACK. A shot at about 250,000 feet in the range of one to four megatons would meet most of the DOD requirements. A shot of at least 40 kilotons at about 100,000 feet would test remote detectability.³⁹⁷

Some thought was given in the latter part of January 1957 to conducting an "emergency" high altitude test shot before HARDTACK, using an existing Army CORPORAL missile and a currently-stockpiled W-7 warhead. This test would provide preliminary information and experience for the HARDTACK shots and give the AEC, as much as a year earlier than expected, some information on the possibility of detecting low-yield nuclear detonations at altitudes of approximately 100,000 feet.

The test would not, however, prove any ability to detect nuclear detonations at substantially higher altitudes. In light of this fact, and because this test might militate against the better-planned and more-thoroughly instrumented HARDTACK high altitude shots, this "emergency" shot was

³⁹⁶Atomic Energy Commission, Proposed Test Plans for CY 1957, Report to the General Manager by the Director of Military Application, undated, but probably early January 1957, pp. 9, 10. The MK 28RE, TX-36-X2, and MK 49 Mod 0 warheads were all design-released in April 1958. ("Timetables of Weapons Events," DOE Albuquerque Operations Office, September 22, 1986.) The 44 kiloton UCRL PLUMBBOB Smoky shot, fired on August 31, 1957, was a test of a new Class "B" Prime bomb mockup (the UCRL TX-41). (Atomic Energy Commission, Declassification of Actual Yield of Shot "Smoky" of Operation PLUMBBOB, Report to the General Manager by the Director of Classification, USAEC, Washington, D.C., June 11, 1957.)

³⁹⁷Department of Defense, Military Liaison Committee, Memorandum for the Chairman, U. S. Atomic Energy Commission, Subject: High Altitude Test for Operation HARDTACK, December 19, 1956.

anceled.³⁹⁸

By late December 1956, there were several objectives for the high-altitude shots:

(a) to provide data on the U.S. detection and intelligence capabilities (for foreign high-altitude detonations).

(b) to investigate the feasibility of very-high altitude proving grounds.

(c) to provide data on design criteria for the vulnerability of U.S. ballistic missile warheads.

(d) to provide data on the design of U.S. ballistic missile defense systems and optimized warheads for use in these systems.

(e) to investigate the SUN LAMP concept which may be described as a proposal for incapacitating crews of manned aircraft flying at approximately 50,000 feet and higher by neutron irradiation from multimegaton yield warheads detonated at altitudes of approximately 100,000 feet and higher.

(f) to provide data to check the reliability of extrapolation of weapon effects from sea level shots to extremely high altitude shots.

The only presently-available stockpiled missile that could carry an atomic warhead to very high altitudes was the Army's CORPORAL. The maximum yield of the currently-stockpiled CORPORAL warhead, the W-7, was only 70 KT. The Army's REDSTONE IRBM, currently under development, would be able to carry a 4 MT W-39 warhead to high altitude. Although the REDSTONE was not yet available, it was expected to be ready in time for HARDTACK.

Since the CORPORAL was now operational and deployed overseas, the Army was prepared to furnish on a few days' notice the number of missiles the AEC required, along with appropriate launching equipment and personnel. Sandia was prepared to

³⁹⁸Atomic Energy Commission, A Study of a Recommendation for an Immediate Very High Altitude Test Shot, Report by the General Manager, January 22, 1957.

modify existing 40 to 70 KT yield W-7 warheads to fuse them for command detonation at high altitude.

In order to test U.S. long-range detection capabilities for high-altitude nuclear explosions, the Air Force wanted a minimum of two tests: a 70 KT yield at 100,000 feet and the maximum yield obtainable which could be detonated above 200,000 feet. If a higher yield warhead were available for the 100,000 foot test, it would be preferred over the lower yield.

Sandia had also prepared a study for two shots of one megaton each, one at 100,000 feet and the other at 250,000 feet; the warheads would be carried aloft by missiles.

In preparation for a series of British tests to be conducted on or after May 1957 in the Pacific near Christmas Island, the Air Force would have enough acoustic detection stations manned and situated so that coverage would be adequate for a U.S. test at or near Eniwetok atoll. Electromagnetic detection stations would also be ready by the spring of 1957.

If necessary, the U.S. could conduct high-altitude tests in the Pacific in early 1957; however, this would conflict with and draw resources away from Operation PILGRIM in Nevada at the same time. Hence, the AEC leaned toward delaying the high-altitude tests until HARDTACK, unless DOD demanded earlier action.³⁹⁹

Late in December, HARDTACK was to include shots in the yield range of a few kilotons to several megatons, and the total number of shots was expected to be about that of REDWING. Post-shot clouds for the larger tests were expected to rise from 90,000 to 110,000 feet, with their bases between 40,000 and 50,000 feet. Debris samples would be required from altitudes between 42,000 and 65,000 feet.

Because of these sampling altitude requirements, and because remote Taongi atoll might be used as a site for several of the megaton tests, Martin B-57Bs and B-57Ds were recommended for use as sampler aircraft. The "D" model could reach higher altitudes than the "B."⁴⁰⁰

³⁹⁹Memorandum for the Chairman dated 26 December 1956 from Brig. Gen. Alfred D. Starbird, USA, Director of Military Application, USAEC, Subject: Comments on Professor E. O. Lawrence's Recommendation for Undertaking a High-Altitude Rocket-Borne Shot without Delay.

⁴⁰⁰Letter dated December 18, 1956 to RAdm. Byron H. Hanlon, Commander, Joint Task Force Seven, from Dr. Roger E. Batzel, Assistant Chemistry Division Leader, UCRL. Not all of the B-

At the end of 1956, the AEC communicated the status of development of new "clean" weapons to Congress:

Studies on the new Class A, B, and C thermonuclear weapons include investigations into the development of "clean" versions of these weapons. Information obtained in the 1956 test series confirmed the feasibility of developing thermonuclear weapons with greatly reduced fission yield and consequent reduction in fallout. There will be some reduction in total yield of "clean" versions as compared with standard versions of the same size and weight.

For example, it is estimated that the new Class B weapon of standard design can have a total yield of [deleted] but its "clean" version would have a total yield of between [deleted] On the other hand, the standard version would have a fission yield of approximately [deleted] while the fission yield of the "clean" version would approximate [deleted]

Continued research and development is expected to result in reducing the total yield of "clean" versions as compared with standard versions.⁴⁰¹

Early 1957: Pressure Builds for a Test Moratorium

At the beginning of 1957, it was becoming increasingly apparent to the AEC that U.S. and world political opinion was strongly favoring an atmospheric nuclear test ban or severe test limitations, at least. Earlier cold war antagonism between the U.S. and the USSR had lessened somewhat, and there was a sentiment that nuclear weapons tests were a form of "saber rattling" that only increased tensions and the likelihood of nuclear confrontation. A unilateral cessation of testing by the U.S., with or without a Soviet reciprocation,

57 flights were at high altitudes: contemporary USAF sampling unit histories record several instances in which pilots were reprimanded for returning to Kwajalein with palm fronds lodged in their aircraft engine intakes and landing gear wheel wells; these branches were collected during unauthorized treetop-level "buzzing" of atoll islands.

⁴⁰¹Proposed Changes to the Program Status Report to the Joint Committee on Atomic Energy, December 31, 1956.

might also be a step on the way to general nuclear disarmament.

On the other hand, development of new weapons was the lifeblood of the labs at Los Alamos and Livermore. Lab scientists and administrators argued that nuclear warheads were the only advantage the U.S. had over the overwhelming conventional forces and manpower of the Soviet armed services.

According to the labs, unilateral nuclear disarmament, or even a reduction in testing, might be perceived by the Soviets as an indication of U.S. weakness and an invitation to general war (many government officials also shared this view). The labs argued that the "minor" health hazards resulting from atmospheric tests was a small price to pay for the benefits of a nuclear deterrent.

The trend toward a test ban was of great import to the weapons development program: future weapons depended directly on firing scaled-down and full-size devices and weapon proof certification before stockpile entry. Without continued testing, the entire U.S. nuclear weapons program would be seriously jeopardized.

The most immediate effect of this troublesome situation was a virtual stampede by the labs at Los Alamos and Livermore to test fire as many devices and weapons as possible before any test ban or test limitation put a damper on their work.⁴⁰²

Late in January, the AEC was still seeking guidance from DOD on the level of emphasis that should be placed on "clean" weapons development and the number and types of "clean" weapons desired for stockpiling.⁴⁰³

The Chairman of the AEC wrote to the Secretary of Defense, advising him of the current status of the "clean" weapons program and estimating future OAD's for various types:

Progress on the design of the new Class "B" has indicated that the same basic design will be close to the optimum for both conventional and clean versions. However, a firm decision would be required at least 18 months in advance of the Operational

⁴⁰²AEC Meeting No. 1265, 2:00 PM, Monday, February 11, 1957, p. 72.

⁴⁰³Atomic Energy Commission, Notes for the 108th AEC-MLC Conference, Memorandum by the Director of Military Application, 23 January 1957.

Availability Date for the conventional weapon in order to produce the clean version on the same date. A decision at an earlier date would materially assist us in our planning.

Our letter ... dated October 10, 1956 stated that we could produce either a conventional or a clean version of the New Class B (bomb) available in February 1959. Subsequent studies have indicated that this estimate is too optimistic to insure the introduction of quality product into the stockpile and to develop a maximum yield weapon capable of an extended interval [deleted] goal) between storage inspections. Following is the revised schedule of availability for both conventional and clean versions of the New Class B weapon:

First Production Unit July 1959
(Training Unit)

First Production Unit Sept 1959
(War Reserve Unit)

OAD November 1959

The above schedule will be reviewed about midway through the development program to see if an earlier Operational Availability Date (OAD) is feasible.

Design studies for the New Class A weapon have not progressed as far as (those) for the New Class B. However, the same considerations are applicable except for the Operational Availability Date which is estimated to be four to eight months later. We estimate that both conventional and clean versions of the New Class A could have the same Operational Availability Date some time during the period March to July 1960.

In the feasibility study of the New Class C weapon, diameters of 30 inches, 35 inches and 40 inches were considered. This study indicates that a clean version of the New Class C weapon of 30 inches diameter is not feasible in the approximate time scale for which the weapon is desired, but clean versions of the 35-inch and 40-inch diameter designs of the New Class C weapon

are feasible.

Inasmuch as the conventional version and the clean version of each of these two designs would be quite similar, we estimate that a clean version and a conventional version of the New Class C weapon, if of the same diameter, can be made operationally available in August 1959, the Operational Availability Date stated in our letter of October 10, 1956.

You will recall that in that letter, we indicated that we could not then provide an estimate of the total yield of the clean version of the New Class C weapon. We are now able to advise you that we estimate the yield of the 35-inch diameter design of the clean version of the New Class C weapon will be [deleted] while that of the 40-inch diameter design will be [deleted]

We hope that the foregoing information, together with other data previously furnished to you, will enable you to advise us of the emphasis we should place on clean weapon development and the number and types of clean weapons desired for stockpile.⁴⁰⁴

Even as the AEC and DOD pushed ahead with plans for large thermonuclear weapons, AEC Commissioner Thomas E. Murray decried what he saw as a serious gap in the current spectrum of U.S. nuclear weapons. Although the U.S. was stockpiling significant numbers of large city-busting hydrogen bombs, the arsenal contained very few small, tactical warheads.

By this time, only a small number of relatively low-yield MK/W-7, MK 8, MK 9, and MK 12 weapons had been stockpiled; both LASL and UCRL were working on potential low-yield 15-inch to 10-inch diameter devices, mainly for air defense applications.

Commissioner Murray stressed his belief in the need for stockpiling more weapons adaptable to limited nuclear war:

⁴⁰⁴Draft Letter to Secretary of Defense from Chairman, AEC, Enclosure "B" to Memorandum for the Chairman through the General Manager dated January 22, 1957 from Brig. Gen. Alfred D. Starbird, Director Of Military Application, USAEC, Subject: Clean Weapons. This letter was sent out with a date of February 8, 1957.

We are today quite ready -- as you can see -- for the so-called "three-hour war" -- this is, a war whose purpose would quite literally be the destruction of the Soviet Union, the annihilation of Soviet military might, accomplished amid a holocaust of the Soviet citizenry.

....

From this fact it follows that total nuclear warfare on the scale described is not a valid military policy, even though we have in hand the weapons with which to wage it. Inescapably, military policy must impose limitations on nuclear warfare. I have constantly insisted that this same principle of limitation, which binds military policy, must likewise bind weapons policy and reflect itself in the kinds and numbers of weapons in our stockpile.

....

I come now to our deficiency in the lower ranges of nuclear weapons. This deficiency seriously cripples our capacity to wage limited wars. I refer to wars that are limited in their objectives. Wars thus limited in their objectives can also be limited in regard to the measure of force used in their prosecution. ...

....

Our existing and projected stockpile does not sufficiently equip us to wage limited war. For this purpose we should need abundant supplies of small weapons, in the very low kiloton range. These supplies, moreover, must be abundant beyond the large numbers we will have to stockpile for the air defense of the United States. But, as the figures show, it is precisely at this lower end of the weapons spectrum that we are deficient. We are least ready for the kind of war for which we should be most ready.

....

My conclusion is that America's capacity for waging limited war, as exhibited in our present and projected stockpile, is unsatisfactory. By comparison, our capacity for the unlikely event, not to say unthinkable event, of total nuclear war is overdeveloped.

....

It is my own conviction that our structure of policy has been stood quite upside down. In saying this, I am indicating that we have made two mistakes. We have permitted a combination of technological and budgetary considerations to assume an undue control over the weapons program. And we have permitted the weapons program itself to assume an equally undue control over military policy.

Although the process began earlier, science and technology assumed much of their present undue influence over both weapons policy and military policy in consequence of the Mike shot in 1952 and subsequent tests. [deleted]

....

And ever since then, our weapons program has been governed increasingly by the concept of what is technologically possible, and correspondingly less by the concept of what is militarily useful. The product itself -- the immense H-bomb -- and the technology of its production took increasing charge of the weapons program and tended more and more to dictate weapons policy. ...

....

... Cost factors, combined with technological factors, are contributing to stand the structure of our policies upside down.

....

At the present moment, sound military policy would require that we put an urgent emphasis on very small weapons. Strangely enough, there is a widespread belief particularly among the public, that we already have these weapons in stockpile. This is not the case. Why then do we not have them? Why have we not established a weapons policy looking to adequate strength in small as well as large weapons? Ultimately the answer is that a program based on such a policy would cost too much.

....

.. Concretely, the demand of the moment is

that we begin to stockpile nuclear strength where nuclear strength is badly needed, and is now badly missing -- in the range of small weapons.⁴⁰⁵

Over the next few years, the quest to make economic and efficient use of fissionable material in the design of small, low-yield "clean" weapons would come to occupy much of the time of many people at LASL and UCRL.

At the end of January, HARDTACK was expected to comprise 31 experiments: five DOD effects tests, 12 LASL shots, and 14 UCRL shots.⁴⁰⁶ These tests might include four to five high-yield surface shots, six or seven smaller surface shots, and eight "small" (low-yield) surface and tower shots.

LASL's shots would include seven barge shots, one barge or tower shot, two surface shots, and two tower shots. UCRL's plans included four barge shots, two tower or barge shots, one surface shot, and one tower shot.

The DOD tests were to include one very-high altitude balloon-borne shot at 90,000 feet (probably at Bikini), one 150-foot deep underwater shot in Eniwetok lagoon, one underwater shot at a depth of 150 or 400 feet six miles southwest of Eniwetok atoll, and two ultra-high altitude (100,000 and 250,000 feet, respectively) missile-delivered megaton-yield bursts, probably at Bikini. Tests at Eniwetok were to be limited to a maximum yield of 5 MT.⁴⁰⁷

Among desired DOD effects tests was the underwater firing of an atomic weapon at a depth of 500 feet to determine the destructive effects of a warhead delivered at this depth to supplement knowledge gained as a result of the 2,000-foot deep WIGWAM shot in May 1955, and to ascertain the potential hazard to the delivery vessel.

⁴⁰⁵United States Atomic Energy Commission, Statement of Commissioner Thomas E. Murray at Hearings before the Joint Committee on Atomic Energy on January 31, 1957.

⁴⁰⁶WT-1682 (EX), p. 63.

⁴⁰⁷Draft Study dated 28 January 1957, Subject: Taongi Atoll, Headquarters, TG 7.1, JTF 7, Los Alamos, New Mexico, p. 3; Minutes of a Meeting of the Informal Eniwetok Planning board on 31 January 1957 attached to letter dated 8 February 1957 to K. F. Hertford, Manager, Albuquerque Operations, USAEC, from Alvin C. Graves, LASL; letter dated 14 February 1957 to RAdm. B. Hall Hanlon, Commander Joint Task Force Seven, from Gaelen L. Felt.

In actual operations, a warhead would be thrown a long distance from a surface ship which would then make a fast escape from the blast area. It was not yet known whether blast or radiation from water waves and spray would be the deciding factor in establishing a safe escape distance for ship-delivered underwater nuclear weapons.⁴⁰⁸ This test became the HARDTACK Wahoo shot.

The AEC provided at this time new and higher total and fission yield estimates for the proposed new "clean" Class "A" and "B" weapons.⁴⁰⁹

At the beginning of February, a 25-shot HARDTACK program was proposed, including five DOD effects shots and 20 weapon development tests divided evenly between LASL and UCRL. Of the latter, five would be barge shots with expected yields in excess of five megatons; five would be barge shots between one and five megatons; five others, yielding between 20 KT to one megaton, would include two barge and three tower shots; and five additional tower shots were expected to yield less than 20 KT. Shots in excess of five megatons would be fired at Bikini.⁴¹⁰ The series was to begin about May 1, 1958 and last for three to four months.⁴¹¹

The development of "clean" U.S. thermonuclear warheads continued apace in early 1957. During the first week of February, the AEC Director's of Military Application reported on the results of a review of the new Class "A," "B," and "C" weapons program with the two weapons laboratories.

In order to meet specifications for higher yields, Livermore would require a time extension for their developmental work. A Class "B" weapon, then being developed as a two-stage device, would be scheduled for availability in November 1959

⁴⁰⁸Memorandum dated 30 December 1955 to Scientific Department Division and Branch Heads from Eugene P. Cooper, Associate Scientific Director, U.S. Naval Radiological Defense Laboratory, San Francisco, Subject: Information and Guidance on Planning for Field Operations 20 and 22.

⁴⁰⁹Letter dated January 31, 1957 to Honorable Reuben B. Robertson, Jr., Deputy Secretary of Defense, from R. W. Cook, Deputy General Manager, USAEC.

⁴¹⁰Memo dated 4 February 1957 to A. C. Graves, Chairman Designate, EPG Coordinating Council, from G. L. Felt, Commander, Task Group 7.1, Subject: Staff Study of the Use of Taongi Atoll.

⁴¹¹Teletype dated 25 February 1957 to Commander Hawaiian Sea Frontier from Commander in Chief, Pacific Fleet.

rather than February 1959, and the Class "A" weapon availability schedule would be changed from June 1959 to March 1960. However, the operational availability date for the LASL Class "C" weapon would remain unchanged.

General Starbird then reviewed a chart showing the long-range weapons schedule based on DOD priorities, with the new ICBM warhead at the top of the list. New Class "A" and "B" weapons were still in the study stage, but Class "B" weapons would go into development as soon as the AEC received instructions as to which of two types to design. The clean MK 21 was under development and was slated for stockpile entry in April 1958.

During the labs' review of the new Class "A," "B," and "C" weapons programs, they had attempted to determine if the operational dates of their "clean" versions could be advanced by disregarding the development of conventional thermonuclear weapons, while still observing DOD priorities. LASL and UCRL had concluded that no time would be saved by doing this.

On the other hand, if conventional weapons development and all high-priority projects except the ICBM warhead were disregarded, the New Class "A" weapon OAD could be advanced from March through July 1960 to November 1959; the Class "B" OAD changed from November 1959 to April 1959; and the Class "C" OAD from August 1959 to January 1959.

Various factors would have to be considered in meeting this schedule, such as the use in some cases of lower-yield designs; the assignment of high priority to nuclear testing, production complex upgrades and expansion, and "clean" weapons; a guarantee that no added applications would be required; adequate manpower and facilities; and increased budget.

General Starbird next reviewed the steps to be followed in the study, development, and production of "clean" weapons. The General also noted that in order to meet current promises for the deployment of new weapons, the AEC needed to obtain as soon as possible the DOD's requirements for maximum permissible diameters for the new Class "A," "B," and "C" bombs.

There was also some discussion of the possibility of advancing the OAD of the "clean" MK 21 to some time before April 1958. General Starbird also stated that DOD withdrawal of its requirement for a 60 MT Class "A" weapon would not advance the OADs of other "clean" weapons because the Class "A" was already scheduled to be deployed last.

By now, it was possible to produce a "clean" weapon yielding the same explosive power as the highest-yield existing "conventional" ("dirty") nuclear weapon. The highest-yield "conventional" hydrogen bomb in the U.S. stockpile at this

time was the 10 MT Class "B" MK 36; like the MK 21, the MK 36 would eventually be stockpiled in both "dirty" and "clean" versions.

The acceleration of the "clean" weapons program and the curtailment of production of "conventional" thermonuclear weapons were the most important matters then facing the AEC. The Secretary of Defense was notified at this time of the operational availability dates of new Class "A," "B," and "C" thermonuclear weapons.⁴¹²

Operation HARDTACK was expected to assist materially with the future development of nuclear weapons with greatly reduced fallout, as a follow-up to the 1956 REDWING tests of "clean" warheads.⁴¹³

The proposed 60 MT Class "A" weapon might be ready for stockpiling by July 1960; however, this date was not intended to indicate a decision to actually stockpile the 25,000 lb. bomb.⁴¹⁴ One UCRL design for a Class "A" bomb was expected to yield 25 to 40 MT during a proposed HARDTACK test.⁴¹⁵

Early in February, Operation PILGRIM had been renamed Operation PLUMBBOB⁴¹⁶ and was expected to comprise 24 shots: two DOD effects tests; 13 weapon development tests (six LASL, seven UCRL), two LASL proof tests of the XW-25 and the XW-34, six "exploratory" tests (two LASL, four UCRL), and one UCRL underground testing technique shot.

By this time, four LASL shots, ranging in yield between two to 60 KT, and two UCRL shots of 10 and 40 KT had been deleted from the PLUMBBOB shot schedule. No test would yield more than 80 KT. The series, the largest yet at NTS, was expected to begin in mid-May and continue for four months.⁴¹⁷

⁴¹²Minutes of AEC Meeting No. 1263, February 6, 1957.

⁴¹³Twenty-Third Semiannual Report of the Atomic Energy Commission, January 1958, USAEC, Washington, D.C., p. 277.

⁴¹⁴Minutes of AEC Meeting 1265, February 11, 1957.

⁴¹⁵Memorandum dated 2 March 1957 to Brig. General Alfred D. Starbird, USA, Director of Military Application, USAEC, from K. F. Hertford, Manager, Albuquerque Operations, USAEC.

⁴¹⁶Letter dated January 22, 1957 to James T. Ramey, Executive Director, JCAE, from Brig. Gen. Alfred D. Starbird, USA, Director of Military Application, USAEC.

⁴¹⁷AEC 944/8, memo by John L. McGruder dated February 9, 1957, title: Test Activities for Calendar Year 1957. At the same

The 1957 Nevada test series was also to include four LASL and one UCRL one-point safety tests, and two safety tests of the plutonium contamination hazards resulting from a one-point non-nuclear detonation of an XW-25 and from the burning of a mass of plutonium.

Nuclear materials to be expended during PLUMBBOB included uranium-233, uranium-235, plutonium, tritium, and lithium enriched to 95% in the Li-6 isotope. The requirement for the latter material implied that there would be some multistage thermonuclear fusion yield from at least one PLUMBBOB test device; however, in response to a press query on March 7, 1957, the AEC's Las Vegas office claimed that "so-called 'H-Bombs' are not fired in Nevada."⁴¹⁸

Aspects of the AEC's nuclear weapons production program were also in high gear by now. Production of plutonium, uranium-235, and lithium-6 during the first half of fiscal year 1957 between July 1956 and January 1957 was at its highest annual rate to date and exceeded amounts forecasted for that period.

Plutonium production set a new record, with new highs in both reactor power levels and separation plant output: the amount of plutonium separated from irradiated uranium during the first half of FY 1957 was 67% greater than that of the corresponding period in the previous fiscal year. Five reactors were in operation at Hanford and several others at Savannah River.

Uranium-235 production was up 65% over the same six-month period in the previous fiscal year.

Production of tritium lagged, due to operational difficulties and the learning curve associated with the pilot plant at the Savannah River facility. Tritium output was only 37% of that forecast for calendar year 1956. Because of this, future

time that the AEC was planning its largest Nevada series, it was concerned about public opinion that the U.S. was intensifying its weapons development program (which was, in fact, exactly what the AEC was doing). (Minutes of AEC Meeting 1265, February 11, 1957.)

⁴¹⁸Atomic Energy Commission, Proposed Test Plans for CY 1957, Report to the General Manager by the Director of Military Application, February 8, 1957; "Note to Editors and Correspondents," Office of the Branch Chief, Atomic Energy Commission, Las Vegas Branch, Las Vegas, Nevada, March 7, 1957. In a February 21, 1957 letter to President Eisenhower requesting approval for the expenditure of "special nuclear materials," there was no mention of either tritium or lithium.

tritium production estimates were reduced.

Tritium to be produced through 1959 was earmarked for use in gas-boosted thermonuclear weapon primaries and in boosted implosion fission weapons.

Full-power operation of the "colex" plants resulted in lithium-6 separation 12% higher than its forecasted rate for the last half of calendar year 1956.⁴¹⁹

At this time, new weapon development programs included the TX-36Y2-X2, TX-43, XW-44, and XW-45; reduced yield versions of the XW-39 and the XW-28 were also underway, in accordance with a DOD request for lower-yield missile warheads.

A number of feasibility studies were also in progress:

Warhead for the Navy SPARROW III air-to-air guided missile

Warhead for a submarine-launched torpedo (Author's note: this became the W-34 for the ASTOR)

Warhead for the Navy's CORVUS air-to-surface radar-homing guided missile (Author's note: this became the W-40)

Warhead for the Army's HAWK surface-to-air missile (Author's note: this became the W-42)

New Class "A," "B," and "C" weapons

Development programs for the XW-15 and XW-21 warheads had been canceled; the XW-39-X1, which was lighter and shorter than the XW-15, was being developed, and the New Class "B" warhead was to replace the XW-21.

At Livermore, California, the establishment and expansion of the new Sandia lab to serve nearby UCRL continued, as did the construction of a new plutonium metallurgy lab at UCRL. The new Rocky Flats plant near Denver, Colorado was 83% completed; it was scheduled to begin fabricating uranium-plutonium pits by June 1957.

Construction of a tritium-deuterium gas packaging plant at Savannah River had begun in May 1956 and was now 39% complete

⁴¹⁹Atomic Energy Commission, Program Status Report to the Joint Committee on Atomic Energy, December 31, 1956, Part II - Special Nuclear Materials, February 8, 1957, pp. 1 - 6, 9.

and scheduled to begin operation by November 1957.⁴²⁰

During an AEC meeting on February 11, LASL director Norris Bradbury described the 10 LASL devices slated for test during PLUMBBOB. These systems fell into two categories: one group of experiments was directed toward advancing the nuclear "state of the art" while a second group comprised specific devices required by DOD and currently either scheduled, or being scheduled, for future production.

In answer to a question during the meeting, Bradbury said that there had been no specific effort at LASL to develop "clean" weapons for test during PLUMBBOB. He pointed out that this phase of weapons development was difficult to pursue except in a full-scale thermonuclear weapons test series which could only be conducted in the Pacific.

However, Bradbury also noted that certain PLUMBBOB tests would contribute to the development of an improved thermonuclear burning system and would thus assist in the design of "clean" weapons.

At this meeting, Dr. Herbert York of UCRL reviewed the 12 UCRL devices to be tested during PLUMBBOB. Five of the 12 were applicable to the development of small nuclear weapons.

Six PLUMBBOB shots had been canceled for a variety of reasons. DOD was no longer interested in one device for which a mockup had been designed. Three other device tests were postponed from PLUMBBOB to HARDTACK. One high-yield shot that was canceled was to have been a repeat of an earlier test, and it was unnecessary to test the device twice. The AEC and UCRL had also concluded that the small weapons test program, as now scheduled for PLUMBBOB, would provide adequate information without testing a 10 KT device originally scheduled.⁴²¹

In the middle of February, the DOD planned to use Army REDSTONE missiles for both the HARDTACK very-high altitude shots. The REDSTONE was scheduled for operational availability in July 1958. The missiles earmarked for HARDTACK were part of the final REDSTONE development program and would double as proof tests of the complete REDSTONE/TX-39 missile-warhead weapon system.⁴²² These rockets were

⁴²⁰Atomic Energy Commission, Program Status Report to the Joint Committee on Atomic Energy, December 31, 1956, Part III - Weapons, February 11, 1957, pp. 4, 5, 8, 9.

⁴²¹Minutes of AEC Meeting 1265, February 11, 1957.

⁴²²Atomic Energy Commission, Very High Altitude Test Shot,

available by this time.⁴²³

At this time, the AEC was advised of how developmental guidance for nuclear weapons was generated within DOD, and why guidance was sometimes difficult to obtain:

The basic reason is the same as that which required a World War and an independent Air Force for full recognition of airplane potentialities. Neither (the) Army nor Navy fully accepted atomic weapons until recently. You are aware of many of the reasons. No service yet fully appreciates the limitless horizons of the atomic scientist, his always accelerating progress, and his growing need for guidance. This is the greatest obstacle now blocking useful long-range guidance from the military.

Since Army and Navy officers only recently began to think seriously of atomic weapons as revolutionary and practicable tools, and have yet to recognize their full potentialities, little thought has been applied to future requirements. For example: SACEUR (Author's note: Supreme Allied Commander, Europe) has never submitted a developmental requirement, and no estimate of desired types or numbers until as recently as May 1955.

This is typical of all commanders, and is a reversal of the procedure by which conventional requirements are generated. It is small wonder that the Pentagon has found it hard to fill the vacuum.

With the existence of such a vacuum it is natural that the one service and the one command which could visualize future requirements (the Air Force and SAC) should channel all development toward the needs of simple air concept -- the largest possible weapon compatible with big bombers. Only strenuous efforts by a few Navy "experts,"

Report to the General Manager by the Director of Military Application, February 18, 1957. The REDSTONE became operational in June 1958.

⁴²³Minutes of Executive Session of AEC Meeting No. 1266, February 20, 1957.

supported by even fewer from the Army, diverted some effort toward putting this yield into smaller packages.

The obstacles facing any developmental guidance between conception and expression are many. In the recent atmosphere of Army and Navy indifference, the originator of any new concept has had difficulty selling it to his own service. Once sold within one service, it must be further sold to the others, and with full agreement on priority of development.

Under existing differences of concept and competition for the stockpile, DOD approval of any guidance is difficult to obtain. I do not believe you are interested in the details by which agreement is attained since the procedures do not affect the answer as they do in the generation of yield requirements.

The situation is improving. The Air Force is regaining a measure of control over SAC, or will shortly. The Army has reorganized and consolidated its atomic weapon and missile responsibilities under one highly competent young general with real authority and top-notch subordinates.

The JCS atomic planning team (Rainbow Team) has assured me that they will inject some long-range "Rube Goldberg" thinking into their next developmental guidance letter to the AEC. They have invited General Starbird to address the JCS planners on the need for DOD guidance. Some time will probably elapse before the AEC feels the change, but it is coming.

In pushing my friends in the Pentagon for long-range guidance, I have tried to highlight the advantages and the potentialities that lie in a "universal" weapon, adaptable to bomb, rocket or missile, [deleted] and of moderate weight and size. I believe we should head in this direction, recognizing that it is a far-off goal whose possibility even the scientist will probably deny at this time.

The scientist who must convert estimates to

realities has always underestimated, and will probably continue to do so. If such a "universal" weapon could be developed, it should replace a large portion of our present family which has already grown too large as we strive for flexibility.

A "universal" weapon would cut required numbers and would be cheap in the long run, even if its nuclear efficiency and yield should be considerably less than optimum. I mention the concept only to emphasize that some goal of this nature should be established as a long-range objective.

There are many other stimulating possibilities which should be drawing military attention, such as; "shaped" atomic explosions; bursts which maximize neutron, x-ray, or other radiation; extremely high altitude bursts; and others which are at the moment only concepts in scientific minds.⁴²⁴

On February 20, the AEC approved the inclusion in HARDTACK of two high-altitude megaton-yield nuclear detonations. These tests were to be conducted no later than May 1, 1958, before other scheduled HARDTACK tests. These two shots would supplement another high-altitude effects test planned by AFSWP. All these shots would provide information on the U.S.'s capability to detect high-altitude nuclear detonations at long distances, and provide effects data required for the design of anti-missile air defense warheads.⁴²⁵ The AEC attached very great importance to the high-altitude tests.

The HARDTACK Shot Schedule: LASL vs UCRL

Later in February, Norris Bradbury wrote to General Starbird and complained about the allocation of HARDTACK shots between UCRL and LASL. The LASL director thought that UCRL was being given too much latitude in their assigned test slots,

⁴²⁴Letter dated February 15, 1957 to Lewis L. Strauss from J. H. Morse, Jr. Emphasis in original. The closest thing to a "universal" warhead in stockpile at this time was the W-7 warhead, which had been adapted to an aerial bomb, an atomic demolition munition, two surface-to-surface missiles, an anti-submarine depth charge, an air-to-surface rocket, and a surface-to-air missile.

⁴²⁵Letter dated 8 March 1957 to Honorable Charles E. Wilson, Secretary of Defense, from Lewis L. Strauss, Chairman, USAEC.

especially those that did not have an immediate bearing on current DOD requirements.

The letter highlighted the tightening bind that the weapons labs were being put into: with public and Presidential desire to limit fallout from testing, each shot was becoming increasingly precious, and tests of new developments that might be applicable to future weapons competed directly with tests of devices soon to enter stockpile:

I have been giving the most serious thought to your teletype to (UCRL director) Herb York and myself dated February 13, 1957 on the general subject of HARDTACK, and there is one point on which I hope very much to be able to persuade you take a somewhat different philosophy. This point has to do with your assumption that out of (the total number of HARDTACK) shots, (half) go to LASL and (half) go to UCRL.

If you examine the list of specific weapons due for stockpiling between HARDTACK and 1960, six are assigned to Los Alamos and three to Livermore (Author's note: among these were the LASL MK/W-28, W-30, W-31, W-33, W-34, MK 39, W-40, and MK 46; and the UCRL MK/W-27, MK/W-41 and W-45). If the (HARDTACK) shot program is evenly divided, then of necessity, our experiments will have to be primarily devoted to these immediate and definite responsibilities.

In contrast, UCRL has considerable freedom to engage in a wide variety of exploratory experiments because their programmatic commitments are presently small.

We are already suffering to some extent from this in PLUMBBOB. As I am sure you are aware, there is a vast difference in the type of experiment proposed therefor by LASL and that proposed by UCRL. Ours, with one or two exceptions are "nose-to-grindstone" programmatic experiments related to things we have to do with specific weapon designs; they are having fun.

[deleted] A preliminary examination of their program (in a teletype dated 1 February from York to Starbird) shows that only a fraction of their 13 to 16 shots have any relevance to their programmatic assignments. In contrast, a first look at

what LASL might do to comply with your message referred to above shows approximately 10 shots are required for our programmatic commitments alone!

The suggestion which I wish to make for your consideration is this: If the total number of (HARDTACK) shots is to be fixed (and I do not disagree with your number), then it should not be divided on an even basis because there are two laboratories, but divided much more like the ratio of programmatic assignments to the laboratories in the two-year period following (HARDTACK).

In other words, if you regard 20 shots as appropriate to the general state of affairs in 1958, I believe that they should be divided (between the labs) much more like 13:7. Otherwise, it seems to me, you take a great chance of seriously curbing exploratory effort at the laboratory with the larger programmatic commitment.

The policy of assigning equal numbers of shots would lead eventually to LASL having no programmatic commitments and thus free to shoot fairly freely.

Conversely, UCRL would be complaining that their programmatic commitments were making it impossible for them to do weapon research shots!

I believe that we should try to avoid such wide swings of the pendulum, and I also believe that UCRL should eventually acquire more specific responsibilities. It is for this reason that I am hanging back on POLARIS and letting UCRL make what promises they wish without LASL complaint. (Author's note: a reference to the UCRL W-47 project; see Volume VII.)

Bradbury also argued that the AEC-imposed fission product limit of HARDTACK, now about nine megatons for each lab, placed unfair constraints on LASL's HARDTACK tests. As if the shot limit were not burden enough, the two labs were also increasingly constrained by how much radioactive contaminants they could inject into the atmosphere:

With regard to your proposed (fission

product) megaton limit (about 9 MT), I also see no particular objection although it has some interesting implications. Presumably it also should be rationed out on roughly the basis of programmatic need.

However, it is small enough so that (tests of) the dirty (Class A Prime bomb) and dirty (Class B Prime bomb) are automatically excluded and therefore must be tested in clean versions and so require only that proportion of the available fission products.

The Class C Prime is in a completely confused state because of the diameter question: At 30" there is no clean version, and probably only a fully-dirty test which would use up about 4 MT of the allowable fission products makes sense; at 35," clean and dirty versions may exist but the prediction of the behavior of the dirty one by shooting the clean (version) will be difficult; and the behavior of the dirty (version) estimated to ± 1 MT.

Thus what the C Prime will need in the way of fission products depends largely on what its diameter is to be and ranges from about 4 MT down to 1/4 MT. [deleted] The remainder of the programmatic devices add only trivial amounts of atmospheric contamination compared to these.

The LASL director pleaded for the allocation of the majority of the HARDTACK tests to the New Mexico lab:

Thus I am led to the conclusion that, not only must the majority of the tests be assigned to LASL but the majority of the (atmospheric) contamination will be required by us as well -- since the dirty A Prime and the B Prime simply cannot get into the picture.

I recognize the difficult position which this puts you in, but I am afraid that a 50:50 distribution of either shots or fission (products) megatons is just not right and that an alternative approach is necessary. I am also afraid that there is no way to reconcile the basic differences in test philosophy between the LASL and

Livermore except for Washington to be arbitrary.

Shortly, in a more formal document, we will describe the general LASL HARDTACK program as seen at this early (actually fairly late) date. I must admit that it will be based on the assumption that we must fulfill our programmatic responsibilities and that we must have some room to explore. Needless to say, we are not likely to shoot for the fun of it; tests are hard enough to design and shoot. However, we don't want to find ourselves in 1960 out of ideas and just redesigning the optimized (Mark) 7 series for the nth time!⁴²⁶

At the end of February, a JCS paper that discussed the 60 MT Class "A" bomb and specific "clean" weapons called for canceling the "clean" TX-21 program.⁴²⁷

The AEC continued to seek guidance from DOD with regard to the latter's requirements for "clean" weapons. During an AEC-MLC conference at the end of February, AEC Chairman Strauss noted that there was an urgent need for a DOD statement concerning its requirements for "clean" weapons, especially the TX-21.

The AEC had informed DOD of its capability to develop low-fission warheads, and the "clean" TX-21 program had reached the point where a decision had to be made regarding its future. In the absence of a DOD requirement for the TX-21, the AEC would continue its development.

Although the Joint Chiefs were studying the likely future need for "clean" weapons, it was not now known when a decision might be reached.

The requirement for a new Class "A" weapon was also discussed at this meeting. Although a March 1956 DOD statement of weapons requirements had indicated a modest need for the new Class "A" in 1959, an October 1956 DOD statement had not included the weapon. At the meeting, Air Force Major General John S. Mills said that requirements for the new weapon were

⁴²⁶Letter dated February 20, 1957 to Brig. General A. D. Starbird, Director of Military Application, USAEC, from N. E. Bradbury, Director, LASL.

⁴²⁷Note to Lewis Strauss dated February 28, 1957 from J. H. Morse, Jr.; "The Military Requirement for a New Class A (60 Megaton) Weapon," April 1, 1957.

part of a larger question for which DOD had not yet formulated a policy.

Mills added that at present, DOD did not have a requirement for the new Class "A" and Strauss agreed that after a careful review of correspondence on the subject, he had also been unable to find any present DOD requirement for the new Class "A."⁴²⁸

At the beginning of March 1957, 31 shots were still being planned for HARDTACK, with DOD effects tests at both Bikini and Eniwetok, the two underwater shots at Eniwetok, the high-altitude shots at Bikini, most UCRL shots at Bikini, and all LASL shots at Eniwetok. At least half of the shots were expected to have yields in the megaton range.⁴²⁹

Early in March, the United Kingdom invited five official U.S. scientific and military observers to witness an airdrop/airburst test of a British thermonuclear weapon shortly after May 10 during Operation GRAPPLE near Christmas Island, a British possession south of Hawaii.⁴³⁰ This was yet another step in the formation of a U.S.-U.K. weapons development and test symbiosis, which would eventually benefit both countries.

Consideration of an Alternate Test Site

As the slate of HARDTACK tests grew in early 1957, the AEC realized that the new series would be significantly larger than REDWING and that it would be impossible to fire the increased number of shots without full dual-atoll capability, which had been demonstrated successfully in 1956.⁴³¹

Some thought was even given to establishing an additional UCRL test site at uninhabited Taongi atoll, 285 miles

⁴²⁸Minutes of 109th AEC-MLC Conference, February 28, 1957.

⁴²⁹Memorandum for Record dated 4 March 1957, Subject: Conference on Support Requirements for Operation HARDTACK.

⁴³⁰Letters dated March 6, March 14, and April 3, 1957 to Carl T. Durham, Chairman, JCAE, from Herbert B. Loper, Assistant to the Secretary of Defense (Atomic Energy). U.S. observers at Operation GRAPPLE included both the Los Alamos and Livermore nuclear weapons lab directors, and a handful of DOD representatives.

⁴³¹Letter dated March 27, 1957, from Brig. Gen. Alfred E. Starbird, Director of Military Application, USAEC, to Chief of Staff, USAF, Washington, D.C., Subject: Cloud Sampling at HARDTACK.

northeast of Bikini atoll; a feasibility study had started in the fall of 1956.⁴³² By mid-February 1957, UCRL had assembled a lengthy and detailed plan for testing at Taongi atoll.

As early as November 1954, in the aftermath of CASTLE, UCRL had asked the AEC to survey possible alternates to Eniwetok and Bikini; the criteria which had to be met included a site located on U.S. territory suitable for tests of high-yield devices and from which fallout should land on either U.S. territory or on "uninhabited and untraveled areas of the world."

Although part of the territory of Alaska north of the Brooks (mountain) Range and east of Pt. Barrow appeared "interesting," the territory on the whole did not look very promising, to say nothing of the weather and forbidding climatic and geographic conditions.⁴³³

About a month later, the AEC stated that although it was "premature and inadvisable" for UCRL to consider the matter of an alternate test site at this time, the AEC intended to pursue the subject on an informal basis with a view toward initiating an intensive study of the problem should it become necessary to do so in the future.⁴³⁴

In the spring of 1955, the AEC had considered several alternate sites in lieu of the PPG. In the Atlantic, the islands of Tristan de Cunha, Gough, and Bouvet were examined; however, all were controlled by foreign governments and did not offer either favorable topographic or climactic conditions.

Bouvet Island was in the South Atlantic Ocean 1,500 miles southwest of South Africa, almost 4,000 miles directly east of Tierra del Fuego in South America, and was owned by Norway. It was very precipitous and offered no good landing

⁴³²Letter dated November 5, 1956 to RAdm. Byron H. Hanlon, Commander Joint Task Force Seven, from G. W. Johnson, Test Division Leader, LASL; Memo to Distribution dated December 27, 1956 from Gaalen L. Felt, HQ JTF 7, Subject: Taongi Concepts; transcript dated 16 January 1957 of conference held on 3 January 1957, subject: UCRL's Proposal on Taongi.

⁴³³Letter dated November 9, 1954 to Brig. General Kenneth E. Fields, Division of Military Application, USAEC, from Herbert F. York, Director, UCRL-Livermore.

⁴³⁴Letter dated December 1, 1954 to H. A. Fidler, Manager, San Francisco Operations Office, USAEC, from Brig. General Kenneth E. Fields, Director, Division of Military Application, USAEC.

beaches. Gough, owned by the British colony of St. Helena, was a volcanic island located in the South Atlantic 230 miles south-southeast of Tristan da Cunha island, about 2,00 miles southwest of the Cape of Good Hope at the southern tip of Africa.

In the Pacific, Lisianski Island in the Hawaiian archipelago, Malden Island, Starbuck Island, and Ducie Island were considered as possibilities. Lisianski was unsuitable for high yield test work because of the potential threat to the other populated Hawaiian islands.

Ducie was eliminated because of unfavorable topography and because it was located very close, only 300 miles, to the United Kingdom's Pitcairn Island (where the HMS BOUNTY mutineers ended up in 1790) in the South Pacific about midway between Australia and northern Chile in South America. Malden and Starbuck were also under British dominion and Malden appeared to be a usable site, although it had many disadvantages compared to the Eniwetok-Bikini area.

Malden and Starbuck were part of the Line Island group about 1,500 miles due south of the Hawaiian chain. The two uninhabited islands were 100 miles apart, between 465 and 540 miles southeast of Christmas Island. Starbuck was a white coral sand island rising only 15 feet at its highest point and about 11 square miles in area. Malden was triangular in shape, rising to a maximum elevation of 30 feet, and about 10 square miles in area. Neither island had extensive boat landing beaches or piers.

The advantages of Starbuck and Malden were the lower nearby human populations, compared to the Marshall Island sites; wind conditions favorable to the lessening of fallout problems; and shorter distances to both Hawaii and the U.S. The chief disadvantages of the area were that it was under British control; had no permanent dwelling, landing, or research facilities; had no nearby weather observing sites on any nearby islands; and Malden lagoon was probably not suitable for barge shots.⁴³⁵

Even Australia was studied as a potential U.S. nuclear weapons tests site; Australia had been the location of a dozen tests of British fission weapons between 1952 and 1957.⁴³⁶ Another solution to the problem was to conduct all

⁴³⁵Atomic Energy Commission, Alternate Test Sites in Lieu of Pacific Proving Ground, Note by the Secretary, May 9, 1955, with attachment, letter dated May 3, 1955 to Chairman, USAEC, from RAdm. C. B. Momsen, USN, Commander, JTF 7.

⁴³⁶"Foreign Nuclear Detonations -- Through December 31, 1986,"

future tests in the open ocean, via airdrops or from shipboard, with effect and sampling data measurements and samples collected by with aircraft, balloons, rockets, buoys, and ships.

Despite these studies, the AEC chose to continue testing the Marshalls until 1959, when the islands would revert to DOD control, forcing the AEC to look elsewhere for test sites in the Pacific.

The full UCRL HARDTACK program was to be 14 shots; of these, five were barge shots with yields ranging from 500 KT to two megatons. One of the proposed tests was to be of a new Class "A" weapon to be fired at full yield. Nine devices were two-stage thermonuclear systems. A total of six tests would be conducted at Taongi, with the remainder at Bikini.

Taongi atoll offered one good advantage over Bikini. The nearest inhabited land west of Taongi was the island of Guam in the Marianas, some 1,300 miles away. There was no populated area to the east for over 1,500 miles; to the north-northwest, Wake Island was 324 miles distant. The nearest inhabitation was at Utirik atoll in the Marshalls, 215 miles to the south-southeast. Because of the unique position of Taongi, northwest of Bikini, the usable firing sector for fallout purposes was much greater than those at Bikini or Eniwetok.

Prevailing wind directions and velocities at Taongi were also more compatible with prolonged testing than were those in the southern Marshalls. Taongi had less rainfall and shower activity than did either Bikini or Eniwetok,⁴³⁷ and there was a better probability of good weather at Taongi during the months of March and April than at Eniwetok or Bikini.⁴³⁸

Late in March, the launch of the two REDSTONE missiles to be used for the high-altitude HARDTACK tests was proposed for Taongi, to lessen chances of a faulty warhead-bearing missile landing on Bikini or Eniwetok or another inhabited atoll.⁴³⁹

U.S. Department of Energy, Washington, D.C. 1987.

⁴³⁷Proposal: It is Proposed that Taongi (Pokaakku) Atoll, Marshall Islands, Be Utilized as a Nuclear Test Site during Operation HARDTACK, 1958, attachment to cover letter dated 19 February 1957 signed by Walter D. Gibbins, UCRL; memorandum to Joe B. Sanders, Test Division, AL00, dated 28 February 1957 from Gaalen L. Felt, CTG 7.1, subject: Comments by Campbell on COPAA-313.

⁴³⁸Teletype dated March 26, 1957 to Dr. Frank Shelton, UCRL, from JTF 7, Pearl Harbor, Hawaii.

⁴³⁹Letter dated March 22, 1957 to Alvin C. Graves, LASL, from

The REDSTONE had never been fired with a nuclear warhead, and the Army's Ballistic Missile Agency desired that such a test be conducted as soon as possible.⁴⁴⁰

As the number of HARDTACK shots increased, so did the total yield of the proposed tests. The 16-megaton total fission yield then proposed for all the HARDTACK shots would be nearly twice that of REDWING; the total fission yield for all REDWING shots was only nine megatons. The two proposed high-altitude shots by themselves would total nearly eight megatons fusion plus fission yield; 4.5 MT of this was fission. On this basis alone, HARDTACK would produce much more worldwide atmospheric radioactive fallout than REDWING.⁴⁴¹

The two high altitude shots approved by the Joint Chiefs included the "ultra high altitude" (UHA) test at 250,000 feet and the "very high altitude prime" (VHA Prime) shot at 100,000 feet. The rockets used for these shots were to be launched from Bikini island and controlled from Enyu island at Bikini atoll; the planned trajectory was to be from west to east and would be confined entirely to the area of Bikini lagoon, with the expended missile stages landing in the water of the lagoon.

Instrumented pods would be ejected from the missile propulsion sections. The missile warhead for the UHA test would be fired during its ascent, and the VHA Prime warhead would be fired during its descent. The UHA pods would be ejected at altitudes of 60,000, 80,000, 150,000 and 220,000 feet; VHA Prime pods would be ejected at 40,000, 50,000, 60,000 and 70,000 feet. Data received by the pods would be telemetered to ground stations for recording.

Each pod was to be about 40 inches long and 10 inches in diameter, with a weight of 150 lbs., of which 70 to 120 lbs. would be instruments. The pods would be attached to the rear portion of the REDSTONE booster unit and would be ejected by small rockets.

The Chrysler Corporation, the REDSTONE manufacturer, would

Brig. Gen. Alfred D. Starbird, USA, Director of Military Application, USAEC.

⁴⁴⁰Memorandum for Chief of Staff dated March 25, 1957 from Col. H. R. Fleming, USA, subject: Staff Visit Report.

⁴⁴¹Memorandum dated April 5, 1957 to Brig. Gen. Alfred D. Starbird, Director, Division of Military Application, USAEC, from Charles L. Dunham, M.D., Director, Division of Biology and Medicine, USAEC, Subject: Comments on Draft Staff Paper "Test Activities for Calendar Year 1958."

provide three missiles between February and April 1958 with the first firing scheduled for May 1958. The UHA shot would be fired first.⁴⁴²

Late in March, the AEC noted that "clean" versions of high-yield thermonuclear weapons other than the TX-21 would not be available for at least three years. The "clean" TX-21C, which might still be ready for stockpiling early in 1958, could provide DOD with an "emergency capability" in "clean" weapons. However, since the JCS had thoroughly reviewed current requirements for a "clean" TX-21, it was unlikely that DOD would reverse its earlier decision to postpone deployment of the TX-21C.⁴⁴³

One shot proposed for PLUMBBOB was to be the U.S.'s first deep underground test. Although shallow subsurface tests had been conducted in Nevada during test series in 1951 and 1955, there had never before been a test specifically designed to capture all fission products underground. Another objective of the deep underground shot was to determine how easily clandestine tests might be detected at long range.

At the end of March, the underground PLUMBBOB shot, then designated Pluto, was renamed Rainier. A hydronuclear test named Venus was also added to the schedule.⁴⁴⁴

By now, PLUMBBOB was up to 28 shots spanning five months. The shots and their associated ready dates were: (Project) 57 (12 April 1957); Boltzmann and Franklin (15 May); Saturn (1 June); Lassen (2 June); Wilson (8 June); Priscilla (15 June); Diablo (25 June); Hood (27 June); Venus, Coulomb A, Pascal A, and Laplace (1 July); Owens (5 July); Shasta (9 July); Kepler and Pascal B (15 July); John (19 July); Newton (25 July); Huygens (1 August); Whitney (8 August); Smoky (19 August); Everest (26 August); Galileo and Coulomb B (1 September); Rainier (3 September); Fizeau (8 September); and Morgan (10 September).⁴⁴⁵

⁴⁴²Memorandum for Chief of Staff dated March 25, 1957 from Col. H. R. Fleming, USA, subject: Staff Visit Report.

⁴⁴³Minutes of the 110th AEC-MLC Conference, March 28, 1957.

⁴⁴⁴"DMA Comments on Operation Order 1-57, Operation PLUMBBOB," March 26, 1957, attached to memorandum dated March 28, 1957 to James E. Reeves, Test Manager, Nevada Test Organization, from Brig. General Alfred D. Starbird, USA, Director of Military Application, USAEC, subject: Test Manager's Operation Order 1-57 Operation PLUMBBOB.

⁴⁴⁵Test Director's Operation Plan, Operation PLUMBBOB, April 1, 1957, p. 8.

The AEC had received by this time a military requirement for a new Class "A" bomb with a yield up to 60 megatons.⁴⁴⁶

At the beginning of April, the weapons labs revealed in more detail their plans for HARDTACK. LASL proposed two prototype tests to meet anticipated requirements for new Class "C" weapons in both conventional ("dirty") and "clean" configurations with estimated Operational Availability Dates of August 1959.

The Los Alamos lab also planned to conduct at least two shots of a new weapon to meet a DOD ICBM warhead requirement, with an estimated war reserve operational availability date of December 1959 and an emergency capability date in October 1959.

LASL also wished to fire one or two shots to provide DOD with a tactical "laydown" bomb with an operational availability date falling in a period between December 1959 and February 1960.

Three LASL exploratory tests would also be conducted, aimed in part at developing methods of obtaining medium-yield thermonuclear warheads. LASL also wished to test a new primary for the Class "C" warheads, and to fire possible candidate devices that might meet other DOD requirements for weapons with operational availability dates of July and October 1958. LASL proposed a total of 14 shots for HARDTACK.

UCRL planned to fire a number of devices for a number of reasons. One test was to satisfy possible DOD requirements for a weapon of a particular yield range; an operational availability date in the March-July 1960 period could be met if the test were successful and if DOD approved.

An exploratory UCRL shot would investigate the possibility of obtaining "clean" high yield, lightweight weapons, while another shot was intended to meet DOD requirements for conventional ("dirty") and "clean" bombs in a specified weight and yield class with an operational availability date of November 1959.

UCRL's proposed exploratory tests at this time included shots intended to lower warhead weight to 600 to 700 lbs. while still obtaining high yield (for the POLARIS warhead); two devices with possible application to the NIKE ZEUS warhead program; another device which might be useful as a defense against ICBM warheads; a warhead for defensive use against

⁴⁴⁶"The Military Requirement for a New Class A (60 Megaton) Weapon," April 1, 1957.

enemy bombers; a device for possible peaceful application; and five exploratory/developmental tests of devices with possible application to all weapons which required warheads with their particular design characteristics. UCRL's planned shot total came to 17.⁴⁴⁷

Early in April, to eliminate confusion, the AEC standardized its thermonuclear weapons designations. A "clean" weapon was one designed to either minimize or greatly reduce the radioactivity of the particulate debris resulting from its detonation. A "conventional" weapon, on the other hand, was one which was designed without special efforts to reduce or minimize fission products or other radiation resulting from its detonation.

At the other extreme was a "salted" weapon, a thermonuclear device which was designed to produce extensive fallout and radiation by the neutron activation of selected isotopes deliberately introduced into the weapon for this purpose. Because of its negative connotation, especially from a publicity standpoint, the AEC wanted to omit entirely the word "dirty" from its weapons lexicon.⁴⁴⁸

During April, the launch site for the high-altitude HARDTACK rockets was moved to Johnston Island, 725 miles west-southwest of Hawaii:

The Secretary of Defense and the Chairman of the Atomic Energy Commission announced today that as a part of Operation HARDTACK, certain tactical missiles with nuclear warheads will be fired from Johnston Island, an isolated land mass in the Central Pacific Ocean. These firings are necessary for the development of suitable weapons for the defense of the United States and the free world.

The missiles to be fired will be of limited range and must be launched from a land mass. To avoid the possibility of danger to

⁴⁴⁷Memorandum for Chief of Staff dated 1 April 1957 from Lt. Col. Albert D. Epley, USA, Chief, Scientific Test Branch, subject: Tentative Schedule for Operation HARDTACK.

⁴⁴⁸Memorandum for K. E. Fields, General Manager, USAEC, dated March 1, 1957, from J. H. Morse, Jr., USAEC, Subject: Standardized Weapon Terminology; memorandum dated April 5, 1957 to K. E. Fields, General Manager, USAEC, from Brig. Gen. A. D. Starbird, USA, Director of Military Application, USAEC, Subject: Standardized Weapon Terminology.

inhabited areas, it has been decided that these firings should not be conducted from Eniwetok and Bikini atolls which lie within (approximately) 100 miles of inhabited islands.

Johnston Island, located some 650 statute miles southwest of the nearest inhabited island of the Hawaiian chain, and some 900 statute miles from the nearest other inhabited island, is in an area in which sea travel and fishing is extremely light, and is best suited, therefore, for these firings. These firings have not been added to the schedule for the current series, but their planned firings transferred from the Eniwetok Proving Ground to Johnston Island for these reasons.

At an appropriate time, a temporary danger area will be established for as limited a time as possible.⁴⁴⁹

In the middle of April, the Air Force's Air Research and Development Command (ARDC) was expressing a desire for a "clean" weapon with a total yield of 50 to 100 KT with only one to five kilotons of fission yield. Such a weapon, it was believed, would go along way toward solving "the many problem of fallout."

If the total fission yield of this device were limited to only one kiloton, then it could be fired as a surface burst in Nevada and obtain sorely-needed data on fallout from such a burst on dry land.⁴⁵⁰

By the end of April, the combined LASL-UCRL-DOD shot program for Operation HARDTACK remained at 31 tests to be conducted during a three-month period, as specified in January. The laboratory experiments were designed to meet weapons development program requirements and to provide a number of "exploratory" shots to assure continued progress in future development. In order to reduce total fallout, several of the laboratory device tests would be conducted with the lowest possible yields required to obtain necessary design data.

⁴⁴⁹Statement by the Secretary of Defense and Chairman, AEC, undated, but ca. April 1957.

⁴⁵⁰Memorandum dated April 17, 1957 for Dr. Libby, Commissioner, USAEC, from Lt. Col. Norair M. Lulejian, USAF, Chief, Nuclear Applications Divisions, ARDC, subject: Measurement of Local Fallout in Atomic Test Operations.

The DOD program consisted of two high-yield high-altitude W-39 detonations (approved in February) at 250,000 and 100,000 feet; a 92,000-foot test firing of a balloon-lofted prototype low-yield anti-missile warhead; and two underwater shots in target arrays to test antisubmarine warheads.

Taongi atoll was still being considered for the two high-altitude shots even though its use as a test site would involve approval by the U.S. Departments of State and Interior, the closure to shipping of a large area of the central Pacific for security, and notification of the closure to the Security Council of the United Nations.

DOD ultimately opposed use of Taongi because of the cost of establishment and operation of a third Pacific testing site, and because of possible adverse public reaction, i.e., political reaction, to a new test site at a time when public opinion leaned toward test limitations.⁴⁵¹

The 31-shot HARDTACK program, an 80% increase over the 17-shot REDWING program, was estimated to cost \$48 million. The AEC's Division of Military Application approved in June a 27-shot program (HARDTACK finally included 15 LASL shots, 15 UCRL shots, and five DOD shots; the shot schedule was revised three times before January 1958, twice again before April 1958 and ten more times during the test series).⁴⁵²

The AEC was still trying to decrease the fission yield portion, the main source of stratospheric globe-circling radioactive fallout, of the proposed HARDTACK tests.⁴⁵³ A number of congressmen favored the recent suggestion by AEC Commissioner Thomas E. Murray that the U.S. not rely solely on "large scale" thermonuclear bombs and should instead develop and produce numbers of small yield nuclear bombs which could be widely dispersed and used on a selective and limited basis in "brush fire" wars. (The controversy over the best way to use fissionable and fusionable materials had been raging since the earliest days of the U.S. hydrogen bomb

⁴⁵¹Letter dated May 22, 1957 to Lewis L. Strauss, Chairman, USAEC, from Herbert B. Loper, Assistant to the Secretary of Defense (Atomic Energy).

⁴⁵²Atomic Energy Commission, Test Activities for Calendar Year 1958, Report to the General Manager by the Director of Military Application, April 29, 1957, pp. 1, 2, 3, 4.

⁴⁵³Statement by the Atomic Energy Commission, press release dated Wednesday, May 29, 1957, USAEC, Washington, D. C.; press release from the Joint Committee on Atomic Energy, dated May 29, 1957.

development program.)

Major HARDTACK fission yield contributors would be a new Class "C" bomb; the two W-39 high-altitude bursts at about four megatons each; and two new ICBM/IRBM warheads, the XW-38 and XW-49. Most of these shots were believed necessary for weapons development or for the high altitude effects program. Many new "clean" bombs would also be tested. Each weapons lab was limited to an original allocation of 20 shots (20 was chosen as representative of the order of magnitude of REDWING); they were specifically requested to consider reduced-yield versions of their high-yield shots, regardless of loss of sorely-needed development information.⁴⁵⁴

LASL activities would be centered at Eniwetok and UCRL activities at Bikini, thus allowing for dual-atoll capability with maximum flexibility and full duplication of diagnostic and monitoring programs and equipment.

Among the devices considered for testing at this stage of HARDTACK planning was a huge 25-megaton "clean" Class "A" aerial bomb, weighing about 25,000 lbs. and carried aloft by a B-52. This weapon, which had been on the AEC's "wish list" since at least mid-1956,⁴⁵⁵ was to be fired only if a development requirement were established by a special committee of the National Security Council composed of the Chairman of the AEC and the Secretaries of State and Defense. If detonated, it would be the largest-yield nuclear test ever conducted by the U.S. Since there was not a current development requirement for such a device, its schedule for test was entirely tentative.⁴⁵⁶

A meeting of the AEC on May 1, 1957, concluded "that in view of the present climate of world opinion, it would be unwise for the total fission yield of Operation HARDTACK to exceed that of Operation REDWING."⁴⁵⁷ In connection with this, it might be possible to invite foreign observers to attend at least one HARDTACK test of a "clean" device, if a simple way could be found to measure the ratio of fission to fusion yield without revealing weapon design information.

⁴⁵⁴AEC Test Activities for CY 1958, April 29, 1957, pp. 5, 6, 9, 10.

⁴⁵⁵"B. United States Weapon Program," USAEC, undated, but ca. June 18, 1956.

⁴⁵⁶Minutes of AEC Meeting 1280, May 1, 1957.

⁴⁵⁷Minutes of Atomic Energy Commission Meeting No. 1280, 3:30 PM, Wednesday, May 1, 1957.

Writing to AEC Chairman Lewis Strauss, Brigadier General Starbird, Director of the AEC's Division of Military Application, stated on May 3:

I feel that we will have to give immediately before HARDTACK some indication as to how clean our weapons are. I do not believe this should be done until just before HARDTACK. We should use it at that strategic time to help overcome opposition to HARDTACK.

My current thought is that the announcement should not give exact percentages as to how clean a design can be but rather make a statement to the effect that "We have found a way to reduce fission debris from very large shots [deleted] This would be somewhat revealing but it would not let others know how clean the weapon is.

... I do believe there is a method whereby (if mutual observation were agreed upon), we could convince these observers through simple instrumentation that our shots are very clean.⁴⁵⁸

If foreign observers were allowed to view U.S. nuclear tests, they were not to be allowed to learn the classified principles of operation of thermonuclear or fission weapons, stockpile size or production rates, or weapon development schedules. They were also not to see any nuclear device.⁴⁵⁹

Consideration of the use of Taongi atoll for HARDTACK shots continued in May. In terms of fallout danger posed to nearby habitations, Taongi was a safer place than Eniwetok or Bikini on which to fire high-yield detonations. The AEC was very concerned about a repeat of the CASTLE Bravo shot debacle in 1954, and the long distances from Taongi to the Marshalls and the Marianas offered distinct advantages.

⁴⁵⁸Memorandum for Mr. Strauss (Personal) dated May 3, 1957 from Brig. Gen. A. D. Starbird, USA, Director of Military Application, USAEC, Subject: The Lawrence-Teller Paper on Safe Management of Nuclear Tests.

⁴⁵⁹Memorandum dated May 6, 1957 to James E. Reeves, Test Manager, Nevada Test Organization, from Brig. Gen. A. D. Starbird, USA, Director of Military Application, USAEC, Subject: Access to Classified Information for Official Observers.

It also appeared that HARDTACK might be shortened by as much three weeks if Taongi were used in addition to the Marshalls. All high-yield tests would be conducted on barges, with diagnostic information collected from shipboard at a safe distance from the atoll. The use of Taongi would require the cooperation and permission of the Departments of Defense and Interior, as well as presidential concurrence and notification of the Security Council of the United Nations.⁴⁶⁰

Use of Taongi, however, required expensive preparations. To blast a channel 30 feet deep, 300 feet wide, and 500 feet long through the atoll reef would cost \$2.6 million, and the labor of 38 men for a year. Of this cost, one million dollars would be for the conventional high explosives alone (apparently no thought was given to using a nuclear explosive to blast a channel). A smaller channel might be opened more quickly at a lower cost.⁴⁶¹

By mid-May, HARDTACK comprised 27 shots: five DOD tests, 12 LASL tests and 10 UCRL tests. This number was expected to change again before the series began.⁴⁶²

At Eniwetok, LASL planned both "dirty" and "clean" tests of a Class C Prime weapon, and two tests of a "Table Leg" prototype "laydown" weapon.

If Taongi were used for HARDTACK, UCRL planned to fire at least four shots at the atoll, including mock-ups of new Class "A" and Class "B" thermonuclear weapons and a new Class "C" warhead. These would include tests of "clean" Class A and Class D weapons, and two tests of a "clean" Class B weapon. All of these shots would have yields lower than five megatons. By contrast, LASL estimated that shots in the SHRIMP crater at Bikini could go as high as 40 MT without endangering the inhabitants of the firing control bunker on Enyu.

Diagnostics planned for these tests included determination of fission yield by radiochemistry, and shipboard measurements of ball-of-fire yield, interstage time intervals, and primary alphas. There might also be hot-spot measurements of all small-yield weapons.

⁴⁶⁰Letter dated May 10, 1957 to Honorable Herbert B. Loper, Chairman, Military Liaison Committee to the Atomic Energy Commission from K. E. Fields, General Manager, USAEC.

⁴⁶¹Memorandum for Record dated 10 May 1957 from Maj. Gen. A. R. Luedecke, USAF, subject: Use of Taongi.

⁴⁶²Teletype message dated May 17, 1957 to Luedecke from Starbird.

Delivery of test devices to Taongi included loading the device onto a barge at Bikini, transporting the barge aboard an LSD to the lee side of Taongi, and towing the barge from the LSD to a passage in the reef into the lagoon. The movement of the barge into the lagoon would have to be at times of slack tide, when there was little current between the ocean and the lagoon.⁴⁶³

A wider shipping channel would have to be blasted out of the lagoon reef to allow the movement of landing craft. Use of Taongi would reduce fallout danger to the Marshalls, and would also reduce field time for UCRL personnel during HARDTACK. It would probably not reduce the overall HARDTACK operational period.⁴⁶⁴

More Controversy over "Clean" Weapons

Late in May, President Eisenhower wrote to Representative Sterling Cole, then chairman of the JCAE. Eisenhower was replying to an earlier letter to him from Cole which raised several questions about the necessity of continued U.S. nuclear testing.

In his letter to Cole, the President offered a defense of continued testing, citing in particular advances in both "clean" and high-yield weapons. Eisenhower also reiterated his previously-publicized statement that U.S. bombs were already "big enough":

... The whole question of testing of atomic weapons has engaged my concern from the time I took office. I have often talked with Lewis Strauss about it, and knowing my views and sharing them fully, he has held testing to the absolute minimum, ruling out shots which he felt could not be justified on the basis of the results sought. As you know, the fall-out (sic) of the last series in the Pacific was only a fraction of that from the 1954 tests.

On the other hand, as you also know far better than almost any other person I can think of, there has been a great

⁴⁶³Memorandum for Record dated 17 May 1957 from Maj. Gen. A. R. Luedecke, Chief, AFSWP, subject: Use of Taongi Atoll.

⁴⁶⁴Memorandum for Chief of Staff dated March 25, 1957 from Col. H. R. Fleming, USA, subject: Staff Visit Report.

enhancement of our defensive strength through the developments which have come from the test program to date. We have increased the numbers, the deliverability, the efficiency and versatility of our weapons.

Most recently, we have learned that certain of them can be made in such a manner that radioactive fall-out is very greatly minimized. The tactical usefulness of this latter development is known to you.

You are correct in your observation that I feel that our large weapons (incidentally, now susceptible of being the relatively the cleanest in the sense of fission product yield) are large enough and that we know we can make them larger on the same principles if ever there were strategic need for them, but I do not think you are correctly informed that there is any approved plan to test larger weapons than were tested in 1954.

The Atomic Energy Commission at any rate has informed me that it has no such plans. If there is ever to be a test of larger weapons, this would have to be submitted to me, evaluated, and cleared by me. Chairman Strauss is opposed to any larger shot than we have had.

Eisenhower also told Cole that tests were required to further develop "cleanliness, which, as of now at any rate, does not yet apply to weapons of small size and yield."⁴⁶⁵ This admission would soon stand in stark contrast to forthcoming claims by the AEC about great progress in the development of small "clean" weapons.

On May 29, the AEC boasted of advances in "clean" weapons:

Progress continues to be made in the Atomic Energy Commission's program of developing nuclear weapons having maximum effect in the immediate area of the target but with greatly reduced widespread radiological fallout.

Such weapons will be come part of the

⁴⁶⁵Letter dated May 27, 1957 to Honorable Sterling Cole, House of Representatives, from President Dwight D. Eisenhower.

United States stockpile of nuclear weapons for various military applications.

The efforts of scientists in the Commission's weapons laboratories are being directed toward the development of such cleaner weapons and the success thus far achieved has convinced us that widespread hazard from fallout is not a necessary complement to the employment of large nuclear weapons.

Laboratory and testing activities have confirmed the important objectives which were stated by the President at his news conference on April 25, 1956, when he stated that this country's testing program was aimed at the development of a variety of weapons to serve a wide range of military uses, including defense against air attack. On that occasion, he stated that there was also the objective of making weapons with reduced fallout.

Chairman Lewis L. Strauss of the AEC, on July 19, 1956, following his return from the REDWING test series at the Eniwetok Proving Grounds, said that "many factors, including operational ones, make it possible to localize to an extent not heretofore appreciated the fallout effect of nuclear explosions."

Dr. Willard F. Libby, member of the Commission, informed the American Association for the Advancement of Science on October 12, 1956, that particular attention was given in the REDWING test series to the problem of large scale fallout.

Dr. Libby said:

"A major effort was made to produce a megaton range weapon with an inherently smaller amount of fallout for a given energy release. This effort was successful."

President Eisenhower stated on October 24, 1956, that "the most recent tests enable us to harness and discipline our weapons more precisely and effectively -- drastically

reducing their fallout and making them more easy to concentrate, if ever used, upon military objectives. Further progress along this line is confidently expected ..."

The Commission's 21st Semiannual Report to the Congress, covering the period July-December 1956, also stated: "Operation REDWING gave important information relating to developing means of reducing fallout from weapons firing."

As recently as May 21, 1957, Commissioner Libby, speaking before the American Society of Military Engineers, said that as a result of weapons development and tests, "the amount of radioactive fallout per megaton of explosive power is very greatly reduced."

He added:

"I believe this is a most important development because it would minimize the potential health hazards to those who are from the scene of the battle if nuclear weapons are used."

The Commission is encouraged by the progress made in the development of weapons having reduced radiological fallout and intends to continue its efforts in this direction.⁴⁶⁶

By now, the "clean" bomb apparently had become just as important to the AEC's public and congressional relations plans as it was to the national security of the United States: the politics of "clean" weapons could be just as powerful as the bombs themselves. Detonations of relatively "clean" devices could be used as an argument for continued atmospheric testing.

This particular press release was not issued without controversy, both within and outside the AEC. In a memorandum dated May 28, 1957 to AEC Chairman Lewis Strauss, AEC Commissioner Thomas E. Murray took exception to the statement that "widespread hazard from fallout is not a necessary complement to the employment of large nuclear weapons."

⁴⁶⁶Hewlett and Holl, p. 398; "Statement by the Atomic Energy Commission," AEC press release dated May 29, 1957.

Murray urged that the statement be temporarily withheld; he complained that he was given only minutes to review the statement before its release, even though the press release ostensibly spoke for the entire Commission.

In addition, Representative Chet Holifield of California, a member of the JCAE, complained on the following day that the committee was not given sufficient notice of the release, and that it violated the AEC's previous practice of giving 24 hours advance notice of such statements (an AEC press release of July 19, 1956 was also issued with even less notice to the JCAE). The latest AEC press statement had been issued in the midst of JCAE hearings on the hazards of fallout.

The battle of the press releases was on again. Holifield also stated:

Testimony before the Joint Committee during the current fallout hearings and in prior executive sessions has indicated that the AEC statement that "the success thus far achieved (in developing cleaner weapons) has convinced us that widespread hazard from fallout is not a necessary complement to the employment of large nuclear weapons" is misleading for the following reasons:

....

(b) It may give the wrong impression to our friends and allies that the use of large-scale megaton bombs under the cleanliness program will not entail widespread radioactive fallout. It is true that fallout is being reduced. But there is and at present cannot be such a thing as an absolutely clean bomb, i.e., no fallout. Moreover, in any large-scale war, it would seem inevitable that weapons of varying degrees of cleanliness will be used, and the inevitable result will be large-scale fallout.

(c) It tends to give the impression that (tests of) so-called clean bombs have little fallout. This is a subject which these hearings will explore further. So far, we have been provided with no evidence to

prove the (Atomic Energy) Commission's assertion. We have no assurance that Russian or British tests are or will be free of substantial radioactive fallout.

This statement ignores the alternative suggestion of (AEC) Commissioner Thomas E. Murray that rather than rely solely on large-scale thermonuclear bombs, we develop and produce numbers of small-yield nuclear weapons, which can be widely-dispersed and used on a selective and limited basis in brush-fire wars.⁴⁶⁷

Just coincidentally concurrent with the skirmish in the press, the AEC on May 29 sent to the JCAE its quarterly progress report for the first three months of 1957. Among other weapon matters, the AEC noted that 70% of the weapons scheduled for production during FY 1957 had been produced and stockpiled by March 30, 1957. Tritium production had been sufficient to meet gas-boosting requirements for these weapons.⁴⁶⁸

Twenty-two test shots were now planned for PLUMBBOB; the first had been fired on May 28, and the series was slated to go through September 10. In addition, seven safety tests were scheduled. These included five one-point detonation shots, to determine whether any nuclear yield would result from accidental explosions. Two safety tests would ascertain plutonium hazards resulting from a one-point non-nuclear detonation of a sealed-pit implosion warhead, and from the burning of plutonium.

The new Sandia facility in Livermore was being enlarged, and a number of AEC production facilities were either under construction or had been recently completed.⁴⁶⁹

By the beginning of June, HARDTACK was planned to include 28

⁴⁶⁷Press Release No. 87 dated May 29, 1957 from the Office of the Joint Committee on Atomic Energy, U.S. Congress, Washington, D.C.

⁴⁶⁸Atomic Energy Commission, Quarterly Progress Report to the Joint Committee on Atomic Energy, March-June 1957, Part III - Weapons, May 29, 1957, p. 1.

⁴⁶⁹Atomic Energy Commission, Quarterly Progress Report to the Joint Committee on Atomic Energy, March-June 1957, Part III - Weapons, May 29, 1957, pp. 3, 8-10.

shots, the largest Pacific test series to date. Both Eniwetok and Bikini would be used as test sites. The HARDTACK test schedule at this time is shown in Table 3-15; shots would be ready for test starting May 1, 1958. Only 12 of the 13 candidate Engebi, Aomon, and Runit tests would finally be conducted.

Table 3-15 HARDTACK Shot Schedule, 1 June 1957⁴⁷⁰

<u>Shot Name</u>	<u>Shot Site</u>
Koa	Engebi island, Eniwetok (predicted yield: 1-2 MT)
Linden	Engebi island, Eniwetok
Elder	" " "
Holly	" " "
Sequoia	" " " (might be "clean" Class C' shot; probably would be last Engebi shot in series)
Varnish	Engebi island, Eniwetok; "clean"
Joshua	Aomon island, Eniwetok (small yield, 60-100 KT; 2- stage experiment)
Tobacco	Runit island, Eniwetok (barge)
Magnolia	" " "
Butternut	" " "
Yellowwood	" " "
Cactus	" " " (surface)
Tamarack	" " " (barge)

⁴⁷⁰General Concept No. 1-57, Operation HARDTACK, Commander, Task Group 7.1, June 1, 1957, pp. 6-8; Notes on Project Officers Meeting, July 30-August 1, 1957, Operation HARDTACK, Headquarters, TG 7.1, JTF 7, pp. 17-20.

Wahoo	Igurin island, Eniwetok (underwater shot, 500' deep)
Umbrella	" " " (underwater at depth of 150'- 200')
Locust	Eninman island, Bikini (originally tower shot, later moved to barge)
Hickory	" " " "
Walnut	" " " "
Buckeye	" " " "
Yucca	Very-high altitude test; balloon suspended; 90,000' HOB
Teak	Ultra-high altitude shot; launched from Bikini; REDSTONE missile; 250,000' HOB
Orange	Very-high altitude shot; launched from Bikini; may be reduced yield; 125,000' HOB
Maple	Namu island, Bikini (later moved to barge)
Aspen	" " "
Fir	" " "
Poplar	" " "
Sycamore	" " "
Redwood	" " "

Under this schedule, shots Poplar, Sycamore, and Redwood would be of such high yield that Bikini would have to be evacuated.

By June 4, yet another draft statement on "clean" weapons was being prepared for possible release by AEC Chairman Strauss. This particular draft had been reviewed by weapons labs directors Norris Bradbury and Herbert York; Bradbury had made a suggestion (that was later overruled) that mention be made of the fact that some yield was lost by using the "clean" rather than the standard geometry.

The proposed statement read in part:

It is from a weapon's fission debris that the major part of the residual radiation comes. Based on tests conducted in Operation REDWING, it is now possible to produce large thermonuclear weapons [deleted]

... The neutrons produced in the fusion reaction can induce somewhat more radioactivity in materials in close proximity to the point of detonation of a thermonuclear weapon than can neutrons accompanying the detonation of a fission weapon of the same yield. However, this increase in induced radioactivity is quite small when contrasted with the over-all decrease in residual radioactivity obtained by reducing the fission yield of a high-yield nuclear weapon.

This means then that the residual radioactivity produced per megaton of yield by such weapons can now be reduced [deleted] Research and development is in progress in an effort to effect a still-further reduction.⁴⁷¹

At a press conference on June 5, President Eisenhower declared that nuclear fallout from large bombs had been reduced by nine-tenths and that with small weapons as well, the push was on "to see how clean we can make them."⁴⁷²

The President also stated that "Our tests in recent years, the last couple of years, have been largely of the -- in the defensive type of armament to defend against attack from the air and, particularly, to make bombs cleaner so that there isn't so much fallout." He reiterated his doubt that the U.S. would ever detonate a weapon larger in yield than those

⁴⁷¹Attachment to memorandum dated 4 June 1957 to K. E. Fields, General Manager, USAEC, from Brig. General Alfred D. Starbird, USA, Director of Military Application, USAEC, Subject: Lab Directors' Opinions on the Possible Public Statement Relative to Cleanliness in Weapons. The fact that portions of this draft are deleted shows that it was still a long way from public dissemination.

⁴⁷²"From Cold War to Nuclear Nostalgia," article by William J. Broad in the New York Times, December 12, 1989, p. B8.

previously tested.⁴⁷³

Near the middle of June, Lewis Strauss replied to a letter dated about two weeks earlier from U.S. Senator Hubert Humphrey, then chairman of the Senate Subcommittee on Disarmament of the Senate Committee on Foreign Relations.

In reply to a question from Humphrey about whether or not the U.S. would require further nuclear tests if U.S. long-range detection techniques were adequate to prevent foreign countries from conducting their own tests, and thus inhibit or prevent the development of nuclear-armed long-range missiles, Strauss said:

Although the U.S. can meet a number of nuclear weapon requirements on the basis of present development, a test cessation would affect the development of improved and new weapon delivery systems, which determine the size, weight, shape, and other requirements of the nuclear warheads.

We are in a period of radical transition as to the weapons systems that are available and are becoming available. The gains made by other nations in their intercontinental delivery systems have increased the technical possibility of surprise attack. These improvements in the delivery systems of other nations necessitate a change in our defensive system.

We require weapons of smaller size, capable of withstanding extreme conditions, instantly ready and tailored to give the desired effect. These new systems generally require a new or modified warhead design. Naturally, in view of the reliance we must place on these nuclear weapons systems, new or modified weapons and design changes must be tested if we are to safely depend upon them in an emergency.

Experience, including that of World War II, proved unfortunately that there is no

⁴⁷³Quotation from Verbatim Transcript of President Eisenhower's June 5, 1957 Press Conference, attachment to letter dated June 12, 1957 to James G. Hagerty, Press Secretary, The White House, from Lewis L. Strauss, Chairman, USAEC; record of phone conversation between General Starbird and G. W. Johnson dated June 5, 1957.

substitute for tests to determine the reliability of a weapon, conventional or atomic. Though there can be very limited extrapolation of present information without testing, a cessation of tests would seriously cripple our development work in this vital defensive area.

....

A ban on nuclear tests, which the U.S. would of course observe, would end shortly the development of significantly improved nuclear warheads and would cut off introduction of improved nuclear weapons systems. Whether a particular long-range missile could be provided with a nuclear warhead would depend upon the state of development of the missile and its nuclear warhead at the time testing ceased.⁴⁷⁴

In a letter to Lewis Strauss dated June 14, 1957, Dr. Hans Bethe, then a consultant to the AEC, outlined a proposed unclassified article for the Bulletin of the Atomic Scientists and sought AEC permission to write and publish the article. Bethe's essay would touch upon the detectability of tests of "clean" devices:

Dear Admiral Strauss:

Thank you very much for giving me so much time in your office last Thursday. You will remember that we discussed a possible article to be written by me for the Bulletin of the Atomic Scientists about the detectability of tests. I will not reiterate my arguments in favor of such an article; they are essentially that the public needs some information in order to form an opinion, and that some public statements have been very misleading.

....

Examination of the samples (of radioactive bomb test debris) will give evidence on:

(a) The fissionable material used in

⁴⁷⁴Letter dated June 11, 1957 to Senator Hubert Humphrey, Chairman, Senate Subcommittee on Disarmament, Committee on Foreign Relations, from Lewis Strauss, Chairman, USAEC, attachment to Memorandum for the Chairman dated June 11, 1957, subject: Response to Senator Humphrey's Letter.

- the test,
- (b) natural uranium used,
- (c) thermonuclear reactions by measure of fast neutron effects,
- (d) the use of hydrogenous material by detection of reactions produced by slow neutrons,
- (e) the use of a fission-fusion-fission assembly.

The latter is extremely important in case a nation announces falsely that it has tested a "clean" weapon. The presence of final fission reaction is of course shown by (the presence of) U-237. (Author's note: the material found by the Japanese in CASTLE Bravo shot debris.)

There is other evidence available in addition to radioactivity. Its is highly relevant to mention these other methods in the article because only by means of these methods can one detect the full yield of clean weapons rather than merely their fission yield. I should like to mention seismic and acoustic detection methods without going into any detail on the method of obtaining the yield from the raw data.

....

Clean bombs are still detectable. While their exact fission yield may be difficult to measure, their total yield can be measured by non-radioactive methods. It can therefore, be ascertained where any nation has evaded a test moratorium which limits the size of permissible test, by exploding a clean weapon with relatively small fission yield.

In this section, I would also point out the desirability of clean weapons. I would state that the last fission stage (in U-238) is left out, with a consequent reduction in total yield for an otherwise similar design. Clearly, I would not touch on the design itself.⁴⁷⁵

⁴⁷⁵Although Bethe later testified before Congress on the subject of test detection, his proposed article was apparently never published in the Bulletin. An article entitled "The Case for Ending Nuclear Tests" by Bethe was

One PLUMBBOB shot, Smoky, now scheduled for August 19 (and finally fired on August 31) was very important to UCRL.⁴⁷⁶ Smoky was to be a test of a boosted primary and a secondary system for the three-stage TX-41 bomb; some thermonuclear yield was expected from the detonation. If this shot were not conducted, Livermore might have to delay completion of the new bomb. A full-yield test of the TX-41 was slated for Operation HARDTACK in 1958.

On June 17, DOD requested the AEC to proceed with the Phase 3 development of a new Class "C" weapon in both "dirty" and "clean" versions with an external diameter not more than 37 inches for both versions.⁴⁷⁷

The emphasis on "clean" weapons testing during HARDTACK was restated on June 24, when AEC Chairman Lewis Strauss and three Livermore weapons scientists held a 40-minute conference with President Eisenhower. Ernest O. Lawrence, Edward Teller, and Mark Mills told Eisenhower that 95% "clean" thermonuclear bombs could be built, and that with further development, fallout would become essentially negligible. They also revealed the possibility of small hydrogen bombs, no larger in yield than "nominal" fission bombs.

During this meeting, Teller told Eisenhower that the U.S. knew how to build "dirty" bombs of almost unlimited size (yield), but that smaller weapons using plutonium still needed to be perfected. Lawrence told Eisenhower that "clean" warheads could be built "down to small kiloton weapons." Teller went on to say that another "six or seven years" would be required to develop entirely "clean" tactical weapons and that the Soviets would, by force of world opinion, also be required to develop "clean" weapons.

published in the August 1960 issue of the Atlantic Monthly magazine; the article did not mention "clean" weapons but did note that in the fall of 1958, "the Russians tested the largest weapons they have ever tested; weapons of many megatons. They increased the yield from their weapons; they made them more efficient in the large- megaton class." (THE ROAD FROM LOS ALAMOS, Hans A. Bethe, Touchstone Books, New York, 1991, p. 50.)

⁴⁷⁶Memorandum dated June 17, 1957 to James E. Reeves, Test Manager, Nevada Test Organization, USAEC, from Brig. General Alfred D. Starbird, Director of Military Application, USAEC, subject: Smoky.

⁴⁷⁷Letter dated June 17, 1957 to Lewis L. Strauss, Chairman, USAEC, from J. B. Macauley, Deputy Assistant Secretary of Defense.

Teller also explained the utility of "clean" thermonuclear explosives for peaceful purposes, including excavation, stimulation of release of oil and steam, and possibly even for weather modification.⁴⁷⁸

When Eisenhower suggested that the U.S. turn over its knowledge on the design of "clean" weapons to the Soviets to encourage them to build their own, Mills, Lawrence, and Teller objected, claiming that U.S. weapons incorporated other technological advances of great values that they did not want to give to the opposition.

Teller was also concerned that if the U.S. stopped testing while the Soviets continued their own development of "clean" bombs, then America would have only a "dirty" arsenal which it would be inhibited by world opinion from using during wartime. Teller also noted that after entirely "clean" weapons were developed, it would be possible to put "additive materials" in them to produce any desired amount of radioactive fallout.

Eisenhower noted that the U.S. could not permit itself to be "crucified on a cross of atoms" and that any proposals for a nuclear test moratorium would have to be made in the context of an international agreement outlawing future wars.⁴⁷⁹

The President also said that if he were asked during his next press conference why the U.S. was not joining unreservedly in a test ban, he would reply that U.S. nuclear tests were intended to "clean up" weapons and thus protect civilians in event of thermonuclear war.

The consensus of the attendees at the meeting was to have the United Nations sponsor and organize a group of international scientists to witness a test of a "clean" device during the forthcoming HARDTACK series. The foreign scientists would be allowed to collect fallout samples, possibly by use of rockets fired through the mushroom cloud, to ascertain and verify the "cleanliness" of the test device.⁴⁸⁰

⁴⁷⁸Shelton, p. 8-15.

⁴⁷⁹Eisenhower's reference to a "cross of atoms" was a paraphrase of a line from a dramatic speech by William Jennings Bryan in 1896 at the Democratic national convention in Chicago, Illinois where Bryan spoke out against establishment of a gold standard then much favored by the Republican Party. Bryan said that the Democrats should not let the working man be "crucified on (this) cross of gold."

⁴⁸⁰"Teller and the 'Clean Bomb' Episode," Katherine Magraw,

Teller also believed that it was imperative for the U.S. to build thermonuclear weapons that limited their immediate effects to the target area and did not dump radioactive fallout on innocent civilians downwind.

It would be politically advantageous to the U.S. if it could portray itself as being more interested and more advanced in the development of "clean" weapons than the Soviet Union. Thus continued testing to develop "clean" weapons, and even their use in combat, could be justified on the basis that "clean" American bombs were more "humanitarian" than "dirty" Soviet bombs.⁴⁸¹

Less than four weeks earlier, Senator Henry M. Jackson of Washington state, then a member of Congress' Joint Committee on Atomic Energy, had met with Mills, Teller, and Lawrence at Livermore; Jackson was

...particularly impressed with the progress they were making in low-yield weapons, the possibility of making them smaller, the possibility of making them cleaner ... the gleam in the scientists' eye of making them almost like Ivory Soap, (but) not quite.⁴⁸²

In public press conferences on June 26 and July 3, Eisenhower confirmed the existence of a "96% clean" bomb and announced that he was thinking of sharing with the USSR and other countries the knowledge of the "clean" bomb. Regarding the latter point, Eisenhower's exact words were:

But I said certain people have questioned the proposition that Dr. Lawrence and Dr.

BULLETIN OF THE ATOMIC SCIENTIST, Vol. 44 No. 4, May 1988, p. 88; Hewlett and Holl, pp. 400-401; memorandum of a Conference with the President, White House, Washington, June 24, 1957, 9:00 AM, document 248 in FOREIGN RELATIONS OF THE UNITED STATES, 1955-1957, Volume 20: Regulation of Armaments; Atomic Energy, John P. Glennon and David S. Patterson (eds.), U.S. Government Printing Office, Washington, D.C., 1990, pp. 638-640; memorandum for General Cutler dated June 26, 1957. Promises of negligible-fallout nuclear weapons made to Eisenhower by Teller, et. al., are reminiscent of the promises of the King's magicians in BARTHOLOMEW AND THE OOBLECK (Theodore Geisel, aka Dr. Seuss, Random House, 1949).

⁴⁸¹Hewlett and Holl, p. 399.

⁴⁸²Hewlett and Holl, p. 399. The comparison of "clean" weapons to "Ivory Soap" resulted in an acerbic Herblock editorial cartoon in the Washington Post in early July 1957.

Teller brought to me, that eventually you could make completely clean bombs, and that even now you are 96 percent clean, that is, you have only 4 percent radioactivity, radioactive fallout, that you did in the original bomb.⁴⁸³

The President also stated that he would at some future time invite foreign countries to send scientific observers and take their own measurements of fallout on the site of the next U.S. hydrogen bomb test.⁴⁸⁴ (This statement led to the so-called "open shot" during Operation HARDTACK Phase I in 1958.)

During his June 26 press conference, Eisenhower told reporters that in four or five years the U.S. would produce "an absolutely clean bomb" with "no fallout to injure any civilians or anyone, any innocent bystanders."⁴⁸⁵

At an AEC-MLC conference on June 27, Lewis Strauss, speaking for the AEC, urged that HARDTACK be accelerated, in light of on-going U.S.-Soviet disarmament negotiations that might lead to a possible test moratorium.⁴⁸⁶

By the end of June, the USS CURTISS had been decommissioned. She had been launched in April 1940, and had seen much service since then during World War II and the postwar years. She was becoming increasingly useless as an atomic weapons test support ship, however, so HARDTACK planners would have to make other arrangements.

At the beginning of July, DOD was resisting the refabrication of stockpiled "dirty" bombs which might be made "clean" by means proven during REDWING.⁴⁸⁷

⁴⁸³Transcript of President Dwight Eisenhower's July 3, 1957 press conference, attachment to memorandum dated August 29, 1957 to Those Listed Below, from Morse Salisbury, Director, Division of Information Services, USAEC, Subject: Draft Statement on HARDTACK.

⁴⁸⁴BULLETIN OF THE ATOMIC SCIENTISTS, Volume 11, 1956, pp. 270, 271. It is highly unlikely that Eisenhower was considering sharing the technology of "clean" weapons; more likely, he probably wanted to prove his "96% clean" comment to a dubious world.

⁴⁸⁵"From Cold War to Nuclear Nostalgia," article by William J. Broad in the New York Times, December 12, 1989.

⁴⁸⁶Minutes of 112th AEC-MLC Conference, June 27, 1957.

Drs. Teller, Lawrence, and Mills, the latter then assistant director of UCRL and one of its top theoretical physicists, received notes at the beginning of July from AEC chairman Lewis Strauss and his aide, Navy Captain J. H. Morse, congratulating the scientists for their "performance" in the June 24 meeting with Eisenhower. (On August 8, Secretary of State John Foster Dulles would tell Lewis Strauss that "the President was in line following the talk he had with Teller and Lawrence.")⁴⁸⁸

Strauss noted that "everything has worked out as we had hoped it would," and Morse said that "everything has gone beautifully" and that Mills had "won a great victory for the country" by dissuading Eisenhower from an immediate test halt.⁴⁸⁹ Regarding "clean" weapons, Morse added his own definition of the term:

You may detect some feeling that clean weapon potentialities have been over-stated and over-simplified recently. I think you did exactly the right thing in statements to the President and to the Press (sic). The situation called for over-selling rather than under-selling, particularly when a simple statement could not possibly cover all the complexities involved.

I explain wherever possible that weapons can be made so clean that radioactive fallout is no longer an important factor in determining their application. Militarily speaking, they are then "completely" clean.⁴⁹⁰

⁴⁸⁷Telegram dated July 2, 1957 (Noon) from the U.S. Embassy in the Kingdom to the U.S. Dept. of State, document 254 in FOREIGN RELATIONS OF THE UNITED STATES, 1955-1957, Volume 20: Regulation of Armaments; Atomic Energy, John P. Glennon and David S. Patterson (eds.), U.S. Government Printing Office, Washington, D.C, 1990, p. 651.

⁴⁸⁸Memorandum for the Files of Lewis L. Strauss dated 8 August 1957.

⁴⁸⁹Letter dated July 3, 1957 to Dr. Mark A. Mills, Assistant Director, Radiation Laboratory, University of California, Livermore, from Lewis L. Strauss; letter dated July 3, 1957 to Dr. Edward Teller from Capt. J. H. Morse, Jr., USN, Special Assistant to the Chairman.

⁴⁹⁰Letter dated July 2, 1957 to Dr. E. O. Lawrence from Capt. J. H. Morse, Jr., USN, Special Assistant to the Chairman.

On July 1, Representative Sterling Cole, a former chairman of the JCAE, took up the Teller-Mills-Lawrence battle cry for continued U.S. testing to refine "clean" weapons, even at the risk of endangering arms control:

To preclude our being backed away from the source of our military strength and protection, we need two things above all else. First, an aggressive and determined program to perfect "clean" weapons which will release a minimum, if any, of radioactive fallout.

This means without reservation that there must be no suspension or termination of atomic tests until the program of 'clean-up' is completed. I must confess that during the past several months, a continuance of atomic tests without restraint or demonstrable humanitarian purpose has been more and more difficult to defend and, therefore, a matter of grave concern to me.

Although the Joint Committee has been aware of considerable progress in the direction of making cleaner weapons, it was not until quite recently that the great success of these efforts was made known. We now have the unhesitating concurrence of all the most competent scientists engaged in the nuclear weapons program that their efforts could eliminate nearly 100 percent of the fallout radiation. Teller, Lawrence, Graves, Bradbury, York, and Mills all seem to agree.

This being so, and certainly none of us have any basis on which to dispute the judgment of these scientists, we have but one course to follow. All humanity, decency, and conscience demand that the tests continue until the weapons can be made "clean" to the end that if and when an (American) atomic weapon is ever used in battle, there will be no victims of innocent, distant civil populations.

This does not imply that the disarmament discussion now in progress in London should be disrupted or modified in regard to a cessation of nuclear testing. We can still agree to a cessation of all further tests

except those which may be needed to prove our success in eliminating radioactive debris. (Author's note: emphasis added.) This kind of limited tests can and should be done with United Nations participation.

Cole also called for an "open" shot during a future test series, at which the "cleanliness" of U.S. weapons could be conclusively demonstrated:

Therefore, that we may demonstrate our good faith in continuing the tests for the purpose of making (weapons) "clean," I propose that our Government (sic) extend an invitation to an appropriately-qualified group of scientists designated by the United Nations, including the Russians, to attend our tests and make the necessary scientific examination of the radiation and see for themselves the success of our efforts.

The scientists who vouch for the achievement of "clean" weapons assure me that such participation by foreign observers will not compromise our vital weapons information or national security in the slightest degree. If the United States sets the pattern for production of "clean" weapons, other nations should follow our example. Certainly, world opinion would soon demand that they do so.

Cole suggested that more information be declassified and made public to convince the world that U.S. nuclear weapons would soon be indistinguishable from conventional non-nuclear weapons in both their accuracy and effects:

The second element in maintaining our national nuclear shield is that we relax our security barriers sufficiently to permit a program of public education through which we can demonstrate not only to the American people but to the entire world that atomic weapons are not as wanton and indiscriminate as they are thought to be.

Our experts, the scientists and engineers, are making constant progress toward reducing the size and yield of atomic weapons, making them more and more precision instruments so that only the immediate military target will be affected or destroyed.

The honest objective of this program of experimentation tests and education will be that we have harnessed atomic energy to provide us with just another "conventional" weapon -- that is, one which uses blast and heat to destroy the target but with no amount of radiation.

Atomic weapons can be discriminating and their effects can be confined to military targets when they are used to that end. Atomic weapons need not be wanton, indiscriminate, or inhuman. They can be made precise and humane as any other weapon.⁴⁹¹

Operation PLUMBBOB had started in Nevada on May 28. By July 5, six shots had been conducted. On that day, shot Hood was fired, a 74 KT UCRL shot on a balloon at 1,500 feet over Area 9 at NTS. The Hood device measured 12.1" in diameter and 42.2" long and weighed 393 lbs.

What made the Hood shot noteworthy was a press briefing about two weeks earlier (on June 25). During the course of this briefing, Dr. Alvin C. Graves, director of NTS testing, stated in response to a newsman's question that "We have said a number of times that here in Nevada we don't test hydrogen bombs; we test fission bombs."⁴⁹²

However, as Graves knew full well, the Hood device was a two-stage fission-fusion "gadget,"⁴⁹³ intended to fill a gap in

⁴⁹¹Remarks of Rep. Sterling Cole (R, N.Y.) in the House of Representatives on July 1, 1957 regarding atomic weapon development and test program. "Graves" was Alvin Graves, test director for the AEC. Not everyone who had access to accurate information about the current status of U.S. "clean" weapons development agreed with Cole. The London talks Cole referred to was the on-again, off-again London Disarmament Conference, which had started in February 1955. (Hewlett and Holl, pp. 296, 297, 382-385, 458-461.)

⁴⁹²Question and Answer Period, Dr. Alvin C. Graves, Press Briefing, Nevada Test Organization, Office of Test Information, USAEC, Las Vegas, Nevada, June 25, 1957.

⁴⁹³"The time interval between primary and secondary was measured ..." (Hood Results, PLUMBBOB Test Bulletin #23, Division of Military Application, USAEC, July 8, 1957); "Primary yield ... total fission yield... Primary alpha ..." (Additional Results - Hood, Diablo, John, PLUMBBOB Test Bulletin #27, Division of Military Application, USAEC, July

300-400 lb. warhead technology. It used a boosted primary and was both the largest atmospheric test, and the first two-stage test, in Nevada. Of its total yield, 70 KT came from fission, and the test was to have been followed by a full-yield test during Operation HARDTACK.

Only 12 weeks later, the AEC's repeated earlier public statements that fusion devices were not tested in Nevada would put the Commission into a most awkward position.

The subject of "clean" bombs came up again during the 54th meeting of the GAC on July 9. The committee noted that there was a great deal of public confusion on the subject, with newspaper stories claiming that an entirely "clean" bomb was possible and that the U.S., in order to minimize its own civilian losses in a nuclear war, would tell the Soviets how to make a "clean" bomb.

AEC Commissioner Harold S. Vance had spoken recently to USAF SAC commander General Curtis LeMay about the military significance of "clean" bombs. A high-priority wartime SAC bomber target was Soviet airfields (as noted in the AEC Operations and Analysis Office study in September 1956); in the event that war seemed imminent, LeMay was ready to strike first if he got the chance.⁴⁹⁴

Since many Soviet airfields were located close to population centers, exploding "dirty" bombs on airfields could produce very serious fallout effects. With "clean" bombs, airfields could -- theoretically -- be attacked without endangering either nearby or distant population centers. Heavy Soviet civilian casualties probably would not have bothered LeMay,

23, 1957.)

⁴⁹⁴According to John Newhouse in WAR AND PEACE IN THE NUCLEAR AGE (Alfred A. Knopf, New York, 1989, p. 280,) LeMay said in September 1957 that "I will know from my own intelligence whether or not the Russians are massing their planes ... for a massive attack against the United States. If I come to that conclusion, I'm going to knock the shit (sic) out of them before they get off the ground." When reminded by his surprised audience that a first-strike, attack-on-warning posture was not then the ostensible national nuclear war strategy of the U.S., LeMay replied, "No, it's not national policy, but it's my policy." In Bundy, p. 321, LeMay is quoted as stating that "believing I could foresee (a Soviet) attack, I was prepared to beat him to the draw and attack all of his bomber and missile bases. ... Of course, I had no authority to order such attacks. All I could do was (to) have the capability and hope that the orders would be given if necessary." Bundy notes that President Eisenhower did not share LeMay's enthusiasm for preemptive warfare.

who had once vowed to turn Russia into a "smoking, radiating ruin" if nuclear war came.

During the GAC meeting, a discussion of current military requirements for "clean" bombs revealed that fact that the AEC was producing the TX-21C, a 17,000 lb. multistage "clean" thermonuclear bomb.⁴⁹⁵

Early in July, the AEC planned to launch a very-high altitude test from Eniwetok, using a balloon to loft the nuclear device. However, balloon trials to date at the atoll had not had good results, leading to the possibility of launching the balloon from shipboard, probably from an aircraft carrier.⁴⁹⁶

A meeting to discuss the proposed "clean" Class "A" bomb was held at AEC headquarters in Washington, D.C. on July 15. In preparation for this meeting, General Starbird, director of the AEC's Division of Military Application, wrote a memo to AEC Chairman Lewis Strauss. This memo touched on a number of subjects, including prospects for 100% fission-free weapons and the military and political effects of "clean" bombs versus "conventional" thermonuclear bombs:

Should we develop a weapon of no fission? I think we should not proceed with development at this time; however, we should continue with research. Our development objective should be to develop a weapon whose fission radioactivity will not be such as to (deposit) serious fallout outside the blast and thermal mortality areas. This means that we can tolerate a percent or two fission yield. It will probably be very costly of materials, difficult and time-consuming to narrow the fission yield from a few percent to a fraction of a percent.

Can an all-fusion weapon induce residual radioactivity? The answer is "yes." In the case of a ground burst, it is estimated that neutrons released from the fusion reaction will induce radioactivity in average soils (to the extent) that radioactivity will be nearly equivalent to radioactivity from a fission weapon (with a

⁴⁹⁵Minutes of the Fifty-Fourth Meeting of the General Advisory Committee to the USAEC, July 9, 10, and 11, 1957, p. 5.

⁴⁹⁶Memorandum to Distribution dated July 11, 1957 from Gaelen L. Felt, Commander, TG 7.1, Subject: Problem Areas 2.

yield equivalent to) 5% of the larger weapon's total yield.

To be specific, if a weapon could be all-fusion, with a 10 megaton yield, and burst at the surface, it could generate residual radioactivity equivalent to 500 KT of fission. In the case of an air burst, the residual radioactivity is less but there is still some from the (irradiated) bomb and delivery vehicle materials.

Will the firing of one large conventional thermonuclear weapon be of a lesser hazard than the firing of an equivalent yield from a small fission weapon? The answer to this is "yes," but this statement when used tends to confuse. The larger number of fission weapons would, of course, have a greater fission yield than the thermonuclear weapon.

On the other hand, if fired at surface level, the larger portion of the fallout from the fission weapons would be very local. The wide-spread fallout effect in a wartime situation (if only small blasts were required to destroy many targets) would be quite small and generally localized in areas near the targets.

The thermonuclear weapon in truth will have a lower fission yield but its fallout (even if fired at ground level) would be spread over a large area, and could be over areas far removed from the target. Certainly if the large weapon were not necessary to do the job, a better way of minimizing fallout dispersal would be to employ a number of smaller weapons.

I therefore believe it undesirable to state in discussions that thermonuclear weapons have a lower percentage of fission yield than small fission weapons. Though a true statement, it can be attacked as unrealistic and as warping the facts.

Starbird also advised Strauss of the possibility that the American public was harboring a mistaken notion, generated mostly by misleading and incomplete press accounts and partly by political rhetoric from the White House, as to how soon the U.S. would have a sizable stockpile of "clean" weapons.

Starbird reiterated the disadvantages incurred, in terms of increased diameter and weight, by the conversion of "standard" thermonuclear warheads to "clean" versions, and the likelihood of only slow improvements in terms of reducing the sizes and weights of "clean" weapons:

Is there danger that the public (from current press comments and editorials) will get a mistaken ideas as to how soon we shall have clean weapons and in what types? I believe that there is great danger that this can occur, and could hurt our interim programs which are necessary until clean versions are available.

- (1) We shall not have any clean weapon in the stockpile until April of 1958, and the weapons then available will be of limited yield for their weights. The stockpiled weapon at that time can be carried by the B-47, B-52, and B-36, but only the latter two can handle it readily. (Author's note: this is a reference to the TX-21C, tested during REDWING.)
- (2) We shall have by early 1960 clean versions of our large thermonuclear bombs. However (unless large sums are then made available immediately for conversion of the stockpile), it will be several years before the large thermonuclear stockpile is all clean. It will require some \$175,000,000 to make the conversion.
- (3) We shall test in HARDTACK a one-megaton clean device. Optimistically, this device could enter the stockpile early in 1961. Here again though, major sums would be required to convert all one-megaton weapons to a clean version and they would be larger in diameter than the Services (sic) would be able to use for many applications. Reduction of size will probably come slowly. (Author's note: emphasis added.)

Starbird also reviewed for Strauss the current short- and long-term statuses of "clean" tactical weapons, a much-desired DOD and AEC program objective:

Following Lawrence's and Teller's appearance here I met for several hours with York. He is the man who must develop the weapon and a man who is generally most optimistic. The earliest he could see a clean tactical weapon is 1961 or 1962, and this would be based on very lucky breaks in PLUMBBOB and HARDTACK (tests).

It would be a fairly large weapon of several hundred pounds and incapable of application to such vehicles as the LITTLE JOHN and the like. It would have a fairly substantial fission yield (about 10 KT). It would be several more years before weapons of less fission yield and greatly-reduced size would be available. (Author's note: emphasis added.) (This generally is not in conflict with what Teller said, but Teller gave the impression that the weapons could be in stockpile when the proposed device was successfully tested. Two or more years are required to build the prototype and start production.)

Starbird commented on the advisability and feasibility of having foreign observers observe results of a demonstration firing of a "clean" U.S. test device. This was one shot that had been proposed for HARDTACK; its purpose was largely for political and propaganda gains. President Eisenhower was a very vocal advocate of such a demonstration:

Can an observer at one of our tests satisfy himself that a shot was clean? If a man wished to be satisfied, he could so satisfy himself. If he wished not to be so satisfied, it would be almost impossible to convince him unless he is given critical design information. Teller and Lawrence spoke of the possibility of firing rockets through the (mushroom) cloud. (I have discussed this procedure with Mark Mills, the real proponent of this, in great detail.)

The method is unproven and many of the better scientists question whether we can get a high degree of accuracy in this manner. International observers could be allowed to see a clean shot and a second equivalent-yield dirty shot. By approaching the two separate contaminated areas, they

could certainly get an idea that one device was cleaner than the other.

However, the true degree of cleanliness could not be given by any simple method unless design information were released. In essence, the observer can be given the impression that one weapon is cleaner than the other. Whether he believes the cleanliness would be of the order of a few percent will be dependent upon the desire to be convinced.

We should be most careful in giving indication that international observers can convince themselves as to the cleanliness of our weapons. About all we can do (unless design information is provided to them) is to let them see that certain weapons are cleaner than others of comparable yield.

Starbird closed his memo with a few remarks about the likely future of "clean" weapons:

In summary, I believe that the recent publicity on clean weapons appears to be affecting the DOD favorably. It has had beneficial public effect. On the other hand, I believe we must be most careful to realize:

- a. Clean weapons will come gradually into the stockpile and over a period of years. We must not let the public be misled into a belief that we have licked radioactivity in an atomic war.
- b. For many years there will be applications which, due to size and weight, will not be able to accommodate the larger dimensions or weight of clean design. Our developmental and test effort through HARDTACK and later test series must be directed in large part to meeting the requirements which cannot be met by clean versions. We must avoid being worked into a position where we are prevented from testing any weapon or device unless it is clean.⁴⁹⁷

⁴⁹⁷Memorandum for Mr. Strauss dated July 11, 1957 from Brig. Gen. A. D. Starbird, Director of Military Application, USAEC,

Later on the same day, a group of officials, including Starbird and Strauss, met with representatives of the Departments of State and Defense to discuss the proposed high-yield Class "A" weapon.

DOD noted that it was important for political and public relations reasons to develop a technique for making a large "clean" bomb with a test that would not exceed the yield of the CASTLE Bravo shot, the largest-yield test explosion conducted by the U.S. A later decision would be made on how many of the new bombs to stockpile.

Military requirements and constraints on the new weapon were:

- (a) Use against "hard" targets in border areas between friendly and hostile territory;
- (b) the greatly increased cost of likely future delivery vehicles;
- (c) the weight limits of [deleted] pounds would probably be reduced even more for future delivery vehicles; and
- (d) concern over retaining a credible U.S. deterrent capability in the face of both a limited defense budget and an increase in Soviet nuclear weapons capabilities.

The AEC claimed that the development of a stockpile of the new Class "A" bomb would probably delay by a year the current program at LASL and UCRL to develop small "clean" bombs. Lewis Strauss noted that the principle of the Class "A" could probably be tested at reduced yield during HARDTACK, but that the desired warhead weight could probably not be obtained using a single reduced yield test.

The firing of one test to ascertain and establish the high-yield principle and technique would probably not postpone small "clean" weapons development, but the stockpiling of a number of the Class "A" bombs would affect the small weapons program.

The State Department wondered why a test of the new Class "A" bomb was required in 1958 when no decision had yet been made whether or not to even stockpile these weapons. State was

Subject: Clarification of Certain Matters Relative to Cleanliness in Weapons. Portions of the text of this memo have been "cleaned up" or paraphrased to make them clearer. Unless otherwise noted, emphasis, where shown, is in the original document.

opposed to any delay of the small "clean" weapons research program, even if the Joint Chiefs of Staff agreed to a delay. In addition, the development of a very high-yield weapon was contrary to recent public statements that President Eisenhower had made, when he had pledged not to develop new high-yield weapons.

The Secretary of State concluded the meeting by asking Strauss to prepare a report as to what might be done by the AEC to test a Class "A" bomb in 1958 without interfering with the small "clean" weapons program and without compromising the President's position on new high-yield weapons.⁴⁹⁸

UCRL Plans for "Clean" Tactical Weapons

Concurrent with Starbird's comments to Strauss, UCRL director Dr. Herbert York was preparing his own letter to Starbird on the Livermore lab's tactical "clean" weapons program. Although York's words bore out vividly Starbird's opinion that York was "a man who is generally most optimistic," the UCRL director was careful to attach a number of caveats to his promises:

The purpose of this letter is to make some preliminary estimates concerning clean tactical bombs. Such estimates can be made with much greater certainty and precision later this year after PLUMBBOB results are available and analyzed.

However, because of the great interest engendered by the very recent recognition of the possibility of producing clean tactical weapons, it seems worthwhile to make such estimates at this time even though they are of necessity very rough and tentative.

Section I below gives a list of some of the possible types of clean tactical weapons in the 10 to 200 KT total yield range. Section II gives a possible set of self-consistent time scales based on the assumption of "good luck" in solving some of the basic design problems inherent in the types

⁴⁹⁸Memorandum for the File, by the President's Special Assistant for National Security Affairs (Cutler) dated July 15, 1957, document 258 in FOREIGN RELATIONS OF THE UNITED STATES, 1955-1957, Volume 20: Regulation of Armaments; Atomic Energy, John P. Glennon and David S. Patterson (eds.), U.S. Government Printing Office, Washington, D.C., 1990, pp. 656-658.

described in Section I.

Section III discusses the importance of cleanliness in tactical bombs, and attempts to describe in a very rough but fairly concise way, the meaning of various degrees of cleanliness as applied to a specific tactical use, namely, the employment of a ground burst against a hardened target in friendly territory or near our own troops.

I. Possible Types of Clean Tactical Bombs

The following is a list of three possible types of clean tactical bombs, including estimated dimensions, weights, total yields, fission yields, and special material costs. Unfortunately, in all cases, the most uncertain figures are probably also the most important, that is, the fission yield and the tritium cost.

The types are listed in increasing order of difficulty. It should be emphasized that no experimental shots have ever been made of any of the types listed below, either in device or prototype form; the characteristics given are extrapolations based on application of the theoretical methods which have proved successful in the past in the design of other types of weapons.

These extrapolations do not at present indicate that such characteristics can certainly be achieved, but rather, that with continued research and development work, characteristics in this general range can quite probably be achieved.

TYPE I: A miniaturized standard type two-stage clean bomb. This type of bomb could probably be made with the following approximate characteristics:

Fission Yield: [deleted, but shown
later to be on the
order of 7% to 50%
of total yield]

Special Materials: [deleted]

<u>Total Yield</u>	<u>Gross Weight</u>
200 KT	1,200 lbs.
100 KT	1,000 lbs.
50 KT	800 lbs.

Diameter: 20" to 24"
Length: 40" to 50"

TYPE II: A miniaturized two-stage bomb
[deleted] and an optimistic
estimate of how low a yield from a
standard primary can be made to
implode the secondary. This type
can probably be made with the
following approximate
characteristics [deleted]

(Author's note: Data same as for
Type I except for fission yield,
which ranges from 2% to 15% of
total yield)

TYPE III: [deleted] This type can probably
be made with the following
characteristics, assuming that
the basic design is feasible.

Fission Yield: [deleted, but
shown later as
ranging from 2%
to 5% of total
yield]

Special Materials: [deleted]

<u>Total Yield</u>	<u>Gross Weight</u>
100 KT	1000 lbs.
50 KT	800 lbs.
20 KT	800 lbs.

Diameter: 20" to 24"
Length: 50" to 60"

II. Possible Time Scales for the Various Types

This section gives a possible set of self-
consistent time for developing the three
types of devices listed in Section I. In
order to make up this schedule, we have
made the following assumptions.

1. The pertinent PLUMBBOB experiments [deleted] will be successful.
2. The "breaks" will be with us at virtually every step in the development program.
3. There will be no net increase in weapons research and development effort, but the program will be given a high priority within the effort available. Some other programs now tentatively scheduled for Phase III will have to be dropped.
4. Each design problem can be solved in regular sequence by continuous extrapolation as the development proceeds, i.e., no new research and development "breakthroughs" will be required.

The following table gives a possible weapons test program for the development of the clean tactical weapons:

	<u>Device Test</u>	<u>Weapon Prototype Test</u>
HARDTACK (Pacific, 1958)	Type I Type II	
Nevada, 1959	Type II Type III	Type I Type II (?)
1960	Type III	Type III (?)

The following table gives a possible joint UCRL-Sandia weaponization program schedule. Other weapons are also included to indicate what other programs can be carried on at the same time as a serious program of tactical clean weapons development. The dates given are the fiscal years in which the indicated weaponization program would begin.

These dates are, of course, very tentative and are intended primarily as examples of what might be done, since they all depend on all of the speculations and estimates above, as well as on DOD-DMA estimates of relative importance and determination of priorities.

FY 1958 Clean Type I (above) XW-38
 (ICBM follow-on), XW-42
 (SPARROW, HAWK warhead)

FY 1959 Clean Type II, small weapon
 (e.g., all-oralloy 17" or other
 diameter, or 45 lb. warhead)

FY 1960 NIKE ZEUS, POLARIS

FY 1960 Clean Type III, ?

Assuming that all of the above time schedules can and will be met, the following is then a list of the dates on which the various types of clean tactical bombs might enter the stockpile:

Type I - Calendar Year 1960

Type II - Calendar Year 1961

Type III - Calendar Year 1963

III. The Meaning and Importance of Cleanliness in Tactical Atomic Bombs

The reason for developing and producing clean tactical bombs is to provide the armed forces with nuclear weapons of low and intermediate yield which can be used in situations where, because of radioactive contamination, a tactical atomic weapon of the present 100% fission type cannot be used.

Perhaps the most important and easily described situation of this type is that in which it is desired to remove or destroy, by means of a ground burst, a hard target, such as a deeply dug-in enemy or an airstrip, in friendly territory or in close proximity to our troops. As indicated by an overall analysis of the recent Army "Sagebrush" exercise (which involved, among

others, simulated ground bursts on airfields), such applications are very dangerous, or, more probably, generally impossible using the present-day atomic weapons.

In the case of high air bursts, in which the fireball is well above the ground, the situation is less bad in the present case of pure fission bombs, since the radioactivity is generally spread out over a very large area and a very large number of tactically-sized bombs can be used without approaching the world-wide "Sunshine Limit." (Author's note: a reference to fallout carried by winds in the stratosphere.)

However, even in this case, there is still the problem of serious localized "rain out" which problem would, of course, be greatly alleviated or completely removed by the use of clean bombs.

None of the bombs listed in Section I are, of course, absolutely clean but all of them represent very large improvements in this respect over the present stockpile. ...
....

Weight Penalty - The weight penalty which must be paid for use of clean bombs rather than the projected fission bombs is a factor of three to five. Thus a 150 KT clean bomb would weigh up to 1,200 lbs. as compared with about 300 for [deleted] fission bomb designed to give about the same yield in the 1962 availability period. It should be pointed out, however, that such clean bombs are as light or lighter than current fission warheads of comparable yield.

For delivery by fighter plane, the difference between 300 and 1,200 lbs. is probably not great, although ground-to-ground rocket capability is another matter.

We believe that the above rough discussion of the meaning and evaluation of cleanliness is sufficient for use as a guide to the development of clean tactical bombs. ...

In ending, we also wish to point out that, in addition to the research and development aimed at achieving the three types of bombs listed in Section I, we are searching, as best we can, for the "breakthrough" which will allow the construction of a bomb having zero fission yield (though the problem of induced radioactivity would still remain for ground bursts). We do not yet know how to make such a bomb, but it does not appear to be impossible.⁴⁹⁹

UCRL went on to test "clean" tactical devices in the Charleston shot of Operation PLUMBBOB, which failed, and the Cedar shot of Phase I of Operation HARDTACK, which also yielded unfavorable results.

After reading York's letter, General Starbird concluded:

As brought out in the letter, Dr. York has outlined some preliminary UCRL-Livermore estimates concerning clean tactical weapons. Investigations in this field have also been conducted by LASL, since mid-1954. The LASL studies indicate that the estimates made by Dr. York may possibly be achievable; however, such estimates would require additional theoretical and experimental effort.

It should be realized, however, that the possibilities outlined in (Dr. York's) letter are based on generally incomplete information available in this field and are of a preliminary nature. Hence, Dr. York's estimates do not represent AEC opinion that these developments will occur. It does appear that, if the development of weapons having these general characteristics does prove feasible, the time scales contained in Dr. York's letter may be somewhat optimistic.

It is hoped that an analysis of the results of experiments currently being conducted in Operation PLUMBBOB will provide more

⁴⁹⁹Letter dated July 11, 1957 to Brig. General A. D. Starbird, Director of Military Application, USAEC, from Herbert F. York, Director, UCRL-Livermore; Atomic Energy Commission, Clean Weapons, Note by the Secretary dated August 23, 1957.

definitive information on the feasibility of development of clean tactical weapons.⁵⁰⁰

Less than a week later, AEC Commissioner Willard Libby noted during an AEC meeting that in view of current disarmament discussions between the U.S. and the Soviet Union in London, it was important to accelerate the development of "clean" tactical weapons.⁵⁰¹

Near the middle of July, Lewis Strauss suggested in a note to Senator Henry Jackson, then a member of the JCAE, that if he were asked about "clean" weapons tests, Jackson should say that "the current (PLUMBBOB) tests are contributing to the overall technology and are part of the program of which the development of clean weapons is an integral part."⁵⁰²

The use of lead still figured prominently in "clean" bomb design, as a tamper around the secondary.⁵⁰³

At the end of July, the AEC requested Presidential permission to expend additional money to re-test modified versions of the PLUMBBOB Lassen and Franklin shot devices which had "fizzled," and because it now appeared possible to test at this time a "clean" tactical device that might be a feasible warhead. The letter to Eisenhower requesting the additional special nuclear material was to specifically cite the latter possibility.⁵⁰⁴

By August, plans for HARDTACK had firmed up enough to project a starting date of May 1, 1958; the AEC sought Presidential approval and outlined the objectives of the test series:

⁵⁰⁰Atomic Energy Commission, Clean Weapons, Note by the Secretary dated August 23, 1957.

⁵⁰¹Minutes of the 1294th AEC Meeting, July 17, 1957.

⁵⁰²Note dated July 19, 1957 sent by Mr. Strauss to Senator Jackson for his use in replying to inquiries.

⁵⁰³"The most significant activity induced in the materials of the 'clean' bomb is Pb-210, which has a half-period of 22 years." (Memorandum for General Starbird dated July 18, 1957, from Cdr. Harry J. Watters, USN, Subject: Residual Radiation Produced in the Air Burst Detonation of a 10 MT, 100 Percent Thermonuclear Device.)

⁵⁰⁴Minutes of AEC Meeting 1298, July 31, 1957; Summary of AEC 944/32, Additional Special Nuclear Material Requirements for Operation PLUMBBOB, USAEC, July 31, 1957.

1. High-altitude firings to test our capability to detect nuclear detonations under these conditions and to obtain effects information applicable to high-altitude missile and counter-missile programs.
2. Several shots necessary for the development of efficient clean weapons in medium and high yield.
3. Proof tests of warheads for the IRBM and ICBM.
4. Developmental and proof tests of several new bombs and warheads for our defense or other tactical use.
5. Exploratory firings to establish principles connected with smaller or cleaner or otherwise improved designs, to include clean, tactical designs.
6. Certain weapons effects tests for the Department of Defense.
7. Further study of the fallout phenomena.⁵⁰⁵

HARDTACK would ultimately test six "clean" designs in a variety of yields.⁵⁰⁶ The complete HARDTACK 27-shot schedule at this time included 12 LASL tests: nine programmatic and three exploratory; 10 UCRL tests: five programmatic and five exploratory, the latter to be selected from 12 possible alternatives; and five DOD effects tests: two underwater shots and three high-altitude tests).⁵⁰⁷ Weather permitting, all UCRL tests would be at Bikini and all LASL tests at Eniwetok.

The two very-high altitude REDSTONE-lofted shots were still expected to contribute more than half of the total HARDTACK fission yield, which was still estimated to be on the same order as that of REDWING; i.e., around nine megatons. In order to more easily track fallout distribution, the high-

⁵⁰⁵Letter dated August 2, 1957 to President Eisenhower from Lewis Strauss, Chairman, USAEC.

⁵⁰⁶Hewlett and Holl, p. 457.

⁵⁰⁷Letter dated July 16, 1957 to Major General A. R. Luedecke, USAF, Commander, JTF 7, from Brig. Gen. A. D. Starbird, USA, Director of Military Application, USAEC.

altitude shots might be "doped" with tracer materials that would generate unique radioisotopes when irradiated during a nuclear explosion. The 100,000-foot high-altitude detonation was to have a four-megaton total yield.⁵⁰⁸

High-altitude sampler aircraft, including U-2, B-57D, and B-50 airplanes, would be used to track distribution of HARDTACK fallout. Existence of the U-2 was still largely unknown outside DOD and the CIA at this time.

Another significant contributor to total HARDTACK fission yield was a planned "dirty" Class C' ("C Prime") shot.⁵⁰⁹

The AEC requested formal Presidential approval for HARDTACK at the beginning of August:

Planning has gone forward, as is necessary, for the next Pacific series, termed Operation HARDTACK, and planned originally to begin about May 1, 1958. HARDTACK is a critical operation to our developmental programs and should include:

1. High-altitude firings to test our ability to detect nuclear detonations under these conditions and to obtain effects information applicable to high-altitude missile and counter-missile programs.
2. Several shots necessary for the development of efficient clean weapons in medium and higher yield.
3. Proof tests of warheads for the IRBM and ICBM.
4. Developmental and proof tests of several new bombs and warheads for our defense and

⁵⁰⁸Letter dated June 26, 1957 to Dr. Alvin C. Graves, LASL, from Duane C. Sewell, Scientific Operations Director, UCRL-Livermore, Subject: Thermal Flux from a 4 Megaton Burst at One Hundred Thousand Feet Elevation.

⁵⁰⁹Letter dated July 26, 1957 to Brig. Gen. A. D. Starbird, Director of Military Application, USAEC, from Alvin C. Graves.

other tactical use.

5. Exploratory firings to establish principles connected with smaller or cleaner or otherwise improved designs, including clean tactical designs.
6. Certain weapons effects tests for the Department of Defense.
7. Further study of fallout phenomena.

In view of the importance of this series to our defense posture and the present state of international relations with respect to testing, it appears desirable that we start HARDTACK as early as possible consistent with achieving the necessary objectives of the operation.

We have reviewed this matter with the Department of Defense and believe that, if the series is limited to essential tests, the series would start about 1 April 1958 and be completed within four months. An estimated 25 shots would be involved. It is not possible at this time to specify the exact devices to be fired or the expenditures of materials involved.

....

... (We) shall submit to you later a more definite listing of devices to be fired and of materials to be expended.⁵¹⁰

Secretary of State John Foster Dulles met with President Eisenhower on August 3 to review HARDTACK plans:

In connection with the discussion of atomic matters, the President said that he had received a communication from Admiral Strauss seeking approval for the next series of tests. The President had been appalled at the number of these tests and was holding up action until I could consider the matter

⁵¹⁰Letter dated August 2, 1957 to The President, The White House, from Lewis Strauss, Chairman, USAEC.

with him.

I also referred to the program we were considering for some very large (yield) tests and said I was concerned lest the matter seem to invalidate the impression the President had given the country that we were on the track of smaller, cleaner bombs, rather than big bombs. The President indicated that he would be reluctant to approve more of the bigger tests.⁵¹¹

On August 7, AEC Chairman Strauss sent President Eisenhower a letter requesting the expenditure of additional special nuclear materials during PLUMBBOB. Reasons cited included the necessity of repeating the Lassen and Franklin shots, and the addition of a test of a "clean" tactical device:

... The (PLUMBBOB) schedule (of February 21, 1957) visualized 24 nuclear tests of devices and seven safety tests of devices of no significant nuclear yield.

Since the start of the series, various modifications in the original schedule have been made, deleting certain shots and adding others. In part, the additions are due to the necessity to fire again modified versions of two earlier shots which did not realize their required yields. One of these (Author's note: the Lassen shot) is an experiment to determine the possibility of eliminating all plutonium from a small diameter warhead.

The other (Author's note: the Franklin test) is the prototype for the warhead for the TALOS missile, soon to be stockpiled. Another shot recently added to the series (Author's note: the Charleston shot) is an experimental firing designed to assist in developing clean tactical weapons.⁵¹²

President Eisenhower's response to the AEC request for approval for HARDTACK was documented in a memorandum by his

⁵¹¹Addendum to Memorandum of Conversation with The President, 4:30 PM, Saturday, August 3, 1957.

⁵¹²Letter dated August 7, 1957 to The President, The White House, from Lewis Strauss, Chairman, USAEC. Eisenhower approved the request on August 8.

son John, an Army major at the time. Present at a HARDTACK review meeting two days later in the White House were Eisenhower and son, Lewis Strauss, Eisenhower's national security adviser Robert Cutler, and Gerard C. Smith from the State Department.

The President was concerned about both the size and the projected duration of HARDTACK; he wondered if the series could be limited to only a dozen shots so as to limit both the expenditure of fissionable materials and the amount of time required for the test series:⁵¹³

Numbers of Weapons - The first question which had disturbed the President with relation to Operation HARDTACK was the number of weapons involved. The President felt that the number 25 was high. Admiral Strauss explained that the number had been arrived at arbitrarily, that he had two laboratories, each of which had requested 20 shots, and had received a request from the Department of Defense for 10 shots.

This figure he had, at first glance, cut in half. Admiral Strauss further explained that he had every intention of shaving down below the mark of 25.

Duration - The next question of concern was the duration of the atomic tests, lasting from 1 May probably through August. The President felt that an excessive amount of time consumed in testing served to magnify our efforts to the world at a time when disarmament negotiations are under way. Admiral Strauss' explanation was a requirement for perfect meteorological conditions for each experiment, particularly for the large yield weapons.

The Admiral pointed out that on occasion he had been required to wait for two weeks for firing a given shot. He expressed the opinion, however, that he would be able to condense the four-month period of time to

⁵¹³Memorandum of a Conference, White House, Washington, D.C., August 9, 1957, document 282 in FOREIGN RELATIONS OF THE UNITED STATES, 1955-1957, Volume 20: Regulation of Armaments; Atomic Energy, John P. Glennon and David S. Patterson (eds.), U.S. Government Printing Office, Washington, D.C., 1990, p. 698.

some extent.

Size - In answer to the President's question on the necessity for tests of large yield weapons, Admiral Strauss had the following to say:

1. AEC and State Department cannot justify a need for the very large weapons. The requirement to test multi-megaton weapons comes primarily from the Department of Defense which is interested in ascertaining the (maximum) size of yield which may be carried in the B-52. The weight of the bomb is estimated at 25,000 pounds.
2. [deleted]
3. Inaccuracy of delivery means can be compensated for in part by increasing the size of the weapon. (The President pointed out, however, that the scaling laws apply on a cube root basis, which would give a 40-megaton weapon a radius of damage only about 1 1/2 times the radius of damage of the 10 megaton weapon.)

Admiral Strauss then mentioned a compromise which he felt he could use to advantage. This compromise would require that no weapon would be used which had a yield larger than that of the 1954 test (15 MT). He further stated that this 15 MT weapon would be far cleaner than the 1954 version.

Altitude - The question came up as to the height at which the high altitude test shots will be burst. Admiral Strauss gave as the top figure 250,000 feet, rocket delivered.

Type - Admiral Strauss estimates that all the weapons tested, with the exception of a few of the smaller ones, will be of the

thermonuclear type.

Public Relations - The main dilemma in conducting tests of this magnitude in 1958, as the President sees it, is that of planning and carrying out extensive tests on the one hand while professing a readiness to suspend testing in a disarmament program on the other.

From much of the world this paradoxical conduct may bring accusations of bad faith. The President is agreed, however, that having gone this far, it is necessary to carry through with HARDTACK. Several measures will be followed to place this decision in the best possible light:

1. Observers from various nations will be invited to witness the shots.
2. Our position in the disarmament talks must be that we cannot suspend testing until a date at least a year from now. Until that time we, as others, will continue to test as necessary.
3. The United States will maintain a flexible attitude in these negotiations for disarmament, and would agree to accept the inevitable inefficiency which will result from a two-year layoff. As to the problem of losing the best scientific talent as a result of the two-year layoff, it was agreed that the bulk of the best talent could be retrieved and the result in organizational setbacks would just have to be accepted. (Author's note: a reference to a proposed two-year international nuclear test moratorium.)
4. The testing schedule will be

kept confidential until more thought can be given as to how to announce it.

5. The time span for the tests will be condensed to the maximum extent within the power of the Atomic Energy Commission.
6. The statement may be made, in which the British will probably join us, that we will put no more radioactivity into the atmosphere than is taken care of by normal decay of that contamination already in the atmosphere.

Conclusions - The President granted authority to continue the HARDTACK tests within the following limitations:

1. Testing of large yield weapons will be accomplished with weapons no larger than that exploded in 1954.
2. The time span for the series of tests will be condensed to the maximum extent.
3. Announcement of the tests will be withheld and the matter will remain confidential until full details of the announcement are worked out.⁵¹⁴

⁵¹⁴Memorandum of Conference with the President, August 9, 1957, 2:30 PM, signed by John S. D. Eisenhower, Major, U.S. Army, also included as document 284 in FOREIGN RELATIONS OF THE UNITED STATES, 1955-1957, Volume 20: Regulation of Armaments; Atomic Energy, John P. Glennon and David S. Patterson (eds.), U.S. Government Printing Office, Washington, D.C, 1990, pp. 699-701. In the original meeting notes, "cube root" is spelled "cube route." The HARDTACK shots were the last Pacific tests that Eisenhower authorized; the next Pacific series, Operation DOMINIC in 1962, was

As part of a program to "acquaint the public with the purposes of (nuclear) testing and the true nature of associated fallout," the director of the AEC's Division of Information Services proposed at this time that a "vigorous" public information program be instituted. Among recommendations for this campaign were:

(There is an erroneous public view regarding fallout.) Scientific confusion regarding the term "clean weapons" has tended to obscure the real military and civilian advantages of developing nuclear explosive devices free or relatively free from fallout.

Many individuals seem to believe that the sole impetus for such development arises from a desire to minimize fallout from weapons testing. (Author's note: the AEC, for its own political purposes, had done little to discourage this attitude.)

(We should correct this misconception by stating that) the development of a "clean" bomb is motivated primarily by military consideration and the possibility of civilian use of nuclear explosives, not solely by a desire to minimize radioactive fallout from testing, which is already extremely low.

....

Basic moves to get the facts to the public would include:

- a. Clarification of the term "clean weapons" by providing a bench-mark to assist the public in evaluation of statements about varying degrees of cleanliness. Without such a reference point, the phrase "95 percent clean," for example, has

conducted during John Kennedy's presidency. Eisenhower also specifically forbade any public announcement by the AEC or DOD of the inclusion of a test a large "clean" bomb during HARDTACK. (Memorandum of a Conference, White House, Washington, D.C., August 9, 1957, document 283 in FOREIGN RELATIONS OF THE UNITED STATES, 1955-1957, Volume 20: Regulation of Armaments; Atomic Energy, John P. Glennon and David S. Patterson (eds.), U.S. Government Printing Office, Washington, D.C, 1990, p. 698.)

no meaning; it simply compounds existing public confusion.

b. Scheduling of at least two clean weapons development shots during the HARDTACK series, along with provisions for observation of the detonations and first-hand post-shot inspections of ground zero by news media representatives and official observers of other nations. This event should emphasize the peacetime utility of clean nuclear explosives.

For this purpose, improvement of the atoll of Taongi has been suggested. Taongi consists of a solid reef surrounding a stagnant lagoon studded with coral heads. If the reef were breached and the coral heads cleared in such a demonstration, the atoll would again support a native population and be useful to shipping and seaplane traffic.

This demonstration would fasten the attention of the world on the peaceful benefits to be expected from the development of clean nuclear explosives from further testing.⁵¹⁵

Earlier in the morning of the same day, Eisenhower had met with John Foster Dulles, Gerard Smith, and Robert Cutler to review the AEC request for a test of a high-yield "clean" bomb during HARDTACK. Eisenhower was very dubious about a requirement to test larger bombs and whether there was a need for a large "clean" bomb.

The President thought that U.S. statecraft was becoming too much a prisoner of the country's nuclear scientists, and he noted that he had stated publicly on several previous occasions that the United States was not going to test any more large bombs and that it was now trying to "clean up" its

⁵¹⁵Attachment to memo dated 8 August 1957 from Morse Salisbury to Col. Kenneth E. Fields, General Manager of the AEC, subject: Draft Report Paper re Public Information Program on Weapons Testing.

small bombs. Nonetheless, Eisenhower did not order that the high-yield "clean" test be canceled, although it was to be conducted at less than full yield.

When Eisenhower suggested that a number of small weapons be used against a target in lieu of a single high-yield weapon, Robert Cutler noted the concern of the Joint Chiefs of Staff over the increased expense of delivery vehicles, both manned and unmanned. The Chiefs also feared a high attrition rate for manned bombers enroute to their targets, negating the likelihood of a large number of warheads successfully reaching their destinations.⁵¹⁶

At this time, the Secretaries of Defense and State and the Chairman of the AEC issued a report in support of the test of the large "clean" bomb during HARDTACK; they were still seeking explicit presidential approval for the shot:

On July 18, 1956, you directed the Secretaries of State and Defense, and the Chairman, Atomic Energy Commission, to make a study and recommendations thereon with respect to a letter addressed to you by AEC Commissioner Thomas E. Murray (regarding the need for high-yield bombs). Mr. Murray stated his conviction that (high yield) bombs we now have in stockpile are large enough and raised two basic questions:

"1. Is a weapon of this size necessary or useful for military purposes?

"2. Would its use be consistent with the dictates of the moral law with regard to the moderate and discriminate use of force in warfare?"

Our recommendations have been postponed until the present to permit a careful review of the military need for high yield weapons, the technical feasibility and time schedules involved in developing such weapons, and the probable political impact should the

⁵¹⁶Memorandum of a Conference, White House, Washington, D.C., August 9, 1957 (Morning), document 281 in FOREIGN RELATIONS OF THE UNITED STATES, 1955-1957, Volume 20: Regulation of Armaments; Atomic Energy, John P. Glennon and David S. Patterson (eds.), U.S. Government Printing Office, Washington, D.C, 1990, pp. 694, 695.

development become generally known.

The Department of Defense and the Atomic Energy Commission conclude that major emphasis in (high yield) weapon development at this time should be placed on reduced fall-out, (high yield) bombs.

Meanwhile, the Atomic Energy Commission has continued to investigate the feasibility of developing very high yield weapons with the objective of determining the upper yield limit of a weapon which would be compatible as to size and weight with the B-52 aircraft.

From laboratory analysis and prior experimentation, the Commission considers it feasible to design a weapon of low fission yield (hereinafter referred to as "clean") [deleted] A broad program of developing "clean" weapons fostered by the Atomic Energy Commission and the Department of Defense is in progress as a consequence of advancing technology in this field and in response to military requirements for applications in which reduced fallout is essential.

"Clean" weapons in medium and low megaton range are under active development and studies are in progress looking toward the development of "clean" weapons in the low kiloton range. [deleted]

A complementary or substitute "clean" weapon (with) the highest yield range corresponding (to) the delivery capability of the B-52 aircraft is not under development. From the military standpoint, there are two factors which make it desirable that the (Air Force's) Strategic Air Command have the capability of delivering with certain of its aircraft a higher yield weapon than is now on hand.

In the first place, such a high yield could be substantially more effective in destroying certain very "hard" Soviet military targets offering direct threats against the United States; targets such as long-range aircraft bases, missile launching sites, submarine pens, nuclear storage sites, and underground command and

communications centers.

In the second place, the increasing complexity and cost of first-line aircraft and the attrition which must be expected from Soviet offensive and defensive action make it necessary to base war plans on the arrivals of reduced numbers of our aircraft over their targets. This factor makes it necessary for our forces to attain maximum unit effectiveness.

It is a matter of serious concern that the effective use of the high yield weapons now in stockpile against such targets (lying) in satellite and border areas could create heavy casualties in adjacent friendly and neutral countries from radioactive fallout.

Yet if we use present weapons but exercise the degree of restraint necessary to protect such populations, we would compromise seriously the effectiveness of our retaliatory attack.

On the other hand, military calculations show that if we substituted very high yield "clean" weapons [deleted] we could decrease the over-all intensity of fallout in adjacent areas by a factor of [deleted] in the case of groundbursts, and by a factor of 30, in the case of airbursts. There is, therefore, a sound military basis for the development of "clean" weapons of the highest yield deliverable by our aircraft.

It is most desirable that we establish definitely what the upper yield limit of a "clean" weapon capable of delivery by the B-52 may be, [deleted]

This answer can be obtained with reasonable certainty if a research and developmental program were carried through the next Pacific test series. (A) decision may be made thereafter as to whether pre-production and production effort would be (started); this decision (would be) based on the then-applying climate of international relations, on the yield possible, and a later estimate of the strategic need for the potential weapon.

The department chiefs stressed the need for secrecy in the

development of the high-yield bomb, warning of unfavorable consequences of early publicity, and suggested that development of the device remain in a research stage only:

Should it become known that continued United States testing is directed in part towards development of a weapon larger than that we now possess, some unfavorable international and possibly national reaction might be expected. Certainly the Soviets could be expected to use this for any propaganda advantage which might be gained.

Therefore, it is not believed necessary nor appropriate that the United States decide now to embark upon a program to develop and stockpile a [deleted] weapon of maximum yield. Rather, we should proceed with our research and early developmental effort to a point where we know the characteristics of the weapon which might be possible.

If information of our course of action becomes known publicly, its effect could be minimized by a frank and factual statement to the effect that research and development is exploratory, that the test devices to be fired will be "clean" and will be of smaller total yield than our firings in 1954 [deleted]

....

In view of the considerations outlined above, it is recommended that:

a. There be a test in the next Pacific test series of a device to determine the technical feasibility and probable maximum yield of a "clean" weapon as large as that which can be carried satisfactorily in our largest bombers of near future years (a weapon of [deleted] and otherwise meeting carrier dimensions).

The planned gross yield of the test device should be as low as possible consistent with securing the required information and in no case

greater than our previous largest Pacific test shots and, of course, far smaller in fission yield (fall-out).

b. If following completion of this test the Department of Defense considers that it requires development of weapons of this or related designs, we will submit to the President our recommendations as to the undertaking of a pre-production development program.

It shall be understood that the authorization recommended in the paragraphs above shall in no way be interpreted as establishing a requirement for a larger weapon than that now in development nor for carrying on a pre-production developmental program for such a weapon.⁵¹⁷

About two weeks later, on August 22, Eisenhower announced publicly that the U.S. would cease nuclear testing indefinitely after October 31, 1958, provided that the nuclear weapons powers, then limited to the U.S., the Soviet Union, and the United Kingdom, could establish an enforceable nuclear weapons production and testing inspection system, and strive to limit nuclear arms production and deployment.⁵¹⁸ To put it mildly, this announcement was not greeted with enthusiasm by the AEC, DOD, or the Los Alamos and Livermore weapons labs.

The next day, Acting AEC Chairman Willard F. Libby notified the chairman of the JCAE of changes to Operation PLUMBBOB. Libby noted that two of the scheduled devices had been detonated with disappointingly-low yields: the XW-30 prototype fired in the Franklin shot and the UCRL all-oralloy device in the Lassen test.

⁵¹⁷Report to The President from the Secretary of State, Secretary of Defense Wilson, and Chairman Strauss, USAEC (undated, but ca. August 9, 1957), document 285 in FOREIGN RELATIONS OF THE UNITED STATES, 1955-1957, Volume 20: Regulation of Armaments; Atomic Energy, John P. Glennon and David S. Patterson (eds.), U.S. Government Printing Office, Washington, D.C, 1990, pp. 701-703, supplemented by a more complete version available from Research Publications in Woodbridge, Connecticut. Emphasis in original.

⁵¹⁸Hewlett and Holl, p. 546.

The AEC now planned to fire a modified design of each device before the end of the test series. This action was necessary because the XW-30 had to meet an OAD of July 1958 for application to the Navy's TALOS air defense missile. The UCRL device offered promise of a small, plutonium-free air-to-air rocket implosion warhead (which led to the W-54).

In addition to these "repeat" shots, the AEC also decided that it would fire two more devices. One was a higher-yield version of a basic-design primary for the TX-41, which was to be produced in both "clean" and "conventional" models. (This became the Smoky shot of August 31.)

The other device was a UCRL prototype of a "clean" tactical weapon, an exploratory device which might lead to the design of a "clean" tactical weapon in the 1,000 to 1,500 lb. weight range with a 20" diameter and a high-fusion percentage yield.⁵¹⁹ (This became the Charleston shot of September 28.)

By this time, planning had started for Operation TRUMPET, the next series of Nevada tests.⁵²⁰

Near the end of August, AEC Commissioner Willard F. Libby requested that the AEC examine the possibility of using either lower-yield or "cleaner" warheads for the high-altitude HARDTACK shots, in order to reduce off-site fallout.⁵²¹

Plans for the "Open" Shot

Early in September, General Alfred Starbird, then AEC

⁵¹⁹Letter dated August 23, 1957 to Carl T. Durham, Chairman, JCAE, from Willard F. Libby, Acting Chairman, USAEC; letter dated August 27, 1957 to General Herbert B. Loper, Chairman, MLC, from Willard F. Libby, Acting Chairman, USAEC.

⁵²⁰Memorandum dated 21 August 1957 to Commanding General, Aberdeen Proving Ground, from Maj. Ralph J. Truex, U.S. Army Ordnance Corps, subject: Objectives and Requirements for Operation TRUMPET; memorandum dated 28 August 1957 to Chief of Ordnance, Dept. of the Army, from Joseph Sperrazza, Deputy Chief, Terminal Ballistics Laboratory, subject: Project Proposals for Operation TRUMPET.

⁵²¹Memorandum dated August 27, 1957 (revised September 4, 1957) to Brig. Gen. Alfred D. Starbird, Director, Division of Military Application, USAEC, and Dr. Charles L. Dunham, Director, Division of Biology and Medicine, USAEC, from Harold D. Anamosa, Acting Secretary, subject: Commission Decision on AEC 952/11 - Fallout Program - HARDTACK.

Director of Military Application, sent a teletype to LASL and UCRL requesting guidance on how to conduct an open shot during HARDTACK that would demonstrate U.S. thermonuclear weapons "cleanliness" without giving away design secrets:

Consideration is being given to sending an international group to observe one or more shots in HARDTACK in an attempt to prove to them our ability to develop and fire clean devices. I discussed this briefly with Bradbury and York at our last meeting in Nevada and asked that they give me some thought as to how this might be done without revealing design information.

One problem of course, is that the information given must be relatively convincing. Also, insofar as is practical without revealing design, it should be capable of giving the best quantitative measure of the degree of cleanliness. Query: given the above problem, what do you believe would be the best way of undertaking the task and what degree of success would you visualize? Would appreciate your earliest thoughts.⁵²²

In response to Starbird's teletype, Darol Froman of LASL reported on likely methods of proving "cleanliness" of weapons to scientific observers without simultaneously revealing weapon design information; this was a prerequisite to any "open shot" to convince foreign powers of U.S. progress in "clean" bombs.

Regarding your teletype on proving cleanliness of weapons to international group, the only really convincing proof must contain believable data on total yield and on either fission yield or thermonuclear yield. A convincing value of total yield can be determined by hydrodynamic techniques such as ball of fire, shock arrival time, or shock intensity without revealing design information.

Past experiments indicate that a convincing value of the fission yield can be measured

⁵²²Teletype dated 9 September 1957 to Darol Froman and Duane Sewell from A. D. Starbird.

by an absolute measurement of the gamma radiation from the ball of fire during the period from one to ten seconds after detonation. Main source of gamma radiation between five and ten seconds is fission fragments since neutrons released to air have disappeared and contribution from activated sodium, etc. is negligible at this time.

....

We guess that a group of half a dozen scientists and technicians could prepare for and carry out this (fission yield) measurement and the total yield measurement by working two or three months before test. ... Measurements of both total yield and fission yield could be performed by an international group or by U.S. scientists under their observation.

One way to prevent design data from being revealed was to deliberately obstruct results of the "open" shot:

If there is a problem connected with deductions about design which might be made by a group which has the total and fission yields and which collects its own radiochemical samples, these deductions could be confused by placing quantities of extraneous materials such as tuballoy near the bomb.

It might also be possible to provide a comparison between a "clean" and a "dirty" bomb, if the U.S. were willing to reveal that it was still developing anything other than "clean" weapons:

There is no reason to limit above observations to a clean device and, if similar measurements were made on a dirty or pure fission bomb, it would provide a kind of check on the absolute calibration of the gamma detectors. However, you may not wish to admit that we are testing anything but clean bombs and we believe the job could probably be accomplished on the basis of a single test with a clean device.

Further complicating plans for an "open" shot at this time was the paucity of suitable test devices:

Present plans for LASL do not seem very

appropriate for the demonstration because of the positions of barges in the lagoon and the types of devices to be tested.

In particular, we are unhappy at the thought of using a previously-untested, relatively small clean device for such a demonstration because it might fail to burn well and give a quite large fraction of fission. Moreover, the fractional cleanliness is expected to be greater for large yields and it is for large yields that cleanliness is important.

Therefore, we suggest a test such as the clean Class B (if Livermore is sufficiently confident of its performance) rather than, for example, the test of the Clean Class C for such a demonstration. Obviously a device with a high degree of confidence in performance will have to be used. ...

At present, LASL is making no preparations to implement or assist in the suggested demonstration.⁵²³

(TO BE CONTINUED)

⁵²³Teletype dated September 10, 1957 to Starbird, USAEC, from Froman, LASL.

VOLUME V INDEX

Note: All page numbers prefixed by "V-"

- 4-
- 4925th Test Group (Atomic),
56, 139, 157
- A-
- A3D (A3), 4, 52, 98, 225
- Aberdeen Ballistic
Laboratory / Aberdeen
Proving Ground, Maryland,
127, 355
- Able shot, Operation
CROSSROADS, 189
- AJ/AJ-1/AJ-2, 4
- Alarm Clock;, 6, 7, 8
- Aluminum, 7, 67, 187, 202
- Anti-Submarine Rocket
(ASROC), 226, 232, 266,
267
- Anti-Submarine Torpedo
(ASTOR, 267, 283
- Apache shot, Operation
REDWING, 116, 128, 133,
139, 143, 146, 150, 156,
158, 164, 180, 184, 195,
209, 214, 215, 219, 223,
224
- ARGUS
See Operation ARGUS, 246
- Armed Forces Special Weapons
Center / Air Force Special
Weapons Command (AFSWC),
103, 123, 143
- Armed Forces Special Weapons
Project, 193, 246, 258,
264
- Armed Forces Special Weapons
Project (AFSWP), 77, 98,
100, 103, 110, 123, 127,
133, 137, 144, 207, 241,
258, 287, 305
- ATLAS D, E, F ICBM, 70, 118,
167, 172
- B-
- Baker shot, Operation
CROSSROADS, 132, 241
- Ballistics, 355
- Beryllium, 56
- Bikini Atoll, 18, 99, 104,
109, 114, 116, 117, 118,
119, 122, 123, 126, 127,
128, 133, 135, 138, 139,
141, 142, 143, 147, 149,
150, 154, 155, 156, 158,
168, 169, 180, 182, 184,
185, 186, 187, 188, 189,
191, 192, 194, 195, 203,
206, 209, 210, 211, 214,
215, 216, 217, 218, 219,
278, 279, 292, 293, 294,
295, 296, 300, 302, 303,
304, 305, 311, 312, 340
- Blackfoot shot, Operation
REDWING, 115, 126, 127,
133, 138, 142, 145, 150,
156, 167, 183, 195, 206,
207, 208, 218
- Boeing Airplane Company,
187, 225
- BOMARC surface-to-air
missile, 228
- Boosting, 55, 59, 112, 137,
145, 211, 257, 310
- Boron, 89
- Bradbury, Norris E., 22, 24,
27, 28, 29, 34, 35, 36,
37, 38, 39, 40, 41, 42,
45, 46, 49, 50, 51, 52,
57, 58, 59, 61, 66, 70,
73, 79, 100, 101, 102,
111, 121, 130, 133, 134,
148, 149, 205, 206, 249,
284, 287, 289, 291, 312,
322, 356
- Bravo shot, Operation
CASTLE, 11, 18, 80, 114,
151, 171, 181, 215, 234,
303, 316, 331
- C-
- Capsules, nuclear, 69, 240
- Charleston shot, Operation
PLUMBBOB, 338, 343, 355
- Chemical and Metallurgical
Research (CMR) Division,
LASL, 212, 214, 216

Cherokee shot, Operation
 REDWING, 97, 99, 116, 127,
 138, 139, 141, 142, 143,
 149, 155, 156, 157, 158,
 165, 169, 180, 182, 183,
 185, 187, 188, 189, 190,
 191, 193, 208, 209, 218,
 224, 225
 Class, 25, 26, 55, 57, 68,
 69, 70, 73, 74, 76, 78,
 79, 80, 107, 117, 160,
 165, 223, 232, 245, 253,
 259, 261, 262, 263, 272,
 274, 275, 290, 291, 298,
 304, 311, 317, 341, 358
 Composite cores, 159, 177,
 230, 246
 Compression, nuclear fuel,
 53, 122, 201, 202
 Copper, 187, 193, 202
 CORPORAL tactical missile,
 239, 246, 269, 270
 CORVUS (XASMN-8) air-to-
 surface missile, 283
 Cryogenics, 100

-D-

Dakota shot, Operation
 REDWING, 146, 151, 156,
 158, 195, 207, 209, 211,
 215, 219, 223, 224
 Dean, Gordon Evans, 34, 44,
 45, 47, 48, 49
 Defense Atomic Support
 Agency, 96, 195
 Deuterium, 20, 53, 55, 71,
 80, 283
 Dirty, 2, 10, 94, 247, 265
 Dog shot, Operation
 GREENHOUSE, 132, 257
 Dulles, John Foster, 146,
 147, 321, 342, 349
 Dummy devices, 186

-E-

EAGLE air-to-air missile,
 187
 EG&G (Edgerton,
 Germeshausen, and Grier),
 180
 Eisenhower, Dwight D., 36,
 53, 54, 74, 75, 80, 99,
 105, 107, 152, 158, 162,

163, 164, 165, 173, 180,
 196, 199, 200, 213, 220,
 234, 265, 266, 268, 282,
 305, 306, 307, 313, 314,
 317, 318, 319, 320, 321,
 325, 329, 332, 339, 340,
 342, 343, 347, 349, 350,
 354
 Emergency Capability, 3, 4,
 7, 52, 75, 147, 297, 298
 Eniwetok Atoll, 8, 31, 95,
 104, 109, 115, 116, 117,
 118, 119, 123, 126, 127,
 128, 133, 135, 138, 140,
 141, 142, 143, 147, 149,
 150, 153, 155, 156, 157,
 165, 169, 176, 180, 181,
 182, 183, 184, 185, 186,
 187, 188, 191, 194, 195,
 200, 202, 203, 206, 207,
 208, 211, 212, 214, 215,
 216, 217, 218, 219, 232,
 251, 254, 261, 264, 265,
 266, 271, 278, 292, 293,
 294, 295, 300, 302, 303,
 304, 307, 311, 312, 326,
 340
 Eniwetok Proving Grounds
 (EPG), 217, 224, 232, 235,
 279
 Erie shot, Operation
 REDWING, 138, 142, 146,
 150, 155, 156, 195, 208,
 209, 218, 223

-F-

Fallout, 10, 11, 16, 18, 25,
 64, 66, 67, 69, 87, 91,
 98, 114, 115, 118, 119,
 122, 124, 125, 137, 141,
 154, 163, 164, 166, 179,
 188, 190, 193, 194, 196,
 197, 199, 200, 201, 202,
 210, 214, 216, 217, 218,
 219, 220, 221, 222, 223,
 225, 227, 228, 230, 231,
 235, 236, 241, 242, 243,
 244, 248, 251, 255, 263,
 265, 266, 272, 281, 288,
 293, 294, 295, 296, 299,
 300, 301, 303, 305, 306,
 307, 308, 309, 313, 317,
 318, 319, 320, 321, 322,

325, 326, 327, 337, 340,
 341, 342, 348, 351, 352,
 355
 Fat Man, 30
 Fermi, Enrico, 35
 Field Command, DASA
 (FCDASA), 127
 Fission, nuclear, 7, 10, 11,
 12, 13, 20, 29, 30, 31,
 34, 36, 40, 55, 58, 59,
 63, 64, 67, 68, 69, 75,
 81, 82, 83, 87, 88, 89,
 94, 98, 103, 104, 108,
 109, 112, 113, 118, 119,
 120, 123, 124, 125, 140,
 141, 144, 147, 151, 154,
 161, 164, 165, 179, 183,
 192, 193, 195, 196, 197,
 198, 203, 204, 205, 214,
 216, 217, 222, 223, 224,
 227, 228, 230, 236, 241,
 244, 245, 247, 248, 253,
 254, 264, 272, 279, 283,
 289, 290, 291, 294, 296,
 297, 299, 300, 301, 302,
 303, 304, 306, 313, 316,
 317, 324, 326, 327, 329,
 333, 334, 336, 337, 338,
 340, 341, 351, 354, 356,
 357, 358
 Fizzle, 60
 Flathead shot, Operation
 REDWING, 116, 128, 138,
 142, 146, 150, 156, 193,
 195, 203, 206, 209, 217,
 218, 223
 Fusion, thermonuclear, 10,
 14, 16, 29, 34, 36, 40,
 54, 55, 64, 66, 67, 68,
 71, 82, 83, 87, 89, 94,
 98, 119, 122, 124, 149,
 151, 152, 154, 164, 165,
 167, 193, 196, 197, 198,
 201, 204, 205, 211, 214,
 215, 216, 222, 223, 224,
 282, 296, 302, 313, 316,
 324, 325, 326, 327, 355
 Fuzing, 18, 19, 95, 241

-G-

General Advisory Committee
 (GAC), USAEC, 9, 19, 28,
 29, 34, 57, 58, 59, 60,

61, 68, 89, 107, 108, 140,
 141, 188, 221, 249, 253,
 325, 326
 GENIE air-to-air rocket, 255
 George shot, Operation
 GREENHOUSE, 88, 161, 189
 Glenn L. Martin Aircraft
 Company, 109, 160, 271
 GMX Division, LASL, 21, 139,
 182
 Gold, 67, 88, 318

-H-

HA shot, Operation TEAPOT,
 18, 112
 Hanford, Washington
 plutonium reactors, 8, 282
 HAWK surface-to-air missile,
 283, 336
 HE (high explosives), 21,
 22, 23, 24, 139
 Helium, 241
 Hickam AFB, Hawaii, 211
 Hiroshima, Japan, 30
 Holmes & Narver, 187, 193,
 202
 Hood shot, Operation
 PLUMBBOB, 324
 Huron shot, Operation
 REDWING, 128, 133, 138,
 142, 150, 156, 182, 183,
 195, 216, 217, 218, 219
 Hydrogen, 10, 28, 29, 30,
 32, 33, 34, 35, 36, 37,
 38, 39, 41, 42, 45, 46,
 47, 49, 55, 81, 86, 87,
 99, 140, 147, 153, 154,
 179, 194, 199, 200, 206,
 212, 215, 220, 221, 224,
 234, 251, 252, 275, 280,
 301, 317, 320, 324

-I-

Implosion, 9, 20, 21, 23,
 30, 55, 56, 57, 61, 66,
 68, 69, 73, 74, 75, 80,
 95, 102, 110, 111, 112,
 129, 135, 144, 145, 160,
 162, 167, 212, 222, 228,
 250, 254, 265, 283, 310,
 355
 Inca shot, Operation
 REDWING, 145, 183

Intercontinental Ballistic
Missile (ICBM), 162, 167,
171, 179, 199, 222, 223,
232, 240, 250, 254, 256,
257, 258, 259, 261, 262,
265, 280, 298, 302, 336,
340, 341

Intermediate-Range Ballistic
Missile (IRBM), 162, 223,
232, 240, 261, 270, 302,
340, 341

Isotope, 9, 12, 14, 18, 89,
165, 166, 180, 282

-J-

J Division, LASL, 103, 104,
208

JCS (Joint Chiefs of Staff),
3, 7, 132, 145, 163, 286,
291, 297

Johnston Island, 235, 299,
300

Joint Committee on Atomic
Energy (JCAE), 15, 17, 35,
38, 48, 76, 108, 109, 111,
137, 144, 145, 151, 152,
155, 162, 165, 172, 180,
189, 194, 207, 217, 220,
234, 251, 257, 259, 266,
272, 278, 281, 283, 284,
292, 301, 305, 309, 310,
319, 322, 339, 354, 355

Joint Task Force (JTF) 7,
98, 99, 115, 126, 141,
143, 180, 184, 186, 194,
203, 207, 208, 210, 212,
216, 222, 278, 293, 294,
295, 311, 340

Jughead, 7

JUPITER IRBM, 246

-K-

Kennedy, John F., 348

Kickapoo shot, Operation
REDWING, 116, 126, 128,
138, 142, 145, 150, 156,
164, 183, 195, 207, 208,
209, 218, 219

King shot, Operation IVY,
319

Kirtland AFB, New Mexico,
116, 126, 157

Kwajalein island, Kwajalein
Atoll, 272

-L-

LACROSSE missile, 226

Lacrosse shot, Operation
REDWING, 128, 133, 138,
140, 142, 143, 149, 155,
156, 158, 183, 185, 186,
190, 193, 209, 215, 218,
219

Landing Craft, Utility
(LCU), 191

Landing Ship Dock (LSD),
186, 305

Lawrence, E.O./Lawrence
Laboratory, Livermore, CA,
95, 146, 239, 271, 303,
317, 318, 319, 321, 322,
329

Laydown, 159, 162, 226, 232,
261, 298, 304

Lead (Pb-203), 32, 47, 65,
67, 72, 98, 105, 122, 152,
172, 193, 198, 201, 202,
205, 206, 212, 215, 216,
247, 263, 289, 320, 339,
355

Linear implosion, 9, 23, 55,
56, 57, 61, 74, 75, 102,
112, 129, 145

Lithium, 6, 8, 9, 18, 20,
53, 56, 71, 80, 89, 137,
147, 164, 180, 201, 229,
282, 283

Lithium deuteride, 18, 164,
180, 201

Lithium-6, 54

Lithium-7, 54, 89

LITTLE JOHN tactical
missile, 226, 232, 329

Lockheed Aircraft/Missiles &
Space Corporation, 246

Los Alamos Scientific
Laboratory (LASL), 2, 8,
9, 10, 11, 15, 21, 22, 23,
24, 25, 26, 27, 28, 29,
30, 31, 33, 34, 35, 36,
39, 40, 41, 42, 43, 46,
47, 49, 51, 55, 56, 57,
60, 61, 63, 64, 65, 66,
68, 69, 70, 76, 79, 86,
87, 90, 95, 96, 97, 98,

100, 101, 102, 103, 104,
 105, 107, 108, 110, 111,
 112, 115, 116, 118, 119,
 120, 121, 122, 126, 127,
 128, 130, 133, 134, 139,
 141, 142, 143, 144, 145,
 146, 148, 149, 150, 153,
 155, 156, 158, 159, 167,
 168, 169, 171, 178, 180,
 182, 183, 184, 186, 187,
 192, 195, 201, 203, 205,
 206, 207, 208, 211, 215,
 217, 218, 222, 228, 232,
 238, 241, 242, 243, 244,
 249, 250, 255, 256, 258,
 259, 268, 275, 278, 279,
 280, 281, 282, 284, 287,
 288, 289, 290, 291, 292,
 293, 298, 300, 301, 302,
 304, 331, 338, 340, 341,
 356, 357, 358
 Low-Altitude Bombing System
 (LABS), 143
 LULU
 see MK 101, 79, 80, 228,
 259, 266, 267, 268

 -M-
 Mike device, Operation IVY,
 120
 Mike shot, Operation IVY,
 29, 31, 42, 103, 277
 Military Air Transport
 Service (MATS), 185
 Military Sea Transportation
 Service (MSTS), 185
 MK 10, 267
 MK 101 (LULU), 267
 MK 12, 275
 MK 15, 19, 52, 58, 75, 97,
 99, 108, 143, 183, 188,
 222
 MK 17, 18, 19, 52, 76, 80,
 109
 MK 18, 30, 32, 58
 MK 21, 19, 52, 75, 80, 94,
 97, 109, 165, 222, 234,
 247, 248, 253, 280, 281
 MK 24, 19, 76, 109
 MK 26, 76
 MK 27, 26, 75, 98, 222
 MK 28, 26, 75, 209, 222, 269
 MK 34, 153

MK 36, 97, 236, 245, 253,
 254, 281
 MK 39, 75, 97, 222, 288
 MK 4, 98, 153, 160, 269, 288
 MK 41, 98, 153, 160
 MK 46, 288
 MK 49, 269
 MK 5, 29, 58, 207
 MK 7, 58, 69, 255
 MK 8, 275
 MK 9, 58, 266, 275
 MK 90 (BETTY), 266
 Mohawk shot, Operation
 REDWING, 116, 126, 128,
 138, 139, 142, 145, 150,
 156, 164, 183, 184, 195,
 208, 209, 211, 212, 213,
 214, 219
 Morgenstern, 60, 94, 95, 193
 Moth shot, Operation TEAPOT,
 56, 73, 112, 114

-N-

National Bureau of
 Standards, 181
 National Bureau of Standards
 (NBS), 80
 NAVAHO (B-64), 69, 227
 Navajo shot, Operation
 REDWING, 116, 128, 138,
 142, 150, 156, 158, 185,
 193, 215, 216, 219, 224
 Naval Ordnance Laboratory,
 White Oak, Maryland, 22,
 23, 24, 139, 268
 Naval Ordnance Test Station,
 Inyokern, California
 (NOTS), 139
 Neutron flux, 11, 67, 205,
 206, 257, 258
 Neutron generator (see also
 Initiator, 93
 Nevada Proving Ground (NPG),
 53, 54, 55, 57, 64, 68,
 110, 111
 Nevada Test Site (NTS), 55,
 56, 57, 74, 108, 117, 120,
 128, 135, 153, 170, 219,
 232, 241, 281, 324
 Nichols, Kenneth D., 66, 90,
 101
 NIKE ZEUS surface-to-air
 missile, 298, 336

Nuclear hand grenades, 161

-0-

Open shot,, 168, 320, 356

Operation ARGUS, 246

Operation CASTLE, 3, 5, 6,
7, 8, 9, 11, 15, 16, 18,
19, 32, 53, 54, 57, 60,
70, 71, 72, 75, 80, 86,
87, 91, 93, 94, 95, 98,
99, 100, 104, 109, 110,
114, 117, 119, 120, 130,
141, 142, 147, 151, 165,
168, 169, 171, 172, 173,
181, 185, 187, 188, 191,
193, 202, 208, 211, 214,
215, 224, 234, 238, 293,
303, 316, 331

Operation CROSSROADS, 99,
168, 189, 241

Operation DIXIE, 20, 54, 56,
66, 78, 100, 101, 102,
103, 108, 117, 118, 135,
242

Operation DOMINIC, 157, 187,
347

Operation GRAPPLE, 292

Operation GREENHOUSE, 85

Operation HARDTACK, 2, 134,
135, 153, 157, 185, 187,
192, 193, 212, 234, 235,
238, 239, 240, 241, 243,
246, 249, 255, 256, 257,
258, 259, 261, 262, 266,
267, 268, 269, 270, 271,
278, 279, 281, 284, 287,
288, 289, 290, 291, 292,
295, 296, 298, 299, 300,
301, 302, 303, 304, 305,
310, 311, 317, 318, 320,
325, 328, 329, 330, 331,
335, 338, 339, 340, 341,
342, 343, 344, 346, 347,
349, 350, 355, 356

Operation IVY, 29, 31, 32,
42, 86, 94, 103, 120, 133,
202, 208, 214, 218, 238

Operation KNOTHOLE, 72, 86,
94

Operation PILGRIM, 2, 135,
167, 238, 241, 243, 249,
250, 254, 255, 256, 259,

261, 263, 264, 265, 268,
271, 281

Operation PLUMBBOB, 268,
269, 281, 282, 284, 288,
297, 310, 317, 324, 329,
332, 335, 338, 339, 343,
354

Operation REDWING, 2, 20,
54, 60, 72, 85, 86, 87,
93, 94, 95, 96, 97, 98,
99, 100, 101, 102, 103,
104, 105, 107, 108, 109,
110, 113, 115, 116, 117,
118, 120, 121, 123, 124,
126, 127, 128, 129, 130,
132, 133, 134, 135, 137,
138, 139, 140, 141, 142,
144, 145, 146, 147, 149,
151, 152, 153, 155, 157,
158, 159, 160, 161, 162,
164, 165, 167, 168, 169,
170, 171, 178, 179, 180,
181, 182, 183, 184, 185,
186, 187, 191, 192, 193,
194, 195, 196, 197, 201,
202, 204, 207, 208, 211,
212, 214, 215, 216, 218,
219, 221, 222, 223, 224,
225, 226, 227, 228, 229,
231, 232, 233, 234, 235,
236, 238, 239, 240, 243,
244, 245, 247, 250, 251,
255, 259, 271, 281, 292,
296, 301, 302, 307, 308,
313, 328, 340

Operation SANDSTONE, 185

Operation TEAPOT, 2, 9, 20,
22, 23, 25, 26, 27, 52,
54, 55, 56, 57, 63, 64,
66, 68, 69, 70, 72, 73,
74, 75, 78, 93, 94, 95,
99, 100, 101, 102, 109,
110, 111, 112, 113, 114,
115, 134, 135, 168, 170,
255

Operation TUMBLER, 257

Operation UPSHOT-KNOTHOLE,
72, 86, 94

Operation WIGWAM, 95, 241,
278

Oppenheimer, Julius Robert,
8, 28, 29, 30, 33, 34, 37,
39, 46, 48, 49, 64

Osage shot, Operation
REDWING, 116, 126, 127,
138, 139, 142, 143, 146,
147, 150, 156, 165, 180,
182, 185, 195, 207, 208,
218, 255

—P—

Pacific Proving Grounds
(PPG), 96, 102, 108, 109,
134, 135, 142, 143, 144,
147, 153, 155, 157, 168,
169, 170, 183, 184, 195,
201, 202, 207, 211, 213,
228, 229, 253, 293, 294
Parachutes, 18, 139, 143,
147, 210
Pawnee shot, Operation
REDWING, 140, 143, 151,
156, 157, 183, 186, 191,
219
PERSHING, 1A, II surface-to-
surface missile, 133
Pit, 20, 55, 66, 93, 94,
112, 134, 135, 143, 182,
216, 217, 283, 310
Plutonium, 23, 75, 125, 128,
159, 164, 180, 193, 201,
204, 224, 229, 260, 282,
283, 310, 317, 343, 355
POLARIS SLBM, 226, 289, 298,
336
Primary stage, 93, 113
PROGRESSIVE case, 1979, 51,
192
Project 56, 128, 153
Pueblo shot, Operation
REDWING, 116, 126, 127,
142
Pusher, 54

—Q—

Quick and Dirty, 94, 95

—R—

Radiation channel, 55, 69,
120
Radiation implosion, 212
REDSTONE surface-to-surface
missile, 69, 239, 246,
270, 284, 285, 295, 296,
312, 340

REGULUS I/II cruise missile,
214, 215
Ruth shot, Operation UPSHOT-
KNOTHOLE, 190

—S—

Sandia Base, Albuquerque,
New Mexico, 96, 195, 207,
268
Sandia
Corporation/Laboratory,
28, 76, 77, 79, 80, 86,
180, 182, 189, 225, 238,
240, 241, 242, 249
Sandia/Special Weapons
Development Board, 164,
207, 225
Savannah River plant, AEC,
8, 80, 282, 283
Sealed pit, 20, 66, 93, 134,
135
Secondary stage, 16, 54, 55,
62, 64, 69, 112, 119, 120,
122, 128, 134, 145, 146,
152, 162, 192, 201, 203,
212, 230, 317, 324, 334,
339
Oralloy (Oy, Oak Ridge
Alloy), 30, 32, 151, 164,
201, 260, 336, 339, 354
Seminole shot, Operation
REDWING, 142, 143, 146,
150, 155, 156, 195, 201,
202, 208, 209, 218, 223
SERGEANT tactical missile,
246
Shawnee shot, Operation
REDWING, 115, 116, 126,
127, 128
Shrimp, 6, 7, 8, 18, 19, 80,
86, 151, 171, 215, 244,
304
Smoky shot, Operation
PLUMBBOB, 269, 297, 317,
355
SNARK (B-62)
intercontinental cruise
missile, 69
Sparkplug, 67
SPARROW air-to-air missile,
283, 336
ADM (Atomic Demolition
Munitions); see also

Medium Atomic Demolition
 Munition (MADM), 232
 Starbird, Alfred Dodd, 97,
 105, 132, 152, 167, 189,
 190, 191, 193, 194, 197,
 199, 201, 203, 204, 206,
 207, 208, 217, 223, 226,
 231, 240, 242, 245, 254,
 271, 275, 280, 281, 286,
 287, 288, 291, 292, 296,
 297, 299, 303, 304, 313,
 314, 317, 326, 327, 328,
 329, 330, 331, 332, 338,
 339, 340, 341, 355, 356,
 358
 Starling, 192
 Strategic Air Command, USAF,
 32, 33, 80, 85, 132, 351
 Strategic Air Command, USAF
 (SAC), 32, 33, 62, 80,
 119, 130, 131, 132, 171,
 188, 285, 286, 325
 Strauss, Lewis L., 7, 8, 15,
 17, 20, 22, 28, 34, 36,
 39, 40, 43, 44, 46, 47,
 48, 49, 50, 51, 54, 58,
 60, 64, 65, 68, 72, 75,
 88, 90, 91, 92, 96, 102,
 108, 109, 110, 111, 121,
 136, 139, 141, 145, 146,
 147, 152, 153, 154, 158,
 165, 168, 193, 196, 199,
 200, 220, 223, 229, 231,
 242, 254, 265, 266, 287,
 291, 292, 301, 303, 305,
 306, 307, 308, 312, 314,
 315, 317, 320, 321, 326,
 327, 328, 330, 331, 332,
 339, 340, 342, 343, 344,
 345, 354
 Submarines, 58, 79, 83, 84,
 133, 218, 226, 241, 254,
 264, 265, 267, 283, 287,
 351
 SWDB (Sandia/Special Weapons
 Development Board, 207,
 225

-T-

TABLEG Committee, 304
 Tactical Air Command, U.S.
 Air Force (TAC), 119

TALOS/TALOS-W surface-to-air
 missile, 20, 228, 260,
 265, 343, 355
 Tamper, 16, 56, 64, 122,
 152, 201, 206, 339
 Task Unit, 100, 147, 180
 Teller, Dr. Edward, 8, 10,
 22, 28, 29, 30, 31, 34,
 35, 36, 37, 38, 39, 41,
 45, 47, 49, 60, 61, 72,
 86, 88, 94, 122, 139, 162,
 196, 197, 206, 220, 225,
 303, 317, 318, 319, 320,
 321, 322, 329
 TERRIER surface-to-air
 missile, 226, 232
 Tertiary stage, 122, 152,
 192, 193, 216
 Tewa shot, Operation
 REDWING, 150, 151, 152,
 156, 164, 184, 193, 195,
 216, 217, 219, 224
 Theoretical Physics (T)
 Division, LASL, 11, 15
 Thorium, 204
 Training shapes, 207
 Transit time, 190, 216
 Travis AFB, California, 185
 Tritium, 8, 54, 55, 71, 93,
 112, 137, 164, 180, 228,
 229, 256, 282, 283, 310,
 333
 Truman, Harry S, 10, 29, 41
 TU (Task Units), 100, 180,
 181, 182, 183, 184, 189,
 212
 Tuballoy (Tube Alloy, Tu),
 12, 357
 Tungsten, 67, 201

-U-

Ulam, Dr. Stanislaw, 31, 35,
 122
 Union shot, Operation
 CASTLE, 46, 62, 99, 158,
 211, 215, 236, 276, 319,
 339, 354
 United Nations, 90, 123,
 124, 152, 199, 301, 304,
 318, 323
 University of California
 Radiation Laboratory
 (UCRL), Livermore, CA, 2,

8, 9, 21, 22, 23, 24, 25,
 26, 27, 29, 31, 33, 35,
 42, 56, 57, 60, 61, 63,
 66, 72, 73, 86, 88, 90,
 94, 95, 97, 98, 100, 101,
 102, 103, 104, 107, 108,
 110, 111, 112, 115, 116,
 118, 120, 121, 122, 126,
 127, 128, 129, 132, 134,
 139, 140, 141, 142, 143,
 144, 145, 146, 149, 150,
 151, 152, 153, 155, 156,
 159, 162, 164, 167, 168,
 169, 171, 180, 182, 183,
 184, 186, 191, 192, 193,
 195, 196, 207, 208, 212,
 214, 216, 222, 226, 228,
 232, 233, 234, 238, 239,
 241, 249, 250, 252, 253,
 255, 256, 261, 263, 268,
 269, 271, 275, 278, 279,
 280, 281, 282, 283, 284,
 287, 288, 289, 292, 293,
 295, 298, 300, 301, 302,
 304, 305, 317, 321, 324,
 331, 332, 335, 338, 340,
 341, 354, 355, 356
 Uranium, 9, 15, 16, 17, 18,
 54, 55, 61, 63, 65, 66,
 67, 87, 112, 125, 129,
 159, 164, 180, 193, 203,
 204, 206, 224, 229, 282,
 283, 316
 Uranium hydride, 9, 61, 129
 US CATAMOUNT (LSD-17), 186,
 191, 203, 211, 216, 217
 USS CATAMOUNT (LSD-17), 186,
 203

USS CURTISS (AV-4), 116,
 181, 182, 183, 185, 186,
 187, 192, 207, 215, 320
 USS ESTES (AGC-12), 185
 USS MOUNT MCKINLEY (AGC-7),
 189, 190
 USSR (Union of Soviet
 Socialist Republics, 7,
 52, 81, 83, 84, 85, 105,
 106, 272, 319

-V-

von Neumann, Dr. John, 9,
 61, 121, 123, 125

-W-

W Division, LASL, 182
 Wasp shot/device, Operation
 TEAPOT, 56, 74, 112, 114
 Weaponization, 5, 27, 97,
 113, 167, 335

-Y-

Yuma shot, Operation
 REDWING, 115, 126, 127,
 133, 138, 142, 145, 150,
 155, 183, 195, 207, 208,
 209, 218, 219

-Z-

Zombie, 6, 7, 99
 Zuni shot, Operation
 REDWING, 116, 126, 128,
 133, 138, 142, 143, 149,
 152, 155, 158, 182, 184,
 187, 189, 191, 192, 193,
 195, 196, 203, 208, 209,
 216, 218, 224

[Back to Document](#)



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Back to Document



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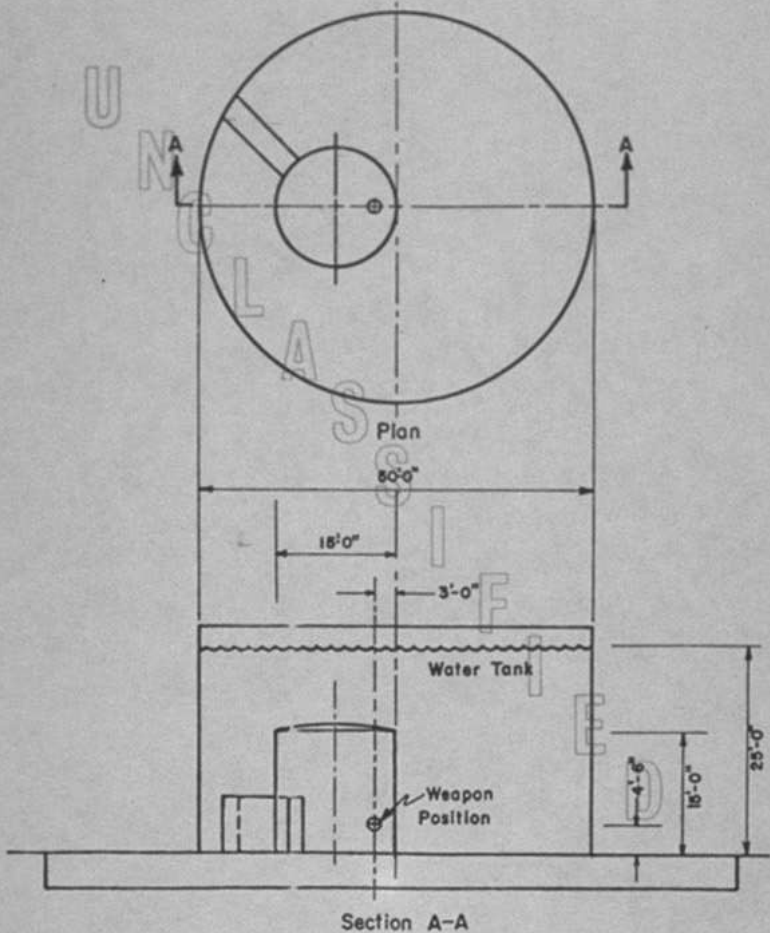


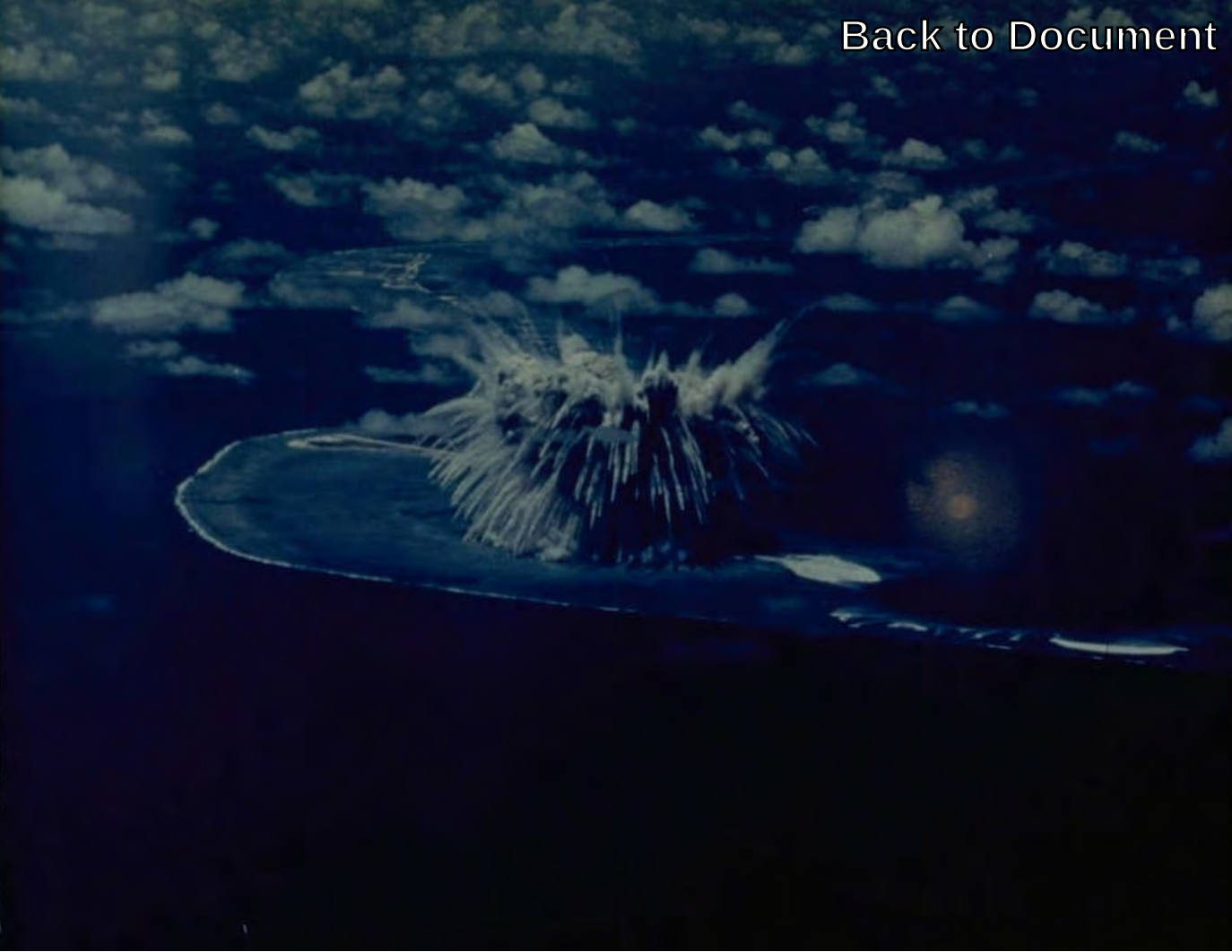
Figure 3.14 Seminole burst position.

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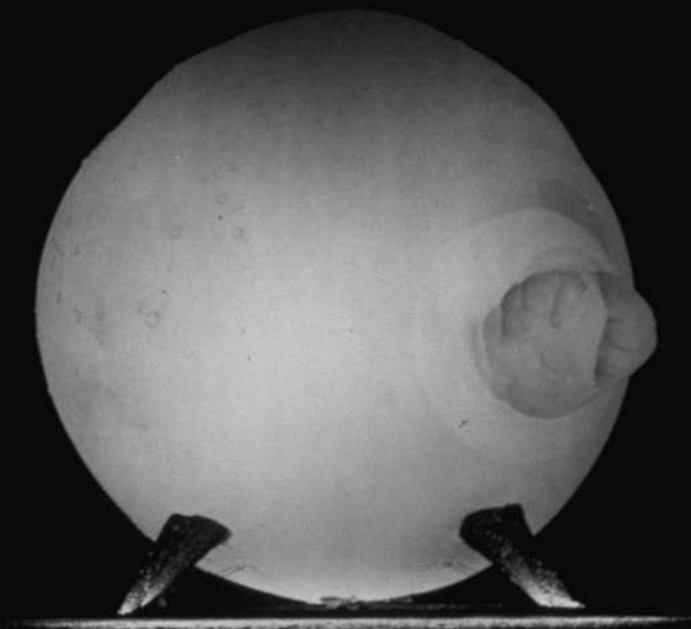


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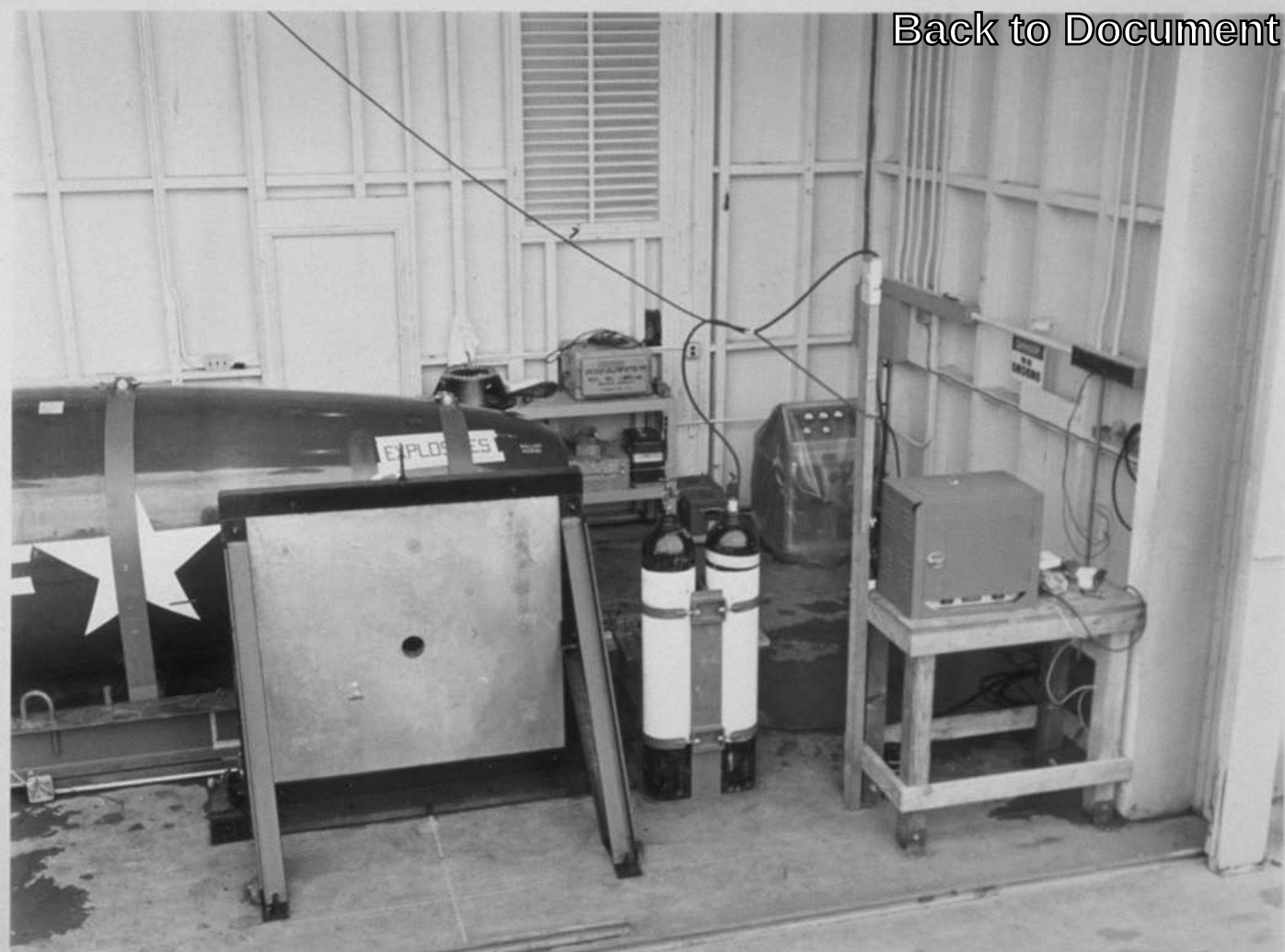


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